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वार्षिक प्रतिवेदन
ANNUAL REPORT 2022

"Recipient of Sardar Patel Outstanding ICAR Institution Award 2021"
(Under large Institute Category)

भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
राजेन्द्रनगर, हैदराबाद - 500 030, तेलंगाना, भारत
ICAR-National Academy of Agricultural Research Management
Rajendranagar, Hyderabad-500 030, Telangana, India

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ICAR

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Contents

ABBREVIATIONS

AAU	Assam Agricultural University
ABF	Agri-Biotech Foundation
ABI	Agri-Business Incubator
ABS	Access and Benefit Sharing
ACP	Awareness Creation Programme
ADG	Additional Director General
AERA	Agricultural Economics Research Association
AESSRA	Agricultural Economics and Social Science Research Association
AGTRI	Advances in Genomics Tools for Rice Improvement
AI	Artificial Intelligence
AICRP	All-India Coordinated Research Project
AICTE	All India Council for Technical Education
a-IDEA	Association for Innovation Development of Entrepreneurship in Agriculture
AIEEA	All India Entrance Examination for Admission
AIMA	All India Management Association
ALTENA	Asian Long Term Experimental Network for Agriculture
AMARA	Association for Management of Agricultural Research and Agripreneurship
ANGRAU	Acharya N. G. Ranga Agricultural University
ANRCM	Academy of Natural Resource Conservation and Management
APAR	Annual Performance Appraisal System
APMC	Agricultural Produce Market Committee
ARGM	Annual Rice Group Meeting
ARS	Agricultural Research Service
ASCI	Administrative Staff College of India
ASC	Agricultural Science Congress
ASEAN	Association of South East Asian Nations
ASRB	Agricultural Scientists Recruitment Board
ASTI	Agricultural Science and Technology Indicators
ATARI	Agricultural Technology Application Research Institute
AUs	Agricultural Universities
B2B	Business to Business
B2C	Business to Consumer
B2F	Business to Farmer
B2G	Business to Government
BARC	Bangladesh Agricultural Research Council
BASIX	Bhartiya Samruddhi Investments and Consulting Services Ltd.
BASU	Bihar Animal Sciences University
BAU	Bihar Agricultural University
BCIL	Biotechnology Consortium India Limited
BIOFIN	Biodiversity Finance Initiative
BIRAC	Biotechnology Industry Research Assistance Council

BMC	Business Model Canvass
CAAST	Centre of Advanced Agricultural Science and Technology
CAGR	Compounded Annual Growth Rate
CAI	Centre for Agri-Innovation
CAT	Common Admission Test
CAU	Central Agricultural University
CBI	Central Bureau of Investigation
CCS NIAM	Ch. Charan Singh National Institute of Agricultural Marketing
CCSHAU	Chaudhary Charan Singh Haryana Agricultural University
CDC	Career Development Centre
CDVL	Centre for Distance and Virtual Learning
CGIAR	Consultative Group for International Agricultural Research
CHRD	Center for Human Resource Development
CIFA	Central Institute of Freshwater Aquaculture
CIFE	Central Institute of Fisheries Education
CIFT	Central Institute of Fisheries Technology
CIIE	Centre for Innovation Incubation and Entrepreneurship
CIL	Coromandel International Limited
CIMMYT	International Maize and Wheat Improvement Center
CIRAD	The French Agricultural Research Centre for International Development
CIWA	Central Institute for Women in Agriculture
CJSC	Central Joint Staff Council
CMAT	Common Management Admission Test
CMFRI	Central Marine Fisheries Research Institute
COLLAge	Centre for Open and Lifelong Learning in Agriculture
CPI	Consumer Price Index
CRD	Completely Randomized Design
CRIDA	Central Research Institute for Dryland Agriculture
CRIJAF	Central Research Institute for Jute and Allied Fibres
CRM	Customer Relationship Management
CSAUA&T	Chandra Shekhar Azad University of Agriculture & Technology
CSIR	Council of Scientific & Industrial Research
CTCRI	Central Tuber Crops Research Institute
CTRI	Central Tobacco Research Institute
DAATTCs	District Agricultural Advisory and Transfer of Technology Centre
DAC&FW	Department of Agricultural Cooperation and Farmers' Welfare
DFID	Department for International Development
DARS	Dryland Agriculture Research Station
DBT	Department of Biotechnology
DC	Data Centre
DCB	Development Credit Bank
DDG	Deputy Director General
DGR	Directorate of Groundnut Research
DIEO	Division for International and Educational Outreach
DKMA	Directorate of Knowledge Management in Agriculture

DRC	Disaster Recovery Centre
DST	Department of Science and Technology
DT	Design Thinking
DU	Delhi University
DWMA	District Water Management Agency
DWRP	Developing Winning Research Proposal
EDP	Executive Development Programme
EEI	Extension Education Institute
e-NAM	electronic National Agriculture Market
ENCORE	Enhancing Coastal And Ocean Resource Efficiency
EoDR	Ease of Doing Research
EP	Emeritus Professor
ERP	Enterprise Resource Planning
ES	Emeritus Scientist
ET	Education Technology
FaaS	Farming as a Service
FADC	Food and Agriculture Division Council
FAI	Fertilizer Association of India
FAO	Food & Agriculture Organization
FCRI	Fisheries College and Research Institute
FDA	Food and Drug Administration
FDC	Faculty Development Centre
FDP	Faculty Development Programme
FET	Field Experience Training
FICCI	Federation of Indian Chamber of Commerce and Industry
FLIs	Farmer Led Innovations
FOCARS	Foundation Course for Agricultural Research Service
FOCFAU	Foundation Course for Faculty of Agricultural Universities
FPC	Farmer Producer Company
FPOs	Farmer Producer Organizations
FSRC	Financial Services Regulatory Commission
FSSAI	Food Safety and Standards Authority of India
FTAPCCI	Federation of Telangana and Andhra Pradesh Chambers of Commerce and Industry
FTAs	Farm Tele Advisors
FTF-ITT	Feed the Future - India Triangular Training
FYM	Farm Yard Manure
GADVASU	Guru Angad Dev Veterinary and Animal Sciences University
GBPUA&T	Govind Ballabh Pant University of Agriculture and Technology
GDD	Growing Degree Days
GEF	Global Environment Facility
GHGs	Green House Gases
GHMC	Greater Hyderabad Municipal Corporation
GIS	Geographic Information System
GKVK	Gandhi Krishi Vigyana Kendra
GoI	Government of India
GPS	Global Positioning System

GST	Goods and Services Tax
HEIs	Higher Education Institutions
HICC	Hyderabad International Convention Centre
HRD	Human Resource Development
HWC	Human-Wildlife Conflict
IARI	Indian Agricultural Research Institute
IAS	Indian Administrative Service
IASRI	Indian Agricultural Statistics Research Institute
ICAR	Indian Council of Agricultural Research
ICFRE	Indian Council of Forest Research and Education
ICM	Institute of Cooperative Management
ICMR	Indian Council of Medical Research
ICRISAT	International Crops Research Institute for Semi-Arid Tropics
ICT	Information and Communication Technology
ICTA-A	Climate Action Transparency Project on Adaptation
ICZM	Integrated Coastal Zone Management
IDDC	International Dryland Development Commission
IDP	Institutional Development Plan
IFPRI	International Food Policy Research Institute
IFS	Indian Forest Service
IGFRI	Indian Grassland and Fodder Research Institute
IGH	International Guest House
IGKV	Indira Gandhi Krishi Vishwavidyalaya
IGNFA	Indira Gandhi National Forest Academy
IIIT	Indian Institute of Information Technology
IIM	Indian Institute of Management
IIMR	Indian Institute of Millets Research
IIOPR	Indian Institute of Oil Palm Research
IIOR	Indian Institute of Oilseeds Research
IIP	Indian Institute of Packaging
IIPM	Indian Institute of Plantation Management
IIRR	Indian Institute of Rice Research
IISER	Indian Institute of Science Education and Research
IISS	International Institute for Strategic Studies
IIT	Indian Institute of Technology
IIITDM	Indian Institute of Information Technology Design and Manufacturing
IIVR	Indian Institute of Vegetable Research
IJSC	Institute Joint Staff Council
ILS	Institute of Life Sciences
IMC	Institute Management Committee
IoT	Internet of Things
IPE	Institute of Public Enterprise
IPR	Intellectual Property Rights
IPTM	Intellectual Property & Technology Management
IRC	Institute Research Council
IRGS	Institute of Research and Graduate Studies

IRM	Insect Resistance Management
ISAP	Indian Society of Agribusiness Professionals
ISAPM	Indian Society of Animal Production and Management
ISB	Indian School of Business
ISO	International Organization for Standardization
IT	Information Technology
ITK	Indigenous Technical Knowledge
ITMC	Institute Technology Management Committee
ITMU	Institute Technology Management Unit
IWELA	Indian Women Excellence and Leadership Award
JAC	Joint Advisory Council
JAU	Junagadh Agricultural University
JNTU	Jawaharlal Nehru Technological University
JNU	Jawaharlal Nehru University
KAU	Kerala Agricultural University
KCCs	Kisan Call Centers
KJWA	Koronivia Joint Work on Agriculture
KPIs	Key Performance Indicators
KVAFSU	Karnataka Veterinary, Animal and Fisheries Sciences University
KVKs	Krishi Vigyan Kendras
LMS	Learning Management System
LSD	Latin Square Design
MABIF	Madurai Agribusiness Incubation Forum
MAFSU	Maharashtra Animal and Fishery Sciences University
MANAGE	National Institute of Agricultural Extension Management
MDP	Management Development Programme
MET	Multi-Environment Trials
MIDH	Mission for Integrated Development of Horticulture
MLA	Member of Legislative Assembly
MoEFCC	Ministry of Environment, Forest and Climate Change
MOOC	Massive Open Online Courses
MORE	Marketing Officers' Resurgence & Excellence
MoU	Memorandum of Understanding
MPUA&T	Maharana Pratap University of Agriculture and Technology
MVIRDC	M. Visvesvaraya Industrial Research and Development Centre
NAARM	National Academy of Agricultural Research Management
NAAS	National Academy of Agricultural Sciences
NABARD	National Bank for Agricultural and Rural Development
NADMP	National Agriculture Disaster Management Plan
NAEP	National Assessment of Educational Progress
NAHEP	National Agricultural Higher Education Project
NAIF	National Agricultural Innovation Fund
NAPCC	National Action Plan on Climate Change
NARC	Nepal Agricultural Research Council
NARES	National Agricultural Research and Education System
NARS	National Agricultural Research System

NASF	National Agricultural Science Fund
NASI	National Academy of Sciences, India
NBA	National Biodiversity Authority
NBA	National Board of Accreditation
NBFGR	National Bureau of Fish Genetic Resources
NBPGR	National Bureau of Plant Genetic Resources
NBSSLUP	National Bureau of Soil Survey and Land Use Planning
NCCT	National Council for Cooperative Training
NCSCM	National Centre for Sustainable Coastal Management
NCDEX	National Commodities Derivatives Exchange
NDDI	Normalized Difference Drought Index
NDRI	National Dairy Research Institute
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
NERC	North Eastern Regional Centre
NET	National Eligibility Test
NF	National Fellow
NFDB	National Fisheries Development Board
NGOS	Non-Government Organizations
NIAM	National Institute of Agricultural Marketing
NIAP	National Institute of Agricultural Economics and Policy Research
NIDHI PRAYAS	National Initiative for Developing and Harnessing Innovations - Promoting and Accelerating Young and Aspiring technology entrepreneurs
NIF	National Innovation Foundation
NIOT	National Institute of Ocean Technology
NIPHM	National Institute of Plant Health Management
NIRD &PR	National Institute of Rural Development and Panchayat Raj
NITI	National Institution for Transforming India
NMSA	National Mission on Sustainable Agriculture
NP	National Professor
NPTEL	National Programme on Technology Enhanced Learning
NRC	National Register of Citizens
NRCB	National Research Centre for Banana
NRCC	National Research Centre on Camel
NRCM	Natural Resource Conservation and Management
NRM	Natural Resource Management
NRRI	National Rice Research Institute
NSTEDB	National Science & Technology Entrepreneurship Development Board
NSTMIS	National Science and Technology Management Information System
NTC	National Training Course
OHQ	Oxford Happiness Questionnaire
OPAC	Online Public Access Catalogue
OPD	Outpatient Department
OUAT	Odisha University of Agriculture and Technology
PARC	Pakistan Agricultural Research Council
PAU	Punjab Agricultural University

PCK	Pedagogical Content Knowledge
PCT	Patent Cooperation Treaty
PDKV	Panjabrao Deshmukh Krishi Vidyapeeth
PDP	Project Directorate on Poultry
DETM	Diploma in Education Technology Management
PGDMA	Post Graduate Diploma in Management-Agriculture
PGDM-ABM	Post Graduate Diploma in Agribusiness Management
DTMA	Diploma in Technology Management in Agriculture
PJTSAU	Professor Jayashankar Telangana State Agricultural University
PME	Priority Setting, Monitoring and Evaluation
PPV&FRA	Protection of Plant Varieties & Farmers' Rights Authority
PRA	Participatory Rural Appraisal
PRT	Primary Response Teams
PVNRTVU	PV Narasimha Rao Telangana Veterinary University
PWB	Psychological Well-Being
QRT	Quinquennial Review Team
RAC	Research Advisory Committee
RAG	Regional Advisory Group
RAWE	Rural Agricultural Work Experience
ReMS	Rashtriya e-Market System
RKVY	Rashtriya Krishi Vikas Yojana
RMP	Risk Management Plan
RRT	Rapid Response Teams
RSM	Response Surface Methodology
RSMs	Retail Sales Managers
RTE	Ready to Eat
RVSKVV	Rajamata Vijayaraje Scindia Krishi Vishwa Vidyalaya
SAARC	South Asian Association for Regional Cooperation
SaaS	Software as a Service
SAC	SAARC Agricultural Centre
SAMETI	State Agricultural Management & Extension Training Institute
SAS	Statistical Analysis System
SAUs	State Agricultural Universities
SBSTA	Subsidiary Body for Scientific and Technological Advice
SCSP	Schedule Caste Sub Plan
SDGs	Sustainable Development Goals
SFAC	Small Farmers' Agribusiness Consortium
SHUATS	Sam Higginbottom University of Agriculture, Technology and Sciences
SKLTSHU	Sri Konda Laxman Telangana State Horticultural University
SKNAU	Sri Karan Narendra Agriculture University
SKUAST	Sher-e-Kashmir University of Agricultural Sciences and Technology
SLAs	Students' Learning Approaches
SLTP	State Level Technical Programme
SMD	Subject Matter Division
SNRM	Strengthening Natural Resource Management
SOC	Senior Officers Committee

SSS	Skilled Support Staff
STEM	Society for Technology Management
SVVU	Sri Venkateswara Veterinary University
SWLS	Satisfaction with Life Scale
SWOT	Strength Weakness Opportunities Threats
TAAS	Trust for Advancement of Agricultural Sciences
TAFE	Tractors and Farm Equipment
TANUVAS	Tamil Nadu Veterinary and Animal Sciences University
TARCMES	Technological Approaches for Resource Conservation and Management for Environmental Sustainability
TBI	Technology Business Incubator
TEFR	Techno-Economic Feasibility Report
TEL	Technology Enhanced Learning
TELAgE	Technology Enhanced Learning in Agricultural Education
TERI	The Energy and Resources Institute
TISS	Tata Institute of Social Sciences
TLSOE	Teaching Learning Strategies of Online Education
TNAU	Tamil Nadu Agricultural University
TNAUUIC	Tamil Nadu Agricultural University University Innovation Cluster
TNFU	Tamil Nadu Fisheries University
TNJFU	Tamil Nadu Dr. J. Jayalalithaa Fisheries University
ToT	Training of Trainers
TPACK	Technology Pedagogy and Content Knowledge
TQIMC	Training Quality Improvement Measures Committee
TRIPS	Trade-Related aspects of Intellectual Property Rights
UAS	University of Agricultural Sciences
UASR	University of Agricultural Sciences, Raichur
UBKV	Uttar Banga Krishi Viswavidyalaya
UK	United Kingdom
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UoH	University of Hyderabad
USA	United States of America
USAID	U.S. Agency for International Development
UTs	Union Territories
VCD	Video Compact Disc
VNMKV	Vasantrao Naik Marathwada Agricultural University
WDP	Wheat Development Programme
WSSV	White Spot Syndrome Virus
WTO	World Trade Organization
ZBNF	Zero Budget Natural Farming
ZTMC	Zonal Agro-Technology Management Centre
ZTMU	Zonal Technology Management Unit

FROM THE DIRECTOR'S DESK

Dr Ch. Srinivasa Rao
Director

It is my privilege to write this prologue for the ICAR-NAARM annual report-2022. NAARM is the nation's premier institute of excellence to build capacity in agricultural research, education and extension education systems, and provide policy advocacy for the National Agricultural Research and Education System (NARES). In the year 2022, ICAR-NAARM successfully conducted 83 programmes reaching 5,885 participants through online and physical modes. These include four leadership development programmes, 45 Need-Based Programmes, 10 workshops/conferences/ webinars, 20 off-campus programmes, two FOCFAU programmes, 10 training programmes for farmers, one winter school and one MOOC programme. Apart from this, ICAR-NAARM also organized 12 Entrepreneurship Development programmes for 1794 students of five agricultural universities towards building a start-up ecosystem. The Academy organized an Induction Course on Project Management and Research Methodology in virtual mode for 26 newly recruited Scientists of ICFRE, Dehradun in February 2022. The Academy also organized a Capacity Building Programme for 42 CJSC Members of ICAR headquarters and Institutes during November 15-19, 2022. A record number of about 3800 students from 52 agricultural/ horticultural colleges/institutions visited the Academy in the year 2022. Despite the concerns, NAARM has been successful in converting the challenges into opportunities by conducting online programmes, and made a tremendous contribution to the NARES with the mantra of 'Reaching the Unreached'.

Agribusiness Management is another important academic activity of NAARM. It gives me immense pleasure to share that all 54 students of PGDMABM 2020-22 (12th batch) were placed successfully in 17 reputed agribusiness companies. The average package of this batch was Rs. 8.32 lakh per annum and the maximum was Rs. 12.56 per annum secured by a girl student. The admission process for the 14th batch of PGDM-ABM (2022-24) was conducted in online mode with the joining of 57 students. New batches of Diploma in Educational Technology and Management (DETM) with 25 candidates and Diploma in Technology Management in Agriculture (DTMA) with 19 candidates were started in 2022. The Academy organized the 4th Graduation Ceremony for the PGDM-ABM programme on May 14, 2022, in a befitting manner. The then Hon'ble Vice President of India, Shri. M. Venkaiah Naidu, as Chief Guest of the Ceremony, awarded degrees to 144 students from four batches. A Three-Member Expert Team from the National Board of Accreditation (NBA) for granting accreditation status to our PGDM-ABM programme, visited the Academy during Dec 16-18, 2022, and were highly satisfied with our work.

The a-IDEA, Technology Business Incubator of ICAR-NAARM launched the AGRI UDAAN 5.0, a much sought-after Food and Agribusiness Accelerator Programme on Aug 6, 2022 with the support of NABARD. Mr Jayesh Ranjan, IAS, Principal Secretary, Department of Industries and Commerce, Government of Telangana was the Chief Guest on the occasion, wherein roadshows were conducted in four locations to promote and create awareness about AGRI UDAAN. Nearly 200 participants attended the roadshows.

ICAR-NAARM, with its renewed role as a Think Tank Policy Research Institute, is continuously undertaking research policy studies on emerging issues and challenges in agriculture. During the year, it has published five peer-reviewed policy/status/strategy papers on diverse aspects like kisan call centres, attracting the best talent to agricultural education, bioinformatics, FPOs and organic farming and a few more are in progress. The academy has published 61 Research Papers, which appeared in national as well as international journals, out of which 17 research papers have a NAAS rating of more than six. I congratulate each one of the NAARM faculty for this accomplishment. The Academy also published Five Books, Nine Book Chapters, 12 technical reports/bulletins, Nine popular articles and 24 conference papers. This year, we have obtained four copyrights and applications for a few more copyrights and trademarks have been filed. We have conducted two IRC Meetings and nine research projects were completed including seven in-house projects and one extramural. Currently, 27 research projects are in operation at the Academy, out of which 18 are in-house and nine are extramural projects funded by GIZ, ICAR, NAHEP, NASF, DBT, DST, NABARD, *etc*. Institute Management Committee (IMC) meeting was also conducted during this year. The Academy has organized an ICAR South Zone sports meet, in which over 750 sports persons participated from 25 ICAR institutes.

For the first time in its journey of 46 years, the Academy bagged the prestigious Sardar Patel Outstanding ICAR Institution Award 2021 for its overall performance. Many of our faculty have been recognized nationally and internationally for their outstanding research contributions. I am confident that the Team NAARM will work continuously towards improving the capacity building and competency enhancement of the human resources of NARES. This will improve the country's farmers' quality of life as well as that of all other stakeholders. I would like to extend my heartiest congratulations to all the scientists and staff members who have contributed immensely to the growth of the Academy and Farming fraternity.

Date: 28-2-2023
Place: Hyderabad

(Ch Srinivasa Rao)
Director

EXECUTIVE SUMMARY

The National Academy of Agricultural Research Management (NAARM) is the ICAR's nodal institution for capacity building, to enhance the competency of professionals at all levels of India's National Agricultural Research and Education Systems (NARES), and to act as a "Think Tank." In 2022, ICAR- NAARM has successfully conducted 83 programmes reaching 5,885 participants through various online and physical modes. These include four leadership development programmes, 45 Need-Based Programmes, 10 workshops/conferences/webinars, 20 off-campus programmes, two FOCFAU programmes, 10 training programmes for farmers, one winter school and one MOOC programme. Further, 12 Entrepreneurship Development programmes were organized for 1794 students of five agricultural universities towards building start-up ecosystem. The Academy organized an Induction Course on Project Management and Research Methodology in virtual mode for 26 newly recruited Scientists of ICFRE, Dehradun in February 2022. The Academy also organized a Capacity Building Programme for 42 CJSC Members of ICAR headquarters and Institutes during November 15-19, 2022. A record number of about 3800 students from about 52 agricultural/horticultural colleges/institutions visited the Academy in the year 2022. The Academy has made substantial contributions to NARES and its stakeholders such as Agricultural Universities, Krishi Vigyan Kendras (KVKs), Farmers' Producers Organizations (FPOs), the Agri-startup ecosystem, and others. The Academy also organized a tailor-made foundation programme for the 81 Assistant Professors of P.V. Narasimha Rao Telangana State Veterinary University, Hyderabad, Orissa University of Agriculture & Technology and various other agriculture universities. The Academy conducted 3 capability building programmes for 68 newly recruited young agribusiness professionals. The Academy established its recognition among industry, which is reflected by successful accomplishments of capacity building of industry professionals.

The Academy has now evolved into a Think Tank of ICAR by providing advisory roles and research policy advocacy in critical areas of importance in agriculture with changing needs. The Academy has produced five policy documents namely Bioinformatics in Agriculture, Attracting the Best Talent to Agricultural Education in India, Strengthening Kisan Call Centres in India, FPOs in India: Creating Enabling Ecosystem for their Sustainability and Agri-startups in India: Opportunities, Challenges and Way Forward. These policy documents resulted from multiple rounds of brainstorming sessions and stakeholder engagements in the various areas, and major policy prescriptions for further improvement of NARES.

During the year, the Academy organized the 4th Graduation Ceremony for the PGDM-ABM programme on May 14, 2022, in a befitting manner. The then Hon'ble Vice President of India, Shri. M. Venkaiah Naidu, graced the occasion as a Chief Guest of the Ceremony and awarded degrees to 144 students from four batches. The academy admitted 57 students for its 14th batch during the academic year

2022-23 whereas the 12th batch of PGDM-ABM got success in employment through campus placement with top agribusiness industries in India. The average package of this batch was Rs. 8.32 lakh per annum and the maximum was Rs. 12.56 per annum secured by a woman candidate. This year, a three-member expert team from the National Board of Accreditation (NBA) visited the Academy during Dec 16-18, 2022 for granting accreditation status to the PGDM-ABM programme and they appreciated the programme and expressed their satisfaction on the conduct of two-year residential programme. Other academic programmes of the Academy, *viz*., Diploma in Technology Management in Agriculture (DTMA) and Diploma in Education Technology Management (DETM) are one-year distance education programmes offered in online mode. Both programmes are being offered in collaboration with the University of Hyderabad (UoH). In these programmes, a total of 44 students took admission in the year 2022. Academy has organized the ICAR South Zone sports meet, and over 750 sports persons participated from 25 ICAR institutes.

On the research front, the academy has taken up various research projects under the six thematic areas namely Agricultural Business Management for Inclusive Growth, Education Systems Management for Enriched Learning Quality and Environment, Enhancing Capacities for Leadership and Governance, Extension Systems Management in Market-driven Environment, Information and Communication Management for Promoting Innovation and Governance, and Mobilizing Science & Technology for Innovation and Sustainable Development. Currently, 27 research projects are in operation at the Academy, of which 18 are in-house and 9 are extramural projects funded by GIZ, ICAR, NAHEP, NASF, DBT, DST, NABARD, *etc*. The meetings of Institute Research Council (IRC), Research Advisory Committee (RAC) and Institute Management Committee (IMC) were conducted regularly during this year.

The Academy, during the year 2022, conducted a few policy workshops/conferences/symposiums in association with other organizations. The 24th Annual Conference of the Society of Statistics, Computer and Applications (SSCA) on Recent Advances in Statistical Theory and Applications - RASTA 2022, was organized during Feb 23-27, 2022. A Brainstorming Workshop on Self-Sustainability in Edible Oils in India was conducted on April 20, 2022. Another Brainstorming Workshop 'Organic Farming in India' was organized in association with ICAR-IIFSR, Modipuram and ICAR-CRIDA, Hyderabad on June 10, 2022. Further, the Academy conducted a National Workshop on Pathways for Effective Implementation of SC Sub-Plan Scheme in ICAR during Aug 8-19, 2022. The Academy is privileged to host the ICAR programme on Laboratory Assessors' Training Course in collaboration with NABL, New Delhi during December 19-23, 2022, in which 24 scientists participated.

Under the SC-Sub Plan programme, the Academy organized six faculty development programmes for the benefit of about 400 faculty members of six agricultural/fisheries/veterinary universities; seven skill development programmes at different KVKs for the benefit of 312 rural women; five programmes for the benefit of 272 farmers on integrated farming systems, value addition of fruits, soil health management and application, use of bio-inoculants in agriculture and backyard poultry farming; and two Entrepreneurial development programmes for 217 agricultural students.

The Academy celebrated its 47th Foundation Day on September 1, 2022, in hybrid mode. Dr H. Pathak, Secretary, DARE & Director General, ICAR was the Chief Guest on the occasion. On this occasion, Annual awards to different categories of staff were presented followed by felicitation of Innovative farmers, best-performing startups, Farmer Producer Organizations (FPOs), Print and electronic media personnel, who have active engagement with NAARM. The International Guest House I & II have been renamed as Haritha IGH-I and Sagarika IGH-II, respectively and a blood donation camp was also organized during the period.

The academy has published 61 research papers in international and national journals, out of which 17 research papers have a NAAS rating of more than six scores. The Academy also published five books, nine book chapters, 12 technical reports/bulletins, nine popular articles and 24 conference papers. This year, Academy obtained four copyrights and applications for a few more copyrights and trademarks have been filed.

The Centre for Agri-Innovation (CAI), located on the campus, provides hand-holding support to agri-startups through a-IDEA, the technology business incubator of the Academy. During the year, 35 incubatees have been inducted. Besides, 20 incubatees who were inducted during earlier years have graduated in 2022. a-IDEA organized several Start-Up-FPO Immersion programmes and training programmes for ABI units. It has launched Nidhi Prayas inviting innovators working on Ideas/POC/Prototypes in the field of agriculture and allied sectors to apply for a "Nidhi PRAYAS" grant up to INR 10 Lakhs. 10 start-ups are selected under Nodhi prayas and provided with grants of up to 10 lakhs. Among others, the Technology Enhanced Learning in Agricultural Education (TELAgE) is a state-of-the-art Agricultural Education Lab, which was established in the Academy to develop digital courses, offer distance and online programmes, conduct virtual classes and support in-house programmes. With its technologically advanced infrastructure, the facility is now being used to offer MOOCs consultancy support to other organizations in developing and hosting digital content. Other neighbouring institutions like MANAGE, PJTSAU and NIPHM have been using this facility for generating their digital output. The facility is now renamed as Centre for Open and Lifelong Learning in Agricultural Education (COLLAge). The Academy celebrated National and International Days such as National Science Day, World Water Day, International Women's Day, International Yoga Day, ICAR Foundation Day, Constitution Day, Agricultural Education Day and National Farmers' Day. The Academy is creating three new infrastructural facilities of New Faculty Building (G+4), Farmers Training Centre (G+3) and Ladies/Girls Hostel. The academy has also upgraded/renovated a-IDEA, MOOC facilities; and a discussion room.

During the year 2022, the Academy signed MoU with Swarna Bharat Trust to work in the areas of mutual interest. Several dignitaries graced different events of the Academy like Shri M. Venkaiah Naidu, former Vice-President of India; Shri Giriraj Singh, Hon'ble Union Minster for Rural Development and Panchayat Raj;

Dr H. Pathak, Secretary, DARE & DG, ICAR; his predecessor Dr T. Mohapatra; Mrs Alka Arora, Additional Secretary & Financial Advisor, DARE/ ICAR; and her predecessor Shri Sanjiv Kumar, Dr Ashok Dalwai, CEO, NRAA; Shri K.V. Shaji, Deputy MD, NABARD; Shri Kamlesh D. Patel, President, Sri Rama Chandra Mission, and Spiritual Guide, Heartfulness Meditation; Prof Appa Rao Podile, Ex-VC, University of Hyderabad; Shri Jayesh Ranjan, Principal Secretary, Department of Industries and Commerce, Government of Telangana; Prof. G. Padmanabhan, Chairman, RAC; Dr Krishna Ella, CMD of M/s Bharat Biotech, International Athlete Shri.N. Ramireddy.

The academy bagged the prestigious Sardar Patel Outstanding ICAR Institution Award 2021 for its overall performance. Further, many of the faculty have been recognized nationally and internationally for their outstanding research work and many were nominated as Member/ Chairman for various committees of ICAR and other organizations.

ICAR-NAARM: An Overview

1

National Academy of Agricultural Research Management (NAARM), Hyderabad was established by the Indian Council of Agricultural Research (ICAR) in 1976. National Agricultural Research and Education System's principal mandate is to enhance capacity in agricultural research, education, and extension education systems, as well as offer policy advocacy. To carry out these tasks, Academy offers a variety of capacity-building programmes for researchers, academics, extension specialists, scholars, students, and other National Agricultural Research and Education System (NARES) stakeholders. The Academy's goal in NARES is to increase both institutional and individual capacity for innovation. Leadership, governance, and innovation are emerging as prerequisites for the transformation of NARES into a more pluralistic innovation system, given the strategic importance of agricultural research in the country's food security and economic prosperity. Through fostering a scientific work culture, the Academy hopes to strengthen the sense of community among agriculture scientists and faculty across the nation. It acts as a catalyst for the NARES' intended change to handle present and upcoming challenges. The Academy has taken advantage of its ability to effectively manage human resources in agricultural research to advance agribusiness management, technology management, and education technology management. In collaboration with the University of Hyderabad, the Academy offers higher education programmes such as the Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) with AICTE approval and accreditation from the National Board of Accreditation (NBA), the Diploma in Technology Management in Agriculture (DTMA), and the Diploma in Educational Technology and Management (DETM) in distance mode (UoH). The Academy's Technology Business Incubator, a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture), has won national recognition for its innovative approach to incubating Agri Startups. Through capacity building, education, research, consultancy, and policy support, the Academy has been working to strengthen the leadership, governance, and innovation capacities of the National Agricultural Research and Education System (NARES), to emerge as an institution par excellence to facilitate and support a culture of dynamic management in agricultural research,

VISION - A global knowledge institution enabling National Agricultural Research and Education System (NARES) adapt to change through continuous innovation.

MISSION - To enhance leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy and policy support.

education, and frontline extension. The Academy is in a good position to act as a think tank for agriculture, putting out plans to improve the National Agricultural Research and Education System and ensure the long-term viability of Indian agriculture.

1.1 Strategic Framework

The Academy built a six-tier strategic framework in accordance with its Vision, Mission, and Mandate to guide various activities towards the defined goals and targets.

These are:

1. **Agribusiness Management for Inclusive Growth**

- Enhance capacities for food and agribusiness management education and research in NARES.
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses.

2. **Education System Management for Enriched Learning Quality and Environment**

- Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES.
- Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments.

3. **Enhancing Capacities for Leadership and Governance**

- Institutionalize a framework for leadership development at all levels of NARES - early, mid-career, senior professional and manager across all functions (research, education, extension).
- Institutionalize good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation.

4. **Extension Systems Management in a Market-driven Environment**

- Enhance the operational, adaptive and generational capacity of front-line agricultural extension systems to address emerging challenges.
- Enhance capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking.

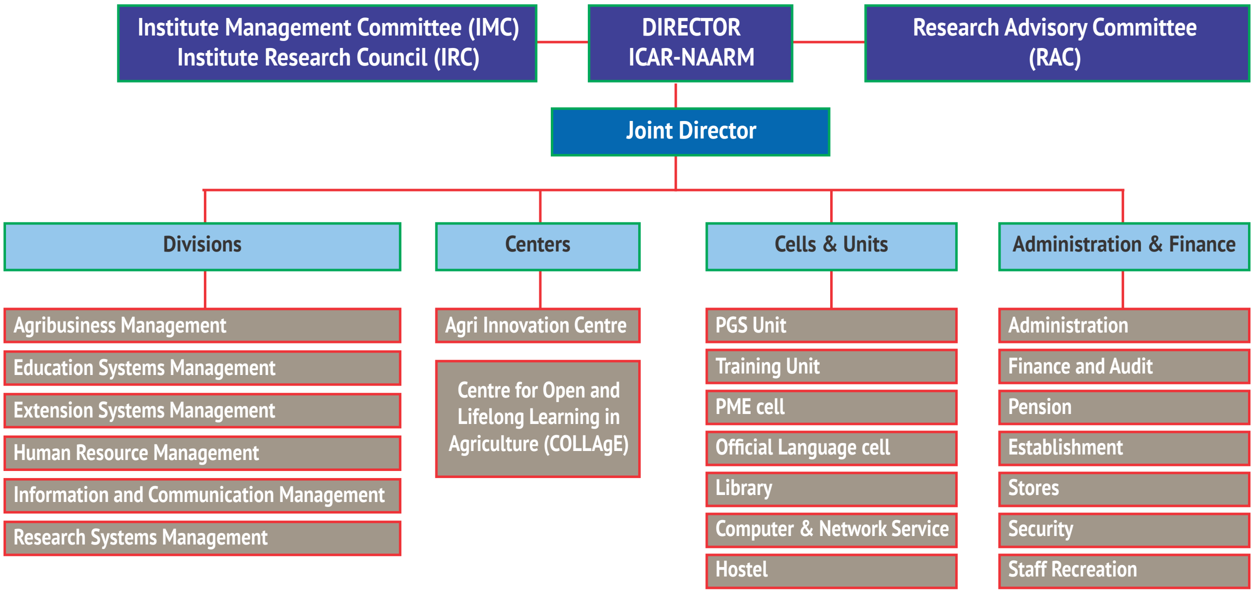
5. **Information and Communication Management for Promoting Innovation and Governance**

- Enhance capacities for integrating ICTs and the Internet of Things (IoTs) in institutional management and governance.
- Enhance capacities for managing data, information and knowledge in agriculture, scientific communication and institutional governance.

6. **Mobilizing Science and Technology for Innovation and Sustainable Development**

- Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change and globalization.
- Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies

1.2 Organization and Management



1.2.1 Research Advisory Committee (RAC)

The Indian Council of Agricultural Research (ICAR) constituted the Research Advisory Committee (RAC) for three years (14-05-2020 to 13-05- 2023).

Prof. G. Padmanaban President NASI and Former Director IISc, Bengaluru	Chairman
Dr B. Venkateswarlu, Former Vice Chancellor, Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani, Maharashtra	Member
Dr Pratibha Jolly, Science and Society Fellow, NASI, Former Principal Miranda House, Delhi University	Member
Prof. V. K. Unni Public Policy Management Indian Institute of Management, Kolkata	Member
Dr R. P. Singh Regional Manager-Operation & Resource, Breeding & Trait Development-Asia Pacific, Bayer BioScience Pvt. Ltd. Madhapur, Hyderabad.	Member
Dr S. L. Goswami Former Director, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad	Member
Dr. Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Member
DDG (Agril. Education), ICAR KAB-II, Pusa, New Delhi	Member
Shri Vallabhaneni Asha Kiran, Vijayawada, Andhra Pradesh	Member
Shri Gone Shyamsunder Rao Seven Hills High School, Mancherial, Telangana	Member
Dr Tavva Srinivas, Principal Scientist ICAR-NAARM, Rajendranagar, Hyderabad	Member Secretary

1.2.2 Institute Management Committee (IMC)

The following officials were nominated as members of the Institute Management Committee of NAARM for a period of 3 years. (24-07-2020 to 23-07-2023).

Dr Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Chairman
Director of Agriculture Department of Agriculture, Government of Telangana	Member
Spl. Commissioner and Director Agricultural Department Department of Agriculture Government of Telangana	Member
Director (Research) Prof. Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad	Member
Shri Gone Shyamsunder Rao Seven Hills High School Mancherial, Telangana	Member
Shri Vallabhaneni Asha Kiran Vijayawada	Member
Dr R. K. Mathur Director, ICAR-IIOPR, Andhra Pradesh	Member
Dr V Tonapi Director, ICAR-IIMR, Hyderabad	Member
Dr. Raghavendra Batta Director, National Institute of animal Nutrition and Physiology, Bangalore	Member
Dr G. Venkateshwarlu Joint Director, NAARM, Hyderabad	Member
Dr Seema Jaggi ADG (HRD), ICAR, KAB-II, New Delhi	Member
Addl. Secy., DARE & FA, ICAR/his nominee	Member
Chief Admn. Officer (Sr.Grade) NAARM, Hyderabad	Member Secretary

1.3 Linkages/Networking/Collaborations

NAARM has a strong strategic and functional network with 113 ICAR Research Institutes and 75 Agricultural Universities (AUs) and plays a key role in developing the capability of these organisations, the agribusiness industry, scientists, and academia. Its network includes numerous institutions at the national and international levels.

1.3.1 National Institutes/ Organizations

1. All ICAR institutes and Agricultural Universities.
2. Various departments of the Government of India like NITI (National Institution for Transforming India) Aayog, Department of Science and Technology (DST), Department of Biotechnology (DBT), Capacity Building Commission, National Science & Technology Entrepreneurship Development Board (NSTEB).
3. The University of Hyderabad, Osmania University, Jawaharlal Nehru Technological University (JNTU), Indian Institute of Information Technology (IIIT), Council of Scientific &Industrial Research (CSIR), Indian Council of Medical Research (ICMR) and various institutions located at Hyderabad.
4. Ministry of Environment, Forest and Climate Change (MoEF & CC), National Board of Accreditation (NBA), Protection of Plant Varieties & Farmers' Rights Authority (PPV&FRA), National Fisheries Development Board (NFDB) and State Biodiversity Boards.
5. Management Institutions located at Hyderabad *viz.* the Administrative Staff College of India (ASCI), Indian School of Business (ISB), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development and Panchayat Raj (NIRD&PR), and Institute of Public Enterprise (IPE); and across the country such as Indian Institutes of Management (IIMs).
6. State Departments of Agriculture of various States.
7. Private Sector Organizations (Agri Input Companies, Service Companies, *etc.*).
8. Non-Government Organizations (NGOs).
9. Other organizations *viz*. National Innovation Foundation (NIF), Federation of Indian Chamber of Commerce and Industry (FICCI), National Bank for Agricultural and Rural Development (NABARD) Centre for Innovation Incubation and Entrepreneurship (CIIE), *etc*.
10. Krishi Vigyan Kendras (KVKs).

1.3.2 International Organizations

1. CGIAR Institutions (CIMMYT, ICRISAT, IFPRI, etc.)
2. World Bank, FAO, UNDP and other United Nations Organizations
3. Department of International Development (DFID, UK).
4. SAARC Agricultural Centre (SAC).
5. NARS of South Asian Countries such as Bangladesh Agricultural Research Council (BARC), Nepal Agricultural Research Council (NARC), Pakistan Agricultural Research Council (PARC), and other NARS Institutions in Afghanistan, Sri Lanka and other ASEAN countries.
6. NARS in other African and South East Asian countries such as Kenya, Mali, Tanzania, Nigeria, Malawi, Liberia and the Philippines.
7. Leading Land Grant and State Universities of USA and other Universities in Europe, Australia and other developing and developed countries.

Thematic Areas: Research, Capacity Building & Education

2

Innovation, governance and knowledge form the fulcrum of the Academy's mission, which is driven by its vision documents (2030 and 2050) as well as those of ICAR. Recognizing the enabling nature of the role of emerging science and technologies across diverse sectors, it voices for their integration into the agricultural value chain. Further, the evolving changes in the agri-education sector shifting focus from 'teaching to learning outcome' signify the requirement to develop newer institutional mechanisms and capacities of the system. The Academy also identifies the crucial need for continuous stakeholder engagement to bring out the effective and efficient performance in the NARES with more transparency and accountability in place. Therefore, the activities of research, capacity building and education have been aligned with the six-tier strategic approach. Through a structured deployment of this strategy, the Academy envisages playing a catalytic role in the planned transformation process of the NARES. The subsequent sections present the annual progress and achievements in research, capacity building and the other activities of the Academy under the six strategic areas during the year 2022.

2.1 Agribusiness Management for Inclusive Growth

With the shift in agricultural development strategies from a pure production-oriented approach to a broader systems perspective, the importance of the agribusiness sector has grown significantly. Seen as an engine for growth, agribusiness and its related industries are receiving increased attention in policies and strategies that aim to promote investments in agro-enterprises and agro-based value chain development. This has prompted a need for a deeper understanding of agribusinesses, agro-industries and agri-food value chains. The theme of agribusiness management for inclusive growth encompasses research, education, capacity building and consulting for developing future leaders in the field of agribusiness. It broadly aims to enhance capacities for food and agribusiness management education and research in NARES and to enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses. Rapid urbanization, rising disposable income of households, expanding and diversifying food demand and globalization of the agricultural economy have given impetus to research and capacity building in the area of agribusiness management. To meet the broader objectives of the theme, the Academy focuses on the following thrust areas for research, capacity building and education:

- *Agribusiness environment: W*ith a major focus on value chain analysis for agri-commodities and services; consumers' demands; price behaviour and forecast analysis;
- *Public policy and impact*: With a focus on market reforms like e-NAM, GST, *etc.*; financial inclusion and its spillover; spread and convergence of crop insurance;

- *Markets and institutional innovation*: Covering market integration of commodities; producer companies and FPOs; inputs and service delivery systems; diversification and intensification of agriculture for high value commodities; business analytics;
- *Nurturing future leaders in agribusiness*: Through imparting quality education in agribusiness management.

2.1.1 Entrepreneurship Development through Farmer Led Innovations- A Case Study in Plantation Sector

The research project on Entrepreneurship Development through Farmer Led Innovations (FLIs) – A Case Study in Plantation Sector under NASF funded project was completed on 31 March 2022. During this period one workshop was conducted during June 10-11, 2022 at the Indian Institute of Plantation Management (IIPM), Bangalore (Fig. 2.1.1.) by inviting all the innovators and assessed their responses for upscaling of innovation. During the workshop, the role of FPOs & NABARD was highlighted in strengthening the innovation ecosystem. Startups were invited to the workshop for providing a way forward in the commercialization of innovations to address the challenges in agriculture.

The project has given significant output in identifying the Cyclic model of innovation as the suitable model for FLIs in the plantation sector. The assessment of 22 potential grassroots innovators was done by developing Entrepreneurial Assessment Index to identify the challenges of innovators in upscaling the innovations. Suitable strategies were prepared based on agri-business climate' analysis, economic potential analysis, and market potential analysis to develop a business plan by the innovators. The SWOT analysis for innovation and agripreneurship was carried out through action research on selected FLIs. The reflection of innovators reported the need for further guidance from scientists of NARES systems for field testing of their innovations, assistance in providing a roadmap for a business plan and funding support.

Figure 2.1.1: Farmer with his innovation

2.1.2 New Market Reforms and e-NAM in India: Effect on Marketing Choices and Price Realization for Smallholders

The electronic-National Agriculture Market (e-NAM), which was launched in April 2016, is being implemented by Small Farmers Agribusiness Consortium (SFAC), a central government agency, with the support of state governments. The purpose of e-NAM is to create a network of existing mandis on a common online market platform as 'One Nation, One Market' for agricultural commodities in India. As on 31st Dec 2022, a total of 1,260 wholesale mandis located in 22 states and three union territories (UTs) have been integrated with the e-NAM platform. These states include Andhra Pradesh, Bihar, Chhattisgarh, Goa, Gujarat, Haryana, Himachal Pradesh, Jharkhand, Karnataka, Kerala, Madhya

Pradesh, Maharashtra, Nagaland, Odisha, Punjab, Rajasthan, Tamil Nadu, Telangana, Tripura, Uttar Pradesh, Uttarakhand, West Bengal, Chandigarh, Jammu & Kashmir, and Puducherry.

The present study aims to examine the effect of policy changes in the operations of major e-NAM mandis in India and to study the price realization by the farmers through e-NAM for different commodities. An online platform for marketing agri-produce can bring transparency, provide better prices and add to the confidence of farmers. However, most of the e-NAM mandis still lack some of the basic elements to operationalize the e-NAM fully. The weakest segment of the mandi is its assaying facility. The mandi's laboratory has 1-2 staff who manually clean and assay sample packets- separating each grain from shrivelled, spoiled/ weeviled grains and foreign matter, by hand. The machines installed at mandis have limited functions, like moisture metre for grains, *etc*. During peak arrivals, sometimes more than 5000 lots of different commodities arrive in a day in any mandi. The staff currently can't assay even 40-50 lots in a day using current methods. Without quality assurance, a buyer from outside won't buy the products online.

To analyze the market arrival and benefits received by the farmers through e-NAM, daily transaction data of all the e-NAM mandis were collected from the e-NAM portal for one full year viz. May 2020 to June 2021. For making the comparison, transaction details from the Agmarknet portal were also collected from non-eNAM mandis for the same period. For further detailed analysis, two states were selected during the reporting period- Telangana and Maharashtra to examine the market arrival trend and modal price of one of the important pulses *viz*. red gram (tur).

From the analysis of the transaction data, it was observed that there was no significant difference in the modal price between e-NAM and non-e-NAM (Fig. 2.1.2.c & Fig. 2.1.2.d). Contrary to it, the daily transaction volume of the same commodity is much higher on the e-NAM platform than non-e-NAM in the same mandis (Fig. 2.1.2.a & Fig. 2.1.2.b). The possible reasons for both trends may be that the mandis are e-NAM enabled, but in absence of efficient and reliable quality assaying equipment, most of the trade of the major commodities are listed on e-NAM. Though the same traders and commission agents participate in the transaction process irrespective of the e-NAM or non-eNAM platform, this leads to an insignificant difference in the price realization by the farmers across both platforms.

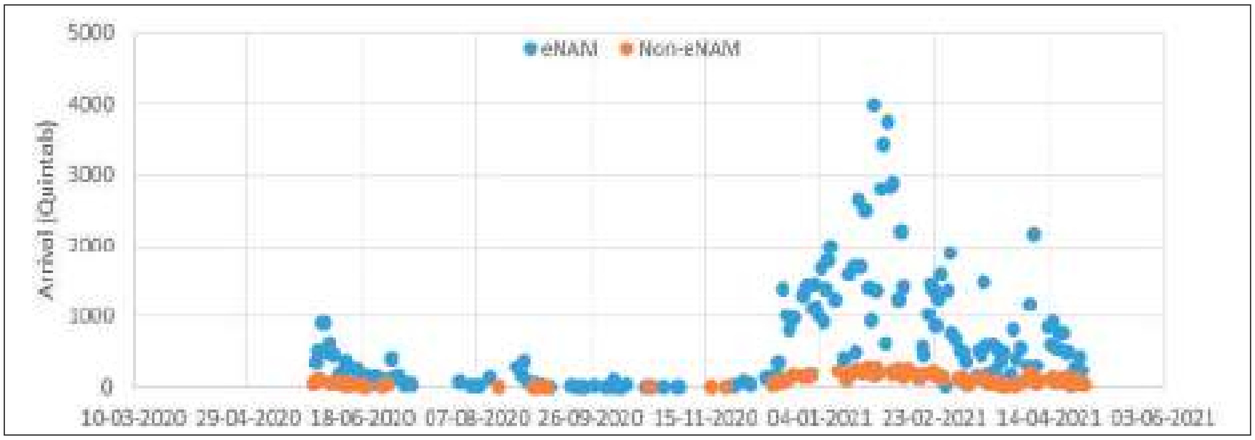


Figure 2.1.2.a: Daily arrival of Red Gram on eNAM and Non-eNAM Platforms at Tandur APMC, Telangana

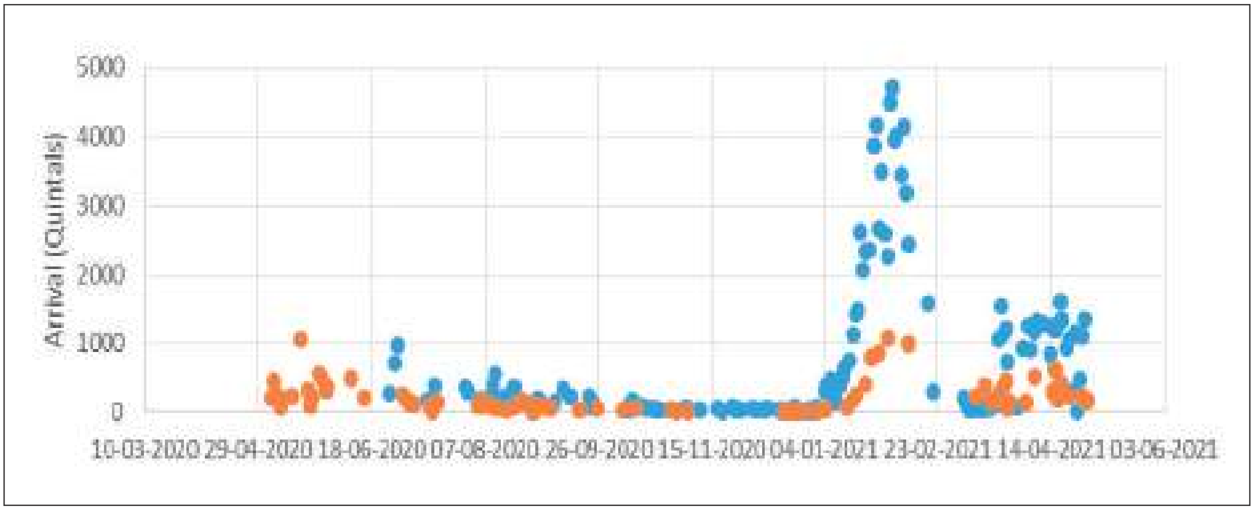


Figure 2.1.2.b: Daily arrival of Red Gram on eNAM and Non-eNAM Platforms at Amravanti APMC, Maharashtra

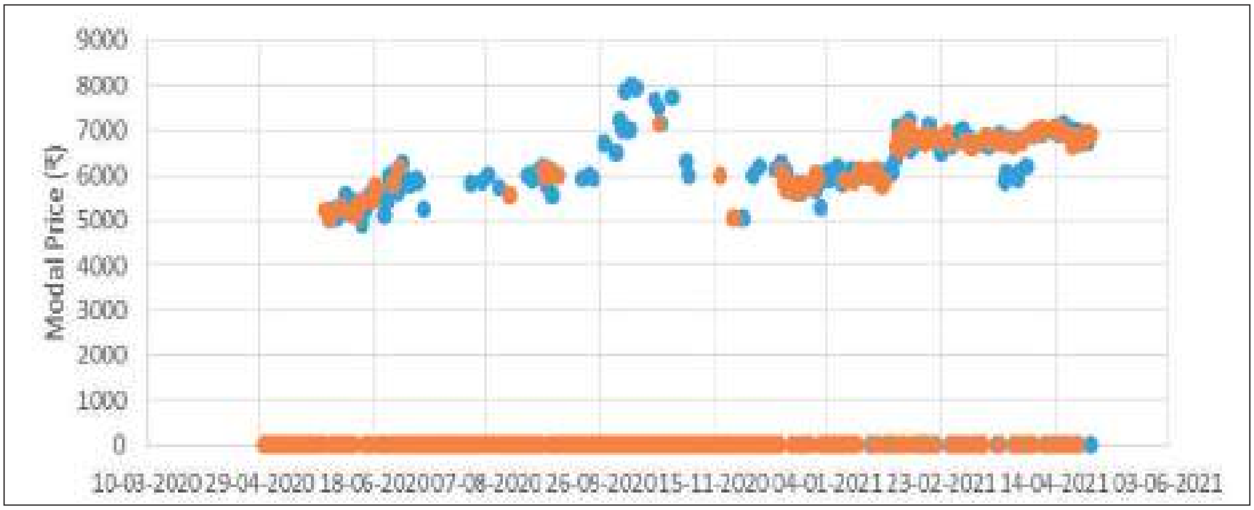


Figure 2.1.2.c: Daily Modal Price of Red gram on eNAM and Non-eNAM Platforms at Tandur APMC, Telangana

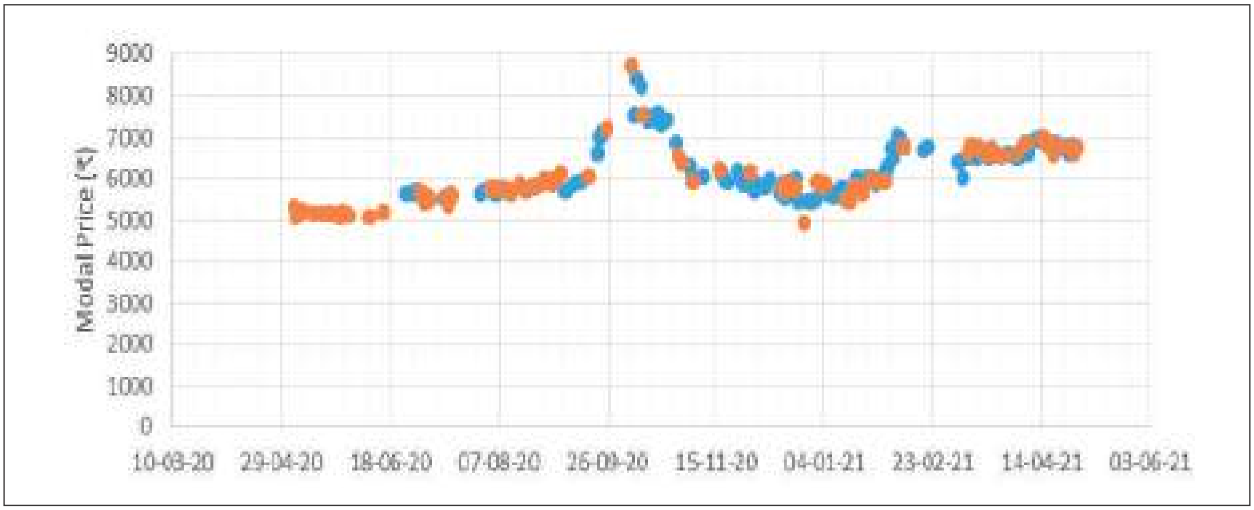


Figure 2.1.2.d: Daily Modal Price of Red gram on eNAM and Non-eNAM Platforms at Amravanti APMC, Maharashtra

2.1.3 Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies/Value-Added Products

Among the top ten technologies identified for preparing the techno-economic feasibility report, the process for the production of freeze-dried sugarcane juice, dehydration of tubers and lac-based fruit coating formulation for Kinnows were studied during the reporting period.

More consumers are opting for freeze-dried fruits, yoghurt melts, snack fruit, vegetable crisps, vegetable powders, and instant soup mixes, thereby driving the market in the region during the pandemic. The freeze-dried Fruits & Vegetables market size exceeded USD 77.6 billion in 2020. The global freeze-dried food market is projected to reach a CAGR of 8.03% during 2021-2026. In this scenario, Freeze dried sugarcane juice is an attractive option to replace synthetic soft drinks. Freeze-dried sugarcane juice retaining its natural qualities and packing will offer good scope for longer shelf life and commercial advantages in packing, storage and transport costs. It can be used as a flavour enhancer of food, beverages, sweets, confectioneries, bakery items, dairy products, weaning food, novel value-added foods and medicinal/pharmaceutical preparations.

The availability of fresh potatoes in far-flung mountainous regions and hostile weather conditions is a big problem, particularly for the Indian defense troops operating there. The supply of dehydrated potatoes is likely to reduce this problem. Commercially viable technologies for the dehydration of potatoes are not available in India so far. ICAR-Central Potato Research Institute, Shimla developed this technology to produce dehydrated potatoes and it signed a MoU with a Hyderabad-based company 'Sunripe' for the commercial promotion of technology. It is expected to reduce the post-harvest losses (PHLs) in potatoes, which are at present very high (about 16 per cent) in India.

The global edible films and coating market size was valued at USD 2.70 billion in 2021 and is expected to expand at a CAGR of 7.7% from 2022 to 2028. North America (First), and Asia Pacific (Second) in the market of edible films and coatings. ICAR-National Institute for Secondary Agriculture, Ranchi developed technology to produce lac-based fruit coating formulation for Kinnow. This technology helps not only to reduce post-harvest losses (less moisture loss and shrivelling) during storage and transportation but also to improve the cosmetic features (shine, colour) of Kinnow. Besides, a draft strategy paper on "Commercialization of ICAR Technologies" was also prepared for the project.

2.1.4 Agri Startups in India: Decoding the Determinants and Success Factors

The agricultural sector, including crops, livestock and fisheries, has been witnessing the emergence of several startups in the recent past. The government has also launched several initiatives for promoting startups. With this background, the study was undertaken with the objectives of investigating the motivations for endeavouring into agri startup; examining the business processes followed by agri startups; determining the factors influencing the success of agri startups; and to come up with cases of agri startups.

Based on details of the business operations, the agritech startups have been categorized into broad categories of agricultural value chain namely startups providing advisory services; capital input manufacturer; delivering consumable input; producing consumable input; farming as a service (FaaS); startups doing innovative agricultural production process; startups making instruments with the internet of things (IoT), sensors, artificial intelligence, software as a service (SaaS); startup providing market information; online market place startup and online produce aggregator. The same is presented in Fig. 2.1.3.a & Fig. 2.1.3.b Here, the sum is more than 100% as some startups are into more than one category of the agricultural value chain. Startups always look for problems of their prospective customers, may it be a farmer or consumers and try to bring out a solution which is technology based and can be scalable. Market linkage, digital agriculture, better access to inputs, farming as a service and financing were new emerging business opportunities in India. It was found that startups making instruments with the internet of things (IoT), sensors, artificial intelligence, and software as a service (SaaS) constitute around 23% of total

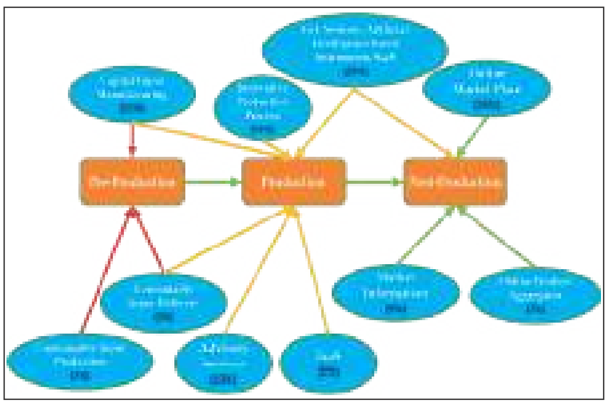


Figure 2.1.3.a: Startup's Presence in Agri-Value Chain

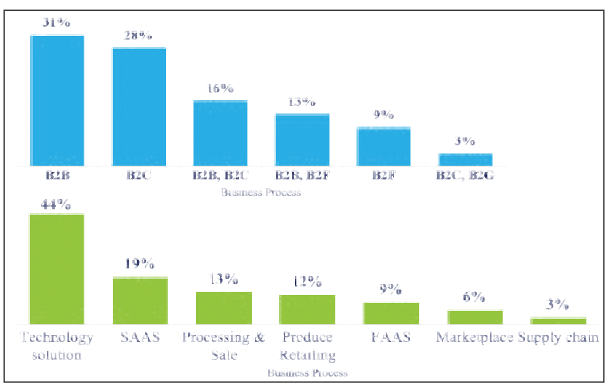


Figure 2.1.3.b: Business Process followed by Sample Agritech Startups

agritech startups which have scaled up. Some of the startups in this category are providing sensor-based instruments which guide in input application, help in soil testing, as well as in dairy operations.

There are 18% of startups which are in the e-commerce and marketplace arena. More than 92% of startups are in the crop and horticulture sector. Very minimal startups are in the dairy/ livestock sector which is 3% of total agritech startups. These are covering areas like fodder delivery, milk testing and IoT solutions for dairy farm operations. The same percentage of startups are in aquaculture with innovative aquaculture production processes.

Looking at the business processes, the businesses have been categorized into B2B, B2C, B2F and B2G (B-Business, C-Customer, F-Farmer, G-Government). It was found that the majority of the start-ups are downstream of the supply chain. It indicates the business potential as perceived by the start-ups. Only 22%, of farmers, are customers (Fig 2.1.3.b). As far as the product is concerned, technology solution is leading with 44% of startups having technology solution for customers.

2.1.5 Financial Accessibility and Occupational Risks of Vegetable Street Vendors in India

The most visible aspect of the informal sector is street vending, which is a global phenomenon. Street vending, like other informal sectors, is distinguished by a low level of income, ease of entry, self-employment, and a large number of participants. Millions of people earn a living selling a wide range of goods and services on the street in cities and towns all over the world. Despite the widespread belief that street vending will decline as economies develop and income levels rise, Amod K (2000) reports that it is increasing in many places. Because of their low level of education and skills, street vendors are frequently unable to obtain regular jobs in the remunerative formal sector. They try to solve their financial problems with their meagre financial resources. They are the primary distribution channel for a wide range of daily consumption products such as fruits and vegetables, ready-made garments, shoes, household gadgets, toys, stationery, newspapers and magazines, and so on. If they were removed from urban markets, it would cause a severe crisis for fruit and vegetable farmers, as well as small-scale industries that cannot afford to retail their products through costly distribution networks in the formal sector.

The study was conducted to know the socioeconomic conditions of the street vendors in Hyderabad. The study includes street vendors of different age groups, from different districts of Hyderabad and those engaged in vending different types of products. The study also accounted for in-migrant vendors. The study focused on knowing almost all aspects with respect to street vending, their personal life, social life and their health-related issues which include operational risk, mental and health conditions, financial product awareness and accessibility.

This survey included 55 female vendors (46 %) and 65 male vendors (54%). It is found that men are more likely than women to engage in street vending. The majority of women engage in street vending to assist their struggling husbands or family members in surviving. Though more men are involved in street vending, they are supported in some way by their families. According to the survey, the number of people engaging in street vending is increasing with each passing year. This is partly related to unemployment and financial difficulties brought on by Covid-19. More migrants are flocking to Hyderabad in search of work, and many of them engage in street vending because it pays better than what they could earn in their home country. Street vending is the primary occupation of 82 per cent of the vendors polled, but some do other jobs to supplement their income, such as auto driving, labour, agriculture, and so on, because street vending does not pay well. Lack of education cannot be considered a reason for people engaging in street vending as most of them had some basic education. Vendors with a pre-university education account for 14% of the total.

Table 2.1.1: Socio-Economic Status of Street Vendors (N= 120)

Socioeconomic Status	Frequency	Percentage
Gender		
Male	65	54
Female	55	46
Age Group		
18-30	38	32
31-50	49	41
Above 50	33	28
Years in vending		
1-5 years	48	40
6-10 years	62	52
Above 10 years	10	8
Primary Occupation as Vegetable Vending		
Yes	98	82
No	22	18
Educational Qualification		
Primary Level	29	24
Mid-Level	34	28
Secondary Level	27	23
Pre University	17	14
Under Graduation	13	11
Reasons for Street Vending		
Health Issues	39	33
Low Investment	39	33
Family Occupation	30	25
Others	12	10

2.1.6 Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India

During the period under report, information on modern value chains was studied, viz. hotels and restaurants, Pista House, and e-Commerce platforms such as Hibachi, Tendercuts and Licious. There are more than 2000 such units in Hyderabad city alone. They buy mutton from slaughter house as well as from butchers who bulk supply to hotels @ Rs. 480 / kg. Hoteliers buy sheep through designated regular traders in Hyderabad city, who in turn send it to butchers and deliver slaughtered mutton pieces and other viscera to the hotels daily. Weekend quantity is 40% more than that of weekdays. Telangana sheep farmers don’t have any direct linkage with them.

Pista House is the maker of Haleem, Biriyani and Kebab in Hyderabad. It acquired a cult-like status for its Haleem which is 100% pure meat with pure ghee, that has a GI tag. They source roughly 100 animals daily and may be 5000 animals during festival days in Hyderabad itself. It was found that Telangana sheep farmers don't have any direct linkage with them. There is a scope for them to form a sheep FPO/FPC so that they can integrate with this modern chain.

Startups like Hibachi, Tendercuts and Licious have their supply chains sourcing the slaughtered animal from traders. Presently, small ruminant farmers are far away from these modern supply chains. Besides, two institutions connected with goat farmers, viz. Goat Trust in Lucknow, a sociopreneurship organization and Theni District Farmers Goat Producer Company Limited, a constituent of Vidiyal, an NGO in Madurai, were studied.

Goat Trust had more than eight years of experience in Goat-based value chains in Uttar

Figure 2.1.4: Small Ruminant Farmer's activities

Pradesh mainly and Maharashtra and Madhya Pradesh also. It promotes the formation of SHGs and establishes linkage with banks for financial intermediation. They get funding support from National Bank for Agriculture and Rural Development (NABARD), Bill Melinda Gates Foundation (BMGF), JBF, National Scheduled Castes Finance and Development Corporation, TRLM. Their major interventions by involving goat garmers are training on goat management such as Commercial Goat Farm Enterprise Training, Community Livestock Business Centre, Bakari Paalak Paathshaala (Shepherd's field School) and Pashu Sakhi (Certified Livestock Nurses). The organization produces many value-added products from goat milk supplied by the farmers and they share a portion of their profit back with them. They also promote the sale of goats from their member farmers on a live-weight basis.

The FPO in Tamil Nadu was formed in 2013 with NABARD support. Presently, 1,234 women goat farmers in Bodi West Hills in Tamil Nadu were the members. Each of these women has contributed an equity share of Rs 1,000. The company was involved in procuring medicines and fodder for their goats and selling matured goats for handsome prices without the interference of middlemen. The profit is shared as a dividend among the members. The women are known in the locality as "voicemail goat farmers". These uneducated women listen to voicemails sent to them through mobile phones and rear the goats in a scientific manner based on the advice they get. Vidiyal, an NGO, provide them with technical support.

It is understood that small ruminant farmers are found still far away from modern efficient markets. The main reason is the absence of any institutional support and innovations in goat farming and markets as in dairy and poultry. Efforts should be made to formalize the relationship/contract between farmers and village traders. FPOs could be a way forward and scope for integrating them into modern value chains in the long run, as the demand for small ruminant meat is always on the rise. Farmers' Association could be promoted, to begin with.

2.1.7 Farmer Producer Companies in India: A Study on their Management Practices and Business Potential

Farmer Producer Companies (FPC) are promoted in India to address some critical issues faced by small and marginal farmers, especially on the marketing front. There is a rapid rise in the number of FPCs registered in the country owing to the support rendered by Central and State Governments. This study is undertaken to analyse the performance and identify the pain points limiting its success. About five FPCs in Tamil Nadu and five FPC in Telangana were studied in detail

Figure 2.1.5: Progression of FPC

and information about 100 farmer members was collected to assess their participation in various activities of the institution. It was found that these FPCs are operating at a very nascent stage of business even after four years of the start of operations. Further, FPCs could exploit only about 6.1 % to 13.6 % of their minimum business potential (MBP). MBP calculates the potential of a business that FPC could attain if all the proceeds of the members' farm businesses, viz., input purchases and output sales, are routed through FPC. It was also observed that less than 10 per cent of members in the sample were directly participating in business activities of FPC and liquidity crunch was a serious issue with 90 percent of FPCs affecting their day-to-day operations and limiting their growth and success. The study identified three stages of growth in the path to success (Fig. 2.1.5)- i) the Early stage where the operation of FPC is at a low level with zero or negative net profits, ii) the Intermediary stage, where FPC expands in many areas of production including processing, opening stores for direct selling and increase in investment, and iii) Maturation stage, where FPC attain specialization in their area of operation and narrow down to few commodities, but scale-up their operations manifolds and moving towards complex distribution network.

2.1.8 Data Visualization in Agribusiness and Agricultural Research

The programme was organized during January 17-22, 2022 with 32 participants including 10 female participants from various ICAR institutes and State Agricultural Universities. The programme covered various visualization tools and techniques which can be used for presenting data in research reports, other publications as well as presentations. It was a hands-on training in which participants were exposed to two popular softwares, namely, R and Tableau Public for generating graphic visuals. Speaking on the occasion, Dr Ch Srinivasa Rao, Director shared the importance of data as well as data visualization and how best these can be used to solve many issues of agricultural science as well as for decision making.

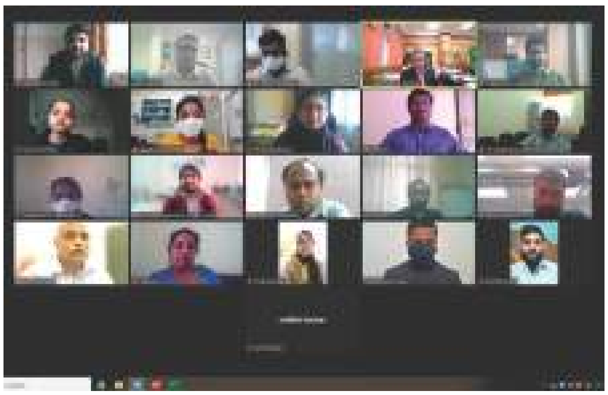


Figure 2.1.6: Director, ICAR-NAARM addressing the participants during valedictory session

2.1.9 Collaborative Capability Building Program for Young Agri-Business Professionals MORE (Marketing Officers' Resurgence & Excellence)

A five day online Collaborative Capacity Building program MORE (Marketing officers' Resurgence & Excellence) for young Agri-business Professionals of M/s Coromandel International Ltd. was organized during February 14-28, 2022. There were 28 participants from all over India. The programme covered various topics which included Farmers Producer Organization, Business Model Canvas, Conflict Management, Marketing Basic & STP environment, Brand management & promotions, Negotiation Skill, Supply Chain Management, Branding & Digital Marketing, Data Analytics & Forecasting. Dr Ch. Srinivasa Rao, Director shared the importance of climate change and a balanced nutrition landscape for the agricultural sector and how the problems of the agricultural sector can be minimized by balanced fertilizer use.

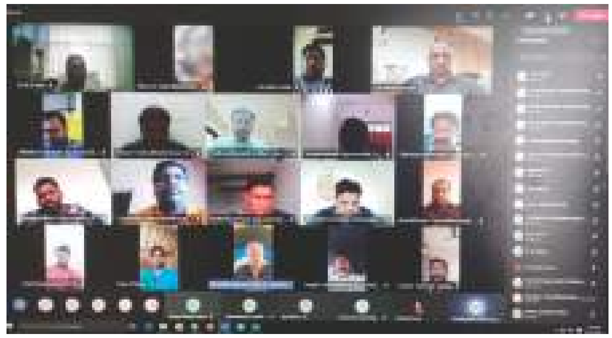


Figure 2.1.7: Coromandel Participants from M/s Coromandel International Limited attending Capacity Building program MORE (Marketing officers' Resurgence & Excellence)

2.1.10 Entrepreneurial Skill Development for Agricultural Graduates

The Academy conducted a training program on Entrepreneurial Skill Development for Agricultural Graduates under SCSP, for the students of Bidhan Chandra Krishi Viswavidyalaya during March 11-13, 2022 in which a total of 100 students including 47 girl students participated.

Delivering the inaugural address as chief guest, Dr Ch. Srinivasa Rao, Director apprised the participants about the role of students in the nation-building and sensitized them for higher education besides motivating them to pursue higher education. Prof BS Mahapatra, Vice-Chancellor, BCKV, Mohanpur, West Bengal presided over the inaugural program and highlighted the need for developing skills to

Figure 2.1.8: Participants of Enterpreneurial Skill Development Training

become a successful entrepreneur. The training program sensitized the students regarding career opportunities, prospects in agricultural education and agri-preneurship. Dr S Goswami, Dean, Faculty of Agriculture, participated as a Guest of Honour and stressed the students to make use of the training to help them pursue their future career avenues. The students have also imparted training on business plan preparation, design thinking and personality development.

2.1.11 Entrepreneurial Skill Development for Dairy Technology Students of ICAR-NDRI, Karnal

A 35-day in-person training programme on "Entrepreneurial Skill Development" for B.Tech, Dairy Technology students of ICAR-NDRI, Karnal was organized at the Academy during June 02 - July 06, 2022. There were 44 participants including eight girls from 3rd and 4th year of B.Tech, Dairy Technology course. The programme was prepared with the objectives of promoting and nurturing entrepreneurship abilities in students; and equipping them with management tools needed for making successful entrepreneurs The sessions were on broad subject areas of agribusiness overview, marketing and business strategy, finance essentials, business & entrepreneurship, business communication skill, analytics skill, personality & HR along with field exposure. Many start-up founders interacted with the participants and shared their experience. Dr Dheer Singh, Joint Director (Research and Academics), NDRI, Karnal thanked the Academy for accepting the request to impart this training to the graduating students and also complimented the students on successfully completing the programme. He also suggested them to apply the learnings in their professional career. Speaking on the occasion, Dr Ch. Srinivasa Rao, Director, ICAR-NAARM emphasised on the role start-ups are playing in nation building and urged the students to explore venturing into start-ups as one of the career avenues for them. Dr G. Venkateshwarlu, Joint Director, ICAR-NAARM highlighted the benefits of such type of training programme and explained the changes in the personality and thinking of students which will help them in building their careers.

Figure 2.1.9: Participants attended 'Entrepreneurial Skill Development for Dairy Technology Students of ICAR-NDRI, Karnal'

2.1.12 MDP on 'Business Plan Development and Accelerating FPOs/FPCs'

The programme was organized during June 13-18, 2022 with 17 participants including two female participants. The MDP was conducted to provide an overview of the key concepts related to FPOs and create awareness about the government schemes promoting FPOs, and develop a practical understanding of scaling up strategies for FPOs through experiential sharing.

The programme was a blend of practical and theory. There was a lso a group formation activity for preparation of business plans in online mode. Each group of participants prepared their own business plan and made presentation. The programme covered all the aspects of FPO, right from mobilization of farmers, FPO formation & registration as producer company, and all aspects of business plan development for FPOs, like business model canvass (BMC), market research, marketing plan, etc. The participants were also made aware about different regulatory compliances needed for producer companies. Resource persons from organizations engaged in FPO financing viz. NABARD and Samunnati Financial Intermediation Ltd.; market linkages and supply chain viz. Ninjacart and NCDEX e Market Ltd were invited. The founder of Sahaja Aharam, an FPO federation involved in organic products also shared the experiences with the participants. The participants were grouped in 7 different teams to practice and prepare their own business plan.

2.1.13 Entrepreneurial Skill Development for Agricultural Graduates

The Academy conducted a training program on Entrepreneurial Skill Development for Agricultural Graduates under SCSP, for the students of Kerala Agricultural University (KAU) during June 20-22, 2022 at College of Agriculture, Vellayani. 57 students consisting of UG & PG and Ph.D. students from College of Agriculture participated and out of which 43 were girls and 14 were boys. Experts from KAU, NAARM, NABARD and DIC, Thrissur handled the sessions. The programme covered topics viz., Preparing Agri-graduates for Entrepreneurship, Credit support for agripreneurship from public sector financial institutions, Innovation process & Enterprise models in agriculture, Entrepreneurship in livestock sector, Digital marketing techniques, Experience sharing by a successful entrepreneur, Pre- entrepreneurial phase analysis, Govt. Schemes for supporting Agri-Startups, Strategies for Promotion of Agri-Clinics and Agribusiness Centre, Managerial skills for entrepreneurship and Entrepreneurial motivation, leadership, interpersonal skills, etc.,

2.1.14 Nurturing Entrepreneurship Ecosystem in Agriculture Universities (For Faculty of Kamadhenu University)

The programme was organized during July 20 – 22, 2022 with 42 participants including seven female participants. To give an understanding of the entrepreneurial ecosystems across major countries, the nuances of establishing entrepreneurial ecosystem in academic institutions. The trained scientists will carry forward the agenda of further awareness and sensitization of other scientists, students and institutions. Scientists and industry experts were involved to give the experience of other organizations.

2.1.15 Entrepreneurial Skill Development for Agricultural Graduates

The Academy conducted a training program on Entrepreneurial Skill Development for Agricultural Graduates under SCSP, for the students of Kerala Agricultural University (KAU) at College of Agriculture, Vellanikkara during June 23-25, 2022. A total of 65 students, consisting of UG & PG students from College of Agriculture, Vellanikkara, College of Forestry and College of Co-Operation, Banking & Management participated in the programme participated and out of which 54 were girls and 11 were boys. Experts from KAU, NAARM, NABARD and DIC, Thrissur handled the sessions. Successful entrepreneurs shared their experiences in agribusiness. The programme covered topics viz., Preparing Agri-graduates for Entrepreneurship, Credit support for agripreneurship from public sector financial institutions, Innovation process & Enterprise models in agriculture, Entrepreneurship in livestock sector, Digital marketing techniques, Experience sharing by a successful entrepreneur, Pre- entrepreneurial phase analysis, Govt Schemes for supporting Agri-Startups, Strategies for Promotion of Agri-Clinics and Agribusiness Centre, Managerial skills for entrepreneurship and Entrepreneurial motivation, leadership, interpersonal skills, etc.,

2.1.16 Training programme on Nurturing the Entrepreneurial Ecosystem in Agricultural Universities (On-line Mode)

The programme was organized during August 23- 27, 2022 with 84 participants including 32 women participants.

Since there is lack of Agri-graduates entering into the Start-up arena and that this training would surely increase the number through the trained faculty members, placement counsellors and nodal officers of agribusiness incubators who will surely act as light houses of knowledge in their respective Universities. The objectives of the Training Programme were to develop an understanding regarding the need and importance of entrepreneurship in agricultural higher education and to learn the ways and means of entrepreneurial ecosystem development in Agricultural Universities. One innovative aspect of this training was to give ample time for brainstorming sessions after each lecture, which gave us a food for thought for our policy paper, which is under process.

2.1.17 BIRD-NAARM Collaborative Programme on 'Interface of Producers Organization with Agri Startups'

The programme was organized during September 1-3, 2022 with 30 participants including two female participants. The main objectives of the programme were to provide adequate exposure of relevant agri-startups to the representatives of FPOs, and to develop a practical understanding of scaling up strategies for FPOs through experiential sharing and exposure visits. The programme was attended by 30 participants which included CEOs and BODs of FPOs, and Bank Officers from 10 different states. The programme was inaugurated by Smt. Susheela Chintala, CGM, Telangana Regional Office, NABARD. During this programme, the participants interacted with five agritech startups including Krishi Tantra and FASAL. They also got the opportunity to visit and interact with two agribusiness incubator centres viz. a-IDEA of ICAR-NAARM and AgHub of PJTSAU. The participants also visited the collection centre of Ninjacart, an agristartup, which is engaged in procuring fresh vegetables from farmers and FPO in Telangana. All the participants expressed huge satisfaction by attending the programme and assured to have progressed in their business in partnership with some of the agristartups.

Figure 2.1.10: Participants attended training program 'Interface of Producers Organization with Agri Startups'

2.1.18 Agri Startups and Entrepreneurial Ecosystem- An Indian Perspective

The programme was organized during November 14 - 19, 2022 with 15 participants including six female participants. The objectives of the programme are (i) To give an overview of the incubators and the incubation ecosystem in India. (ii) To train the participants with adequate knowledge and skill for starting and managing a startup. (iii) To expose the participants to the challenges and strategies for agri startups. The training was implemented through a total of 26 sessions including lectures and practicals completed by eminent experts on various aspects of entrepreneurship. An entrepreneurial field trip was arranged to incubation centers and technology parks. 15 participants from eight states viz., Gujarat, Andhra Pradesh, Telangana, Karnataka, Uttar Pradesh, Maharashtra, Jharkhand and Kerala attended the training program. The entire course content was designed to provide a comprehensive understanding of Agri startups and the entrepreneurial ecosystem.

Figure 2.1.11: Dr Ch.Srinivasa Rao, Director, ICAR-NAARM and Mr. Kalidas Pramanik, EVP, Coromandel International Ltd. addressing the participants during inaugural session

2.1.19 Collaborative Capacity Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (Phase –I)

The programme was organized during December 12 – 16, 2022 with 18 participants from M/s Coromandel International Ltd.. The objective of the programme is to enable participants to benchmark their select soft skills, provide opportunities to practice and develop soft skills through structured exercises, and inculcate strategies to leverage soft skills for personal and professional excellence. This programme had a mix of lectures, case study and management games. Eminent researchers and corporate officials delivered invited talks in this programme to share their experience. The participants were also exposed to the latest developments in the field of agronomy, crop protection, weed science and agricultural chemicals. The topics include design thinking, effective communication, experimental designs, business analytics, team building and personality development, marketing and understanding and delivering to the expectations of the employer. The participants gave a score of 4.3 to 4.9 for the feedback on a scale of 1 to 5 expressing high level of satisfaction upon attending this programme.

2.1.20 Programme on Collaborative Capacity Building Programme for Young Agri-Business Professionals New Entrants Assimilation Training (Phase –I)

The programme was organized during December 19-23, 2022 with 16 participants from M/s Coromandel International Ltd. The objectives of the training programme were (i) To enable participants to benchmark their select soft skills.

Figure 2.1.12: Participants of NEAT Phase-I Training

(ii) To provide opportunities to practice and develop soft skills through structured exercises.
(iii) To inculcate strategies to leverage soft skills for personal and professional excellence

2.1.21 Soft Skills and Entrepreneurship Development

The programme on 'Soft Skills & Entrepreneurship Development' sponsored by the College of Agriculture, Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth under SCSP scheme was organized during December 19-23, 2022 with 90 participants including 43 female participants. The training program was. This training programme was meant for the students of CoA, Dr. BSKKV who are pursuing graduation and post-graduation degrees belonging to different years. The objective of the program was to expose the participants to Soft Skills and Entrepreneurship in Agriculture. Total 90 participants nominated by the college attended this training program including 43 female participants. During the programme, the participants were given insights on improving different soft skills and entrepreneurial abilities through series of lectures and interactions.

Figure 2.1.13: Participants attending an online training program on Soft Skills and Entrepreneurship Development

2.2 Education Systems Management for Enriched Learning Quality and Environment

The Division was extensively involved this year in overall educational research, policy, and technology-driven capacity-building initiatives. Three projects were completed and one project is started in this year.

The Project "AgAcademy: A Framework for Scaling up e-learning in Agriculture highlighted the cloud-based instance creation and management for e-learning and knowledge management using customized learning path construction along with a copyrighted version for the creation of a digital ecosystem for foundation programmes of the Academy. An externally funded project on "Entrepreneurship Development through Farmer-led Innovations (FLIs) in Plantation Sector" came up with a comprehensive report on Commercially viable Farmer-led Innovations in the plantation sector with its techno-economic feasibility and business climate. Another concluded project on "Students' Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis" identified the existing student learning approaches in the current educational ecosystem.

The Division has continued its efforts in offering a Diploma in Educational Technology Management with an intake of 25 students in the ongoing 4th batch. Another new MOOC was offered on “Digital Assessment and Evaluation Methodologies” attracting 1731 participants, besides generating a revenue of Rs. 6.11 Lakhs through a nominal certification fee. A new training workshop on Course Management through MOOCs” as per the suggestion given by the Education Division of ICAR during the Vice Chancellors’ meet was conducted to guide AUs in developing and managing courseware for MOOCs. Besides MOOCs, the division provided 9 capacity-building programmes on a customized basis to the stakeholders of NARES on contemporary aspects of education.

2.2.1 Students’ Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis

An applied research project was taken up with the objectives: to analyse Students’ Learning Approaches (SLAs) among agriculture and allied sciences for enhanced learning outcomes; to analyze the changes in Students' Learning Approaches over time; to investigate the factors affecting Students’ Learning Approaches for effective learning outcomes, and to develop strategies for enhancing the quality of learning outcomes in agriculture education. By using the instrument ASSIST (Approaches and Study Skills Inventory for Students (Tait, Entwistle & McCune (1998), the data were collected from 2161 students, representing 30 State Agricultural Universities in 18 states to analyse the student learning approaches. The study also analysed the quality of teaching for better learning outcomes by adopting a Course Experience Questionnaire (Ramsden, 1991; McInnis et al., 2001) which comprises five components - Clear Goals, Good Teaching, Appropriate Assessment, Generic Skills, Appropriate Workload and the data were collected from 180 final year students. The Student Learning Approaches should be considered in the development and implementation of the curriculum and also in designing assessment methodologies. The factors affecting the quality of teaching are to be given due weightage in the quality teaching.

Based on the findings the following strategies for enhancing the quality of learning outcomes are proposed.

Curriculum: (i) Rigorous curriculum matching input goals with outcome behaviors and should be student-centric. (ii) Engagement of all stakeholders in the process of curriculum development. (iii) Flexible, interdisciplinary approach in curriculum development. (iv) Timely revision of curriculum (regular intervals). (v) Focus on balancing theory-practice creating a curriculum that promotes student employability & entrepreneurship. (vi) Relevance of curriculum to the context. (vii) Development, Implementation, Monitoring and Evaluation is a continuous process and every stage has its significance.

Faculty Competency: (i)Competency mapping at regular intervals based on the stage of the career. (ii) Building the competencies through (Faculty Development Programmes) FDPs at the entry-level and (Management Development Programmes) MDPs at middle levels and (Executive Development Programme) EDPs at senior levels. (iii) Continuing Professional Education for enhancing the efficiency of the faculty. (iv) Periodic performance evaluation of faculty.

Linkages: (i) Academia – Government - Industry Linkages for providing an effective and vibrant, challenging learning environment for the students. (ii) Regular Interaction, and exchange of faculty between the HEIs for cross-learning and enrichment of knowledge and skills. (iii) Industry-oriented training and education. (iv) International collaboration.

Facilitative Teaching–Learning: (i) Provision for need-based, skill-oriented education to the students. (ii) Curriculum should explicitly focus on clear goals and learning outcomes. (iii) Flexible curriculum to be adopted based on the requirement/choice of the students. (iv) Development of soft skills / generic skills through integration in the curriculum and teaching methodologies and blended learning and collaborative approaches. (v) Placement cells / Career Development Cells to be established at the college/university level. (vi) Academic rigor and standards are to be given focus. (vii) Appropriate workload and objective assessment are to be defined and accordingly course load (credits) are decided for each programme.

Resources: (i) Access and availability of resources including digital resources, OERs, etc. to all students to bring equity and equality among students. (ii) Infrastructure development needs to be given priority, apart from human resources, for enhancing the quality of education. (iii) Regular recruitment of technical and non-technical staff. (iv) Functional autonomy and effective leadership are the key to the success of HEIs. (v) One of the objectives has been integrated as an activity into the National Agricultural Higher Education Project on ‘Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education’.

2.2.2 Ag-Academy: A Framework for scaling up e-learning platform

A full-fledged Ag-Academy is developed and available at http://agacademy.co.in/, in which methodology for developing a customized learning path construction for each and every individual end-user has been developed and implemented using collaborative filtering and content-based filtering technologies. The platform suggests a learning path based on the background education of the end user and the rating of the various courses present in the platform. As a part of this project, 6 platforms and 4 methodologies have been developed as listed below.

Platforms: (i) Experimenting with Moodle was implemented at https://uasbnaarm.com/ for the participants of a training programme. (ii) Cloud-based Image warehouse implemented at http://pgdma.in/imagewarehouse/ (iii) Implemented Moodle at a training programme organized for PJTSAU at https://pjtsaunaarm.com to find its suitability and robustness. (iv) Multiple cloud-based instances for better implementation of Teaching - Learning process is implemented for the ERP course. Cloud-based ERP Instances & CRM Software for Small and Medium Agro

Figure 2.2.1.a.: Home page of platform

Figure 2.2.1.b.: Dashboard of platform

enterprises were implemented. (v) Developed a Digital platform and implemented LMS at http://agacademy.co.in/ for providing customized learning path construction. (vi) Vinutna: A digital ecosystem for foundation courses is developed at http://111focars.in/ (Copyrighted)

Methodologies: (i) Sharing of resources even after the completion of the training programme through https://uasbnaarm.com/ & https://pjtsaunaarm.com/ is implemented. (ii) Methodology for improving the effectiveness of Teaching Learning process by using cloud-based independent and individual instances is developed and implemented by creating an individual website for each student. https://pgdma.in/anu/, https://pgdma.in/damini/.... (iii) Methodology for developing a Customized Learning Path Construction for each and every individual end-user has been developed and implemented using Collaborative filtering and content-based filtering technologies. (iv) Methodology of integrating several open source software for providing a digital ecosystem of services to the foundation courses where participants stay longer than a month.

2.2.3 Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education

The salient achivements of the project during the year are viz., (i) Total number of trainings/ Workshop organized are 14. (ii) Total number of beneficiaries (Female) are 1447(495). (iii) Number of students awareness Workshops organized on Soft skills and Entrepreneurship are 04. (iv) Number of external advisory panel meetings coordinated is 01. (v) Monitoring of CDC activities, 79 workshops conducted at 05 SAUs and 6434 students benefitted. (vi) Monitoring of FDC activities, 05 workshops conducted at 01 SAU and 350 faculty benefitted. (v) Monitoring of ICAR-DRC Centre. (vi) Facilitating Global Collaboration among Agricultural HEIs. (vii) Collecting Alumni Information from SAUs. (viii) Updating the information of network of inspired teachers of NARES. (ix) Creation of a database of abroad trained scientists of NARES. (x) National Symposium on Agricultural Higher Education by Private Universities. (xi) Web portals and databases for CDC/FDC monitoring systems were developed. (xii) Updating of the network of inspired teachers in all disciplines across India. (xiii) Mapping of AUs alumni abroad (collecting and providing data to IASRI for uploading on the national portal). (xiv) Network meetings in India and abroad with identified institutions for MoUs. (xv) Facilitate global collaborative partnerships for identified Universities based on the need. (xvi) Implement global collaborative partnerships for minimum 02 selected universities through faculty and student exchange, short-duration studies, etc. (xvii) Consultations with the Indian Agriculture University Association and National Academy of Agricultural Sciences. (xviii) Creation of awareness among the students on soft skills, innovativeness, and entrepreneurship -

Organization of Student Awareness Programmes (5 nos). (xix) Maintenance of Disaster Recovery Centre. (xx) Coordination of External Advisory Panel of globally renowned educational experts. (xxi) Pilot testing of a model for nurturing academia-industry linkages

2.2.4 Virtual Review meeting of Career Development Centers and Faculty Development Center

The programme was organized on January 07, 2022 with 32 participants including 05 female participants, to monitor the progress of CDCs and FDCs under NAHEP Component 2.

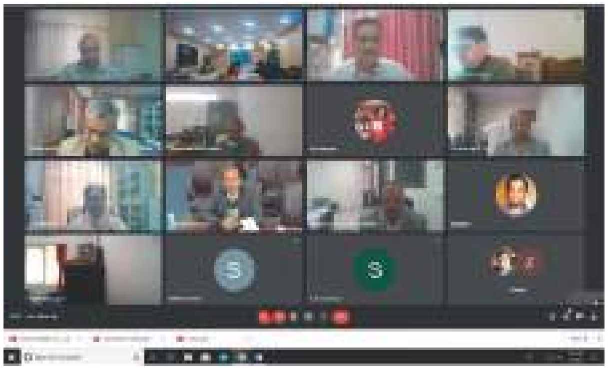


Figure 2.2.2: Participants during virtual review meeting of career development centers and faculty development center

2.2.5 National Training Programme for the Faculty Members of RAJUVAS, Bikaner (Online)

The programme was organized during February 4-11, 2022 with 60 participants. An Online Faculty Development Program was conducted by the Academy for the Faculty members of RAJUVAS, Bikaner. A total of 60 participants including 1 Professor and 59 Assistant Professors participated in the program. This program provided an indepth understanding of enhancing the competencies of the faculty. The program consisted of a blend of lectures, self exploration instruments, experiential learning, and group discussion. The program provided an excellent opportunity for mutual interaction and information sharing among the participants.

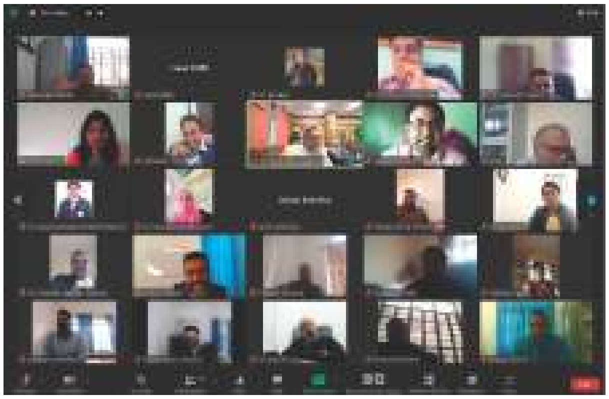


Figure 2.2.3: Participants of the RAJUVAS, Bikaner

2.2.6 Faculty Development Programme on Teaching Competency Enhancement through Innovative Methods

The programme was organized during March 15-19, 2022 with 48 participants including 24 female participants. The objectives of the programme are (i) To apprise the trainees with teaching-learning concepts in depth and about the latest educational policies. (ii) To demonstrate & provide hands-on experience with various e-learning technologies and learning management systems. Faculty Development Programme on "Teaching

Figure 2.2.4: MoU Signing with Dr YSR Horticultural University

Competency Enhancement through Innovative Methods" is organized for Dr YSR Horticultural University Faculty, which spanned over 5 days at ICAR-NAARM in physical mode. The programme was designed into 3 sections of 2,2 & 1 day each which covers 1). Concepts of Teaching/ Learning and Micro Teaching 2). Electronic learning and Learning Management System 3). Sessions related to policies. The programme was attended by 48 participants (24 male & 24 female) belonging to 10 disciplines. All the resources required for the programme were shared in digital mode, which includes a digital learning platform at naarm-ysrhu.in which is used for the hands-on practical session as well.

2.2.7 FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of KVAFSU

The programme was organized on May 17-21, 2022 with 35 participants including six female participants. An off-campus training programme on "FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of Karnataka Veterinary, Animal and Fisheries Sciences University (KVAFSU), College of Fisheries, Mangalore is sponsored by ICAR-NAARM. The aim of the programme is to impart knowledge to the faculty on the contemporary trends in education technology and management The programme was organized at the College of Fisheries, Mangalore under the Sub-plan of ICAR-NAARM. Twenty-five participants from seven constituent colleges, Krishi Vigyan Kendra of KVAFSU attended the programme which included six women and belonging to 15 disciplines. The program covered the topics: Perspectives and Challenges for Agriculture, modern methods of teaching/learning to oral communications, Cyber security, Importance of AI and IoT applications in Agriculture, Science communication, Bioinformatics and its Applications in agriculture, IPR issues in agriculture, New Vistas in Technology Enhanced Learning, Psychology principles of teaching and learning, Overview of Teaching Management and Innovative Teaching Methods for Quality Education etc. The participants were engaged mostly through discussions, demonstrations, and personality development analyses. Design Thinking in Research Project Formulation and Implementation, Building the Right Perspectives in Science Communication and Counselling and Mentoring of Students. Resource Generation through Research, Software in open source and freeware environments like a whiteboard, and Cloud-Based Digital Teaching, Recent trends in ET – ICTs, Social Media, etc, and Developing Entrepreneurial skills among faculty were demonstrated.

2.2.8 Foundation Course for Newly Recruited Faculty of PVNR Telangana Veterinary University

The programme was organized during May 18 to June 9, 2022 with 25 participants including 13 female participants. The objective of the programme is to enhance the faculty competency and to improve the teaching, research, and extension proficiency of the newly recruited faculty of PVNRTVU. The modules on

Figure 2.2.5: Foundation Course for Newly Recruited Faculty of PVNR Telangana Veterinary University

Teaching Management, Research and Extension Management, Human Resource Management, Administrative, and Financial management were covered during the programme with hands-on experience. Apart from these modules other important topics viz. Positive Thinking, Goal Setting for Professional Excellence, Health Management & Orientation to Yoga, Climate Change and Livestock Production, IP Informatics and Technology Management, Bioinformatics tools in Veterinary Teaching and Research, Orientation to One Health, Achieving Excellence by Young Faculty, Development of Personal Website using Open Source Software, Power of MS Office- Word, Power of MS Office- Excel were also covered. The program encompassed all aspects that newly recruited faculty are expected to be equipped with. The sessions were handled by 22 NAARM faculty and 15 guest faculty as resource persons spread over 64 sessions and 126 hours of instruction.

2.2.9 FDP on Competency Enhancement in Agricultural research and Education for the Faculty of SVVU

The programme was organized on May 24-28, 2022 with 35 participants including 12 female participants. An off-campus training programme on "FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of Sri Venkateshwara Veterinary University (SVVU) under SCSP. The programme aims to impart knowledge to the faculty on the contemporary trends in education technology and the programme was organized at the Centre for continuing Veterinary Education and Communication (CCVEC), SVVU under the Sub-plan of ICAR-NAARM. Thirty-five participants from four constituent colleges, two polytechnic, Krishi Vigyan Kendra of SVVU attended the programme which included 12 women and belonging to 15 disciplines. The program covered the topics: Perspectives and Challenges for Agriculture, winning research proposal, modern methods of teaching, Importance of AI and IoT applications in Agriculture, Science communication, Bioinformatics and its Applications in agriculture, Design thinking IPR issues in agriculture, New Vistas in Technology Enhanced Learning, Psychology principles of teaching and learning, Overview of Teaching Management and Innovative Teaching Methods for Quality Education, etc. The participants were engaged mostly through discussions, demonstrations, and personality development analyses. Design Thinking in Research Project Formulation and Implementation, Building the Right Perspectives in Science Communication and Counselling and Mentoring of Students. Resource Generation through Research, Software in open source and freeware environments like a whiteboard, and Cloud-Based Digital Teaching, Recent trends in ET – ICTs, Social Media, etc, and Developing Entrepreneurial skills among faculty were demonstrated.

2.2.10 Course Management through MOOCs

The programme was organized during June 6-10, 2022 with 35 participants including 10 female

Figure 2.2.6: Participants of Course Management through MOOCs Training

participants. The objectives of the programme are (i) To sensitize practices of digital content development for online courses (ii) To build and manage the online courses through MOOCs mode. This programme was conducted as per the suggestion received from ICAR Education Division during the Vice-chancellors meet held in April 2022. The training workshop was exclusively designed to sensitize and enhance the capacity of AUs in conducting MOOCs on their own starting from content production to course certification. A maximum of two from AUs were invited to attend the workshop so that they can go back as trainers for their AUs.

2.2.11 Foundation Course for Newly Recruited Faculty of Odisha University of Agriculture and Technology (OUAT)

A Foundation Course for the Newly Recruited Faculty of Odisha University of Agriculture and Technology (OUAT) was held at NAARM, Hyderabad, during 17 to 30, June 2022. Twenty (20) newly recruited faculty of OUAT participated in the programme out of which 10 were females.

The freshly recruited Assistant Professors and scientists belonged to crop, agriculture, veterinary and community science disciplines. Most of the participants (16) were from colleges; 4 were from

Figure 2.2.7: Snapshot of the participants

research stations of OUAT. The modules covered during the training programme include Research, Education & Extension Systems; Teaching Basics, Methods & Technologies; Innovative teaching methods, Effective communication skills, Student Mentoring & Motivation; Health & Lifestyle Management, Role of assistant professor in improving quality education and the serving to Odisha farmers, Climate Change and Agricultural Production, Achieving Excellence by Young Faculty, Positive Thinking,

Winning research Proposals Experimentation Basics, Research Methodology, Project Management Tools, Data Analysis; HRM – Higher education opportunities; Computers, Bioinformatics, AI and Cloud Computing GIS & ICTs; Power of word and Excel, Organic Farming, IPR, and IP Informatics Agri Startups: Prospectus and Challenges.

Dr A. Padma Raju Former Vice Chancellor, of ANGRAU, upon the inauguration of training, stressed to do the research on the consumption of dairy products- A Critical study. Processing of milk and dairy products imports and highlighted the exports of fish - problems, and implications. Production of rice - A way to diversify and double the farmer's income. Fortification of products- Value addition, vermicompost, poultry, biofertilizers and the unsuitability of organic farming and natural farming and pollution of ground water. Finally, advised the participants not to spoon-feed the students and involve them in studies in such a manner that, they have to go to the library and refer books, etc.

2.2.12 FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of MPUAT

The programme was organized on July 19-23, 2022 with 35 participants including 2 female participants. An off-campus training programme on "FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of Marana Pratap University of Agriculture Technology (MPUAT), Directorate of Extension Education, MPUAT, Udaipur is sponsored by ICAR-NAARM. The aim of the programme is to impart knowledge to the faculty on the contemporary trends in education technology and management. The programme was organized at the Directorate of Extension Education, MPUAT under the Sub-plan of ICAR-NAARM. Thirty-five participants from six constituent colleges, Krishi Vigyan Kendra of MPUAT attended the programme which included two women and belonging to 15 disciplines. The program covered the topics: Perspectives and Challenges for Agriculture, winning research proposal, modern methods of teaching, Importance of AI and IoT applications in Agriculture, Science communication, Bioinformatics and its Applications in agriculture, Design thinking IPR issues in agriculture, New Vistas in Technology Enhanced Learning, Psychology principles of teaching and learning, Overview of Teaching Management and Innovative Teaching Methods for Quality Education, etc. The participants were engaged mostly through discussions, demonstrations, and personality development analyses. Design Thinking in Research Project Formulation and Implementation, Building the Right Perspectives in Science Communication and Counselling and Mentoring of Students. Resource Generation through Research, Software in open source and freeware environments like whiteboard, and Cloud-Based Digital Teaching, Recent trends in ET – ICTs, Social Media, etc, and Developing Entrepreneurial skills among faculty were demonstrated

2.2.13 MOOC on Digital Assessment and Evaluation Methodologies

The programme was organized during September 1-30, 2022 with 1731 participants including 732 female participants. The objectives of the programme are (i) To understand the concept of various assessment and evaluation tools (ii) To develop effective digital assessment methodologies for quality education. This is a newly designed MOOCs on enhancing digital capacities for evaluation while sensitizing about the basic principles of evaluation. Besides the online exam, participants had to use a technology-based evaluation tool to create a quiz.

Figure 2.2.8: A course page on MOOC

2.2.14 FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of RVSKVV

An off-campus training programme on "FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of Rajmata Vijayaraje Scindia Krishi Visas Vidyalaya (RVSKVV), College of Agriculture, Indore under SCSP to impart knowledge to the faculty on the contemporary trends in education technology and management was organized at the College of Agriculture, Indore under the Sub-plan of ICAR-NAARM. Forty participants from five constituent colleges, Krishi Vigyan Kendra of RVSKVV attended the programme which included six women and belonging to 12 disciplines. The program covered the topics: Perspectives and Challenges for Agriculture, winning research proposal, modern methods of teaching, Importance of AI and IoT applications in Agriculture, Science communication, Bioinformatics and its Applications in agriculture, Design thinking IPR issues in agriculture, New Vistas in Technology Enhanced Learning, Psychology principles of teaching and learning, Overview of Teaching Management and Innovative Teaching Methods for Quality Education etc. The participants were engaged mostly through discussions, demonstrations, personality development analyses Design Thinking in Research Project Formulation and Implementation, Building the Right Perspectives in Science Communication, and Counselling and Mentoring of Students. Resource Generation through Research, Software in open source and freeware environments like a whiteboard, and Cloud-Based Digital Teaching, Recent trends in ET – ICTs, Social Media, etc, and Developing Entrepreneurial skills among faculty were demonstrated.

2.2.15 Training Programme on Soft skills Development for Agri-Students [for Students of Sri Konda Laxman Telangana State Horticultural University] (Online mode)

The programme was organized during September 20-24,2022 with 64 participants including 46 female participants. The objectives of the programme are (i) To enable participants to benchmark their select soft skills. (ii) To provide opportunities to

Figure 2.2.9: Participants of Training Programme on Soft skills Development for Agri-Students

practice and develop soft skills through structured exercises. (iii) To inculcate strategies to leverage soft skills and entrepreneurship skills for personal and professional excellence. The programme was designed to cover the aspects of soft skills, Oral Communication, Group dynamics, and Entrepreneurship skills. The programme attended by 64 students belongs to UG and PG Programmes of Sri Konda Laxman Telangana State Horticultural University. MS Teams and Zoom Platforms were used for imparting training in Online mode. Progressive and experiential learning strategies were followed to allow participants to progress in learning from known to unknown and immediately apply the learned KAS through structured exercises (individual and team). A learning log was used to track the summative and formative evaluations respectively. Participants were organized into small groups for fostering cooperative and peer learning. Overall, it was an insightful learning experience for the participants, resource persons and coordinators alike. Innovations included Digital group photographs, Participant's Videos on Oral Communication Exercises.

2.2.16 Foundation Course for Faculty of Agricultural Universities (Online)

The programme was organized during December 1-21, 2022 with 56 participants including 41 female participants. A 21 days' course on

"Foundation Course for Faculty of Agricultural Universities (FOCFAU)" in online mode was organized. The objectives of the Training Programme were to orient the participants to the national and international agricultural scenario, to enhance the knowledge, skills and attitude (KSA) of the participants in various aspects of education, the teaching-learning process and to expose the participants to research and technology transfer processes. Sessions were also conducted on human capital, administrative and financial management for the newly joined faculty of State Agricultural Universities. The trainees were asked to make a short teaching session which was recorded and feedback on their strengths and avenues for improvements were discussed. Participants of the training programme representing Himachal Pradesh and Punjab.

2.2.17 Laboratory Assessor's Training Course

The programme was organized during December 19-23, 2022. The Laboratory Assessor's Training Course was organized in collaboration with NABL, Gurgaon for 24 scientists from 11 ICAR Institutions and one from ICAR Head Quarters. The ICAR has initiated Laboratory Assessor's Training Course for its scientists mainly to develop an in-house pool of assessors such that it facilitates accreditation of ICAR laboratories with international standards i.e. ISO/IEC 17025:2017. The NABL team under the leadership of its Chief Executive Officer organised 5 days of training at the Academy.

Figure 2.2.10: Participants of Laboratory Assessor's Training

2.3 Enhancing Capacities for Leadership and Governance

The Human Resource Management Division has adopted a strategic approach to the effective and efficient management of people in the organization so that they contribute to the organization's ability to obtain a competitive edge. This will help in maximizing employee performance in the service of an employer's strategic objectives. The Division participates in the creation and execution of the ICAR's human resource policies and procedures. Human Resource Management deals with issues related to Performance Management, Organizational Development, Safety, Wellness, Employee's motivation, Teamwork and training individuals and groups on different soft skills required for individual and organizational development in ICAR. The HRM Division's primary goal is to maintain better workplace relationships through the application and evaluation of policies, development, procedures, and programmes related to human resources. Leadership and management development are two important key areas of HRM. The Strategic framework of the thematic area is to enhance capacities for leadership and governance by

- Institutionalize a framework for leadership development at all levels of NARS - early, midcareer, senior professional and manager across all functions (research, education, extension)

- Institutionalize good governance in NARS by building capacities for effective management of institutions, performance assessment, accountability and developing strong organizational value systems and work culture that promote innovation

The Human Resource Management Division places a strong emphasis on management development for the improvement of individual and team leadership abilities. An organization's objectives and goals can be accomplished by enhancing individual and organizational performance, which is typically accomplished by performing a training programme termed a management development programme (MDP). The training aims to raise an employee's skill levels and has a longer-term perspective to prepare them for upcoming challenges. The major objectives of these include effectiveness, learning outcome, and performance of participants in their professional careers. There is a real need for leadership development programmes that aim in developing creative solutions for both current and future issues in both society and public administrations. A leadership development programme was started to monitor how senior managers reacted to systemic achievements, failures, and results. One of the most popular ways to improve managerial and leadership abilities inside an organization is through management development programmes. When it comes to building high-performance organizations, leaders and managers are seen as a very potent group. Management Development is a systematic awareness, ability, attitude and behaviour process that helps in the development of future leaders.

2.3.1 Happiness and Psychological Well-Being of Scientific Personnel in NARES

This study was conducted to investigate the personality traits of agricultural scientists to find out their relationship with well-being attributes and to predict their happiness and life satisfaction. Data was collected from 622 respondents (424 males, 198 females; mean age 41.8 ± 6.3 years) by administering Big-Five Inventory (BFI-10), Oxford Happiness Questionnaire (OHQ), Satisfaction with Life Scale (SWLS), and Psychological Well-Being (PWB-18) questionnaires. Findings revealed significant gender differences in neuroticism and openness, favouring female scientists. Such gender differences were not observed in their scores on happiness, life satisfaction, and PWB. Most of the scientists reported higher levels of happiness (86.2% moderately to very happy), life satisfaction (68.9% satisfied to extremely satisfied), and PWB (57.4%). With age, extraversion, agreeableness, and conscientiousness increased, whereas neuroticism and openness decreased. Similarly, environmental mastery, positive relations, and total PWB improved with age. Except for neuroticism, personality traits were positively correlated with happiness, life satisfaction, and PWB. The regression model on happiness captured 63.7% of variation by considering personality traits (neuroticism being the strongest contributor) and traits of PWB (self-acceptance and environmental mastery being the largest contributors). In the model for life satisfaction, neuroticism, among the personality traits, self-acceptance, environmental mastery, positive relations, and purpose in life, among the PWB, captured 35.3% of the variation in life satisfaction. The implications of these findings about the workplace environment are discussed.

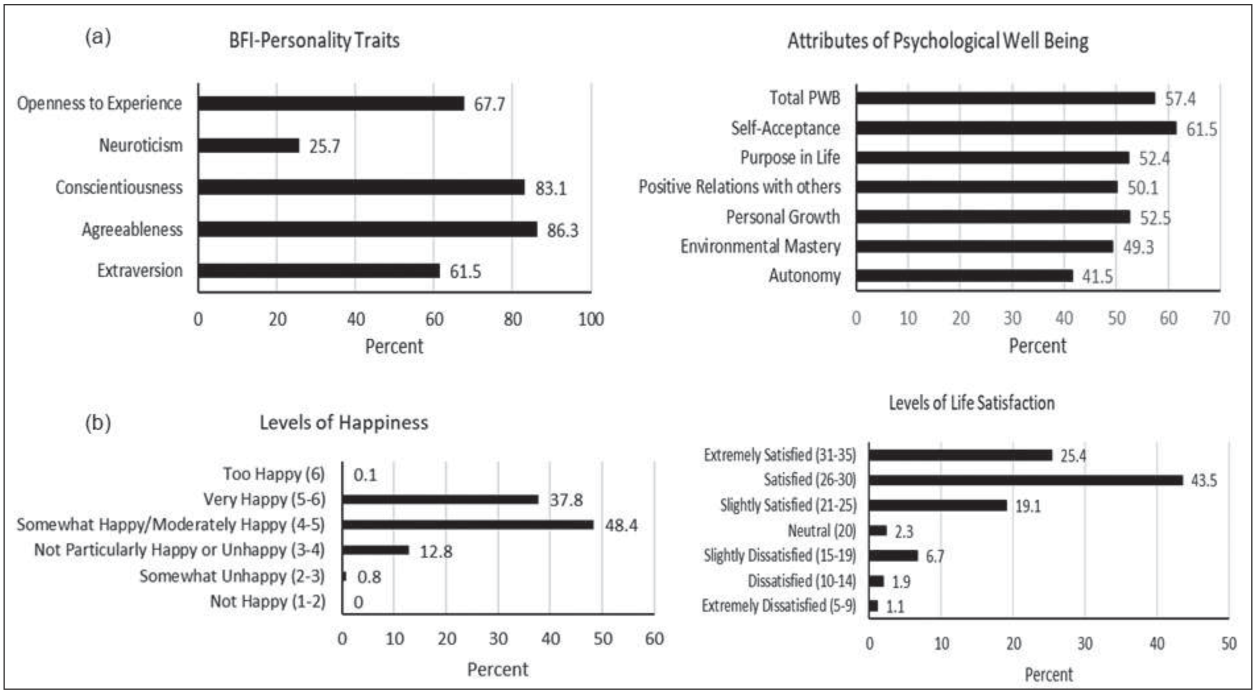


Figure 2.3.1 (a) Respondents with above average scores in BFI – personality traits and attributes of psychological Well-being. (b) Distribution of levels of Happiness and Life Satisfaction

Table 2.3.1: Regression analysis of the associations between personality traits and psychological well-being with happiness and satisfaction with life

Variables	Happiness			Satisfaction with Life		
	b	SE	*p*	b	SE	*p*
Intercept	1.565	0.191	< 0.001	13.718	2.157	< 0.001
Gender (Male)	-0.008	0.035	0.821	-0.610	0.392	0.120
Age	-0.001	0.002	0.644	0.009	0.019	0.640
Personality traits						
Extraversion	0.026	0.008	0.002	-0.156	0.095	0.102
Agreeableness	0.024	0.010	0.019	-0.130	0.117	0.269
Conscientiousness	0.036	0.010	0.001	0.111	0.116	0.341
Neuroticism	-0.051	0.008	< 0.001	-0.198	0.094	0.035
Openness to Experience	0.023	0.009	0.010	-0.119	0.099	0.230
Attributes of Psychological Wellbeing						
Autonomy	0.009	0.005	0.087	0.110	0.057	0.051
Environmental Mastery	0.035	0.005	< 0.001	0.271	0.061	< 0.001
Personal Growth	0.023	0.006	< 0.001	-0.011	0.067	0.869
Positive relations with Others	0.022	0.005	< 0.001	0.126	0.056	0.026
Purpose in Life	-0.001	0.004	0.890	-0.209	0.051	< 0.001
Self-Acceptance	0.063	0.005	<0.001	0.620	0.062	<0.001
F	78.97			25.29		
p	< 0.001			< 0.001		
Adjusted R^2	0.637			0.353		

2.3.2 Identification of Best Practices developed by the Leadership in Scientific and General Administration in NARES

16 best practices in the area of functional autonomy; 5 best practices in the area of financial autonomy; 11 best practices in the area of training and career development; 20 best practices in the area of Leadership, Mentoring and Counselling; 4 best practices in the area of Work-life balance; and 19 best practices in the area of Organizational climate and culture are identified. Identified 11 different undesirable practices that need to be rectified. The suggestions for Futuristic Best Practices were also made to start a system of collection and distribution of monetary help to the families of researchers affected by pandemic and other unforeseen circumstances. Based upon input given by NARES, 75 best practices were identified (Practiced by Leadership in Scientific and General Administration of NARES during their respective period and it was recognized by different stakeholders) and simultaneously 11 undesirable practices have been identified that need to be avoided for the better work environment.

2.3.3 HR Interventions and their influence on Institutional Innovations

The elements like Participative Decision Making, Reward Systems, Team Spirit, Delegation, Training & Development, Career Planning and Development, Motivation, Remuneration, and Talent Management, were major HR intervention influences for Institutional Innovation in ICAR. A model is fit for HR interventions and institutional innovation with knowledge management practices (KM Creation, storage, transfer and application) as mediation and Innovative climate as moderation effects. Recognition programmes, develop leaders to "horizon scan", Determine which positions contribute disproportionately to innovation value, break down internal silos and foster idea sharing, Design the organization (structures, processes, roles, capabilities, and so on), new working habits, Institutional venturing, Selective hiring, extensive and well-designed training, self-managed teams and decentralized decision making, and talent management influences for Institutional Innovation.

2.3.4 Competency Enhancement program for Effective Implementation of Training Functions by HRD Nodal Officers of ICAR

The programme was organized during February 21-23, 2022. The Academy organized a competency enhancement program for the effective implementation of training functions by HRD Nodal Officers of ICAR in virtual mode. A total of 63 participants (06 female) participated in this program. The HR activities of the institutes under ICAR are handled by HR nodal officers who handle HR activities including that training function. ICAR plans for competency enhancement of the training nodal officers in the process of training starting for training need analysis to train impact assessment, to enable them to handle training function effectively and efficiently and to provide an opportunity for the HRD nodal officers, who are in the ICAR. The main

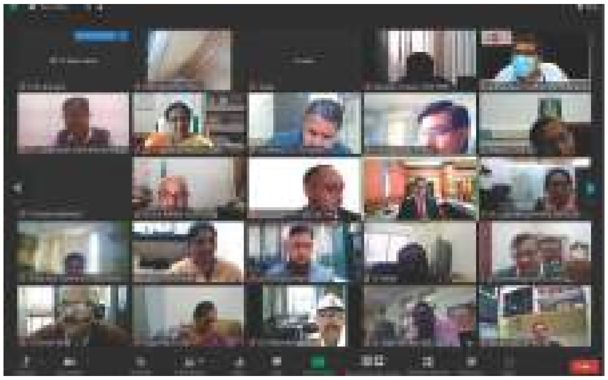


Figure 2.3.2: HRD Nodal Officers Training programme

purpose of the training program was to identify the competencies and competency framework of the scientific cadre for agricultural research and extension who are engaged in agricultural research and education (including extension education) whether in physical, statistical, biological, engineering, technological or social sciences discipline. This cadre also includes persons engaged in planning, programming and management of scientific research.

2.3.5 Executive Development Programme on Leadership Development

The programme was organized during May 9-14, 2022 with 15 participants including 01 female participant. The objective of the programme is to develop leadership capacities, competencies and skills to improve organizational efficiency by adapting change and innovation management in the broader system of the National Agricultural Research and Education System, Managing human resources for enhancing organizational performance; Science, Management & Organizational Efficiency; Building Strength-based Organizations; Self-evaluation of leadership qualities, Best Leadership Practices, Psychometric assessment of happiness and wellbeing etc. are included. Apart from these exclusive sessions Positive Thinking; Happiness and Well-being are also included in the programme. Special Sessions on internalizing startup ecosystem into ICAR were organized and all the ICAR institutes had MoUs with a-Idea of NAARM, NAHEP Workshop "A Dialogue on Fostering Collaboration for Quality Agricultural Education between Actors of Agriculture Research and Education" was also organized to develop collaboration between private, public universities and ICAR institutions. A half-day workshop was organized at ASCI with a focus on the sessions on Raising Extra-Budgetary Resources by Leveraging Assets and 'The Essentials of Being a Leader' are also organised. The programme was attended by 15 RMPs representing 11 ICAR institutes and ICAR HQ.

2.3.6 Management Development Programmes (MDPs) on Leadership Development (A pre-RMP programme)

Two pre-RMP programmes were organized during June 14-25, 2022 and December 12-23, 2022 at ICAR-NAARM, Hyderabad. A total of 36 participants attended the Management Development Programmes (MDPs) on Leadership Development (A Pre-RMP Program) at ICAR-NAARM. This 12- days programme provides an opportunity for the pre-RMPs, who are in the "middle zone", to develop a smooth and

Figure 2.3.3.a: Participants of MDP on Leadership Development

Figure 2.3.3.b: Participants of MDP on Leadership Development

gradual transformation to step into a future leadership position. This program exposes the participants to different aspects of leadership challenges and competencies needed in the present context, imparts knowledge and skills required towards technology management, information and knowledge management, administrative and financial management. A total of 57 sessions in both programmes were handled by senior resource persons from the Academy. The participants were exposed to the leadership challenges and experiences shared by the renowned guest faculty from ICAR and the corporate sector.

2.3.7 Executive Development Programme on Leadership Development

The Executive Development Programme on Leadership Development was organized at the Academy during Jul 4-9, 2022 for 18 Research Managers representing: Directors – 15; Assistant Director General – 02; Deputy Director General-01 across the country spanning from Arunachal Pradesh to Andaman & Nicobar Islands. The programme sessions included Cross Learning from other Scientific Organizations viz., ISRO, CSIR & NIAS; HRM outbound exercises viz., Collaboration, Leadership attributes and Team building; Features of Administrative and Financial Management in ICAR and Happiness and well-being with Yoga classes and Heart-based Meditations for Spiritual Transformation.

Shri Hemang Jani, Secretary, Capacity Building Commission, New Delhi, Govt of India, was the Chief Guest of the Valedictory Programme, highlighted the need for demand-driven capacity building programmes, democratizing the training with the flexibility of anywhere and anytime mode.

2.3.8 Training Workshop for Vigilance Officers of ICAR Institutes

The programme was organized during August 24- 26, 2022 with 20 participants. The objective of the programme is to Enhance the competencies of the Vigilance Officers to provide support for vigilance matters in the Institute. The participations were nominated by ICAR, New Delhi. They were sensitized to different vigilance management activities such as preventive vigilance, duties and responsibilities of vigilance officers, disciplinary procedures recruitment process, RTI and purchase procedures, etc. The panel discussion was arranged in which the presentation was made by the participants on different topics relevant to Vigilance matters.

Figure 2.3.4: Participants of EDP on Leadership Development

Figure 2.3.5: Training Workshop for Vigilance Officers of ICAR Institutes

2.4 Extension Systems Management in Market-driven Environment

Timely dissemination of agricultural innovations through outreach and educational services is vital for farmers' empowerment. It is a core research management competency that requires concerted inputs in terms of policy, research and capacity building besides other support systems. It is here that effective management of extension systems assumes critical importance within the larger domain of agricultural research management. Agricultural extension systems at national, regional, state and district levels have played significant role in the growth of agricultural and allied sectors and farmers' welfare. However, the changing context of agriculture needs innovative approaches and methods to manage extension systems to align their efforts towards sustainable livelihood security of millions of peasant families. Sustainable livelihood security and farmers' empowerment through management of frontline extension systems is one of the six thematic areas of the Academy's strategic plan for achieving its vision and mandate. This thematic area is addressed through research and policy, capacity development and outreach services. The core functional areas under this theme are extension policy and management, institutional innovations, knowledge management, including digital extension and gender mainstreaming. Brief account of the work done and achievements pertaining to research and policy, capacity development and outreach activities during the period under report is presented below.

2.4.1 Development of a framework for gender mainstreaming in the development schemes of Agriculture sector in India

The study aims at developing methodological framework for assessment of gender attribution in developmental schemes of agriculture across ministries which would result in formulation of a status report on mainstreaming gender in agriculture sector and in turn recommend reforms in identified government schemes for gender mainstreaming.

The five step framework for Gender budgeting framework considered for the study is as follows:

Figure 2.4.1: Five step framework for Gender budgeting

Table 2.4.1: **Gender budget allocation for different Ministries**

Programmes with 100% allocation for women (20 No.)		
Sl. No.	**Ministry**	**No. of Programmes**
1.	Ministry of Rural Development	02
2.	Ministry of Women and Child Development	17
3.	Ministry of Development of Northeastern Region	01

Programmes with 30% allocation for women (38 No.)		
Sl. No.	**Ministry**	**No. of Programmes**
1.	Ministry of Rural Development	02
2.	Ministry of Women and Child Development	10
3.	Ministry of Development of Northeastern Region	01
4.	Ministry of Agriculture & Farmers' Welfare	20
5.	Ministry of Fisheries, Animal Husbandry Dairying	02
6.	Ministry of Labour & Employment	03

The methodological framework adopted for selection of ministries is as follows:

Selection of 53 Ministries (Based on programmes for women related to agriculture). Narrowing down to 8 Ministries (Based on primary agricultural development programmes for women). (i) Ministry of Agriculture & Farmers' welfare. (ii) Ministry of Rural Development. (iii) Ministry of Food Processing Industries. (iv) Ministry of Environment, Forest & Climate change. (v) Ministry of Labour & Employment. (vi) Ministry of Fisheries, Animal husbandry and Dairying. (vii) Ministry of Development of North Eastern Region. (viii) Ministry of Cooperation.

2.4.2 Management and Impact Assessment of Farmer FIRST Project

This research project has been implemented since 2017 with funding support from Indian Council of Agricultural Research, New Delhi. ICAR - NAARM is a partner institution in the project management consortium, mandated to support management of Farmer First Project (FFP) through capacity building, documentation, information showcasing and impact assessment. During this period, Onsite field visits were made to FFP villages of ICAR-NIANP, Bengaluru; ICAR-CIFA, Bhubaneswar; ICAR-NRRI, Cuttack and SKUAST-Jammu. The project team conducted focused group discussions with farmers & FFP teams and documented good practices for scaling up. Project team facilitated scouting five farmers (from FFP villages) and awarded them with "Innovative Farmers' Award" on the occasion of 47th Foundation Day of ICAR-NAARM on 1st September 2022.

2.4.3 Impact Assessment of SC Sub Plan (SCSP) implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries

The SC population of our country is socially and economically very backward as compared to the other categories of population. About 71% of SC farmers are agricultural labourers.As a constituent of ICAR, NAARM has been implementing various activities and interventions under SCSP for the target beneficiaries from SC farmers, including youth and women in selected districts such as Nagar Kurnool and Mulugu districts in Telangana state by adopting the villages having at least '50% and above' SC population since 2019. Interventions in the form of providing Agri-inputs and imparting skill development programmes to enhance their income and livelihood. Any such intervention is expected to bring about the desired impact after a considerable period. More than 10 Skill development training programmes have been organised for the farmers and farm women such as Integrated farming system, Backyard Poultry, Jute bag making, bio inoculants, Soil testing, crafting in mango, exposure visit to the Centre of Excellence for horticulture, *etc.* Primary data were collected with the help of interview schedule on socio economic status of beneficiaries based on the selected parameters. Critical inputs were demonstrated among the sub plan beneficiaries. ICAR-National Academy of Agricultural Research Management, Hyderabad conducted a two-day National Workshop on "Pathways for Effective Implementation of SC Sub-Plan Scheme in ICAR" during 18-19th August, 2022 in hybrid mode. The objectives of the workshop were to deliberate on issues and challenges faced in effective implementation of SCSP across ICAR institutions and to identify consensus-based operational reforms and good practices for effective implementation of SCSP scheme. During the workshop, brainstorming sessions were held on three thematic areas namely; challenges faced in implementation, good practices, and operational reforms. The points emanated from the brainstorming sessions were synthesised in the plenary session held on 19th August, 2022. Out of the total 62 participants representing 21 ICAR institutes implementing SCSP, 34 attended in-person. A meeting was also convened with finance officials in headquarters so as to implement SCSP in a more effective manner. ICAR-NAARM has come out with recommendations after focussed deliberations and cross learnings from other ministries to evolve innovative reforms within the ambit of the prescribed administrative and financial procedures and guidelines for the effective implementation of SCSP in ICAR.

2.4.4 Doubling Farmers' Income with IFS: Training Organised for Farmers

The Academy organised a training programme on Doubling Farmers' Income with Integrated Farming Systems (IFS) under SCSP during Jan 4-6 and 10-12, 2022. The programme was carried out in collaboration with National Academy of Agricultural Sciences (NAAS), Hyderabad Chapter and Professor Jayashankar Telangana State Agricultural University at Krishi Vigyan Kendra, Palem, Nagarkurnool for the farmers of Vanguronipally, Kulmulonipally and Agraharam Potlaplly villages from two districts *i.e* Mahbubnagar and Nagarkurnool of Telangana State. Total 90 farmers were trained on the aspects related to Integrated Farming System for doubling the farmers' income. In view of the pandemic situation, the programme was conducted in blended mode consisting both online and in-person sessions.

Figure 2.4.2: Participants of Doubling Farmers' Income with IFS training programme

Dr Ch. Srinivasa Rao, Director participated as a Chief Guest and lucidly explained the farmers on how to double the income by using different Integrated Farming System technologies based on the availability of resources with the farmers. The farmers were also informed about the opportunities for natural farming and Dr Rao stressed on the need and the importance of this training programme for the farmers.

2.4.5 Winter School on Advances in Social Science Research and Evaluation

A 21-days online ICAR sponsored Winter School on Advances in Social Science Research and Evaluation was organized by the Academy during Jan 25 – Feb 14, 2022. Dr Ashok Kumar Singh,

Figure 2.4.3: Winter School on Advances in Social Science Research and Evaluation

Deputy Director General (Agricultural Extension), ICAR, New Delhi inaugurated the programme on Jan 25, 2022. This programme was organized with the core objective of imparting advanced knowledge and skills in conducting impact-oriented research and evaluation in social sciences in general and agricultural extension in particular. Forty-two participants representing NARS institutions from 15 States and one Union Territory participated.

2.4.6 Training on Bakery (Cakes, Muffins & bread) as an Enterprise for Rural Women

The Academy organized a training programme on Bakery (Cakes, Muffins & bread) as an Enterprise under SCSP in collaboration with Professor Jayashankar Telangana State Agricultural University (PJTSAU) at Rythu Vedika, Amrabad, Nagarkurnool for the rural women/ girls of Vanguronipally and Kulmulonipally villages during Feb 17-19, 2022. A total of 40 women farmers were trained on the aspects related to

Figure 2.4.4: Enterprise for Rural Women training programme

Bakery (Cakes, Muffins & bread) as an Enterprise. Dr M Balakrishnan, Chairman, Sub-Plan Committee, ICAR-NAARM, during his inaugural speech, briefed the rural women about SC Sub-Plan Programme and importance of training on bakery products for their livelihood. The programme was successfully completed under the guidance of Dr T. Supraja, Professor, Head of Department (Foods & Nutrition), PGRC, PJTSAU.

2.4.7 Training programme on Value addition of Fruits

The programme was organized during 24 – 26 February and 09 – 11 March, 2022 with 40 women farmers from the ICAR-NAARM adopted villages under sub-plan in collaboration with College of Community Science, PJTSAU at KVK- Palem and KVK-Adilabad, respectively. Dr Ch Srinivasa Rao, Director participated (On-line mode) as a

Figure 2.4.5: Training programme on Value addition of Fruits

Chief Guest and lucidly explained the farmers on how to double the income by using different horticultural crops based on the availability of resources with the farmers. The farmers were also informed on the opportunities for natural farming. Technologies for value addition of fruits, demonstration on 'fruit squash, introduction of ingredients- procedure- demonstration- do's & don'ts - nutritional values, hands on training on fruit squash preparation, practical training on 'preserves, fruit rolls, preparation of jam and role of food additives in preservation formed the core areas of training.

2.4.8 Skill Development Training for Women Farmers on Jute Bags

The Academy, in association with NAAS-Hyderabad Chapter organized a training programme on Enhancing Livelihood of Women

Farmers with Skill based training on Jute bag making under SCSP during Mar 16-20, 2022. The programme was carried out in collaboration with Krishi Vigyan Kendra, Ghantasala, Krishna (Dist), Andhra Pradesh for the farmers of Kuchhikayalapudi, Ramanapudi, Ghantasala and Thadepalli villages from Krishna districts. A total of 32 women farmers were trained on the aspects related to jute bag making.

Dr K Jhansi, Programme Coordinator, KVK, Ghantasala highlighted the role of women farmers in the nation building and sensitized them towards uses of jute fibre bags instead of plastic bags. Shri V Asha Kiran, IMC and RAC member, ICAR-NAARM, graced the programme as guest of honour, appreciating the efforts of the Academy, called upon the farmers to make use of this training programme. Delivering the inaugural address as a Chief Guest, Dr Ch Srinivasa Rao, Director and Convener, NAAS-Hyderabad Chapter distinctly explained about the harmful effects of usage of plastic bags and plastic materials, its effects on soil health, human health and as well as environmental pollution. Delivering the valedictory address as Chief Guest, Dr T Neeraja, Dean of Community Science, enlightened the trainees towards making different kinds of jute gunny bags as an income generation activity apart from their regular agricultural activity.

Figure 2.4.6: Skill Development Training for Women Farmers on Jute Bags

Dr G. Venkateshwarlu, Joint Director presided over the valedictory programme and spoke on the remarkable progress made in agriculture by women farmers and also the need to preserve the natural resources by using all the biodegradable materials. He also highlighted that jute bag making can boost up family income as it involves low cost investment but gives higher economic returns.

2.4.9 Extension Programme Planning – A Systems Perspective

The programme was organized during March 22 – 26 March, 2022 with 24 participants including 12 female participants. The objectives of the programme were (i) To enrich the participants with adequate knowledge on extension programme and programme planning. (ii) To teach the participants on the preparation of simple and manageable extension projects. (iii) To train the participants on monitoring and evaluation mechanisms in extension programme planning. The programme was implemented through a total of 16 sessions including lectures and practical sessions by eminent experts on extension programme planning. The course provided a comprehensive understanding of the project management cycle in extension programme planning including the various mechanisms involved in monitoring and evaluation. There were a total of 24 participants from various states and union territories *viz.*, Uttar Pradesh (1), Karnataka (4), Tamilnadu (1), Delhi (2), Punjab (2), Puducherry (5), Rajasthan (3), Maharashtra (3), Telangana (2) and Uttarakhand (1) representing ICAR, State Agricultural Universities, KVKs, Research scholars and students who virtually participated in the training programme.

2.4.10 Workshop on Mainstreaming Farmers' Innovations in Plantation Crops

A workshop on Mainstreaming Farmers' Innovations in Plantation Crops was organized by the Academy at IIPM, Bangalore during Jun 10-11, 2022. The objective of the workshop was to create a network among the farmer-innovators and scope for commercialization and upscaling. A total of 15 farmer-innovators from Karnataka, Kerala, Tamil Nadu and Maharashtra participated in the workshop. The invited innovators presented their innovations' present status in commercialization and their expectations on upscaling of the innovations. Three innovators demonstrated the innovations and received appreciation from others. The procedures for patent applications and varietal protection were delivered by IP experts. The existing startup support schemes and programmes of GoI were also presented. There was an experience sharing session with one of the startups namely; AgVerse Technologies Pvt. Ltd., Bangalore. Dr G Venkateshwarlu, Joint Director, ICAR-NAARM and Dr Sudha Mysore, CEO, Agrinnovate India Ltd., ICAR, New Delhi addressed the participants.

Figure 2.4.7: Participants of Workshop on Mainstreaming Farmers' Innovations in Plantation Crops

2.4.11 Skill development Training on Soil testing

The programme was organized on 27, July 2022 with 40 participants including 5 women participants. As a part of the SC Sub-Plan implementation, ICAR-NAARM organized one-day training programme on Soil testing in collaboration with KVK Yagantipalle, AP on 27.07.2022 for the farmers of AP, to create an awareness on the importance of soil testing and soil health for crop productivity enhancement. The training programme covered importance of soil health, critical role of crop residue addition, locally available organic materials to recycle into the soil, mulch cum manuring and importance of soil organic carbon in improving use efficiency of added fertilizer nutrients besides drought mitigation, soil sampling procedure, testing and interpretation of data. Soil health cards were also distributed with the fertilizer recommendations. Around 40 farmers from seven villages of district Kurnool, AP participated in the programme.

2.4.12 National Workshop on "Pathways for Effective Implementation of SC Sub-Plan Scheme in ICAR"

The programme was organized on August 18-19, 2022 with 20 participants including 10 women participants at ICAR-National Academy of Agricultural Research Management, Hyderabad in Hybrid mode. The objectives of the workshop were to deliberate on issues and challenges faced in effective implementation of SCSP across ICAR institutions and to identify consensus-based operational reforms and good practices for effective implementation of SCSP scheme. In the workshop, brainstorming sessions were held on three thematic areas namely; challenges faced in implementation; good practices; and operational reforms. The points emanated from the brainstorming sessions were synthesized in the plenary session held on 19th August 2022. A total of 62 participants representing 21 ICAR institutes implementing SCSP attended the programme of which 34 attended in-person.

2.4.13 Skill development training programme on "Management of Horticulture Fruit Crops & Nurseries"

The Academy organized a three-day training programme on Management of Horticulture & Nurseries under SCSP in collaboration with Krishi Vigyan Kendra, Ramagirikhilla, Peddapalli (Dist) affiliated to Sri Konda Laxman Telangana State Horticultural University in JNTUH College, Ramagirikhilla during Sep 3-5, 2022. A total of 205 farmers were trained including 102 women farmers on the aspects related to Management of Horticulture & Nurseries.

During the inaugural programme, Dr A Kiran Kumar, Director of Extension, SKLTSHU highlighted the present status of production and productivity of horticulture fruit crops in our country. He also emphasized the good practices for management of horticultural nurseries for higher production of fruit crops. Shri. Gone Shyamsunder Rao, RAC & IMC Member, ICAR-NAARM stressed upon the need to develop skills for becoming an entrepreneur and briefed about how to develop a market linkage, appropriate packaging and storage methods for value added fruit products.

Delivering the valedictory address, Dr Sridhar Reddy, Vice-Principal, JNTUH, Ramagirikhilla urged all the participants to become a job provider by establishing different horticulture based enterprises. Dr A Srinivas, Head & Programme Co-ordinator, KVK, RGK was also present.

Figure 2.4.8: Skill development training programme on Management of Horticulture Fruit Crops & Nurseries

2.5 Mobilizing Science and Technology for Innovation and Sustainable Development

Research in agriculture and allied sectors brings out technologies that help and sustain the national agricultural research system. Therefore, vibrant research systems ensure the relevance of research institutions. Imparting innovative ideas and bringing in a new milieu in the agricultural research system through the integration of advancement in information technologies plays a critical role in better governance and online management by developing digital products and services, which will greatly improve decision-making and reduce the redundancy in research. Besides generating and disseminating the same to the stakeholders, ensuring appropriate management of agricultural technology through the application of management fundamentals for efficient management of research outputs becomes essential. Further management of Intellectual Property also takes center stage for gaining a competitive advantage. In order to achieve the objectives under this theme, the Division of Research Systems Management has carved out the following three niche areas in which the mandated activities of research, academic and capacity-building activities are carried out: (1) Research Management and Policy Advocacy; (2) Technology and Innovation Management; (3) Research in the context of Food Security, Sustainability and Globalization.

2.5.1 Strategies for Strengthening National Water Policy

The study objectives are (i) To critically understand the existing water policy ecosystem in India. (ii) To identify the issues in the implementation of the water policy. (iii) To suggest needful revisions in the present policy especially w.r.t. water for irrigation. Brought out a draft policy paper by organizing a brainstorming workshop the draft paper was reviewed by three experts and the revised draft is under finalization. Some of the major recommendations of this policy paper are (i) Water Budgeting at the Panchayat level to be made mandatory for development plans approval and implementation. (ii) Water conservation is made as a movement like Clean India Mission (*Swachh Bharat Abhiyan*), a health campaign, etc. (iii) Water auditing and awards and rewards for panchayats and other local bodies to be done on an annual basis, where assessment to be more in a social audit manner by the neighbouring communities and other stakeholders. (iv) Promote more micro-irrigation rather than medium and macro irrigation projects. (v) Involve corporates in funding irrigation and water conservation projects through their Corporate Social Responsibility (CSR) funds. (vi) Introduce the Indian Environment Service (IEnS) with an emphasis on water, climate, and other related environmental issues. The IEnS cadre is to be posted at the district level. A dedicated District Water Management Agency (DWMA) that was introduced in the state of Andhra Pradesh to be started in all states, and this agency be made responsible for all water management in the district, rather than making it work only for rainfed areas and watershed project implementation

2.5.2 Economic Analysis of Food prices and Policies for the Development of Indian Agriculture

Estimated effects of trade price transmission on major Indian agricultural crops such as cereals, pulses, and oilseeds and published in World Food policy. The research findings have important implications to be considered in designing agricultural price policies and programmes to improve farmer-to-market linkages besides enhancing/ doubling the farmer's income. The study findings have important inferences to be considered in designing agricultural price policies and schemes to protect domestic producers, to boost exports and foreign earnings of agricultural commodities in India. The study guides producers, exporters, and importers for price signals of agricultural commodities in the market. Indian welfare is possible only through the farmers' welfare. If farmers are empowered, the nation would become self-sufficient. The study strongly suggests that price insurance and price loss programmes as well as direct income approaches like developed nations may be implemented for Indian Farmer's welfare. The findings can be useful to the government in designing price fixing mechanisms and monetary policy, distortion of prices, and control of inflation.

2.5.3 Dynamics of Agriculture and Food Systems in Different Agroecosystems of India

Planning to organize a workshop to develop a conceptual methodology for assessing agri-food system in India

2.5.4 IP Management and Transfer/ Commercialization of Agricultural Technologies Scheme

Copyrights Filed are 7. Trademarks Filed are 2. IP support is provided to other ICAR institutes such as ICAR-IIOR, ICAR-IISWC, and BIRAC BIG Grantees. Hand holding to a-IDEA incubates and startups. Organized one capacity-building programme for scientists/faculty involved in agricultural research

2.5.5 Profiling Resources Devoted to R&D in Agriculture Sector

Under this project, as a part of collaborative research undertaken by NAARM with the Department of Science and Technology, New Delhi International Food Policy Research Institute (IFPRI), Washington the data pertaining to (i) human resources deployed (ii) financial resources used (iii) research focus and (iv) research outputs were collected from all ICAR institutions and Agricultural Universities for the period from 2018-19 till 2020-21. So far, collected 70 percent of the data and uploaded it in the DST and NAARM Data Drive. An interactive Dashboard to draw ready inferences from the datasets was developed. The GoI has released the S&T Indicators tables and the R&D Statistics 2020-21, wherein the data collected by NAARM for the agricultural sector have been integrated. Through this project, NAARM was part of a national initiative for collecting S&T data for the farm sector and thus, aid in evidence-based policy-making in India. The Academy also developed a prototype for the Dashboard jointly with one of the IT firms empaneled with NAARM, to aid in data-driven decision-making in the domain of agricultural research.

2.5.6 Developing and Pilot Testing Specialized Knowledge Products on Human Wildlife Conflict (HWC) Mitigation for Agriculture and Veterinary Sector in India.

In order to accomplish the objectives of the project, various activities were carried out during the reporting period. Field Missions to Kodagu and Gorumara Landscapes: The ICAR-NAARM-GIZ team organized field visits to panchayats in the Kodagu district of Karnataka for interaction with farmers and pre-testing data collection instruments for the proposed desk study on crop/livestock loss assessment methodology and vulnerability assessment.

The field mission to Gorumara Forest Landscape, West Bengal was organized to develop, pilot test and understand local perspectives and perceptions of various stakeholders on different aspects of HWC mitigation and validate the checklist prepared for data collection. Conducted Focus Group Discussions (FGDs) at 7-range RRTs and 8 gram panchayats with Forest and Panchayat Officials, Rapid Response Teams (RRTs), farmers, and villagers.

These field missions resulted in understanding the different mitigation measures adopted, the existing cropping pattern, the value chain analysis of different commodities, and the potential markets of the alternate crops. Trainings conducted and participated by ICAR-NAARM-GIZ: Team participated and also organized training programmes at all three landscapes where the project has been working. Participated in the National Workshop to finalize the structure and contributors for the Implementer's Toolkits/HWC-NAP and Guidelines in New Delhi at the Ministry of Environment, Forest and Climate Change (MoEF&CC) jointly by MoEFCC and GIZ under the Indo-German project on 'Human-Wildlife Conflict Mitigation' in India. The team participated in a four-day 'GIZ-Dale Carnegie training of trainers (ToT)' on the effective delivery of trainings using participatory methods.

Conducted training sessions on 'Holistic Approach to Human-Wildlife Conflict Mitigation" for Gram Panchayats and Potential Members of Community-level Primary Response Teams (PRT) at Kodagu, Gorumara, and Haridwar-Rajaji landscapes. Introduced modules of one health and HWC mitigation 'Taking a One Health Approach to Human Wildlife Conflict Mitigation' in the CBP of the Academy senior officials including ADGs and Directors of the institutions under ICAR. This resulted in concrete ideas towards increased forest-agriculture-veterinary cooperation on issues like the national dialogue on One Health, engagement of Krishi Vigyan Kendras for HWC mitigation awareness, integration curriculum on HWC and One Health into the curriculum of undergraduate students of agriculture universities.

Identified training needs at different landscapes need for introducing alternate crops and options of value additions and improving livelihoods. Case studies and good practices of HWC mitigation measures following the species guidelines under the HWC-National Action Plan are documented. Business plans for alternate crops such as jute, lemongrass, citronella, dragon fruit, etc. are identified and are being developed.

2.5.7 Launching Workshop on Human Wildlife Conflict Mitigation for Agriculture Sector

The Academy organized a virtual launch workshop on January 24, 2022 for the multi-institutional, transdisciplinary project on "Developing and Pilot Testing Specialized Knowledge Products

Figure 2.5.1: Launching Workshop - Human Wildlife Conflict Mitigation for Agriculture Sector

on Human Wildlife Conflict Mitigation for Agriculture and Veterinary Sector in India" implemented in technical collaboration with GIZ, German Agency for International Cooperation. The project is targeted to develop and pilot test knowledge products including policy briefs, training materials, and research survey report to facilitate cross-sector cooperation on human-wildlife conflict mitigation between the forest and agriculture sector in India.

Dr. Ch. Srinivasa Rao, Director, in his presidential address, explained the functional networks of the Academy with diverse stakeholders that enhance the reach of the knowledge products developed under this project. Dr. Neeraj Khera, Team Leader- Human Wildlife Conflict Mitigation Project, GIZ-India explained the progress already made under the national project and the specific outcomes expected from the ICAR-NAARM-GIZ study component. Dr. G. Venkateshwarlu, Joint Director, expressed hope that this project will serve as a platform to demonstrate the spirit of collaboration among diverse stakeholders. Dr. SB. Barbuddhe, Director, ICAR-NRC-Meat pointed out the importance of one health being undertaken under this project.

2.5.8 MDP on Intellectual Property Valuation and Technology Management

The Academy, in association with a-IDEA (TBI of ICAR-NAARM) conducted an online Management Development Program on Intellectual Property Valuation and Technology Management during January 18-22, 2022. The participants included 20 scientists/ faculty from ICAR institutes and State Agriculture Universities. The training program covered various aspects of IPRs, IP valuation, and technology management viz., entrepreneurship and grass root innovations, technology management strategies, risk

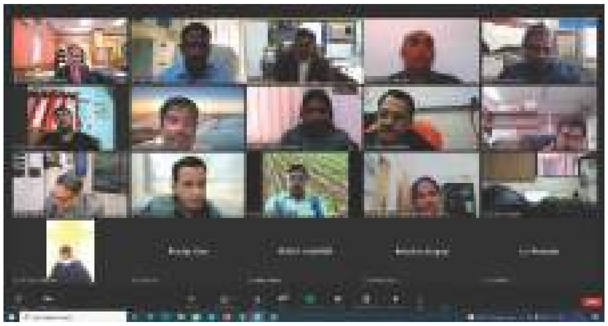


Figure 2.5.2: Valedictory Program of MDP on Intellectual Property Valuation and Technology Management

assessment, business planning, etc., including current practices of innovation management in the ICAR system. The training also dealt with the other aspects of Intellectual Property like copyright issues in academic research, IP informatics, valuation of Intellectual Property, potentials of Geographical Indications, etc.

Dr. Ch. Srinivasa Rao, Director, while addressing the valedictory function, highlighted on the importance of innovations and technology management in research organizations and impressed upon the participants to disseminate and implement the learnings from this training.

2.5.9 Design Thinking in Research and Education for faculty of SKLTSHU, Rajendra Nagar

The programme was organized during February 15-19, 2022 with 15 participants including 09 female participants. The program was sponsored by the College of Horticulture, Sri Konda Laxman Telangana State Horticultural University, Hyderabad. Fifteen young faculty members and research scholars of the college participated in the program. The major objectives of the program were to build design thinking perspectives among the researchers and educators and develop competency for the implementation of the research idea and delivery of education with empathy, involving clients and key stakeholders.

In his valedictory address, Dr. JC Katyal, Former DDG (Education), ICAR threw light on various aspects of design thinking to translate innovation to action with validation through the active participation of stakeholders. Dr. Ch. Srinivasa Rao, Director, in his presidential address during the valedictory session, spoke on the efforts of ICAR-NAARM in enhancing the quality of research and education and advised participants to take their career as an opportunity to serve in farming.

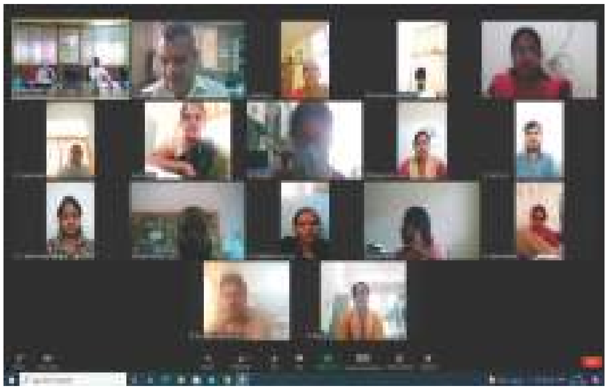


Figure 2.5.3: Dr. JC Katyal, Former DDG (Education) addressing the Valedictory in the Program on Design Thinking in Research and Education for faculty of SKLTSHU

2.5.10 Recent Advances in Organic Farming Research

The programme was organized during February 22 – 26, 2022 with 49 participants including 15 female participants, to give an overview of recent innovative approaches and strategies followed in the organic production system and to appraise the different organic production technologies towards higher profitability and sustainable production.

Figure 2.5.4: Dr.Ch.Srinivasa Rao, Director, ICAR-NAARM addressing in the program on Recent Advances in Organic Farming Research

2.5.11 Management Development Programme (MDP) on Design Thinking in Agricultural Research and Education

The programme was organized during May 23-28, 2022 with 31 participants including 09 female participants. The major objectives of the programme were to 1) Build design thinking perspectives among the researchers and educators, and 2) Develop competency for the implementation of the research idea and delivery of education with empathy, involving clients and key stakeholders.

2.5.12 Project Management & Research Methodology

The programme was organized during July 07 – 18, 2022 with 26 participants including 8 female participants. The online training programme was organized for newly recruited/promoted scientists of the Indian Council of Forest Research and Education (ICFRE), Dehradun. The programme dealt with various aspects of project management. It was designed in two

Figure 2.5.5: Dr.Ch.Srinivasa Rao, Director addressing in the program on Project Management & Research Methodology for newly recruited ICFRE scientists

sections of 5 days each covering topics on 'Project Management & Science Communication' and 'Research Methodology'. The first module covered various aspects of project management and communication skills viz., design thinking, building research concepts, monitoring and evaluation terminologies, etc. The second module dealt with research methodology and various statistical concepts viz., correlation, regression, multivariate analysis, and cluster analysis were covered beside the detailed practical sessions on the design of experiments.

2.5.13 Intellectual Property Valuation and Technology Management

The programme was organized during July 17 – 22, 2022 with 17 participants including 2 female participants. The main objectives of the programme include raising the level of awareness and knowledge about IPR issues and the importance of integrating IPR in academic research plans. The programme also intended to develop an understanding of the valuation of technological assets and various aspects of technology commercialization. The programme included various sessions on different aspects of intellectual property, technology commercialization, technology valuation, innovation management, and new product development. A practical session on patent search using various databases was also conducted.

Figure 2.5.6: Dr.Ch.Srinivasa Rao, Director addressing trainees of Intellectual Property Valuation and Technology Management Program

2.5.14 Training Programme on 'Developing Winning Research Proposals'

The programme was organized during July 19 – 21, 2022 with 32 participants including 18 female participants. The training programme was conducted for the faculty of PJTSAU, Hyderabad. The main objective of the programme was to significantly raise the level of knowledge among academicians and researchers about writing quality research proposals. Various theoretical and practical sessions on the subject to cover the various aspects of research proposal writing were conducted. Speakers delivered the sessions on topics including elements of the research proposal, design thinking, building research concepts, use of project log frame writer, M&E indicators, tips on writing winning proposals, etc. The programme included practical sessions also on writing project proposals using Research Concept Writer.

Figure 2.5.7: Faculty of PJTSAU in the Training Programme on 'Developing Winning Research Proposals

2.5.15 Training Program on Developing Winning Research Proposals (DWRP)

The programme was organized during August 17- 19, 2022 with 30 participants including 15 female participants. Three-day training program on DWRP was organized for the Faculty of PJTSAU, Telangana. Thirty faculty representing Assistant and Associate Professors attended. The session on Project Management Techniques, Research Concepts, Elements of Research Proposal, Use of Log Frame Writer, and Design Thinking were covered besides practical presentations on new project proposals by the participants.

Figure 2.5.8: Faculty of PJTSAU in DWRP

2.5.16 Training on Developing Winning Research Proposals (for the faculty of College of Horticulture, Rajendranagar Sri Konda Laxman Telangana State Horticultural University, Hyderabad) (Online)

This programme was organized during September 05-09, 2022 with 15 participants including 09 female participants. The program aimed at developing research proposal writing skills focusing on good project design, rationality, and stakeholder needs. The other aim was also to explain and give hands-on experience in project log-frame, priority setting, monitoring & evaluation (M&E) through specially designed web-based open access software.

2.5.17 Management Development Programme on Developing Winning Research Proposals

The programme was organized during September 12-17, 2022 with 33 participants including 09 female participants. The programme objectives were to develop research proposal writing skills focusing on good project design, rationality, and stakeholder needs and to explain and practice project log-frame, priority setting, monitoring & evaluation (M&E) through specially designed web-based open access software.

Figure 2.5.9: Trainees of MDP on Developing Winning Research Proposals

2.5.18 Agricultural System Modelling for Natural Resource Management

The programme was organized during September 19 – 24, 2022 with 16 participants including 03 female participants. This program was aimed at giving an overview of recent modeling approaches and strategies followed in agriculture, providing hands-on learning of different crop and soil models and the use of crop different crop models in natural resource management and agriculture sustainability.

Figure 2.5.10: Trainees of Agricultural System Modelling for Natural Resource Management

2.6 Information and Communication Management for Promoting Innovation and Governance

The Open Data policy of the Government of India calls for data sharing for promoting innovation. Opening up Data from Government sources for the public good has resulted in newer innovative solutions which are easily adopted and spread thanks to ICT tools. For a Research Organization, managing research data from its creation to archiving for future use has become of paramount importance to use it effectively and efficiently to answer increasingly complex research questions which require data from multiple sources. Research Data Management calls for ease of data sharing within as well as between organizations. ICAR Research Data Management policy envisages managing data at the individual or at project team level initially and storage and dissemination through centralized repositories. Another off-shoot of the implementation of the policy is monitoring and better governance both at the institutional and organizational levels.

The use of the Internet of Things (IoT) and artificial intelligence have shown tremendous potential for farming and are steps towards digital agriculture. Developing tools for extracting information from data through advanced statistical tools is in constant demand from agricultural scientists and Academy's thrust is also in this direction. Geospatial analysis can provide a better understanding of the use and management of natural resources such as groundwater. Skill development in Bioinformatics is another area to meet the requirements of different stakeholders ranging from research scientists to students.

Thus to promote innovation in research data management and better governance and provide necessary tools and skills, Academy is conducting capacity-building programmes in data analytics, Bioinformatics, Geo-Spatial Analysis, artificial intelligence, etc. The research programmes of the division are also aimed to achieve these broad objectives.

2.6.1 Mega Environment Analysis Tools for AICRP Trials

The project envisages developing analytical tools for AICRP Trials. During the previous year, 50 trials data were identified for developing prediction models for individual genotypes by incorporation of location and environment effects. Using growing degree models for phonological traits such as days to 50% flowering were estimated. Based on the available weather information, environment indices are being calculated and tested for possible use in the model. These include average temperature during the growing season, vegetation indices for nearby areas, etc. The environment indices are rescaled from 0 to 1 with 0 as the worst site and 1 as the best site. The comparisons of different environment indices are under study.

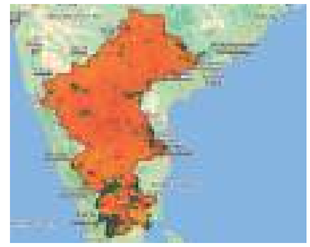
2.6.2 Comprehensive Geo-Spatial Analysis of Environmental Change in Farming

High-resolution satellite imageries of Greenhouse gases emissions are available and can be processed through various tools. The available data on agricultural activities and other activities along with remote sensing imageries can be used to estimate Greenhouse gases. The objectives of this project is to identify and prepare a pertinent agriculture spatial database for the study area of Agro-Climatic Zone-X and to apply geospatial analytics to estimate atmospheric greenhouse gases emissions and environmental

Jan-Jun 2019
Image Count: 2534

July-Dec 2019
Image Count: 2572

Jan-Jun 2020
Image Count: 2308



July-Dec 2020
Image Count: 2531

Jan-Jun 2021
Image Count: 2513

July-Dec 2021
Image Count: 2503

Jan-Jun 2022

Figure 2.6.1: Methane Concentrations observed in the study area

pollutions from secondary sources and remote sensing imageries and the impact of crop pattern and productivity. It is expected that suitable strategies for climate-resilient agriculture can be found based on these estimates.

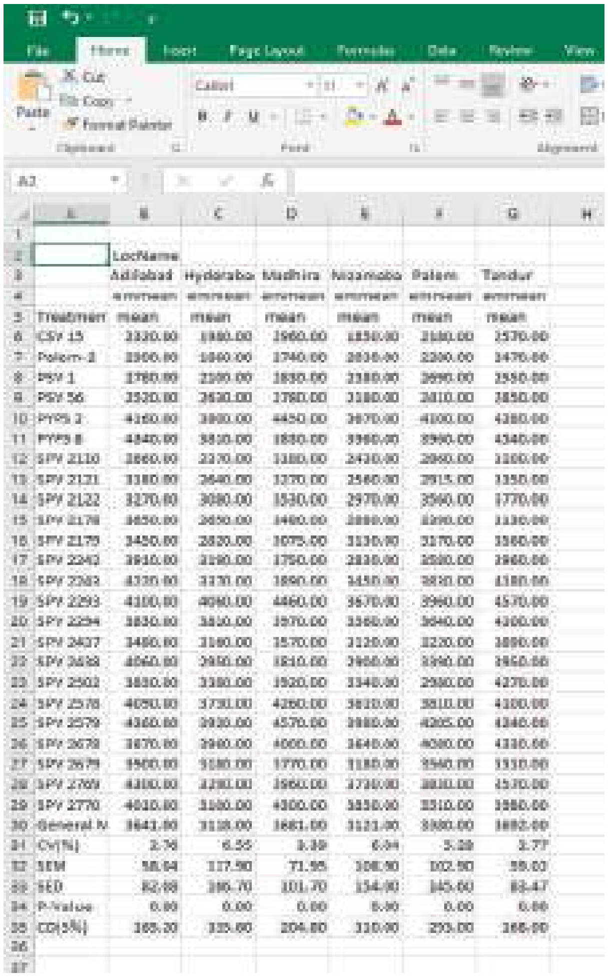
Sentinel-5P OFFL CH4 dataset provides offline high-resolution imagery of methane concentrations. These datasets are processed on the google earth engine platform for extracting the six-month cumulative values of methane emission for the study area of Agro-Climatic Zone-X. Some of the processed images are shown here. Further analysis is in progress.

2.6.3 Bioinformatics analysis for Identification and characterization of superior haplotypes for important traits in field crops

The whole genome sequencing of many plant genomes resulted in massive amounts of sequence data that are now available in public databases. Processing and interpreting huge genomic datasets, as well as getting functional insights into plant genomes is very crucial. Rice (*Oryza sativa* L.) is the major staple food for over and above half of the world's population. The sequencing data creates a "digital library" of the 3,000 accessions, allowing natural genetic differences in rice to be identified. Despite the fact that several high-yielding rice varieties are developed each year, the yield potential of these cultivars is strongly influenced by biotic and abiotic stressors. Heat stress and low P tolerance have become severe hazards to rice production among many abiotic stresses. The project is aimed to characterize superior haplotypes using 3K rice genome panel and to develop haplotype-specific molecular markers.

2.6.4 KRISHI- ICAR Research Data Repository for Knowledge Management

The project is aimed to operationalize ICAR data management policy and the targets assigned during the current year analyzing the use of different quality checks for AICRP data and implementing the same in the AICRP Information system; creation of off-line analysis



	LocName					
	Adilabad	Hyderabo	Madhira	Nizamabo	Palem	Tandur
	amnmean	amnmean	amnmean	amnmean	amnmean	amnmean
Treatmen	mean	mean	mean	mean	mean	mean
CSV 15	3320.00	1980.00	1960.00	1830.00	2180.00	2570.00
Polom-2	2900.00	1860.00	1740.00	2030.00	2200.00	3470.00
PSV 1	2780.00	2100.00	1830.00	2380.00	2690.00	2550.00
PSV 56	3520.00	2630.00	1780.00	3180.00	3410.00	3850.00
PYPS 2	4160.00	3800.00	4450.00	3670.00	4100.00	4280.00
PYPS 8	4840.00	3810.00	1830.00	3960.00	3990.00	4340.00
SPV 2110	2860.00	2270.00	1180.00	2430.00	2960.00	3200.00
SPV 2121	3180.00	2640.00	1270.00	2540.00	2915.00	3350.00
SPV 2122	3270.00	3080.00	1530.00	2970.00	3560.00	3770.00
SPV 2178	3650.00	2650.00	1460.00	2880.00	3290.00	3130.00
SPV 2179	3450.00	2820.00	1075.00	3130.00	3170.00	3560.00
SPV 2242	3910.00	3190.00	1750.00	2830.00	3580.00	3960.00
SPV 2383	4320.00	3170.00	1890.00	3450.00	3830.00	4180.00
SPV 2293	4100.00	4080.00	4460.00	3670.00	3990.00	4570.00
SPV 2294	3830.00	3810.00	3570.00	3360.00	3640.00	4200.00
SPV 2437	3480.00	3180.00	1570.00	3120.00	3220.00	3890.00
SPV 2438	4060.00	2950.00	1810.00	2900.00	3390.00	3950.00
SPV 2502	3830.00	3380.00	1520.00	3340.00	2980.00	4270.00
SPV 2578	4050.00	3730.00	4260.00	3810.00	3810.00	4100.00
SPV 2579	4360.00	2930.00	4570.00	3980.00	4205.00	4340.00
SPV 2678	3870.00	3940.00	4000.00	3640.00	4080.00	4330.00
SPV 2679	3900.00	3180.00	1770.00	3180.00	3540.00	3310.00
SPV 2769	4300.00	3290.00	3960.00	3730.00	3830.00	4570.00
SPV 2770	4010.00	3160.00	4300.00	3850.00	3310.00	3980.00
General M	3641.00	3118.00	1681.00	3121.00	3380.00	3882.00
CV(%)	2.76	6.55	3.39	6.04	5.28	2.77
SEM	58.64	117.90	71.95	108.90	102.90	59.02
SED	82.88	166.70	101.70	154.00	145.60	83.47
P-Value	0.00	0.00	0.00	0.00	0.00	0.00
CD(5%)	165.20	335.60	204.60	310.00	293.00	168.00

Figure 2.6.2: A snapshot of the result table

tools to enable researchers for in-depth analysis. At present in AICRP Trials, to assess the quality of data of individual centers, Coefficient of Variation (CV) based criteria and in some cases, General Mean based criteria are used. For Yield traits, whenever an individual center CV exceeds 20% (or as fixed), the center data is not considered for estimating the zonal/all-India means. Similarly, the individual center General Mean falls below the long-term average of center/zone, and the center data is not considered. However, based on the available literature, the use of heritability of different traits at individual centers is an alternative to assess the quality of data. The heritability of individual traits at each center can be estimated assuming the genotype effect is random. The heritability values of those traits which are not affected by the environment are expected to be high and this can be used to assess the quality of the data. The same has been incorporated in the SAS Macro used in AICRP analysis and has been tested. Besides, as a test feature, bi-plots of replication vs yield traits at individual centers have been produced in offline mode. Any deviations among the replications in the bi-plot analysis show quality issues in that locations. The same facility is expected to be incorporated into the AICRP Analysis module after testing.

Apart from automated analysis of individual trials, many off-line analysis, particularly pooled analysis over years can be undertaken for in-depth analysis. To facilitate such analysis, R-functions have been developed. The same functions can be used for analyzing past data or any experiments conducted over multiple environments or seasons or both. The first R function, METImport() can import multiple Excel files saved in a particular folder and convert them to the format required for further analysis. The next function Location_Analysis() provides a tabular report after analyzing the individual centers data for all traits. The tabular report includes zone-wise, and location-wise treatment means and associated statistics including P-value, CD, SEM, etc. for all the traits in an excel file. The function also saves the data in the R dataset for pooled analysis. The third function, L2 Analysis() provides the pooled means and associated statistics assuming centers within zones have different Mean Square Errors. The tabular report combining these two outputs in a single sheet is currently under development. These two functions were shared with the participants of the workshop on the Analysis of Multi-Environment Trials conducted by the Academy.

2.6.5 Establishment of Skill Development in Bioinformatics infrastructure facility (SKILL-BIF) on Agricultural Biotechnology at NAARM

The DBT-funded project is aimed at imparting skills in Bioinformatics to agricultural researchers and students through training programmes and hands-on workshops. Apart from the training activities, studies in collaboration with other ICAR-Institute scientists are also undertaken. During the year 2022, one training programme on "Application of Bioinformatics in Agricultural Research and Education" from November 15-24, 2022 was conducted at SKILL-BIF, ICAR-NAARM. A workshop on "Application of Bioinformatics in Agricultural Research and Education" from 14-December 16, 2022 was also organized at SKILL-BIF, ICAR-NAARM. The transcriptomic analysis of *Gallus gallus domesticus* (chickens) mechanism controlling the expression of genes involved in ovarian development and differentiation has been completed in collaboration with ICAR-Directorate of Poultry Research. The study

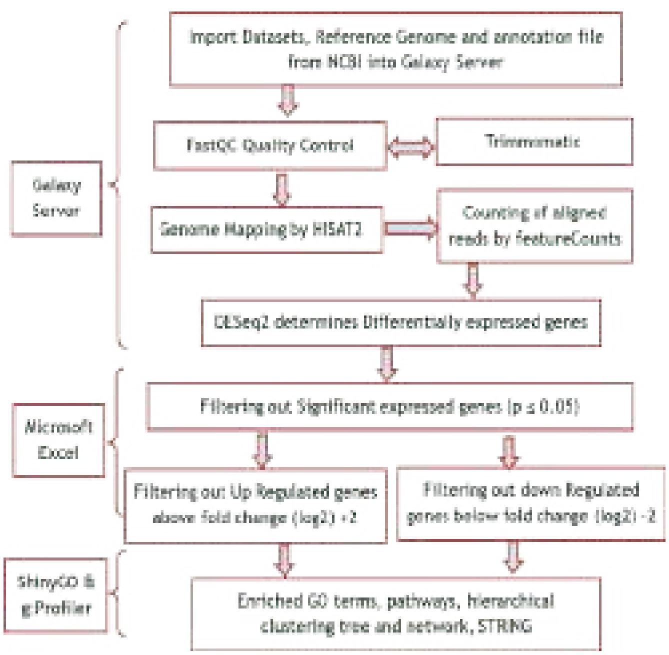


Figure 2.6.3: Schematic representation of Transcriptomic analysis workflow

identified genes responsible for the regression of chicken right ovary during the embryonic and post-hatch period. The datasets available from the public domain (https://www.ncbi.nlm.nih.gov/) were used for studying the differential gene expression in the chicken right ovary during Embryonic 6th-day (E6), Embryonic 12th-day (E12) and Post hatch 1st day (D1). The complete study was carried out using a web-based Galaxy server (https://usegalaxy.org/), ShinyGO 0.76 (http://bioinformatics.sdstate.edu/go/), and g:Profiler (https://biit.cs.ut.ee/gprofiler/gost) to perform the differential gene expression and functional enrichment analysis. The procedure includes a quality check of the raw data, trimming of poor-quality reads, mapping to *Gallus gallus* reference genome (GRCg7b), quantification of transcripts, and statistical testing for differentially expressed genes. The differentially expressed genes are further processed for enrichment analysis. In addition, Genome-wide identification of lncRNAs in plants and the development of the Bioinformatics Resources Portal have been completed; database development on SSR markers of Chicken and Duck is in progress.

2.6.6 Training Programme on Impactful ICT Applications and Technologies in Agriculture (Online mode)

The programme was organized during January 03-07, 2022 with 20 participants including 05 female participants. The objective of the programme is (i) To apprise the scientist trainees about existing ICT Applications by Startups, Cloud Computing, Artificial Intelligence, and IoT tools and technologies for Agriculture. (ii) To demonstrate various ICT technologies which help in rapid application development by participants. The course was designed to sensitize

Figure 2.6.4: Valedictory programme

and apprise the NARES Researchers/Faculty on the use of Information and Communication Technology (ICTs) Specifically Artificial Intelligence technologies in their research domain. All the resources required for the programme were shared in Online mode, which includes an ecosystem of digital resources required for the hands-on practical session as well. The programme was designed into 2 sections of 2 & 3 days each which covers 1). Concepts of ICTs, Artificial Intelligence, Machine Learning, Deep Learning & IoT 2). Presentation by various Startups explaining about their digital platforms that are currently implemented for various stakeholders. The programme is attended by 20 participants from 13 states and belongs to 12 agricultural disciplines.

2.6.7 Training Programme on "Analysis of Experimental Data using R" (Online mode)

The programme was organized during January 17-22, 2022 with 132 participants including 28 female participants. The objective of the training programme was to orient the researchers with the basics of statistical tools and techniques mostly utilized in agricultural research.

Participants were exposed to various advanced analytical techniques and hands-on training for the analysis of data using open-source, powerful and popular R software. The topics covered include "Principle of Experiments for Conducting Research; Exploratory Data Analysis; Introduction to R and Data Management, Correlation, Regression and Diagnostics; Analysis of Experimental Data using Basic as well as advanced designs, Pooled analysis, Multivariate Data Analysis using Principal Component Analysis (PCA), Response Surface. Emphasis was given to the analysis of data using appropriate statistical techniques and interpretation of the results. Data from real-life studies and simulated data were utilized to explain various statistical techniques. Trainees were provided with necessary datasets to analyze by appropriate techniques they have learned. Sufficient time was given to the participants to discuss specific issues/problems. Since the programme was conducted online, the softcopy of the presentations of lectures by various faculty and day-wise recordings on various topics as per the timetable was prepared consisting of lecture materials, hand-outs, datasets, and codes were provided to all the participants.

2.6.8 Training Programme on 'Geo-Spatial Analysis using QGIS & R' (Online mode)

The programme was organized on February 14-19, 2022 with 45 participants including 12 female participants. During the programme,

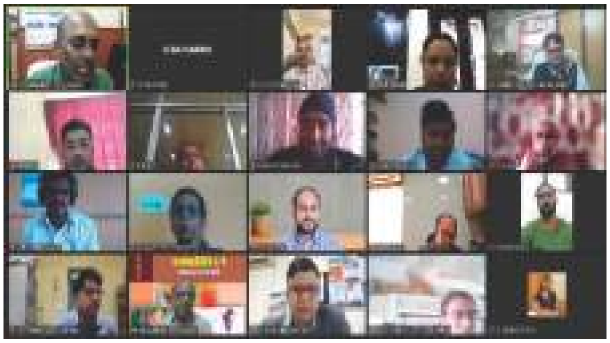


Figure 2.6.5: A snapshot of the virtual training session in the Training Programme on 'Geo-Spatial Analysis using QGIS & R'

the participants were provided with various advanced geo-spatial analytical techniques and hands-on training for the analysis of both spatial and attribute data using open-source softwares like QGIS, SAGA-GIS and R. The programme was aimed at orienting the researchers with the twin goals of understanding and applying geospatial technologies and developing a decision-support framework for managing agricultural production systems for sustainability using open-source data and softwares.

2.6.9 Data Visualization using R

The programme was organized on April 09-11, 2022 with 105 participants including 31 female participants. The objectives of the program were to expose the participants to R & R Studio and to equip them in creating impactful data

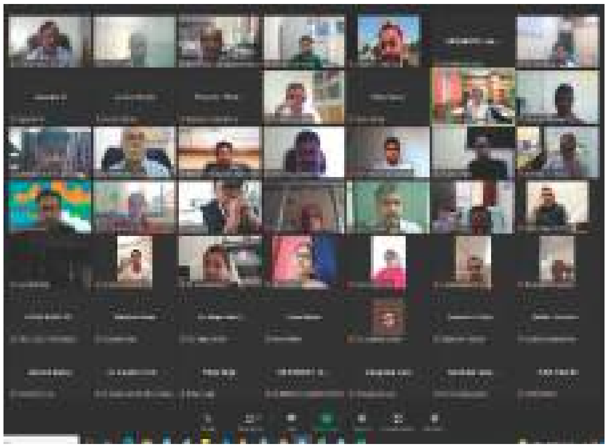


Figure 2.6.6: Virtual training session in the Training Programme on Data Visualization using R

visualization using various R packages. The content of the workshop included hands-on sessions on the Basics of R, Data Management/ wrangling, Visualization of different types of data viz., panel data, spatial data, text data and developing R Shiny Applications. Several R packages such asggplot2, dplyr, tidyr, tmap etc were employed and the applicability of various functions available in these packages was taught to the participants.

2.6.10 Training Programme on Statistical Data Analysis for the Faculty of PJTSAU (Online Mode)

The programme was organized on April 06-08, 11-12, and 18-21, 2022 with 483 participants. The training programme on Statistical Data Analysis for the Faculty of PJTSAU was organized by the Academy on request from the PJTSAU. The major objective of the training programme was to refresh the knowledge of Statistical reasoning of all faculty members involved in teaching, research and extension and equip them in planning, analyzing and interpreting data collected through experiments, surveys and observational studies.

483 participants from 19 specializations were nominated by the university for the training programme and the participants were divided into 4 groups. The number of trainees in each group varied from 87 to 188. The participants' designations ranged from Senior Professors to Assistant Professors. The training programme consisted of a one-day orientation for all the participants followed by two-day modules based on the broad subject matter groups.

The topics covered during the training programme are research methodology,

Figure 2.6.7: Virtual training session in the Training Programme on Statistical Data Analysis for the Faculty of PJTSAU

experimental planning, preparation of data files for different statistical analyses, design and analysis of experiments using different layouts (CRD, RBD, Split Plot etc.), Factorial Experiments, Transformations of data, repeated measurements ANOVA, Pooled Analysis, G X E interactions, Response Surface Designs, Analysis of Covariance, Special designs like Alpha Lattice, Augmented, Cluster Analysis, Principal Component Analysis, Multiple Linear Regression, Regression Diagnostics, Exploratory Factor Analysis, Structural Equation Models, Multi-Dimensional Scaling and Impact Assessment. All these topics were introduced with an example followed by when to use it, assumptions involved, data file preparations, how to do the analysis and interpretations. Some exercise datasets were also provided to the participants for practice and their own interpretations. For each group about one to one and half hours were allotted as an open discussion on specific problems related to their research.

The resource materials consisting of presentations, Example Data Sets and Codes were shared during the training programme. All sessions were recorded and made available along with the resource materials through PJTSAU computer center as a knowledge resource for the trainees.

2.6.11 Advances in web and mobile application development

The programme was organized on August 02-06, 2022 with 39 participants including 12 female participants. The five-day Capacity Building Training program on "Advances in web and mobile application development" was organized in virtual mode. The objective of the programme was to raise awareness and build essential skills in mobile app development.

Figure 2.6.8.a: Dr. Ch. Srinivasa Rao, Director addressing the participants of the training programme on Advances in web and mobile application development

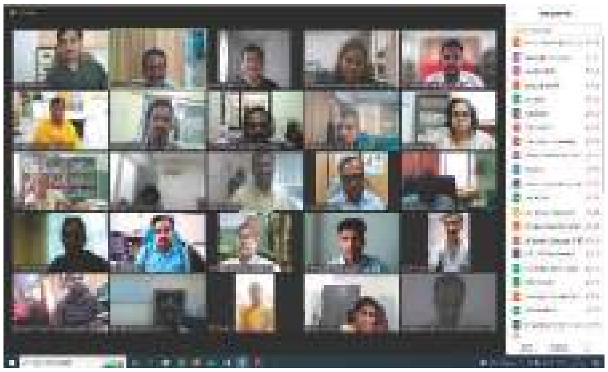


Figure 2.6.8.b: Virtual training session in the Training Programme on Advances in web and mobile application development

During the virtual training, the participants were provided with theory and hands-on practical sessions on an overview of the Android platform, Java programming, Android Studio installation, development of Android applications, database connectivity with android apps, and insight into no-coding technologies for android app development during the virtual training.

2.6.12 Research Design and Statistical Analysis for Rallis India R&D

The programme was organized on August 08-12, 2022 with 25 participants. The training programme was organized as a custom-made for the R & D team of Rallis India. The programme covered an introduction to R, visualization of data and basic data analysis, planning and analyzing

Figure 2.6.9: Participants of the training programme on Research Design and Statistical Analysis for Rallis India R&D

different types of experiments including split, strip plot designs, pooled analysis, bi-plot, use of transformations, cluster, and principal component analysis.

2.6.13 Training Workshop on "Response Surface Methodology" (Online Mode)

The programme was organized during August 18-20, 2022 with 22 participants. The objectives of the training workshop on Response Surface Methodology (RSM) are to make the participants to understand the various stages of experiment planning and data analysis towards optimizing input variables through response surface methodology. The RSM methodology is a powerful tool to optimize input variables and several RSM designs are available. The participants were trained in designing and analyzing the RSM designs such as Central Composite and Box-Behnken. The optimization through a canonical analysis was explained through examples. The workshop was conducted in an online mode comprising lectures as well as hands-on self-exercises. Some advanced issues such as the use of screening designs for a large number of factors and the steepest method for moving towards optimum levels. Open Source R software was used in the training workshop.

2.6.14 Training Workshop on Analysis of Multi-Environment Trials (Online mode)

The programme was organized during November 3-4 & 7-8, 2022. A total of 43 trainees participated in the programme. The training workshop which is being conducted second year in a row aims to sensitize and train scientists, particularly those working in coordinated trials, on various analytical tools and options for the analysis of data collected from Multi-Environment Trials (MET). The workshop was conducted in online mode. Initially, the concept of multi-environment trials, the categorization of MET data into different levels, and various analysis options was explained to the participants. To facilitate the analysis, R functions which were developed as a part of the KRISHI project were shared with the participants. The use of Bi-Plot and AMMI analysis to explore and identify the patterns in G x E interactions was introduced and participants were encouraged to try the tools on their own data. The use of Different types of selection indices was one of the other topics covered in the workshop. The workshop also provided a brief introduction to R as all examples/codes were in R.

2.6.15 Application of Bioinformatics in Agricultural Research and Education

The programme was organized under the (SKILL-BIF), DBT Project during November 15-24, 2022 with 17 participants including 06 female participants representing 10 research disciplines from 10 ICAR Institutions and two Agricultural Universities. The main aim of the programme is to create awareness with respect to the basic tools and techniques used in bioinformatics in agriculture. This training programme was conceived with the idea to provide insight to the participants about the enormous potential and application of Bioinformatics and its tools management in the domain of agriculture. They were also provided with theory and hands-on

Figure 2.6.10: Participants of the training programme on Application of Bioinformatics in Agricultural Research and Education

practical sessions on various Bioinformatics software and tools on the different topics related to the emerging requirements of Bioinformatics Tools in the agriculture and allied sectors.

2.6.16 Workshop on Current Trends in Agricultural Bioinformatics

The programme was organized on December 14-16, 2022 with 19 participants including 09 female participants. In view of globally increasing importance of bioinformatics, a Workshop on Current Trends in Agricultural Bioinformatics was organized the SKILL-BIF, ICAR-NAARM during December 14-16, 2022. This workshop focusses on changed scientific scenario in the postgenomic era by exposing the agricultural scientists to the latest computational biology tools for biological research. In addition to expert deliberations, the workshop also covered theory and hands-on practical sessions on various software and tools on various topics related to emerging requirements of bioinformatics tools in agriculture and allied sectors.

Figure 2.6.11: Participants of the Workshop on Current Trends in Agricultural Bioinformatics

2.6.17 Training programme on "Analysis of Experimental Data" (Online mode)

The programme was organized during December 19-21 and 26-28, 2022 with 90 participants including 32 female participants. The objective of the training programme is to orient the researchers with the basics of statistical tools and techniques mostly utilized in agricultural research. Data analysis using R was shown to the participants and BlueSky Statistics software. The topics covered include Principles of Experiments for Conducting Research; Exploratory Data Analysis; Introduction to SAS and Data Management, Correlation, Regression and Diagnostics; Analysis of Experimental Data using Basic as well as advanced designs, Pooled analysis, Multivariate Data Analysis using Principal Component Analysis (PCA), Factor Analysis, Cluster analysis and Ridge regression. Emphasis was given to the analysis of data using appropriate statistical techniques and interpretation of the results. Data from real-life studies and simulated data were utilized to explain various statistical techniques. A softcopy of the training manual on various topics as per the timetable was prepared consisting of lecture materials, hand-outs, datasets, and codes was provided to all the participants. There was also a session on the communication of research results wherein research papers published using various statistical techniques were discussed and distributed to the trainees and results from a few were discussed during the session.

2.7 Research Projects (Completed & On-going)

2.7.1 Completed Research Projects during 2022

S. No.	Project Code	Project Title	Team	Duration	
				From	To
In-House					
1	AGED/NAARM/SIL/ 2017/009/00129	Developing A Framework for Enhancing Agricultural Research Management Efficiency	Ch Srinivasa Rao, P Krishnan, I Sekar, SK Soam, B Ganesh Kumar, K Kareemulla, VV Sumanth Kumar	April 17	March 22
2	AGED/NAARM/SIL/ 2017/013/00133	Assessment and Development of Competency Framework for Agricultural Research and Extension Scientists	K H Rao, Alok Kumar, RVS Rao, P Ramesh, BS Sontakki, SK Soam, Ch Srinivasa Rao	April 17	March 22
3	AGED/ NAARM/SIL/ 2018/002/00141	Ag-Academy: A Framework for Scaling Up E-Learning in Agriculture	VV Sumanth Kumar, P Krishnan, SK Soam, GRK Murthy, NS Rao	April 18	June 22
4	AGED/ NAARM/SIL/ 2018/012/00151	Students' Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis	D Thammi Raju, P Ramesh, Bharat S Sontakki, P Krishnan	October 18	March 22
5	AGED/ NAARM/SIL/ 2019/008/00162	Economic Analysis of Food prices and Policies for the development of Indian Agriculture	MB Dastagiri, B Ganesh Kumar, A Dhandapani	June 19	July 22
6	AGED/ NAARM/SIL/ 2020/003/00169	Financial accessibility and occupational risks of vegetable street vendors in India	PC Meena, Sanjiv Kumar	June 20	May 22
7	AGED/ NAARM/SIL/ 2020/005/00171	Happiness and Psychological Well-Being of Scientific Personnel in NARES	P Ramesh, RVS Rao, BS Yashavanth	June 20	May 22
8	AGED/ NAARM/SIL/ 2021/001/00177	Social Media in Agriculture Marketing : A study on its role & potential	N.S. Rao, P.D. Sreekanth, S. Ravichandran, M. Balakrishnan, Surya Rathore, P. Venkatesan, M Ramesh Naik	January 21	December 22
Extra-mural Projects					
1	AGED/NAARM/SOL/ 2019/002/00156	Entrepreneurship Development through Farmer Led Innovations - A Case Study in Plantation Sector	S Vinayagam, GRK Murthy, B Ganesh Kumar, P Venkatesan	April 19	March 22

2.7.2 Ongoing Research Projects

2.7.2.1 In-House Projects

S. No	Project Code	Project Title	Team	Duration	
				From	To
1	AGED/NAARM/SIL/ 2020/004/00170	Identification of Best Practices Developed by the Leadership in Scientific and General Administration in NARES	R.V.S. Rao, P. Ramesh, Alok Kumar, K. Kareemulla, S. Senthil Vinayagam, S. Ravichandran, D. Thammiraju and Ch. Srinivasa Rao, G. Venkateshwarlu	July 20	February 23
2	AGED/NAARM/SIL/ 2022/004/00184	Motivational Factors Driving Scientific Personnel: Identification of Strategies for Scientific Excellence	P. Ramesh, R.V.S. Rao, B.S Yashavanth, G. Venkateshwarlu	April 22	April 24
3	AGED/NAARM/SIL/ 2020/001/00167	New Market Reforms and E-NAM in India: Effect on Marketing Choices and Participation and Price Realization for Smallholders	Ranjit Kumar, Sanjiv Kumar, A. Dhandapani, K. Srinivas, P.C. Meena, KH Rao	June 20	March 23
4	AGED/NAARM/SIL/ 2020/002/00168	Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies/ Value Added Products	Srinivas Tavva, B. Ganesh Kumar, G.R.R.K. Murthy, K. Kareemulla, Sudha Mysore	June 20	May 23
5	AGED/NAARM/SIL/ 2018/009/00148	Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India	B. Ganesh Kumar, Sanjiv Kumar, D. Thammi Raju, P.C. Meena, Tavva Srinivas, A. Dhandapani	October 18	Septem-ber 23
6	AGED/NAARM/SIL/ 2018/011/00150	Farmer Producer Companies in India: A Study on their Management Practices and Business Potential	N. Sivaramane, Ranjit Kumar, P.C. Meena, Bharat S. Sontakki, Vijay Avinashilingam, P. Venkatesan, Tavva Srinivas	October 18	March 23
7	AGED/NAARM/SIL/ 2019/012/00166	Agri Startups in India: Decoding the Determinants and Success Factors	Sanjiv Kumar, K. Srinivas, Ranjit Kumar, Umesh Hudedamani, K.H. Rao, Senthil Vinayagam, Surya Rathore	June 19	May 23
8	AGED/NAARM/SIL/ 2020/010/00176	Development of a Framework for Gender Mainstreaming in the Development Schemes of Agriculture Sector in India	Surya Rathore, P. Krishnan, Lipi Das (ICAR-CIWA), Veenita Kumari (MANAGE), B. S. Yashavanth	January 21	March 23

S. No	Project Code	Project Title	Team	Duration	
				From	To
9	AGED/NAARM/SIL/ 2022/001/00181	MOOCs in Digital Agricultural Education-strategies to Enhance Effectiveness, Inclusivity and Reach	G.R.R.K Murthy, V.V Sumanth Kumar, S Senthil Vinayagam, P Supriya	April 22	March 25
10	AGED/NAARM/SIL/ 2020/007/00173	Mega Environment Analysis Tools for AICRP Trials	A. Dhandapani, Yashavanth B S	June 20	May 23
11	AGED/NAARM/SIL/ 2021/003/00179	Impact Assessment of SC Sub Plan (SCSP) Implemented by Agricultural R & D Institutes on Socio-economic Status of Beneficiaries	M. Balakrishnan, B Ganesh Kumar, P Venkatesan, M Ramesh Naik, B S Yashavanth, P Supriya, N Sivaramane, Vijay Avinashilingam, D Tammi Raju, P C Meena, B S Sontakki	July 21	June 24
12	AGED/NAARM/SIL/ 2022/002/00182	Comprehensive Geo-spatial Analysis of Environmental Change in Farming	P D Sreekanth, D. Thammi Raju, P. Venkatesan, V.V Sumanth Kumar, M Ramesh Naik	February 22	February 24
13	AGED/NAARM/SIL/ 2022/003/00183	Bioinformatics analysis for Identification and characterization of superior haplotypes for important traits in field crops	P. Supriya, Satendra Kumar Mangrauthia, M Balakrishnan, R M Sundaram, Umesh H, Papa Rao (ICAR-DGR)	February 22	February 25
14	AGED/NAARM/SIL/ 2021/002/00178	Strategy for Strengthening National Water Policy	K. Kareemulla, S. Ravichandran, K.V. Rao, P.S. Brahmanand	January 21	March 23
15	AGED/NAARM/SIL/ 2020/009/00175	Dynamics of Agriculture and Food Systems in Different Agroecosystem of India	Mude Ramesh Naik, Umesh Hudedamani, Sivaramane N, BS Yashavanth, N. Subash, Monoranjan Mohanty, Nishant K. Sinha, Ch Srinivasa Rao	January 22	Decem-ber 24
16	AGED/NAARM/SIL/ 2022/005/00185	Good Governance and Institutional Performance in NARES	Alok Kumar, P Ramesh	Novem-ber 22	October 24
17	AGED/NAARM/SIL/ 2022/006/00186	Development of implementa-tion framework for National Education Policy -2020 in Ag-ricultural Education	D Thammi Raju, GRK Murthy, P Ramesh	October 22	Septem-ber 25
18	AGED/NAARM/SIL/ 2022/007/00187	Identification of similarity in research studies using machine learning	B S Yashavanth	October 22	Septem-ber 25

2.7.2.2 Externally Funded Contract Research Projects (Sponsored/Collaborative/Grant-in-Aid)

S. No.	PME Code	Project Title	Team	Budget (In Lakhs)	Spon-sored by	Duration	
						From	To
1	AGED/NAARM/CCL/2021/004/00180	Developing and Pilot Testing Specialized Knowledge Products on Human Wildlife Conflict Mitigation for Agriculture and Veterinary sector in India	Ch Srinivasa Rao, P Krishnan, Tavva Srinivas, G. Venkateshwarlu, Bharat S Sontakki, Ranjit Kumar, Sanjiv Kumar	87.10	GIZ	September 21	May 23
2	-	Distance Education Courses on Extension Management for Agricultural Professionals	N. Sandhya Shenoy (Emeritus Professor)	21.00	ICAR Emeritus Professor	January 22	January 25
3	AGED/NAARM/COP/2017/001/00121	Management and Impact Assessment of Farmer FIRST Project	P Venkatesan, B S Sontakki, N Sivaramane	114.05	ICAR	March 17	March 22
4	AGED/NAARM/SOL/2016/002/00112	Establishment of Agri-Business Incubation (ABI) Centers under XII Plan Scheme for NAIF	S. Senthil Vinayagam, H. Umesh, B. S Yashavanth	79.15	NASF	January 16	March 23
5	AGED/NAARM/COP/2019/001/00155	Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education	S. K. Soam, D. Thammi Raju, N. Srinivasa Rao, Alok Kumar, V.V. Sumanth Kumar, Surya Rathore, Sanjiv Kumar, M. Balakrishnan, S. Senthil Vinayagam, B.S. Yashavanth	1704.17	ICAR NAHEP	February 19	December 22
6	AGED/NAARM/SOL/2015/007/00110	KRISHI- ICAR Research Data Repository for Knowledge Management	A. Dhandapani, S.K. Soam	76.00	ICAR	November 15	March 26
7	AGED/NAARM/SOL/2019/005/00159	Establishment of Skill Development Bioinformatics Infrastructure Facility (Skill-BIF) for Agricultural Biotechnology (DBT)	M. Balakrishnan, S.K. Soam, P. Supriya	44.46	DBT	December 18	March 22
8	AGED/NAARM/SOL/2018/013/00152	Profiling Resources devoted to R&D in Agriculture Sector	M. Balakrishnan, P. Krishnan M. Ramesh Naik, H. Umesh, S.K. Soam	20.52	DST	August 20	July 21
9	AGED/NAARM/SOL/2008/003/00067	IP Management and Transfer/Commercialization of Agricultural Technologies Scheme	Umesh Hudedamani, N. Sivaramane, Ramesh Naik, Sanjiv Kumar	19.15	ICAR	April 14	March 22

2.8 Educational Programmes

2.8.1 Project Reports Submitted by 12th Batch (2020-22) PGDM-ABM Students

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
1	PGDM-ABM-2001	A J Anupriya	Study of Vegetable Collection Center -A Path to Sustainability, Maddur (Telangana)	Dr Tavva Srinivas
2	PGDM-ABM-2002	Ajasr	Sustainable Rice Production Technologies for Food Security and Market Opportunities for different Paddy Segments	Dr Ch. Srinivasa Rao
3	PGDM-ABM-2003	Akhil C	Better Life Farming: Unlocking Farming Potential From Smallholders	Dr Sanjiv Kumar
4	PGDM-ABM-2004	Akshaykumar D	Financial Accessibility and Occupational Risks of Vegetable Street Vendors in Hyderabad	Dr PC Meena
5	PGDM-ABM-2005	Ambala Anila	Perception Analysis about Application of Artificial Intelligence (AI) in Agribusiness.	Dr SK Soam
6	PGDM-ABM-2006	Amrit Kaur	Assess the Effectiveness of Sale Campaign Approaches for Arize Hybrid Rice	Dr Alok Kumar
7	PGDM-ABM-2007	Ankit Kumar	Analyzing Food Demand and Purchase Behavior of the Target Population in Patna	Dr S Senthil Vinayagam
8	PGDM-ABM-2008	Apoorva Marandi	Product Development on Nutrient Use Efficiency in Madhya Pradesh	Dr M BalaKrishnan
9	PGDM-ABM-2009	Ashish Ranjan	A Study on Consumption Pattern of Various Types of Major Dry Fruits and Nuts in Patna City	Dr P Venkatesan
10	PGDM-ABM-2010	Barsha Agrawal	Egg Industry in India: An Overview	Dr Sivaramane N
11	PGDM-ABM-2011	Bhanu Shree C	Consumer Preference towards Online Grocery Shopping in Krishnagiri district of Tamil Nadu	Dr P Krishnan
12	PGDM-ABM-2012	Bharatha Vinaykumar	Evaluation of Polyhouse Farming in Ranga Reddy District of Telangana-An Economic Analysis	Dr A Dhandapani
13	PGDM-ABM-2013	Chichchula Mythreyi	A Study on Understanding Coffee Consumption Behaviour in Andhrapradesh	Dr D Thammi Raju
14	PGDM-ABM-2014	Divya	Consumer Behaviour Analysis for Egg Industry in Haryana	Dr Umesh Hudedamani
15	PGDM-ABM-2015	Gaikwad Kirti Balkrishna	Frozen Food Segment – A New Paradigm Shift in India	Dr B Ganesh Kumar
16	PGDM-ABM-2016	Gangula Akshitha Reddy	Study and Plan of Replacement of Traditional Crops with Alternative Crops in Rajasthan.	Dr N Srinivasa Rao
17	PGDM-ABM-2017	Gangula Soundarya	Women Entrepreneurs in Agri-Business	Dr Surya Rathore
18	PGDM-ABM-2018	Hasika Yeruva	Effectiveness of Agri Input Dealers in Providing Extension Services as Perceived by Farmers of Andhra Pradesh	Dr Bharat S Sontakki

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
19	PGDM-ABM-2019	Ippili Bhaswini	Competitive Analysis and Market Potential Assessment of Aquafeed in Andhra Pradesh	Dr VV Sumanth Kumar
20	PGDM-ABM-2020	Jagannadham Kireeti	An overview of Bio-Pesticides: Importance and Reasons for their Low Demand and Consumption in Andhra Pradesh	Dr GRK Murthy
21	PGDM-ABM-2021	Janakiramachan-dran R	Impact Analysis of NBFC Intervention in Tamil Nadu	Dr RVS Rao
22	PGDM-ABM-2022	Jayasurya R	Market Analysis of Palm Oil in Indian Edible Oil Industry	Dr Ranjit Kumar
23	PGDM-ABM-2023	Kaithoju Vivek	Value Chain Mapping of Cotton in Rangareddy District	Dr Surya Rathore
24	PGDM-ABM-2024	Kaleru Nikhil	Value Chain Analysis of Turmeric	Dr Mude Ramesh Naik
25	PGDM-ABM-2025	Kalla Pardha Teja	Analysis of Production Costs and Determine the Best Cultivation Method for Tomato and Watermelon	Dr P Ramesh
26	PGDM-ABM-2026	Keerthana Menon	Assessment of Amul Products Diversification	Dr K Kareemulla
27	PGDM-ABM-2027	Koka Lalitha	Identification of Products From Potato Waste Accessing their Market Potential	Dr KH Rao
28	PGDM-ABM-2028	Korlagunta Yagana Sri	Value Chain Analysis of Mango in Andhra Pradesh	Dr S Ravichandran
29	PGDM-ABM-2029	Kotha Prudhvi	International Trade Scenario of Tomato	Dr MB Dastagiri
30	PGDM-ABM-2030	Masku Nishanth Reddy	Demand Generation for Verdesian Life Sciences International's(VLSCI) Select Portfolio, among Chilli and Cotton Growers of Guntur, Andhra Pradesh	Dr PD Sreekanth
31	PGDM-ABM-2031	Medi Saiprakash	Financial Accessibility and Occupational Risks of Vegetable Street Vendors in Telangana	Dr PC. Meena
32	PGDM-ABM-2032	Megha Raju	Sustainable Agricultural Practices Towards Zero Carbon Emissions: Way Forward	Dr I Sekar
33	PGDM-ABM-2033	Nalli Sandeep	Consumer Behaviour Analysis on Ready-to-Cook Foods by Elite Indians	Dr BS Yashavanth
34	PGDM-ABM-2034	Nidhin. J. K	Comparison of Conventional Farming and Integrated Farming System (IFS) Model in Increasing Farmers Income	Dr Tavva Srinivas
35	PGDM-ABM-2035	Nivas Bharathi P	Value Chain Analysis of Paddy and Pulses in Tamil Nadu	Dr Sivaramane N
36	PGDM-ABM-2036	P Prasanth Kumar	Cost of Cultivation and Returns under Seed Production of Vegetables and Flowers	Dr Sanjiv Kumar
37	PGDM-ABM-2037	Pallavi S	Seaweed Cultivation and Supply Chain Development in India	Dr S Ravichandran

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
38	PGDM-ABM-2038	Panimalar T	Farming at Auroville: Organic Production and Agribusiness by Multicultural Community	Dr SK Soam
39	PGDM-ABM-2039	Pasala Satya Praveen	Value Chain Analysis of Wild Ber and Jatropha in Udaipur Belt of Rajasthan	Dr Alok Kumar
40	PGDM-ABM-2040	Pramod G B	Production and Marketing Potential of Eggs in Karnataka	Dr S Senthil Vinayagam
41	PGDM-ABM-2041	Rajender Kumar	Value Chain Analysis of Sugarcane in Mandya, Karnataka	Dr M BalaKrishnan
42	PGDM-ABM-2042	Renukareddy	Dealer Survey on Products of Kisankrafts Ltd. in South Karnataka Region	Dr P Venkatesan
43	PGDM-ABM-2043	Rohan S H	Protection to Production- Regarding Nematode Management in Pomegranate by Focusing Crop Program	Dr VV Sumanth Kumar
44	PGDM-ABM-2044	Sagar Satpathy	Price of Potato Seeds & Farmer's Accessibility to Internet in States of North India	Dr P Krishnan
45	PGDM-ABM-2045	Shaik Ayathullah Siddique	Understanding Digital Payments, their Role and Adoption in India	Dr A Dhandapani
46	PGDM-ABM-2046	Shivani Chidambar Huddar	Digitizing Agricultural Value Chain for Atmanirbhar Bharat	Dr D Thammi Raju
47	PGDM-ABM-2047	Shivangi Gupta	Market Analysis of Fortified Foods	Dr Ranjit Kumar
48	PGDM-ABM-2048	Shrija	Study on Farm Mechanization Scenario and Farm Machinery Purchase Behaviour of Farmer Producer Companies	Dr B Ganesh Kumar
49	PGDM-ABM-2049	Sourabh Katake	Market Analysis of Pomegranate in Sangola	Dr N Srinivasa Rao
50	PGDM-ABM-2050	Sunil J Malagimani	Establishment of Environmental Science Product Portfolio in Better Life Farming Segment	Dr BS Yashavanth
51	PGDM-ABM-2051	Sushma S	Effectiveness of Agri Input Dealers in Providing Extension Services as Perceived by the Farmers of Karnataka	Dr Bharat S Sontakki
52	PGDM-ABM-2052	Thimmareddy Bharath Kumar Reddy	Market Potential Analysis of Pomegranate in Domestic and International Market	Dr Ch. Srinivasa Rao
53	PGDM-ABM-2053	Vasanth Kumar K M	Analysis of Price Malady in Tomato Marketing - A Case study	Dr GRK Murthy
54	PGDM-ABM-2054	Yerramsetti Sai Mohana Kavya	Effective Hyper-Local Marketing Channel for Gardening Plants and Tools	Dr RVS Rao

2.8.2 Summer Internship of 13th batch (2021-23) PGDM-ABM Students

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
1.	PGDM-ABM-2101	Abhishek R	Farmers Perception on 2 phosphatic fertilizer of Coromandel International Limited.	Dr Umesh Hudedamani Dr Binay Parida
2.	PGDM-ABM-2102	Abinav E Basi	Market Research on Non-Basmati Rice Varieties	Dr S Senthil Vinayagm
3.	PGDM-ABM-2103	Alana Raphy	Identification of potential prospects for Hyperspectral Imaging-based Decision Support System (DSS) in Tea plantations.	Dr PC Meena
4.	PGDM-ABM-2104	Amrut Basavaraj Salimath	Pomegranate procurement in vegrow	Dr Ch. Srinivas Rao
5.	PGDM-ABM-2105	Ankireddy Mahathi	To understand black thrip infestation and control measures taken by farmers in chilli crop	Mr. Ramesh Naik Mude
6.	PGDM-ABM-2106	Anmol Mahajan	Identification of potential prospects for Hyperspectral Imaging Based Decision Support System in Tea Plantations	Dr Ranjit Kumar
7.	PGDM-ABM-2107	Anshika Sharma	Study of evolving Agri marketing opportunities in Bihar	Dr PD Sreekanth
8.	PGDM-ABM-2108	Apoorva	Study and Analysis of Customer Returns in Modern Trade and Value Loss in General Trade and Provide Solutions for the Same.	Dr A Dhandapani
9.	PGDM-ABM-2109	Arem Bhanu Prakash	Comprehensive study on the warehousing of Agri Commodity	Dr VV Sumanth Kumar
10.	PGDM-ABM-2110	Ashutosh Upadhyay	Study of Agriwarehousing and Commodity financing in Indore (Madhya Pradesh)	Dr Alok Kumar
11.	PGDM-ABM-2111	Asifpasha F	Sourcing of Pomegranate	Dr Sanjiv Kumar Sh. Rajeev S
12.	PGDM-ABM-2112	Attravanam Venkatesh	To understand the reasons for EROS volume downfall	Dr Ranjit Kumar
13.	PGDM-ABM-2113	Balram Patidar	Target market mapping and finding market potential	DrBS Yashavanth Mr. Amyn Abdul Aziz Pirani
14.	PGDM-ABM-2114	Banala Pavan Reddy	Study of agrochemical market in tomato, drivers and key trends in india, Understand business model for FPOs,FPCs and recommend GTM Strategy	Dr Surya Rathore
15.	PGDM-ABM-2115	Bavisetti Sai Satya Lakshmi Sruthi	Procurement and Identifying Challenges in Marketing Systems of Raw Cashew Nuts through Farmers Producer Company in Andhra Pradesh	Dr D Thammi Raju
16.	PGDM-ABM-2116	Bheemana-pally Pranay Kumar	Mapping Potential Areas for Procurement of Chilli in Prakasham (A.P.)	Dr P Ramesh
17.	PGDM-ABM-2117	Bijender	Demand generation and brand awareness of S-4001 in Eastern UP	Dr Tavva Srinivas

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
18.	PGDM-ABM-2118	Bute Sneha Tulshiram	To study yhe potential of warehousing and collateral management in to the market of Maharashtra,	Dr Ranjit Kumar
19.	PGDM-ABM-2119	C Anand	District profile and market survey of plant protection industry in Bellary district of Karnataka	Dr Bharat S Sontakki
20.	PGDM-ABM-2120	Chethan C S	Assessment of entering into last mile warehouse financing through whatsloan digital platform and Assessment of entering into dark store warehousing in bangalore market	Dr SK Soam
21.	PGDM-ABM-2121	Chintalapudi Ramana Sairam	Market Analysis of Aquaculture Feed, Chemicals and Machinery in Andhra Pradesh	Dr P Venkatesan
22.	PGDM-ABM-2122	Chirra Naveen Kumar	Study on livelihood dimension of Tribals in Telangana	Mr. PC Meena
23.	PGDM-ABM-2123	Dhadwe Devashish Bhimrao	District Profiling and Market survey of Plant Protection Industry in Washim (Maharashtra).	Dr B Ganesh Kumar
24.	PGDM-ABM-2124	Gaurav	Assessment of Cost of Cultivation of Paddy and Wheat and Field Demonstration of Gracia insecticide in Punjab, Haryana and Uttar Pradesh	Dr GRK Murthy
25.	PGDM-ABM-2125	Gourineni Prathima	Understanding black thrips infestation and control measures taken by farmers in chilli crop	Dr NA Vijay Avinashlingam
26.	PGDM-ABM-2126	Gupala Mounika	Comprehensive study on agri warehousing commodities	Dr K Kareemulla
27.	PGDM-ABM-2127	Illamparithi S	Study of edible oil market and consumer behaviour analysis in Tamil Nadu to know the brand awareness and barriers for trials of Fortune Sunflower Oil	Dr G Venkateshwarlu
28.	PGDM-ABM-2128	Jammula Sravani	Installation of Spice Powders Processing Plant	Dr M Balakrishnan
29.	PGDM-ABM-2129	Jaware Shweta Sanjay	Business trends of corn in Western Maharashtra for grain/fodder/silage and opportunities for Syngenta	Mr. Joydeep
30.	PGDM-ABM-2130	Jayasurya	Procurement Strategies and Price Behaviour of Mangoes in Chittoor district of Andhra Pradesh	Dr B Ganesh kumar
31.	PGDM-ABM-2131	Jyoti Singh	Development of customer portal and Mobile app in coordination with Go Green IT team	Dr S Ravichandran
32.	PGDM-ABM-2132	Kalthi Alekhya	A Comprehensive study focusing market research export feasibility and detailed value chain of onion and chilli	Dr M B Dastagiri
33.	PGDM-ABM-2133	Kantamureddy Devi Pravallika	Market Research on aqua feed , machines and chemicals	Dr Umesh Hudedamani
34.	PGDM-ABM-2134	Karthik A S	Sourcing Tender Coconut in Karnataka	Dr S Senthil Vinayagam
35.	PGDM-ABM-2135	Krishnasree S	Study of Supply chain of Non Basmati Rice Varieties and Identification of Potential Plant Locations for the Company to Expand its Product Portfolio.	Dr A Dhandapani

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
36.	PGDM-ABM-2136	Mallikarjun Kamanalli	Processing Hot Pepper in Karnataka: HM Clause Opportunities Analysis	Dr RVS Rao
37.	PGDM-ABM-2137	Mhatre Srishti Ankush	Sustainable Sugarcane Production	Dr PD Sreekanth
38.	PGDM-ABM-2138	Nancy Gupta	Value based pricing of key crops	Dr B Ganesh Kumar
39.	PGDM-ABM-2139	Nehal Ahuja	Study of agrochemical market of grapes in Maharashtra	Dr VV Sumanth Kumar
40.	PGDM-ABM-2140	Nikhil Mund	Study of Agri-Warehousing and Commodity Financing in the State and recommend the company to enter into the state in next financial year.	Dr Ch. Srinivasa Rao
41.	PGDM-ABM-2141	Nithya Vinod	Study of evolving agri marketing opportunities in Bihar	Dr Alok Kumar
42.	PGDM-ABM-2142	Palakuri Sree Vidya	A Comprehensive Study on Warehousing finance and Insurance of Chilli Cold storages	Dr BS Yashavanth
43.	PGDM-ABM-2143	Polepaka Parimala Paul	Developement and Execution of Aquaconnect Partner App	Dr Surya Rathore Mr. Ramlingam Sivasailam
44.	PGDM-ABM-2144	Pragya Srivastava	Understanding herbicide application trend and scope in Soybean	Dr Sanjiv Kumar
45.	PGDM-ABM-2145	Pulugujju Swetha	Procurement and Identifying Challenges in Marketing Systems of Raw Cashew Nuts through Farmers Producer Company in Andhra Pradesh	Dr Tavva Srinivas
46.	PGDM-ABM-2146	Ramneni Manognya	Study of global philanthropic market supply and demand trends for the past three years	Dr Bharat S Sontakki
47.	PGDM-ABM-2147	Rohini G	Impact Study on Paramfos Sambhrama Scheme implemented in Karnataka	Dr P Venkatesan
48.	PGDM-ABM-2148	Shaik Bepari Mohammed Junaid	Issues and Challenges in Mango Procurement in Andhra Pradesh and Telangana	Dr D Thammi Raju
49.	PGDM-ABM-2149	Shaik Sameer	Process of Raw Cashew Nuts through Farmers Producer Company (FPC) in Andhra Pradesh	Dr GRK Murthy
50.	PGDM-ABM-2150	Shwetha .P.	Credit System for FPOs in Input Procurement	Dr P Supriya
51.	PGDM-ABM-2151	Simrandeep Kaur	Development and execution of Aquaconnect farmer application	Dr NA Vijay Avinashlingam
52.	PGDM-ABM-2152	Sourabha Kumar Naik	Overview of Gram Flour Industry in India and Survey of Market and the Users.	Dr K Kareemulla
53.	PGDM-ABM-2153	Sourav Kumar Dubey	Understanding the Pre and Post-emergence herbicides application trend and scope in Soybean	Dr N Srinivasa Rao
54.	PGDM-ABM-2154	Sreelakshmi Murali Nair	Analysis of crop protection trends in Pomegranate in Maharashtra & Gujarat	Dr M Balakrishnan
55.	PGDM-ABM-2155	Sreenitha S	Assisting in getting International and Domestic funds for Swades Programmes	Dr Sivaramane N

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
56.	PGDM-ABM-2156	Sukhpreet Singh	Demand Generation and Brand Awareness of NK7328 in eastern U.P.	Dr S Ravichandran
57.	PGDM-ABM-2157	Umang Kumar Shyam	Increasing trend and market analysis of biologicals	Dr RVS Rao

2.8.3 Students (M.Sc. & Ph. D) Guided by NAARM Faculty as Major Guide/Supervisor

Sl. No.	Name of the faculty	Student Name	Title of the thesis	Programme Name (M.Sc./Ph.D)	University / Institute Name
1	Dr M. Balakrishnan	D. Karunakaran	Development of Algorithm for prediction of fish yield in reservoirs using Artificial Neural Network	Ph.D.	Bharathiar University, Coimbatore
2	Dr Bharat S. Sontakki	Rajitha, B.	Effectiveness of Agri. Input Dealer(s) in Providing Extension Services as perceived by the Farmers of Telangana State	M. Sc. (Agri) in Agricultural Extension Education	PJTSAU, Hyderabad
3	Dr K. Kareemulla	Ch. Shekhar	Comparative Economic Analysis of Sweet Orange and Mango in Anantapuramu District of Andhra Pradesh	M.Sc. (Ag)	ANGRAU, Guntur

3 NAARM as a Think Tank of ICAR

3.1 Policy Research and Advocacy

3.1.1 Bioinformatics in Agriculture

Agricultural Bioinformatics in association with biotechnology has potential application to address longstanding issues in agriculture. Increased use of applications of bioinformatics in agriculture was due to the latest developments in functional genomics, Next-Generation Sequencing (NGS) tools, Whole-Genome Sequencing (WGS), Data mining techniques, genotyping assays, Genome-Wide Association Studies (GWAS). Sequencing-based trait mappings require high-performance computing (HPC) facilities and expertise. The exponential growth of data in commensurate with the demand for visualization, integration, analysis, prediction, and management helped breeders, biotechnologists, and agriculture scientists to address the pertinent issues. However, many agricultural researchers are still unfamiliar with the available methods, databases and bioinformatics tools, which may lead to misinterpretations or lack of informational opportunities. The potential use of bioinformatics is the availability of various tools (for storage, retrieval, analysis, and annotation of results) and large genomic resources to a vast pool of agricultural scientists. This is all possible only when data are analyzed using appropriate bioinformatics tools. However, the major issues concerned are the lack of infrastructure facilities like HPC, computational tools, lack of opportunities for training for skill up-gradation, lack of large-scale research initiatives at the national level, etc. Capacity building of National Agricultural Research and Education. The creation of a separate Bioinformatics department/center in colleges under State Agricultural Universities (SAU)/Central Agricultural University (CAU)/ institutes would strengthen the research and development. The proactive step is to fund all the Bioinformatics departments/centers for at least 5 years by ICAR to establish/strengthen requisite infrastructure for its sustainability. The databases, software, and bioinformatics tools generated/developed by ICAR institutes/SAUs are to be integrated into national repositories created centrally at an institute identified for the purpose. The protection of databases and Intellectual Property Rights (IPR) is the utmost critical issue. The implementation of the much-needed strategies requires support from the NARES to reap the benefits of the newly emerging science in agricultural education.

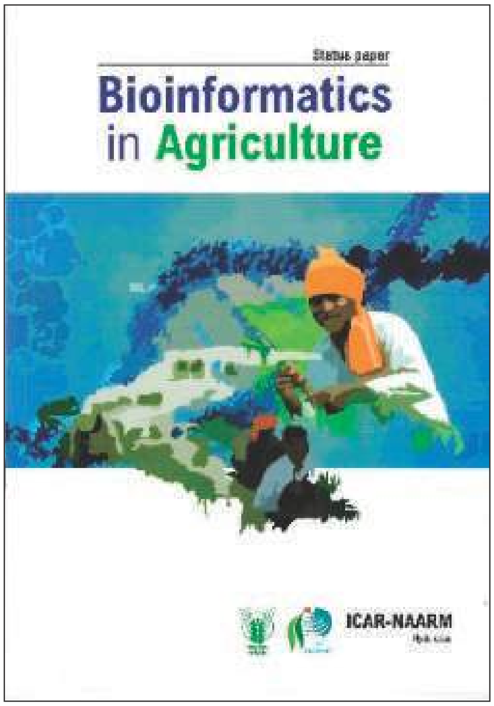


Figure 3.1.1: Bioinformatics in Agriculture Status Paper

The use of bioinformatics in agricultural research and education can happen through certain proactive steps of NARES supported by exclusive budgetary allocation, the creation of infrastructure, and curriculum changes.

3.1.2 Attracting the Best Talent to Agricultural Education in India

Feeding a global population of about 10 billion people by 2050 would necessitate a drastic shift in how food is produced, processed, exchanged, and eaten. Improvements in global, regional, and local food systems are needed to feed the expanded population so that decent employment and livelihoods of producers and actors are ensured (https://www.fao. org/food-systems/en/). Agricultural research, policy support, and institutional innovations have significant potential for reshaping agriculture to meet future demand for food and eliminate hunger. The agriculture sector contributed 57.70 percent to India’s GDP in the 1950s, since then it has been on the downside and it currently contributes about 19.19 percent (MOSP: http://mospi.nic.in/). However, with economic growth, a change from an agrarian-centric economy to industry and service-sector-oriented economy is a natural evolution all over the world and India is no exception. However, in India, agriculture, agribusiness, and agri-based industry have a unique and strategic importance in contributing to job creation, food and nutrition security, and broad-based economic development as the sector still supports nearly 50% of the population. Recent developments in Indian agriculture indicate stagnant or declining factor productivity causing concern for scientists and planners. Among many factors causing this decline, natural resource degradation, slow pace of new innovations, and human resources not matching the challenges, are important.

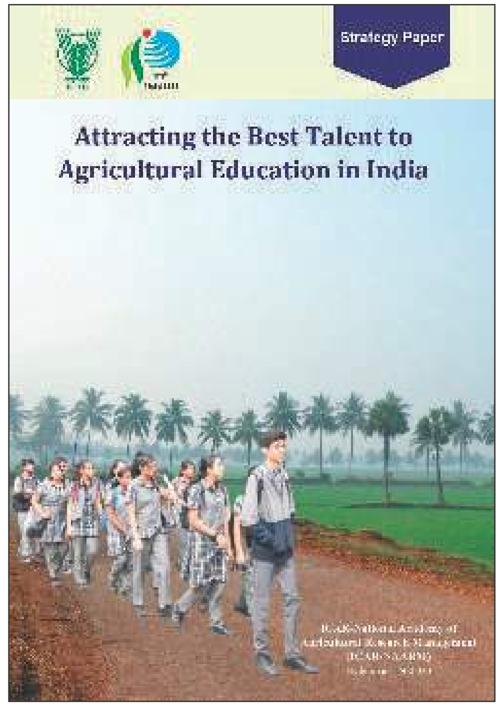


Figure 3.1.2: ‘Attracting the Best Talent to Agricultural Education in India’ policy paper

3.1.3 Strengthening Kisan Call Centres in India

Ministry of Agriculture & Farmers’ Welfare, Government of India, with an aim to improve awareness and knowledge efficiency of farmers, developed a comprehensive Information and Communication Technology (ICT) strategy to reach out to farmers in an easy and better way and also for better planning and monitoring of schemes. Such ICT strategies include websites/portals (Agmarknet, eNAM, Farmers’ Portal, Kisan Sarathi, Soil Health Card Portal, Crop Insurance etc.), use of mobile apps (Kisan Suvidha, Pusa Krishi, Agrimarket, Crop Insurance etc.), use of basic mobile telephony (SMS, IVRS, OBD, USSD) and provision of personalized information through Kisan Call Centres (KCC).

Globally, lack of information has been one of the factors hindering the growth of the agricultural sector. Farmers, for example, may lack information on how to respond to new pests and illnesses, or they may be unaware of which local market offers the best price for their produce (Aker & Mbiti, 2014). Furthermore, they may not have access to an independent data source, such as a public extension agent, and must rely on an input provider representative. Due to the lack of access to information, the output is reduced, input costs are increased, dangerous substances are misused, and profitability is reduced. (Aker & Mbiti, 2014; Muto & Yamano, 2009). In India, only 6.8 % of the farmers receive extension support. As an answer to this issue, the Department of Agriculture & Cooperation, Ministry of Agriculture, Govt. of India launched Kisan Call Centers on January 21, 2004 across the country to deliver personalized extension and advisory services to the farming community. This initiative attempted to answer farmers' questions over the phone in their native language/dialect. There are call centers for every state which are expected to handle traffic from any part of the country. These call centers handle questions relating to agriculture and associated industries. At present, the KCC services are managed from 21 locations spread across the nation. All KCC locations are accessible by dialing a single nationwide toll-free number 1800-180-1551 through landline as well as mobile numbers of all telecom networks from 6.00 A.M. to 10.00 P.M. on all 7 days a week including holidays. The IFFCO Kisan Sanchar Nigam is the execution partner of KCCs.

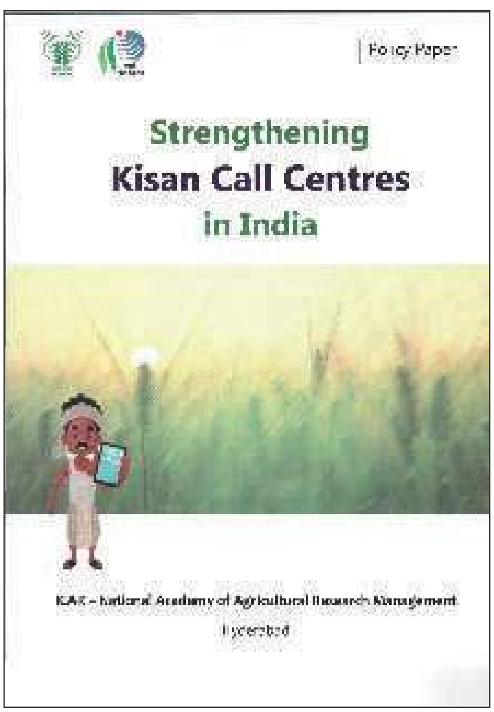


Figure 3.1.3: 'Strengthening Kisan Call Centres in India' policy paper

3.1.4 FPOs in India: Creating Enabling Ecosystem for their Sustainability

Farmers Producers Organizations (FPOs) are considered to be one of the most imminent tools of intervention for the upliftment of the farmers' condition in India. When more than 85 percent of the farmers are smallholders, it becomes quite challenging for them to access modern production technologies, access and use the market information for their advantage, transact the commodities in the input or output market on their own terms, and ultimately keep their farming profitable. The collectivization of farmers through FPOs helps in bringing the economy of scale in different on-farm as well as off-farm activities at all three stages- pre-production, production, and post-production levels. The Government of India is promoting the formation of FPOs through different programmes & schemes, since the last decade. The organizations like SFAC and NABARD have specific schemes to promote the formation of FPOs in India. Currently, there are more than 10,000 FPOs in India registered under different legal structures. The majority of them are registered as a company under the Companies Act, while the second largest category is under the Cooperative Societies Act of respective states. Some are also in the form of Society or Trust. The Producer Companies are designed on a ‘mutual assistance principles’ and ‘patronage’ basis, to bring together desirable aspects of the cooperative and corporate sectors for the benefit of primary producers, especially small and marginal farmers. In several case studies/literature, success stories have been documented demonstrating the benefits realized by the farmer-members from the FPOs in the form of reduced input cost and market linkage as aggregation allows better bargaining power for the FPOs. Advisory services and value-added services offered by the FPOs have also benefitted the farmers in timely decision-making. However, there exist many challenges for the long-term growth and viability of FPOs or when the scale of operation increases. The majority of the FPOs are new and small in terms of the number of producer-members and their own equity capital, which makes them less attractive for funding by the financing agencies. Lack of understanding about the business plan, lack of funding support for many FPOs, less importance towards hiring of skilled persons with managerial abilities, poor governance structure, difficulty in regulatory compliances, and marketing difficulties need proper attention to provide an enabling ecosystem for the FPOs. In 2019, the Government of India has announced the formation of 10,000 new FPOs on a cluster basis under Central Sector Schemes (CCS). These FPOs get supported by the promoting agency (NABARD/ SFAC/NAFED/NCDC) in terms of technical, managerial and financial aspects. However, the challenges with the existing FPOs and/or many other FPOs formed outside the ambit of CCS remain the same. They are not able to move to the next stage of their lifecycle to become an economically viable business unit to create value for their shareholders and society at a large. To ensure the economic viability and long-term sustainability of FPOs, the policy

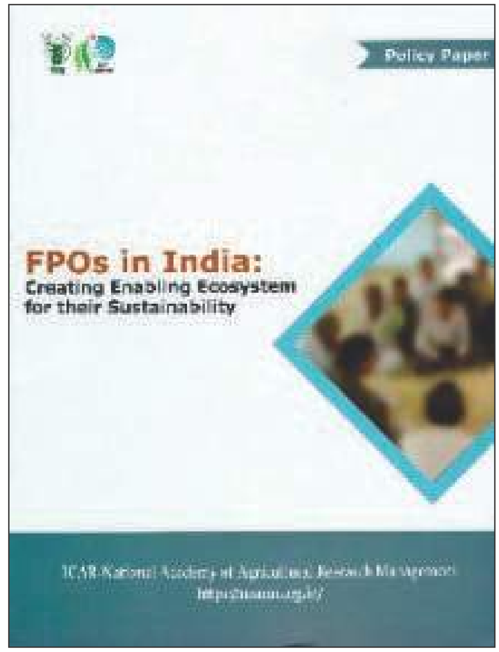


Figure 3.1.4: ‘FPOs in India: Creating Enabling Ecosystem for their Sustainability’ policy paper

paper offers the following recommendations: 1). It is proposed to set up a National Board of FPO (NBFPO) on the pattern of MSME and different commodity boards which can promote, impart necessary skills, provide network and monitor the progress of all the FPOs. 2). There should be a provision of an assured Seed Fund as a one-time grant for all the registered FPOs on pre-decided criteria. It may be done by creating Special Purpose Vehicle (SPV) with a corpus fund entrusted with the NBFPO. 3). Customized financial products may be created with the banks for the FPOs to enable them easy access to short-term loans at a subsidized rate. The funding may be done based on the cash flow of the FPOs, rather than existing provision of asset-based financing. 4). Linking of each FPO with KVK/ agricultural university/ development institutions: To provide continuous technical support and guidance, the FPOs should be closely linked to any agricultural university/ KVK/ICAR institutes or any other development organization in the region, which may act as an

Incubation/ Facilitation Centre for the FPOs. 5). For regulatory compliances, a pool of empaneled chartered accountants (CAs) in each state shall be created, who can serve the FPOs for regulatory compliances, the expenses of which may be borne by the government during the initial 3-5 years. 6). Setting up common post-harvest infrastructure facilities at the district level would encourage rural micro-entrepreneurs and reduce the burden on individual FPOs to have their own infrastructure at the initial stage. 7). The marketing problems can be addressed to a certain extent by extending preferential treatment to the FPOs in schemes like MSP-based procurement, relaxing norms of APMC for transacting agricultural commodities, Mid-day Meal Scheme, *Poshan Abhiyan*, etc.

3.1.5 Agri-startups in India: Opportunities, Challenges and Way Forward

Globally, entrepreneurship culture has been gaining considerable importance in recent years. The transition from controlled and license-regulated public company-oriented economies to knowledge-based ones has encouraged individuals to start businesses. The term startup denotes the early stage of an enterprise integrating science-based innovations that can scale it up. In short, all startup founders are entrepreneurs, but not all entrepreneurs are startup founders. Startups are emerging as a vehicle for realizing social purposes, a means of actualizing technological prowess, the wings of creative imagination that transport the purposeful dreamers to a land of their creation, and preferred routes to riches. Startups are considered to be the next growth engines to take the Indian economy to the size of US$5 trillion and to realize the dream of *Atmanirbhar Bharat*. India has the second-largest cultivated area, with a large number of cultivators (146 million), mostly (86%) smallholders. The sector is facing several complex problems and challenges. The tiny landholdings increase transaction costs, making it challenging to adopt several modern technologies essential for efficient resources. Even a lack of information about soil, weather, market, etc., hinders crop planning and diversification. It also leads to a smaller scale of production. Consequently, a small marketed surplus makes it non-remunerative for producers and buyers to engage inefficient price discovery and trade. These hold for the livestock production system as well. The climate change scenario further aggravates these challenges. The cumulative effect of these is lower household income. The latest Situation Assessment Survey (SAS) conducted by the NSSO in 2019 shows that the income from crops for rural households is even less than wage earnings. On average, agricultural families earn Rs. 816.5 per person per month from farming. On the other hand, urban consumers, whose absolute number and average per capita income have been rising, are demanding foods of better quality in different forms (a variety of the products fresh and processed) and at reasonable prices. Apart from domestic demand, there is a massive upsurge in the export market for quality agricultural products, including organic food products. It is evident that during the COVID pandemic period, when all other sectors of the economy languished, the farm sector has sustained its growth momentum. The share of agriculture in the gross domestic product (GDP) reached almost 20% in 2020-21, for the first time in the past 17 years (Economic Survey 2020-21). The sheer size of the agricultural sector offers umpteen opportunities for evolving an efficient market-based agrarian ecosystem. Several new ventures are registered with Startup India. As of November 10, 2021, more than 1.73 lakh startup ventures had registered, of which the Department for Promotion of Industry and Internal Trade (DPIIT), Government of India has recognized 58,650 startups. Considering the problems and challenges in agricultural development, several startups have filled the gap in agri-food value chains. Currently, there are 7241 agri-startups, of which the DPIIT recognizes only 2605. There are also 2594 startups in the food & beverage industry. However, compared to the importance of agriculture, the number of agri-startup looks much less in number. During the last ten years, a total of 80+ startups reached a valuation of more than $1 billion, commonly known as “Unicorns”. Further, a whopping 38 startups have already made it to the list within 11 months of 2021. But agri-startups somehow remained in the shadow. More surprisingly, most high-growth agri-startups have been founded by young professionals with engineering and management backgrounds. On the other hand, about 45,000+ students graduate (under-graduation and post-graduation) every year from 75 agricultural universities (ICAR, 2017). This raises a few critical concerns: How to make agricultural education and research organizations hubs for developing and nurturing scalable business ideas? Should the course curriculum of agricultural universities be rejigged to trigger students’ interest to take up entrepreneurship as a career? What kind of policies and infrastructures in the National Agricultural Education & Research System (NARES) would promote agri-startups in India? What capabilities and resources are needed for Agribusiness Incubators to identify and nurture high-quality and scalable agri-startups? Are there any policy hurdles in setting up startups and their scaling up? Against this backdrop, the National Academy of Agricultural Sciences (NAAS) organized a brainstorming session (in hybrid mode- online and offline) on 5th November 2021. The experts shared their views and experiences on enabling an environment for scaling up agri-startups.

3.1.6 Mainstreaming of Agricultural Higher Education by Private Universities in India

Under NAHEP Component-2, a dialogue on mainstreaming of agricultural higher education by private universities has been initiated and a National Workshop on “Agricultural Education in Private Universities in India: Let’s Listen to Stakeholders” has been planned. As the first step, a Pre-Workshop Meeting was organized on Mar 25, 2022 at Krishi Anusandhan Bhavan –II, New Delhi under the Chairmanship of Dr. RC Agrawal, National Director, NAHEP & DDG (Agricultural Education), ICAR, New Delhi. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM; Dr. Rajender Parsad, Director, ICAR-IASRI; and Dr. Prabhat Kumar, National Coordinator, NAHEP component-2 participated in the deliberations. The Vice-Chancellors and Deans of Shobith University, Lovely Professional University, Malla Reddy University, IFTM University, GLA University, Quantum University and Anurag University also participated. Dr PS Pandey and Dr Seema Jaggi, ADGs from the Education Division, and other senior officials of ICAR and NAHEP participated in the meeting.

Figure 3.1.5: Group photo of participants attended National Workshop on “Agricultural Education in Private Universities in India: Let’s Listen to Stakeholders”

3.1.7 A Dialogue on Fostering Collaboration for Quality Agricultural Education

The Academy organized a brainstorming workshop on "A Dialogue on Fostering Collaboration for Quality Agricultural Education between Actors of Agriculture Research and Education" at Hyderabad on May 10, 2022. This programme was organized to discuss the collaboration among private and government institutions for quality agricultural higher education. The workshop was organized as part of the World Bank-sponsored National Agricultural Higher Education Project Component-2A.

Forty-Five participants including several Deans from SAUs and Private Universities, Directors from ICAR and SAUs participated in the brainstorming workshop. Dr. Ch. Srinivasa Rao, Director, NAARM and Dr. G. Venkateshwarlu, Joint Director, NAARM guided the whole process for highly successful outcomes. Dr. AK Singh, Director, IARI, New Delhi; Dr. Praveen Rao, Vice Chancellor, PJTSAU, Hyderabad and Dr. Neeraja Prabhakar, Vice Chancellor, SKLTSHU, Hyderabad were the prominent speakers at the event. Dr SK Soam highlighted the role of private universities and provided the objective of NAHEP explaining the expectations to improve agricultural higher education in the country.

Figure 3.1.6: Participants attended the workshop on "A Dialogue on Fostering Collaboration for Quality Agricultural Education between Actors of Agriculture Research and Education"

3.1.8 Building Academia-Industry Partnerships through Alumni

The Academy organized a workshop on "Building Academia – Industry Partnerships through Alumni for Quality Agricultural Higher Education" on May 14, 2022. The workshop aimed at nurturing a platform for networking and developing the fraternity among Alumni and the participants get an opportunity to communicate with alumni students and get enlightened on how the interactions between the higher educational system & industry promote professional career and technology exchange can be done by alumni meets. Dr Surya Rathore educated the gathering about establishing a partnership with agricultural HEIs, promoting CDC at Agricultural Universities, the competence of the curriculum standards, strengthening linkages with industries and raising the quality of education to global standards. Dr. V V Sumanth Kumar urged the participants about how to strengthen Alumni Networks and explained the importance of the KVC ALNET portal "Krishi Vishwavidyalaya Chhatr Alumni Network" and the role of connecting the agri alumni globally with the portal.

Figure 3.1.7: Participants attended the workshop "Building Academia–Industry Partnerships through Alumni for Quality Agricultural Higher Education"

3.1.9 National Symposium on Agricultural Higher Education by Private Universities

The Academy organized a National Symposium on "Mainstreaming of Agricultural Higher Education by Private Universities in India" on Sep 29, 2022 under NAHEP, Component- 2. Dr RC Agrawal, DDG (Edn), ICAR, and National Director of NAHEP inaugurated the symposium and highlighted the need for transformation of agricultural education in the country in view of the National Agricultural Education Policy-2020. The one-day symposium deliberated on four important themes viz. equal opportunity in education and employment, quality assurance in agricultural education, enabling environment for quality education and accreditation in the context of national education policy.

Among 99 participants, 42 were from 29 private universities from 10 states (AP, Telangana, UP, MP, Punjab, Uttarakhand, HP, Odisha, Maharashtra, and Tamil Nadu). The delegates discussed the problems and prospects of agricultural higher education at private universities. The prominent speakers were Shri Kunwar Shekar Vijendra, Chancellor, Shobhit University; Dr Rupa Vasudevan, Chancellor, BEST Innovation University, AP; Dr Atul Khosla, Founder and Vice Chancellor, Shoolini University, HP; Prof. Raghuvir Singh, Vice Chancellor,

Figure 3.1.8: Participants attended National Symposium on "Mainstreaming of Agricultural Higher Education by Private Universities in India"

Teerthankar Mahaveer University, UP; and Dr Supriya Pattanayak, Vice Chancellor, Centurion University of Technology and Management, Odisha. Shri BL Meena, IAS (Retd), Chairman, BEST Innovation University; Shri Ravi Verma, Vice Chairman, MNR University represented educational groups.

Most other participants were Pro Vice Chancellors, Deans and Directors from private universities. Dr. Anuradha Agrawal, National Coordinator (Component 2), Dr. Sudeep, Principal Investigator (Component 2), and several senior faculty and officers from IASRI and NAARM also participated in the event in various capacities.

Dr Ch Srinivasa Rao, the Director, highlighted the importance of quality education and the accreditation process. He also extended an invitation for the capacity building of faculty of private universities at the Academy.

Centres of Excellence

4

4.1. Centre for Agri-Innovation (CAI)

The Centre for Agri-Innovation of ICAR-NAARM Comprises of Technology Business incubator namely a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) which is supported by ICAR-NAIF (National Agriculture Innovation Fund), Department of Science & Technology (DST), National Bank for Agriculture and Rural Development (NABARD) and Department of BioTechnology (DBT). a-IDEA was previously supporting BioNest an Agri- Bio-incubator for Agri-Biotechnology Startups that was backed financially by BIRAC, DBT and GoI, in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad.

a-IDEA nurtures agricultural innovations and entrepreneurship in the fields of agriculture and allied sectors in India. Since its inception in the year 2014, a-IDEA, has backed startups covering as many as 14 domains of agriculture and allied sectors such as dairy, fisheries and animal husbandry. Through its initiatives and programmes, a-IDEA focuses on sensitizing, ideating, incubating and accelerating the innovative early-stage startups/ entrepreneurs that are scalable to become fiercely competitive food and agri-business ventures through capacity building, one on one mentoring, networking and advisory support. The startups backed by a-IDEA have been addressing pressing issues concerning the farming community and those that have a large impact on the lives of farmers, rural and urban stakeholders besides being disruptive in nature.

At Present, the key thrust areas of agriculture innovations across the value chain in agriculture on which a-IDEA is emphasizing on:

Figure:4.1.1: a-IDEA focus areas

4.1.1 Agri Udaan 4.0

a-IDEA, the TBI of ICAR-NAARM, organized Demo Day under Agri Udaan 4.0 on 21 May 2022. This is the fourth food and agribusiness accelerator program initiated by a-IDEA supported by NABARD. The objective of this program was to catalyze food and agribusiness startups and provide them with a unique platform to pitch

Figure:4.1.2: Agri Udaan 4.0

their ideas before investors for scaling up to the next level by raising funds.

The event was attended by 10 startups selected out of 200 applications under Agri Udaan 4.0 and 25 Investors from different parts of the country. This program has helped startups to network with investors to raise the funds and cross-learning. There was a pre-demo day during 19-20 May 2022 where these startups were groomed to effectively present their ideas/ products.

Table No. 4.1.1: Agri Udaan 4.0

S. No	Startup Name	No. of Investors Showed Interest
1	BMH Transmotion, Maharashtra	05
2	Fermentech Labs, Uttarakhand	13
3	Farm Sathi, Andhra Pradesh	08
4	Chimertech Pvt Ltd, Tamil Nadu	12
5	Genflow AI, New Delhi	05
6	Borlaug Web Services, West Bengal	08
7	Khetigaadi, Hyderabad	05
8	Urban Plants, New Delhi	03
9	FarmOR Agrisolutions, Hyderabad	03
10	Inventohack Innovations, Bhopal	08

4.1.2 Agri Udaan 5.0

Agri Udaan, Food and Agribusiness Accelerator 5.0 is the 5th edition of the flagship accelerator program of a-IDEA, NAARM. It brings on board a range of diverse partners to create an inclusive and collaborative ecosystem focusing on catalyzing scale-up stage Food and Agribusiness startups through rigorous mentoring, industry networking and investor pitching. It is a unique platform for scale-up stage innovators, entrepreneurs and startups in Food and Agribusiness sectors to showcase their products/services and to receive valuable inputs from Mentors, Incubators, R&D Institutions, Agribusiness Industry, and Investors.

Figure:4.1.3: Agri Udaan 5.0

AGRI UDAAN 5.0, the food and agribusiness accelerator, was launched on the 6th Aug 2022 at Hotel Radisson Blu, Hyderabad. Mr Jayesh Ranjan, IAS, Principal Secretary, Department of industries and commerce, Government of Telangana was invited as the Chief Guest. Mr Inigo Arul Selvan, GM, NABARD, Hyderabad; Mr Srinivas Rao Mahankali, CEO, t-hub, Hyderabad and Dr Sudha Mysore, CEO, Agrinnovate India Ltd., New Delhi were the Guests of Honours.

Agri Udaan 5.0

The application for AGRI UDAAN 5.0 was opened on the 6th of August and the date for submission of the application was given on the 22nd Oct 2022.

Following the launch of AGRI UDAAN 5.0, Roadshows were conducted in 4 locations to promote and create awareness about AGRI UDAAN. (i) NDRI-Bangalore on the 6th Sep 2022. (ii) VAMNICOM, Pune on the 6th Sep 2022. (iii) The Spring Club, Kolkata on the 7th Sept 2022. (iv) Anveshan Foundation, Delhi on the 23rd Sep 2022. Nearly 50 participants attended each of the roadshows. The top 14 startups cohort of Agri Udaan 5.0, was selected from a pool of 149 applications.

4.1.3 Agri Udaan Cohort Startups of Agri Udaan 5.0

Table 4.1.2: Agri Udaan Cohort Startups of Agri Udaan 5.0

S. No	STARTUP NAME	Domain	PRODUCTS/SERVICES
1	Inav Agro Mech private limited, Hyderabad	Farm Mechanization	We INAV AGRO MECH have started with an aim to provide a great alternative (Battery operated ride on type multipurpose agriculture toolbar) for the problem faced by the small and marginal farmers of India through our innovations and new products.
2	Anitra Tech Pvt. Ltd, Hyderabad	Livestock E-Tradind	Anitra, a full-stack livestock company, that monetizes your livestock by providing end-to-end services, we provide monetization of livestock, Animal Traceability, Health Assistance, Feed & Nutrition, Logistics for easy & effective monetization of livestock.
3	CisGEN Biotech Discoveries Private Limited, Tamil Nadu	Cattle Artifical Insemination	CisGEN Biotech Discoveries Pvt. Ltd. is first of its kind animal health start-ups in India working towards the development of novel products in veterinary field, medical device for animal productivity enhancement and services for other unmet needs in one health.
4	Organikness, Hyderabad	Organic E-Marketplace	Organikness is a two-sided marketplace to connect healthy lifestyle seeking consumers with authentic organic & natural products producers/ sellers on a proprietary and centralized online platform.
5	Marut Dronetech Private Limited, Hyderabad	Drone-Tech	Multi-utility drones with seven different applications. An all weather, all season drone providing best RoI to the user. A software fully integrated with SoPs for autonomous operations over different crops.
6	Jivanah Platform Services Pvt Ltd, Karnataka	Farm-Mechanization	A tractor driven Single Row Potato Combine Harvester that has cutting edge “separation technology” for soil, soil clods and weeds / halum (leafy materials and twigs). Offers a 10x harvesting efficiency Vs manual method.
7	Contrivation Labs Pvt Ltd, Karnataka	Chilli-Calyx Seperation	An Agritech Startup developing advanced automation solutions/farm mechanization for harvest and post harvest produce. Calyxtract, fully designed and developed by Contrivation Labs is a fully automated solution to separate the Calyx and Stem from the Chili pods using image processing, advanced control algorithms and electro-mechanical and pneumatic actuators.

8	Bee Basket Enterprises Pvt. Ltd, Maharashtra	Bee Pollination	Bee Basket provides pollinators such as Sting-less Honeybees on rent for precision cross-pollination to produce quality seed production and increase agricultural yield through our tech-enabled platform "Paragan".
9	Agriotics Technologies Pvt LTD, Maharashtra	Micro Climatic Data Collection Device	We at Agriotics aid farmers by giving Insights into native farms. The insights include microclimatic data along with crop risk intimation and advisories. The timely advisories in turn help the farmers take preventive measures and also help them manage their farms remotely and efficiently.
10	Univia Private Limited, Gujarat	Agri-Input Marketplace	Univia is the fastest growing B2B2C AgriTech company of Gujarat. We are an Agri Input (Seeds, Pesticides, Fertilizers, Hardware) PHYGITAL marketplace aiming to focus on Western India Agriculture Market. Agri Input Market in Western India is 15000 + Crores.
11	Bhojpatta Agripreneur Private Limited, Bihar	Solar Dryer	Our company works on Zero Carbon Emission Dryer and its manufactured products. such as sorting, grading, grinding, packaging, and selling
12	AgriVijay (Renewagri Om Ecommerce Pvt Ltd), Maharashtra	Renewable energy equipment marketplace	AgriVijay is India's first Marketplace of Renewable Energy products for farmers & rural households bringing all products in solar, biogas, thermal, wind & electric under one roof and with Energy Advisory approach where farmers energy needs are understood coupled with waste availability at their end before products are recommended, sold and deployed along with abating GHG/CO2 emissions mitigating Climate Change aligned with United Nations SDG's becoming Energy Independent along with increased savings & income.
13	Attaware Biodegradable Private Limited, Delhi	Edible Cutlery	Edible Cutlery and Tableware From Jaggery and Grains. Non Baking Technology, Net-Zero Pollution, 100% Natural, no water used in Production
14	Auspice Social Networks OPC Pvt Ltd, Uttar Pradesh	Spices	Auspice is a purpose driven for-profit social enterprise. All our products are crafted by persons with autism with limited use of machinery. We do not employ non autistic people for our food processing activities. Autistic workers at our facility are engaged in food processing activities like the cleaning, sorting, grading, grinding, blending, packaging and labeling of spices, herbs and seasonings.

Figure:4.1.4: Cohort Startups of Agri Udaan 5.0

4.1.4 Aggnite 2.0

A National Level Ideation Competition named "Aggnite" was conducted by a-IDEA to motivate the A National Level Ideation Competition named "Aggnite" was conducted by a-IDEA to motivate the students towards entrepreneurship in agriculture and to educate them on Startup ecosystems and entrepreneurship skillsets. Three awardees Mr. Sathriyan.M, TNAU, Coimbatore, Ms. Ranjitha S Betageri, UAS, Bagalkot and Dr. Umesh Kumar Jaiswal, University of Veterinary and Animal Science, Hisar received cash awards from Mr. Jayesh Ranjan IAS, Principal Secretary, Government of Telangana on 06 August 2022.

Figure 4.1.5: Aggnite 2.0

4.1.5 Memorandum of Agreement

a-IDEA signed a Memorandum of Agreement with 28 ICAR Institutes and six other organizations with the aim to support startups by organizing joint events, extending lab facilities, co-networking etc. The details are given below.

Figure 4.1.6: MoA with ICAR institutes

Table 4.1.3: MoA with ICAR institutes

S. No	Name of The Institute	Date of Execution
1.	ICAR-Indian Institute of Rice Research, Hyderabad	24 Feb 2022
2.	Crescent Innovation and Incubation Council, Chennai	
3.	RR Animal Health Care, Hyderabad	
4.	AIC, NITTE Incubation Center, Karnataka	
5.	AIC - Mahamana Foundation for Innovation and Entrepreneurship, IIM-BHU, Varanasi	
6.	ATARI, Bengaluru	14 May 2022
7.	ICAR-NINFET, Kolkata	
8.	ICAR-NIPB, New Delhi	
9.	ICAR-DWR, Jabalpur	
10.	ICAR-NCIPM, New Delhi	
11.	ICAR-NRC on Camel, Bikaner	
12.	ICAR-DCFR, Bhimtal, Uttarakhand	
13.	ICAR RC for ER, Umiam, Meghalaya	
14.	ICAR-CRIDA, Hyderabad	
15.	ICAR-CIAE, Bhopal	

S. No	Name of The Institute	Date of Execution
16.	ICAR-CICR, Nagpur	08 July 2022
17.	ICAR-MGIFRI, Motihari	
18.	ICAR-IASRI, New Delhi	
19.	ICAR-NDRI, Karnal	
20.	ICAR-IISS, Mau	
21.	ICAR-NRC on Meat, Hyderabad	
22.	ICAR-CIPHET, Ludhiana	
23.	ICAR-NRC on Yak, Dirang	
24.	ICAR-CCARI, Ela, Old Goa	
25.	ICAR-CIARI, Port Blair	
26.	ICAR-IGFRI, Jhansi	
27.	ICAR-IIVR, Varanasi	
28.	ICAR-CIFE, Mumbai	
29.	ICAR-NRC Pomegranate, Sholapur	
30.	ICAR-CMFRI, Kochi	
31.	NACL Industries Limited, Hyderabad	04 Nov 2022
32.	Sri Konda Laxman Telangana State Horticultural University (SKLTSHU), Hyderabad	
33.	ICAR-Sugarcane Breeding Institute (SBI), Coimbatore	
34.	ICAR-National Research Centre on Grapes, Pune	

4.1.6 Cohort Networking & Capacity Building Programmes

a-IDEA Organized COHORT Networking and Capacity Building Program for the incubatees of a-IDEA to develop networking among startups, and train them on aspects like financial management, effective pitch making and pitching process, the concept of digital marketing and branding etc. The details are given below.

Figure 4.1.7: Cohort Networking & Capacity Building Programmes

Table 4.1.4: Cohort Networking & Capacity Building Programmes

S. No	Program	Dates	No. of Startups attended
1	1st COHORT	March 10-11, 2022	15
2	2nd COHORT	April 21-22, 2022	25
3	3rd COHORT	June 22-23, 2022	22
4	4th COHORT	Nov 02-04, 2022	25

4.1.7 Joint Sensitization Workshops under ABI

The Agri-Business Incubation (ABI) Centre of ICAR-NAARM, Hyderabad organized seven Joint sensitization programmes for the benefit of startups, students and staff of ABIs in collaboration with Agri-Business Incubation Centre, ICAR ABI Center. The details are given below.

Table 4.1.5: Joint Sensitization Workshops under ABI

S. No	In collaboration with	Date	No. of startups attended
1	ICAR-RC, NEH, Umiam	10 Feb 2022	61
2	ICAR-CIRCOT, Mumbai	28 Feb 2022	70
3	ICAR-RCER, Patna	15 March 2022	43
4	ICAR-CPCRI, Kasaragod	23 April 2022	55
5	NRC on Pig, Guwahati	31 May 2022	30
6	SBI, Coimbatore	21 Oct 2022	63
7	ICAR-NDRI, Karnal	18 Nov 2022	73

Figure 4.1.8: Joint Sensitization Workshops under ABI

4.1.8 Immersion Programmes

a-IDEA, Technology Business Incubator of ICAR-NAARM, Hyderabad organized seven FPOs/FPCs, Farmers, Startups Immersion Programme to connect the FPOs/ FPCs and the farmers with the startups supported by a-IDEA. The details are given below.

Figure 4.1.9: Inauguration Immersion Programmes

Table 4.1.6: Immersion Programmes

S.No	Location	Date	No. of Startups attended	No. of FPOs attended
1	Guwahati, Central Assam	03 March 2022	8	12
2	Nalbari, Lower Assam	04 March 2022	8	12
3	ICAR-RCNEH, Shillong	05 March 2022	10	43
4	Warangal, Telangana	04 May 2022	22	50
5	ICAR-NAARM, Hyd	16 July 2022	15	53
6	Vijayawada, AP	21 Sep 2022	18	30
7	UHS Campus, Bangalore	30 Dec 2022	20	90

4.1.9 Sensitization Programmes on Entrepreneurship Development in Agriculture

a-IDEA, the Technology Business Incubator of ICAR National Academy of Agricultural Research Management (NAARM), Hyderabad organized to sensitize the students on entrepreneurial importance, schemes etc. The following sensitization Program on "Entrepreneurship Development in Agriculture" in online mode for faculty and students was organized.

Table 4.1.7: Sensitization Programmes on Entrepreneurship Development in Agriculture

S. No	Date	Name of the University	No. of Participants
1	27 Jan 2022	Maharana Pratap University of Agriculture and Technology, Udaipur	441
2	26 Feb 2022	University of Agricultural Science, Raichur	150
3	02 July 2022	Arunachal University of Studies (AUS), Namsai, Arunachal Pradesh	77
4	12 Aug 2022	ICAR- Central Island Agricultural Research Institute (CIARI), Port Blair	200
5	05 Sep2022	Sardarkrushinagar Dantiwada Agricultural University, Sardarkrushinagar, Gujarat	531

4.1.10 Meetings

4.1.10.1 General Body Meeting

The 10th General Body Meeting of a-IDEA was organized on 14 March 2022 under the Chairmanship of Dr. Ch. Srinivasa Rao, President, a-IDEA. During the meeting, Dr. S. Senthil Vinayagam, CEO & Principal Scientist of a-IDEA presented the progress of a-IDEA from 1 March 2021 to 13 March 2022. During the meeting, amendment of society members, seed support policy, the financial status of a-IDEA, publications, finalization of Charted Account, Status of Seed Fund and Grants were discussed. Dr. Ch. Radhika Rani, Associate Professor & Head, NIRD, Hyderabad; Mr. Y.K Rao, CGM, TSRO, NABARD (Virtual); Mr. Devasis Padhi, CGM-OFDD, NABARD, Mumbai (Virtual); Dr. Manish Diwan, Head, SP & ED, BIRAC (DBT Representative) (Virtual); Heads of Divisions and faculty from NAARM participated in the meeting.

Figure 4.1.10: 10th General Body Meeting of a-IDEA

4.1.10.2 PRAYAS Monitoring Committee Meeting

The NIDHI-PRAYAS Monitoring Committee (PMC) Meeting was held on 23-24 March 2022 for the selection of startups under the NIDHI PRAYAS scheme. During the meeting, the committee selected 10 startups/ Individual innovators out of 21 shortlisted candidates under the 3rd Call.

4.1.10.3 Seed Support Management Committee Meeting

The Seed Support Management Committee Meeting of a-IDEA was held virtually on 22 March 2022 under the Chairmanship of Dr. Ch. Srinivasa Rao, President, a-IDEA. During the meeting, the progress of Seed funded startups was presented. During the meeting, matters regarding the status of startups which have received seed funds from aIDEA, the next phase of investment to selected startups, the sale of a-IDEA share to other investors and BIRAC Seed Fund Call and other upcoming proposals were discussed.

4.1.10.4 Project Monitoring and Review Committee (PMRC) Meeting

PMRC meeting was organized to review the progress of activities conducted by a-IDEA with the support of NABARD. The details are given below.

Table 4.1.8: Project Monitoring and Review Committee (PMRC) Meeting

S.No	PMRC	Date
1.	6th PMRC Meeting	04 January 2022
2.	7th PMRC Meeting	21 April 2022
3.	8th PMRC Meeting	28 June 2022
4.	9th PMRC Meeting	28 October 2022

4.1.11.5 Catalytic Capital Support Management Committee (CCSMC) Meeting

The Catalytic Capital Support Management Committee (CCSMC) Meeting was conducted on 20 December 2022 under the Chairmanship of Director, NAARM. Fund sanctioned by NABARD under the Seed support program Catalytic Capital Fund (CCF) to support startups in the area of Agri-tech and Food tech areas under the valley of death. The same has been committed to 5 startups after three rounds of presentation and pitching by 39 startups. The committee has gone through the detailed presentations of the nine startups. A total of 78 applications have been received by a-IDEA ICAR NAARM.

4.1.11 Participation

4.1.11.1 World Organic EXPO

a-IDEA participated in World Organic Expo during Feb 25-27, 2022 in New Delhi organized by NNS Media Group and a-IDEA was awarded 2nd prize under the category of promoting Agri-business. For the expo, a-IDEA booked one stall for showcasing the activities of a-IDEA. Incubatees of a-IDEA also displayed their products and technologies. The a-IDEA stall was visited by faculties, Scientists, researchers, aspiring entrepreneurs, students, women groups and other exhibitors during which the institutional activities of a-IDEA were briefed to them. Received Award (2nd Prize) for promoting Agri-business.

4.1.11.2 Biotech Startup Expo, New Delhi

a-IDEA participated in Biotech Start-up Expo on June, 9-10, 2022 at Pragati Maidan, New Delhi organized by BIRAC. BIRAC allocated one stall for a-IDEA, ICAR NAARM and 4 stalls for our Incubatees to display their products and technologies. a-IDEA entrepreneurs displayed their products at their respective stalls. The startup incubated at our Incubation centre got good exposure and connections during the expo. Four startups viz., Healventure Biosciences

LLP (Nidhi Prayas Grantee), Shrimp Hoard Technologies (Birac BIG Grantee), Fermentech Labs Pvt. Ltd. (Nidhi Prayas Grantee), Dharaksha Ecosolutions Private Limited (Agri Udaan) showcased their products during the expo. Faculties, Scientists, researchers, aspiring entrepreneurs, students, women groups and other exhibitors visited a-IDEA Stall. Institutional activities of the a-IDEA Incubation Centre were briefed to all visitors.

4.1.11.3 DST Startup Utsav at New Delhi

a-IDEA participated in DST's Startup Utsav held on 12 August 2022 at Dr. Ambedkar International Centre (DAIC), New Delhi organized by DST, New Delhi on the occasion of 75 years of independence "Azadi Ka Amrit Mahotsav". The event also commemorated the completion of 40 years of NSTEDB and the completion of five years of NIDH. a-IDEA had fruitful discussions with Dr. Anita Gupta, Adviser & Head (Innovation & Entrepreneurship), NSTEDB, DST; Dr. H K Mittal, Former Advisor and Member Secretary, NSTEDB, DST; Dr. Poyni Bhatt, CEO, SINE, IIT, Bombay and Ms. Kanan Dhebar, Senior Project Manager, SINE, IIT, Bombay; Dr. S. Chandrashekhar, Secretary, Department of Science and Technology, Govt. of India; Dr Sapana Kaushik, Department of Science and Technology, Govt. of India and Dr Ajai Chowdhry, Co-founder HCL Technologies and Chairperson FICCI Startup Committee.

4.1.11.4 Buyer-Seller Meet at Marriott Hotel & Convention Centre, Hyderabad

a-IDEA team and 10 startups attended Buyer-Seller Meet at Marriott Hotel & Convention Centre, Hyderabad, Telangana on 30th September 2022 organized by NABARD and Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH under the Indo-German development cooperation project 'Capacity Enhancement for Sustainable Agriculture and Sustainable Aquaculture (C-SASA)'. Dr. Hara Gopal Yandra, GM, NABARD, TS, Sri Hanumant K. Zendage, IAS, Special Commissioner of Agriculture, Government of Telangana TCS Reddy MD & CEO, APMAS, Mr. Deepak Chamola, GIZ, Mr. Madhu Murthy, APMAS, Mr. Somasundaram, DGM, NABARD, TS, Mr. Deepak Chamola, GIZ were the guests for the event.

4.1.11.5 Local Startup Meet (LSM), Hyderabad

Team a-IDEA comprising of Dr. Pooja S. Bhat, Ms. Caroline K and Mr. Vishwa Phansal participated in the Local Startup Meet (LSM) organized by Entrepreneurship Cell, IIT Kharagpur at T-Hub, Hyderabad on October 9 2022, and learnt about the startup ecosystem and networked with eminent entrepreneurs through a series of Workshops by Mr. D. Sunil Reddy (Founder, Dodla Dairy) and Mr. Vihari Kanukollu (Co-Founder - CEO Urbankissan).

4.1.11.6 Agri Startup Conclave & Kisan Sammelan 2022, New Delhi

a-IDEA Team and six startups i.e., MLIT Sol Pvt Ltd, Krishivan Technologies Pvt Ltd, Agri Feeder Agricultural Services Pvt Ltd, AUMSAT Technologies LLP, Chimertech Pvt Ltd, Animal ICU (Verdant) participated in Agri Startup Conclave which was held at IARI Pusa, New Delhi on October 17-18, 2022 in which it's a great opportunity for networking and collaboration, 300+ Agri-startups were invited from different Agri Incubation centres from all across the country.

The Government of Odisha, jointly with FICCI organized the Odisha Investors' Meet on 17 October 2022 at Hotel Taj Krishna, Banjara Hills, Hyderabad. Mr. Naveen Patnaik, Chief Minister of Odisha, Mr Pratap Keshari Deb, Minister - Industries, MSME & Energy, Govt of Odisha; Mr Suresh Chandra Mahapatra, Chief Secretary-Government of Odisha; Mr Hemant Sharma, Principal Secretary, Industries Department, Skill Development & Technical Education and Chairman-IDCO and Chairman-IPICOL, Government of Odisha, Mr. Manoj Kumar Mishra, Secretary, E&IT Department, Dr. Omkar Rai, Executive Chairman, Start-Up Odisha, Mr. Arun Chawla, Director General, FICCI and Senior Officials from the industries and department attended the conclave. Dr. S. Senthil Vinayagam, CEO, a-IDEA & Principal Scientist and Mr Siddharth, Senior Manager attended the meeting.

4.1.12 Call for Applications

4.1.12.1 KRISHIBOOT 2.0 Call for Incubation

The application for KRISHIBOOT 2.0 was opened on 31 March 2022 and the last date for application was 30 April 2022. The total number of applications received was 107. The first level of screening was done by the internal a-IDEA team members based on the eligibility criteria. Based on these 40 Startups were selected for the second round of evaluation. The final round of evaluation was done through virtual mode by the external jury members for two days. In the final round of evaluation, 14 out of 40 startups are selected for the Incubation program.

Table 4.1.9: KRISHIBOOT 2.0 Call for Incubation

S. No	Name of the Startup	Domain	One liner
1	Increeks Private Limited, Telangana	Precision/Smart Agriculture/ IoT/AI/ Sensor/Technology, Supply /Value Chain	AgriTech Partners for providing technology solutions to make agri-value chain operate at higher equilibrium.
2	InstaCoCo, Tamil Nadu	Agritech	Fresh coconut water vending machine
3	VertoX Labs Private Limited, Odisha	Soil, Water & Weather, Fisheries, Supply/Value Chain, Others	Plashbot is one of a kind aquatic weed harvester that removes aquatic weeds
4	Sattvika Agrowcraft Foundation, West Bengal	Animal Husbandry/ Dairy	Probiotics along with plant extract which have properties for immune system stimulant.
5	My Millets, Telangana	Food Technology	Nutritious snack for all age groups, especially children
6	Naturex Technologies LLP, Telangana	Precision/ Smart Agriculture /IoT/AI/ Sensor Technology, Fisheries	Precision farming, hydroponics, climate smart agriculture by leveraging the power of technology and data.
7	Dhisathi Farm Robotics Pvt Ltd., Telangana	Agri Tech	Products revolutionize traditional ways of farming. Makes you self-reliable with fully Autonomous-Solar Powered Farm Bots.
8	One Place Velocious Private Limited, Maharashtra	Food Technology, Supply / Value Chain, Agritech, Precision/ Smart Agriculture / IoT/AI/Sensor Technology	Processing, cold storage freezing supply chain aided by robotics and data science, automation
9	Crop Intellix Private Limited, Andhra Pradesh	Precision/ Smart Agriculture /IoT/ AI/Sensor Technology, Agri- Inputs, Agritech, ICT & Advisory services	Provision of end to end solution for Agri stakeholders

10	Nano Composite Technologies Private Limited, Andhra Pradesh	Agritech	Nano fertilizer
11	Agro Origin Pvt Ltd, Assam	Food Technology	Healthier substitute of snacks through omni channels for marketing products viz., Retail and e-commerce.
12	Organikness, Telangana	Supply / Value Chain-E commerce platform	Organikness is an online marketplace to buy and sell organic, natural and sustainable products.
13	Chimertech Private Limited, Tamil Nadu	Animal Husbandry/ Dairy	One Stop IoT integrated mobile application based cloud platform for Disease management and breeding solutions in dairy industry
14	Aumsat Technologies LLP, Maharashtra	Precision/Smart Agriculture/ IoT/ AI/Sensor Technology,Soil, Water & Weather	Precision-driven, satellite-based, AI-enabled hydrological analysis for locating, predicting, and forecasting groundwater resources

4.1.12.2 KRISHIBOOT 3.0 Call for Incubation

The application for KRISHIBOOT 3.0 was opened on 26 October 2022 and the last date for application was 26 November 2022. The total number of applications received was 100. The first level of screening was done by the internal a-IDEA team members on 29 November 2022 based on the eligibility criteria. Based on these 51 Startups were selected for the second round of evaluation. The final round of evaluation was done on the 8th & 9th of December 2022 through virtual mode by the external jury members for two days. In the final round of evaluation, 21 out of 51 startups are selected for the Incubation program.

Table 4.1.10: KRISHIBOOT 3.0 Call for Incubation

S. No	Name of the start-up	Domain/ Sector	One liner
1.	Farmigai Agro Industries Pvt. Ltd., West Bengal	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	AI/IoT enabled integrated farming systems
2.	GreenGrahi Solutions Pvt. Ltd., Uttarakhand	Food technologies and fisheries	Food technologies and fisheries
3.	Renewagri Om Ecommerce Pvt Ltd., Maharashtra	Post-Harvest Technology, Farm Machinery and Equipment	Marketplace of Renewable Energy products for farmers & rural households
4.	Shri kamadenu panchagavya udyog., Himachal Pradesh	Waste to wealth, Rural innovation	Manufacturing eco-friendly products with cow dung
5.	Mallikarjun Sajjan, Karnataka	Dairy technology	Device to detect milk adulteration

6.	Farmoid Robotech Pvt Ltd., Karnataka	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	Farmoid offers a unique combination of IoT and AI in the form of precision farming. Providing real-time monitoring of crops, analyzing the data, and offering the best advice to the farmers, resulting in an increase in quantity and quality of yield.
7.	khadyam speciality foods Pvt ltd., Telangana	Food Processing & Value Addition, Soil & Water Conservation, Supply Chain Management	Creating marketplace for farmers by cultivation of climate resilient crops
8.	Foo Foods India Pvt ltd., Kerala	Food Technology, Post-harvest Processes & Value addition, Fisheries	making fisheries & poultry based ready to serve traditional foods using thermal processing technology
9.	X BOSON AI Pvt Ltd., Kerala	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	X BOSON AI PVT LTD to carry on the business of designing, development, customization, implementation, maintenance, testing and dealing in Information technology and Artificial Intelligence Solutions and to render consultancy, advisory and all such related services as are required by the customers in relation to the above areas.
10.	Applied Genomics Pvt. Ltd., Tamil Nadu	Agritech	Indian Biotech services company with expertise in the application of genomics to address various biological problems.
11.	Jeevamrut Pvt Ltd., Telangana	Food Processing & Value Addition, Soil & Water Conservation, Supply Chain Management	technology driven new age organization focused on food supply chain founded by experienced Professionals with cross-country diversed experienced in new age fields like Ecommerce, IT, Food Supplyc chain, Project Management
12.	Udvaban Technologies Pvt Ltd., Odisha	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	We make innovative and advanced IoT based Bee box monitoring system called Ezibees
13.	YayJay Digital Works LLP, Telangana	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	involved in developing a aggregator for floriculture and florist trades.
14.	Nyasta Gramojvala Solutions Pvt Ltd., Telangana	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	We provide the high quality, reliable, cost effective and high-performance irrigation solutions to empower farmers and agriculture.
15.	TechAdvent Mobility Pvt. Ltd., Maharashtra	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	TechAdvent aims to start manufacturing electric tractors.
16.	Khethari Agri Tech (OPC) Pvt. Ltd., Telangana	Sustainable inputs	Market place for all organic inputs
17.	Vakapi Vajra Srikavach Greentech Pvt.Ltd, Telangana	Green energy, water technology	working on surface technology to reduce carbon footprint and energy of rice mills. Enhance air quality and ensure health of workers. Enhance productivity and revenues by ensuring required standards without any negative impacts

18.	Rosys Virtual Solutions Pvt Ltd., Andhra Pradesh	Precision/ Smart Agriculture/ IoT/AI/ Sensor Technology	designing, developing and producing cutting edge technologies/ solutions in Artificial Intelligence, IoTs, Drones, Integrated Safety and Security Systems that finds a firm place in digital era.
19.	Physis Living Pvt. Ltd., Uttar Pradesh	Waste to wealth	manufacture bio-based, toxic free, natural products using agricultural waste to solve numerous environmental challenges and chronic illnesses.
20.	GENFLOW AI, Delhi	Animal husbandry	Providing Tele medicinal services for dairy farmers
21.	Fins And Tails Agritech Pvt Ltd., Telangana	ICT/IoT in Aquaculture	technology company creating sustainable aquaculture to bring fresh shrimp and fish direct to the consumer through neighborhood farms

4.1.13 Graduated Startups under KRISHIBOOT

S. No	Startup Name	Year of Induction	Domain	One liner
1	Isaayu Farms Pvt Ltd, Telangana	November 2021	Vertical Farming	Green Health through Hydroponics
2	Datair Technology Private Limited, Karnataka	November 2021	Precision Agriculture	Smart Farming Data through IoT
3	Inventohack Innovations Pvt Ltd, Madhya Pradesh	November 2021	Smart /Precision Agriculture	Conducting Quick Soil Nutrients Test to provide detailed report with predictions, suggestions and dilution of crop specific microbes/ fertilizers to make farmers increase profits/production organically
4	FarmOR Agri Solutions Pvt. Ltd., Telangana	November 2021	Sustainable inputs	One stop agri inputs ecommerce store for Retailers, FPOs, Nurseries.
5	SocialWell Technologies (OPC) Pvt. Ltd., Orissa	November 2021	Agri Tech	Automatic Business Generation Plan app for FPOs
6	Water N Spices Foodsz Pvt Ltd.,	November 2021	Food & Beverages	Automatic Food Cart – App based IoT enabled
7	Quick Solution Project Consultancy Pvt Ltd, West Bengal	November 2021	Food & Beverages	Next Generation Herbal Immune Booster
8	Microbial Research Lab Pvt Ltd, Telangana	November 2021	Agri-Biotechnology	Converting to organic farming by using soil micro-biota for higher crop yield.
9	Navshali Innovations Private Limited, Pyotam, Haryana	November 2021	Water & weather Technology	Alkaline Water with a Pure Indian Ayurveda Essence
10	Visron Pvt Ltd, Rajasthan	November 2021	Precision/ Smart Farming	Premium quality drone with Unique Airframe Design
11	ELAI Agritech Pvt Ltd, Karnataka	November 2021	Precision /Smart Farming	On-farm Tech for Yield Management

12	Farm Pool Pvt. Ltd. (Farm2Families), Delhi	November 2021	ICT & IoT in Agriculture	Aggregate Platform for Farmers to directly connect with Urban Families
13	Vattam Agro and Dairy Industries Pvt. Ltd., Tamil Nadu	December 2021	Sustainable Inputs	B2B Platform for Small & Medium E-Tailers
14	Starskill Consultancy Pvt. Ltd. (Animal ICU), Rajasthan	December 2021	Animal Husbandry	Animal ICU world's largest livestock healthcare B2B2C tech platform for Cattle, Credit and Care.
15	Pasidi Panta Pvt Ltd, Telangana	November 2021	Sustainable inputs	Disrupting the agriculture supply chain through an integrated digital platform
16	Contrivation Labs Pvt. Ltd., Karnataka	November 2021	Farm Mechanization	Calyxtract - Calyx and Stem negatively impact the Red Chilis
17	MLIT Sol Pvt Ltd, Telangana	November 2021	Animal Husbandry	Development of Artificial Intelligence based system for precision livestock farming
18	RR Animal Health Care, Telangana	December 2021	Animal Husbandry	Development of equine immunoglobulin IgG-F ab 2 hyper immune serum
19	Celebrating Farmers Edge International Pvt Ltd, Maharashtra	December 2021	Innovative Food Technology	New Eco system for a completely new range of unique value added products of sugarcane
20	Amvicube Private Limited, Karnataka	December 2021	ICT/IoT in Agriculture	Paddy quality tester

4.1.14 BIRAC BIG

The 20th call for applications for BIRAC-BIG Grant was announced on 1 January 2022. In this Call, a-IDEA received 81 innovative proposals exclusively from the agriculture and allied sector. In light of the opening of the new call, a series of four awareness webinars were organized in collaboration with following institutions various institute like CIIC, Chennai; AIC, AAU, Gujarat etc. Two startups are selected for the 20th Call of BIRAC BIG

Table 4.1.11: Startups selected for BIRAC BIG

S. No.	Name of the Startup	On liner
1.	Quantrello Digital Pvt. Ltd, West Bengal	Blockchain SaaS for enhancing visibility to Agri Supply chain
2.	Applied Genomics Pvt. Ltd, Tamil Nadu	Genotyping technologies to support seed industries

The 21st call for applications for BIRAC-BIG Grant was announced on 01 July 2022 and a-IDEA received 84 applications for further processing. Nine webinars were organized in collaboration with various institutes like ICAR-CPCRI, AIC NITTE etc. A total of 12 startups are selected for the final round.

4.1.15 NIDHI PRAYAS

NIDHI-Promotion and Acceleration of Young and Aspiring technology entrepreneur (NIDHI-PRAYAS) - Support from Idea to Prototype, is the scheme under NIDHI, aimed at addressing the gap in the very early stage of an idea/proof of concept funding. Rs. 1.2 Cr funding is given for one cycle valid for 30 months and 10 startups could be supported with Rs. 10 Lakhs each.

The 3rd call for NIDHI PRAYAS was opened on 25 November 2021 and the last date for application was 10 January 2022. The total number of applications received was 191. Based on eligibility criteria, 114 applications were shortlisted in the 1st round. In the 2nd round of selection, these 114 applications were assigned to two technical evaluators and one commercial/ business expert and 21 applicants were shortlisted based on the average percentage (cut-off marks was 65%). In the final round of selection, 10 out of 21 startups were selected by the PRAYAS Monitoring Committee on 24 March 2022 after which the 10 startups were onboarded and due diligence was carried out for the same.

Table 4.1.12: Startups selected for NIDHI PRAYAS

S. No	Name of the Startups	One Liner
1.	Ayurnosh, New Delhi	Gluten free and Vegan Snacks
2.	Fishy Farmers Pvt. Ltd., Telangana	Low Profile Auto Pneumatic Bead Filter for Shrimp
3.	Dr. Praneeth Juvvi (Individual), Karnataka	Vacuum fryer for mini and small-scale producers
4.	Sparrotronics Pvt Ltd, Karnataka	Smart Farming with Drone (BHADRA)
5.	Invenics Innovations Pvt. Ltd., Odisha	Portable Solar Agro-Storage Solution
6.	Alphaev Technologies Pvt. Ltd., Odisha	Multipurpose Electric Farm Utility Vehicle
7.	Assurity Techsolutions Pvt. Ltd., Odisha	Portable solar UV milk sterilizer
8.	Fermentech Labs Pvt. Ltd, Uttarkhand	Production and purification of high yield prebiotics
9.	Quick Solution Project Consultancy Pvt. Ltd., West Bengal	Healthy Herbal Tea
10.	Navshali Innovations Pvt. Ltd., Haryana	Water Based Ayurvedic Beverages

4.1.16 NABARD SEED Fund 1st Call

a-IDEA, ICAR-NAARM announced the 1st call for applications under NABARD CCF Investment support on 15 Sep 2022. In this Call, a-IDEA received 78 applications from various focus areas of Agri, Food, Rural and allied sectors, which have been verified with the documents and shortlisted 39 by the internal team members. The second level of screening was done by the external jury members during 1st week of Dec 2022, where the startups presented online/ offline mode and the committee shortlisted 10 applicants for the final round of presentations. In the final selection round, five startups were selected by the NABARD CCSMC Committee Members to receive investment from a-IDEA on 20 Dec 2022.

Table 4.1.13: NABARD SEED Fund 1st Call

S. No	Name of the Company	Amount Sanctioned (in INR Lakh)
1.	Renewagri Om Ecommerce Pvt Ltd, Maharashtra	25
2.	GreenGrahi Solutions Pvt. Ltd, Delhi	25
3.	Genflow AI, Delhi	15
4.	Foo Foods India Pvt Ltd, Kerala	25
5.	Starskill Consultancy Pvt Ltd, Rajasthan	10

4.1.17 NIDHI SSS SEED Fund 3rd Call

a-IDEA, ICAR-NAARM announced the 3rd Call for applications under NIDHI SEED FUND (DST Funded) from eligible incubatees/startups with the last date of applications to receive on 10 September 2022. This investment is meant to support start-ups in the focus area of Agriculture & the Allied Field. In this response, a-IDEA have received 51 applications which have been done preliminary screening and shortlisted 30 applications. The second level of screening was done by external jury members, where the startups presented online/offline and the committee shortlisted 11 applications out of 28 applications. In the final selection, four startups were selected by the 12th SSMC Committee Meeting Members on 16 Nov 2022.

Table 4.1.14: NIDHI SSS SEED Fund 3rd Call

S.No.	Name of the Company	Amount Sanctioned (in INR Lakh)
1.	Innentohack Innovations, Madhya Pradesh	25
2.	Organikness, Hyderabad	10
3.	Farmor Agri Solutions Pvt Ltd, Hyderabad	20
4.	Wingrow Agritech Producer Company Ltd, Maharashtra	25

4.1.18 BIRAC SEED Fund 4th Call

a-IDEA, ICAR-NAARM announced the 4th call for applications under the BIRAC SEED FUND (Bio Seed) from eligible incubatees/startups with the last date of applications to receive on 31 March 2022. In response, we have received and reviewed 58 applications. Based on the preliminary screening, 40 applications are shortlisted for pitching online. The second level of screening was done by Inter team members, where the startups presented online and the committee shortlisted 15 applications out of 40 applications. In the final selection, five startups were selected by the 11th SSMC Committee Meeting Members on 10 August 2022 to invest.

Table 4.1.15: BIRAC SEED Fund 4th Call

S. No	Name of the Company	Amount Sanctioned (in INR Lakh)
1.	Organikness, Hyderabad	15
2.	ELAI AgriTech, Bangalore	15
3.	Vattam Agro and Dairy Industries, Chennai	20
4.	Genflow AI, Delhi	10
5.	Khadyam Speciality Foods, Hyderabad	20

Glimpses of the Events at a-IDEA

Memorandum of Agreement with ICAR Institutes

AgriUdaan 5.0 Launch

Shri. Shaji K V Deputy Managing Director, NABARD visits a-IDEA, ICAR-NAARM

Dr Ashok Dalwai, CEO, NRAA visits a-IDEA, ICAR-NAARM

Padma Bhushan Dr Krishna Ella visits a-IDEA, ICAR-NAARM

PMRC Meeting of NABARD Project

COHORT Networking and Capacity Building Program

Visit of NBA Team

Glimpses of the Program Promotions through Webinars

Glimpses of the Program Promotions through Webinars

Glimpses of the call for applications

4.1.19 Accolades for a-IDEA, Incubatees

- Animal ICU received Techno fund of Rs. 15 lakhs from Government of Rajasthan.
- Animal ICU participated in iStart Rajasthan Digifest from 19th-20th August 2022, wherein Honourable CM of Rajasthan Shri Ashok Gehlot visited their expo unit and praised their efforts to control Zoonotic outbreak and cattle based ecosystem.
- Aumsat Technologies LLP represented India at IWA World Water Congress & Exhibition which was held during 11 -15 September 2022 at Denmark hosted by IWA and Denmark Technical University.
- Aumsat Technologies LLP, Maharashtra selected for Maharashtra Startup Week 2022 which is a platform for startups to showcase their innovative solutions and they can gain work orders of up to Rs. 15 lakhs.
- Aumsat Technologies LLP, Maharashtra for being selected to participate & won among top 7 global winners which was organised by United Nations World Food Program 48th Innovation Boot camp in ensuring water security for 3800 small holder farmers from 21st to 30th November 2022
- Aumsat Technologies LLP, Maharashtra for being awarded the Japanese Space Agency Award on 15 December 2022 for making the groundwater infrastructure of India resilient and sustainable.
- Cowbit technologies Private Limited was selected as top 75 startup Innovation under market linkage and animal health category.
- Cowbit Technologies Private Limited was selected as top startup Innovation under market linkage and animal health category in the Digital India Week 2022 held in Gandhinagar from July 4 to 6, which was inaugurated by the Hon'ble Prime Minister of India, Shri Narendra Modi.
- Fermentech Labs Private Limited is bestowed with "Young Innovative Entrepreneur Award" at Assam Biotech Conclave, organised by Guwahati Biotech Park at IIT Guwahati. Ms. Umme Habiba from Fermentech collected the award in the distinguished presence of Dr. Himanta Biswa Sarma (CM of Assam), and Shri Dharmendra Pradhan.
- Fermentech Labs Pvt. Ltd won Social alpha and DBS bank flagship 'Techtonic challenge' and raised Rs 44 Lakhs.
- Krishivan Technologies under the functional domain of Agri-solutions and Agri-advice has been selected for the UPJA-2022 Incubation program with a grant of INR 25 lakh by Pusa Krishi, ICAR-IARI out of 334 applications across India.
- Navshali Innovations Private Limited received patent for an invention "Improved Multi-Stage Water Purifier".
- Nimble Growth Organics which focuses on Agrisupply chain of fruits & vegetables with production traceability has been conferred the ABI Front Runner Award on 27th May, 2022 by Agribusiness Incubation Centre, ICAR Research Complex for NEH Region, Umiam, Meghalaya during the National Conference on "Agri StartUps: Prospects, Challenges, Technologies and Strategies"
- Nimble Growth Organics Pvt Ltd, Karnataka one of our incubatee for being selected and awarded among Top 50 Startups of the VentuRISE Global Start-up challenge - "Invest Karnataka 2022" Organized by TiE Banglore, TiE Invest Karnataka, TiE Global Angels Amazon on November 2-4, 2022.

- WaterN Spices Foodz Pvt Ltd (e-panipurikartz) is selected to participate at "World of Hospitality", a business event for Horeca professionals.

Awards:

- On the occasion of NAARM 47th Foundation Day on 01 Sep 2022 the following startups received best Startup Award-2022.
- Mr. Rahul Saria, Nimble Growth Organics Pvt. Ltd., Bengaluru under KRISHIBOOT
- Mr. Tarun Jami, Green Jams Buildtech Pvt. Ltd., Haryana under Nidhi Prayas
- Dr. Vishal Singal, Temperate Technologies Pvt. Ltd., Hyderabad under BIRAC BIG
- Mr. Chetan Soni, Aumsat technologies LLP, Mumbai won "MANAGE Samunnati Agri-Startup Awards 2022" on 14th September 2022.
- Animal ICU won Best Emerging Startup Award under Agricultural Category at the Business Excellence Award Ceremony organized by Global India Business Forum, which was facilitated by H.E Dr. Roger Gopaul (High Commissioner) High Commission for the Republic of Trinidad and Tobago on 12 July 2022.
- AYURNOSH was awarded ABI Women Power Award 2022 by Agribusiness Incubation Centre, ICAR Research Complex for NEH Region, Umiam, Meghalaya during the National Conference on "Agri Start Ups: Prospects, Challenges, Technologies and Strategies" (AGRiPACTS 2022) held on 27th May, 2022 in Manan.

4.2 Centre for Lifelong Learning in Agricultural Education (COLLAgE)

The centre which has been carved out of the efforts under the Niche Area project on Technology Enhanced Learning has been spearheading the activities of Educational Technology that support various online programmes like MOOCs and Distance Education programmes like PGDETM. The Centre actively supported conducting the online assessments for PGDETM programmes.

4.2.1 MOOCs in Digital Agricultural Education -Strategies to Enhance Effectiveness, Inclusivity and Reach

A new project is initiated to comprehensively study the Massive Open Online Courses (MOOCs) about enhancing their effectiveness, reach and inclusivity, given the fact that MOOCs adoption has risen tremendously post *Covid*. The objectives of the study are (i) To analyze factors affecting course completion rates in MOOCs. (ii) To develop the metrics to measure the effectiveness of MOOCs. (iii) To study the perception and impact of MOOCs on learners and their completion patterns. (iv) To evolve strategies to enhance participation in MOOCs.

A preliminary study was conducted on the effect of Covid on MOOCs adoption patterns in comparison with those offered during pre Covid period. The data harvested from conducting 9 MOOCs at the Academy from 2016 to 2021 were considered to analyze the learner profiles, participation patterns and performance in evaluations like Quizzes and assignments. The MOOCs conducted from May 2020 to December 2021 were considered Post Covid period. The comparative studies on pre and post-Covid MOOCs adoption pattern indicate that average

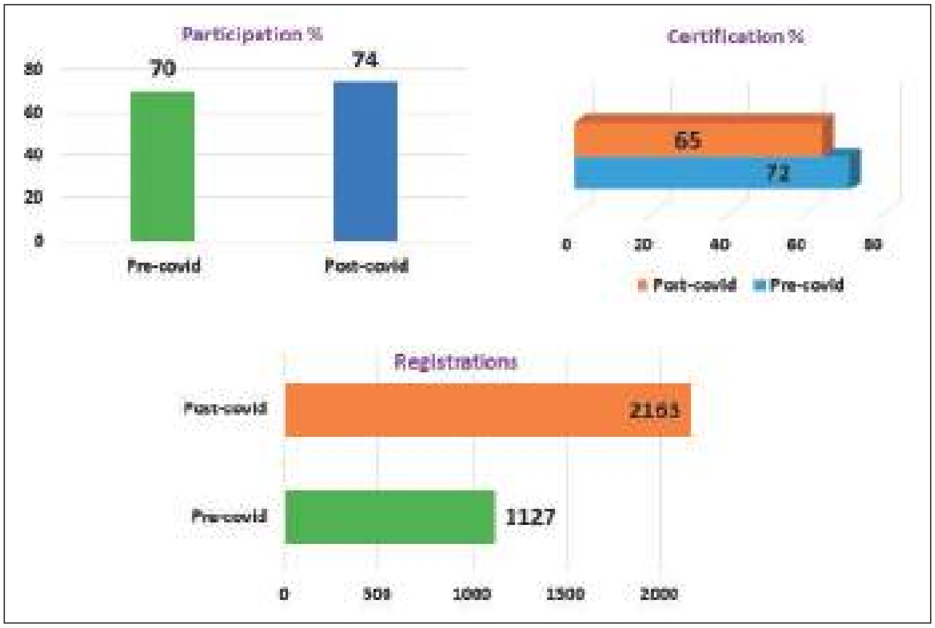


Figure 4.2.1: MOOCs participation pattern during Pre and Post Covid period

registrations for MOOCs have almost doubled while the average participation have increased by 4 per cent. However, average certifications were reduced by 7 per cent (Possible reasons: Enhanced evaluation mechanism and larger learner sizes who are likely to comprise more non-serious learners).

Learner profiles like age and subject domain are almost the same for pre and post-Covid MOOCs. Participant activity (measured as hits/ participant) increased by 27 per cent during Covid. Assignment submissions remained the same while the average marks in the final evaluation were reduced by 4 per cent.

4.2.2 Massive Open Online Courses (MOOCs)

Keeping in view the surge in demand for online learning and to use of technology in education, a MOOC on Digital Teaching Techniques was organised by the Academy during Dec 1-31, 2021. The evaluation and certification were completed in February 2022. The course was offered through e-learning portal of the Academy with 9 online digital modules with associated activities like assignments, discussions and quizzes. The program touched upon the aspects of exploring and using digital techniques and resources for enhancing quality of pedagogy and learning. The program was registered by 1870 participants from the entire country. Out of these, 1469 participants actively participated in the course of whom 801 participants successfully completed the course for certification.

Figure4.2.2: Massive Open Online Courses (MOOCs)

Training and Capacity Building

5

5.1 Training Programmes Attended by ICAR-NAARM Employees

Name	Programme	Duration (in days)	Organizers
Dr S Senthil Vinayagam	Three-day Capacity Building Program for Incubation Executives	October 13-15, 2022	I – Create, Ahmedabad, Gujarat Organized by ISBA.
Dr G R Ramakrishna Murthy	Training Programme on e-Learning Content Development for iGoT Karmayogi Platform (Online)	January 20-21, 2022	Institute of Secretariat Training and Management (ISTM)
Dr D Thammi Raju	Training Programme on e-Content Development and hosting the modules on the i-GOT Karmayogi Platform	January 20-21, 2022	Institute of Secretariat Training and Management (ISTM)
	GIZ -Dale Carnegie Training of Trainers Effective delivery of training using participatory training methods	April 26-29, 2022	Kodagu, Karnataka
Dr Surya Rathore	DoPT – GOI-sponsored 5 days State Level Course on Direct Trainer Skills (DTS)	April 25-29, 2022	Dr Marri Channa Reddy Human Resource Development Institute of Telangana, Hyderabad
Dr Bharat S Sontakki	Training of Trainers Programme on "Effective delivery of training using participatory training methods" organized jointly by GIZ India and Dale Carnegie under the Indo-German Cooperation Project on Human-Wildlife Conflict Mitigation in India (HWC)	April 26-29, 2022	Coorg Cliffs Resorts and Spa, Coorg, Karnataka, India
	GIZ-Dale Carnegie Online 'Coaching Connect' as a follow-up to the Training of Trainers Workshop attended during April 2022 at Kodagu	November 10, 2022	GIZ
	Workshop on Developing Implementer's Toolkits for Framework on Human-Wildlife Conflict Mitigation National Action Plan (HWC-NAP) and Supplementary Guidelines" organized by GIZ India under the Indo-German Cooperation Project on Human-Wildlife Conflict Mitigation in India (HWC)	June 7-8, 2022	Indira Paryavarana Bhavan, MoEFCC, New Delhi
Dr P Venkatesan	Analytical Techniques for Impact Evaluation Methods	April 25-30, 2022	Institute of Agricultural Sciences, Banaras Hindu University (BHU) organized by the International Food Policy Research Institute

Name	Programme	Duration (in days)	Organizers
Dr NA Vijay Avinashilingam	Managing Technology Value Chains for Directors and Division Heads	July 25 – 29, 2022	Administrative Staff College of India, Hyderabad
Dr MB Dastagiri	IFPRI: "Affordability of Healthy Diets in South Asia: Challenges and recommendations"	December 15-16, 2022	New Delhi, IFPRI
	Wiley Training and Webinars on "Essentials of Data Sciences" organized by Wiley publishers	August 24, 2022	Wiley publishers
Dr Umesh Hudedamani	Effective delivery of training using participatory training methods	April 26-29, 2022	GIZ India and Dale Carnegie under the Indo-German Cooperation at Coorg, Karnataka, India
Dr M Balakrishnan	Online Training on Bioinformatics, Genomics, and Next Generation Sequencing data analysis	December 12-30, 2022	Nextgenhelper, New Delhi
Dr P Supriya	Training On Application of Bioinformatics in Agricultural Research and Education	November 15-24, 2022	ICAR-NAARM
Dr Sanjiv Kumar	ICAR sponsored a 21-days winter school on 'Advances in Social Science Research and Evaluation'	January 25 to Feb 14, 2022	ICAR-NAARM
Dr Alok Kumar	Training Need Analysis	July 11-16, 2022	Department of Personnel & Training (DOPT), Govt. of India at Dr MCRHRDIT, Hyderabad
Mrs G Aneeja, CTO	The Prevention of Sexual Harassment Act-2013	January 20, 2022	NCD INDIA
	Public trust in news media during Covid-19, and implications for communication professionals	February 01, 2022	Global Alliance for PR and Communications Management
	Global communication and its dynamics in PR and Advertising	February 19, 2022	Public Relations Council of India, Pune Chapter
	Bridging the gap between Gen Y and Z	April 16, 2022	PRCI, Hyderabad Chapter

5.2 Capacity Building Programmes Organized by ICAR-NAARM

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Partici-pants	Programme Type
1.	Training Programme on Impactful ICT Applications and Technologies in Agriculture (Online Mode)	January 3-7, 2022	Dr VV Sumanth Kumar	20	Need-Based
2.	Off-Campus Training Programme on Integrated Farming Systems (IFS) for the Benefit of Farmers of Adopted Villages at KVK, Plaem (under Sub-Plan) (In-Person Mode)	January 4-6, 2022	Dr M Ramesh Naik	45	Off-Campus

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Partici-pants	Programme Type
3.	Off-Campus Training Programme on Integrated Farming Systems (IFS) for the Benefit of Farmers of Adopted Villages at KVK, Palem(under Sub-Plan) (In-Person Mode)	January 10-12, 2022	Dr P Vijender Reddy	45	Off-Campus
4.	Training Programme on Analysis of Experimental Data (Online Mode)	January 17-22, 2022	Dr S Ravichandran	133	Need-Based
5.	Training Programme on Data Visualization in Agribusiness and Agricultural Research (Online Mode)	January 17-22, 2022	Dr Sanjiv Kumar	32	Need-Based
6.	MDP on Intellectual Property Valuation and Technology Management (Online Mode)	January 18-22, 2022	Dr Umesh Hudedamani	17	Need-Based
7.	ICAR Sponsored Winter School on "Advances in Social Science Research and Evaluation" (Online Mode)	January 25 to February 14, 2022	Dr P Venkatesan Dr Bharat S Sontakki Dr N Sivaramane	40	RC/Summer/ Winter School-21days
8.	Sensitization Programme on "Entrepreneurship Development in Agriculture" students of MPUAT, Udaipur (a-iDEA) (Online Mode)	January 27, 2022	Dr Vijay Avinashlingam	190	Need-Based
9.	"Faculty Development Programme" for the Faculty Members of RAJUVAS, Bikaner (Online Mode)	February 4-11, 2022	Dr RVS Rao	60	Need-Based
10.	Induction Training On "Project Management & Research Methodology" For the Newly Recruited/Promoted Scientists of the Indian Council of Forest Research and Education (ICFRE), Dehradun (Online Mode)	February 7-18, 2022	Dr Umesh Hudedamani	26	Need-Based
11.	Off-Campus "Livelihood Enhancing Skill Development Training for Rural Women" On Eco-Friendly Bag making at KVK, Yagantipalle (In-Person Mode) (Under SCSP)	February 7-18, 2022	Dr P Supriya	20	Off-Campus
12.	Training Programme on "Geospatial Analysis using QGIS & R" (Online Mode)	February 14-19, 2022	Dr PD Sreekanth	43	Need-Based
13.	Training Programme on "Design Thinking in Research and Education" for College of Horticulture, Hyderabad (Online Mode)	February 15-19, 2022	Dr SK Soam	15	Need-Based
14.	Off-Campus Bakery as Enterprise: Training Organized for Women Farmers by ICAR-NAARM at Rythu Vedika, Amrabad (Mandal), Nagarkurnool, Telangana (Under SCSP) (In-Person Mode)	February 17-19, 2022	Dr M Balakrishnan	40	Off-Campus
15.	Training Programme on "Competency Enhancement Programme for Effective Implementation Training Functions by HRD Nodal Officers of ICAR" (Online Mode)	February 21-23, 2022	Dr RVS Rao	63	Need-Based

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Partici-pants	Programme Type
16.	Training Programme on "Soft Skills &Entrepreneurship Development" for students of College of Agriculture, Dr BSKKAV, Dapoli (Online Mode)	February 21-26, 2022	Dr BS Yashavanth	91	Need-Based
17.	Training Programme on Recent Advances in Organic Farming Research (Online Mode)	February 22-26, 2022	Dr M Ramesh Naik	50	Need-Based
18.	Coromandel training program NEAT Phase-II (Online Mode)	February 23-25, 2022	Dr Ranjit Kumar	30	Need-Based
19.	24th Annual Conference of Society of Statistics, Computer & Applications (RASTA - 2022) (Online Mode)	February 23-27, 2022	Dr Ch Srinivasa Rao	160	Conference
20.	Off-Campus Skill Development Training Programme on "Value Addition to Fruits" at KVK Palem, Nagar Karnool (Under SCSP) (In-Person Mode)	February 24-26, 2022	Dr M Ramesh Naik	21	Off-Campus
21.	Sensitization Programme on Entrepreneurship Development in Agriculture for students of the University of Agricultural Sciences, Raichur (a-iDEA) (Online Mode)	February 26, 2022	Dr Vijay Avinashlingam	150	Need-Based
22.	Training Programme on "Enhancement of Teaching Competencies of Agri Faculty" (Online Mode)	March 1-5, 2022	Dr D Thammi Raju	20	Need-Based
23.	Training Workshop on Data Visualization using R (Online Mode)	March 9-11, 2022	Dr BS Yashavanth	80	Need-Based
24.	Off-Campus Training Programme on Value Addition to Fruits for Farm men and Women at KVK Adilabad (under SCSP sub plan) (In-Person Mode)	March 9-11, 2022	Dr M Ramesh Naik	20	Off-Campus
25.	Cohort Networking & Capacity Building Program for the startups/incubatees of a-IDEA (In-Person Mode)	March 10-11, 2022	Dr S Senthil Vinayagam	20	Workshop
26.	Training Programme on Entrepreneurial Skill Development for Agricultural Graduates for students of Bidhan Chandra Krishi Viswavidyalaya (BCKV) at Mohanpur, West Bengal (under SCSP sub plan) (In-Person Mode)	March 11-13, 2022	Dr M Ramesh Naik	108	Off-Campus
27.	Training Programme on Faculty Development Programme on "Teaching Competency Enhancement through Innovative Methods" for Faculty of Dr YSRHU (In-Person Mode)	March 15-19, 2022	Dr VV Sumanth Kumar	48	Need-Based
28.	Off-Campus Skill Training on Jute Bag making for Farm Women, at KVK Ghantasala, Andhra Pradesh (under SCSP sub plan) (In-Person Mode)	March 16-20, 2022	Dr M Ramesh Naik	32	Off-Campus
29.	Training Programme on “Extension Programme Planning – A Systems Perspective” (Online Mode)	March 22-26, 2022	Dr Vijay Avinashlingam	24	Need-Based

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Partici-pants	Programme Type
30.	Training Programme on "Statistical Data Analysis for the Faculty of PJTSAU" (Online Mode)	April 6-21, 2022	Dr A Dhandapani Dr S Ravichandran Dr N Sivaramane Dr BS Yashavanth	483	Need-Based
31.	Workshop on Self-Sustainability in Edible oils in India (Online Mode)	April 20, 2022	Dr K Kareemulla Dr MB Dasthagiri	69	Workshop
32.	Executive Development Programme on Leadership Development (In-Person Mode)	May 9-14, 2022	Dr G Venkateshwarlu Dr D Thammi Raju	15	EDP on LD
33.	Off-Campus FDP on Competency Enhancement in Agricultural Research and Education for Faculty of Karnataka Veterinary, Animal and Fisheries Sciences University (KVAFSU) College of Fisheries, Mangaluru (Under Sub-Plan)	May 17-21, 2022	Dr M Balakrishnan	50	Off-Campus Programme
34.	Foundation Course for Newly Recruited Faculty of P.V. Narasimha Rao Telangana Veterinary University (PVNRTVU) (In-Person Mode) FOCFAU	May 18 to June 9, 2022	Dr D Thammi Raju Dr T Srinivas	25	FOCFAU
35.	Management Development Program on"Design Thinking in Agricultural Research and Education" (Online Mode)	May 23-28, 2022	Dr SK Soam	31	Need-Based
36.	Off-Campus FDP on Competency Enhancement in Agricultural Research and Education for Faculty of Sri Venkateswara Veterinary University (SVVU), Tirupati (Under Sub-Plan)	May 24-28, 2022	Dr M Balakrishnan	35	Off-Campus Programme
37.	Training Programme on "Entrepreneurial Skill Development" for 3rd and 4th year B.Tech, Dairy Technology students of National Dairy Research Institute NDRI, Karnal (In-person Mode)	June 2 to July 6, 2022	Dr Sanjiv Kumar Dr PC Meena	44	Need-Based
38.	Training Programme on Course Management through MOOCs (In-person Mode)	June 6-10, 2022	Dr GRK Murthy	35	Need-Based
39.	Brainstorming Workshop on Organic Farming in India (In-person Mode)	June 10, 2022	Dr M Ramesh Naik Dr Umesh Hudedamani	37	Workshop
40.	MDP on Business Plan Development and Accelerating FPOs/FPCs (In-person Mode)	June 13-18, 2022	Dr Ranjit Kumar	17	Need-Based
41.	MDP on Leadership Development (a Pre-RMP Programme) (In-person Mode)	June 14-25, 2022	Dr P Ramesh Dr Alok Kumar	20	a pre-RMP
42.	Management Development Programme for Newly Recruited Senior Scientists & Heads of Krishi Vigyan Kendras (In-person Mode)	June 15-29, 2022	Dr P Venkatesan Dr NA Vijay Avinashilingam	30	Need-Based
43.	Off-Campus HRD-Training to Skilled Support Staff (SSS) at KVK's Tuljapur, Babhleshwar – Maharashtra – State	June 16-19, 2022	Dr Laxman M Ahire Dr M Shekar reddy	33	Need-Based
44.	Foundation Course for Newly Recruited Faculty of Odisha University of Agriculture and Technology (OUAT) (In-person Mode) under NAHEP FOCFAU	June 17-30, 2022	Dr M Balakrishnan Dr M Ramesh Naik	20	Need-Based

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Partici-pants	Programme Type
45.	Off-campus Training Programme on Entrepreneurship Skill Development for Agricultural Graduates of KAU at the College of Agriculture, Vellayani (Under SCSP Sub Plan)	June 20-22, 2022	Dr M Balakrishnan	54	Off-Campus
46.	Off-campus Training Programme on Entrepreneurship Skill Development for Students of KAU at the College of Agriculture, Vellanikkara (Under SCSP Sub Plan)	June 23-25, 2022	Dr M Balakrishnan	54	Off-Campus
47.	Executive Development Programme on "Leadership Development" Batch - II (In-Person Mode)	July 4-9, 2022	Dr G Venkateshwarlu Dr D Thammi Raju	18	EDP on LD
48.	Training Programme on "Developing Winning Research Proposals" for the Faculty of PJTSAU (In-Person Mode)	July 5-7, 2022	Dr P Venkatesan	31	Need-Based
49.	Training Programme on "Developing Winning Research Proposals" for the Faculty of PJTSAU (In-Person Mode)	July 19-21, 2022	Dr Umesh Hudedamani	32	Need-Based
50.	Off-Campus Faculty Development Programme on "Competency Enhancement in Agricultural Research and Education" for faculty of Maharana Pratap University of Agriculture and Technology (MPUAT) Udaipur, Rajasthan (Under SCSP Sub Plan)	July 19-23, 2022	Dr M Balakrishnan	35	Off-Campus
51.	Off-campus Training Programme on "Nurturing Entrepreneurship Ecosystem in Agriculture Universities" for faculty of Kamadhenu University at Amreli, Gujarat	July 20-22, 2022	Dr K Karemulla	30	Off Campus
52.	Off-Campus Training programme on Soil testing under SC-Sub plan for the Farmers in collaboration with KVK Yagantipalle, AP (Under SCSP Sub Plan)	July 27, 2022	Dr M Balakrishnan	50	Off-Campus
53.	Training Programme on "Advances in Web & Mobile Application Development" (Online Mode)	August 2-6, 2022	Dr N Srinivasa Rao	39	Need-Based
54.	Training Programme on "Research Design and Statistical Analysis" for the R&D Team of Rallis India (In-Person Mode)	August 8-12, 2022	Dr A Dhandapani	25	Need-Based
55.	Training Programme on "Developing Winning Research Proposals" for the Faculty of PJTSAU (In-Person Mode)	August 17-19, 2022	Dr S Senthil Vinayagam	30	Need Based
56.	National Workshop on "Pathways for the Successful Implementation of SC Sub-Plan Scheme in ICAR" (In-Person Mode)	August 18-19, 2022	Dr M Balakrishnan Dr M Ramesh Naik	65	Workshop
57.	Training Workshop on Response Surface Methodology (On-line Mode)	August 18-20, 2022	Dr A Dhandapani	22	Workshop

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Partici-pants	Programme Type
58.	Training Programme on Nurturing the Entrepreneurial Ecosystem in Agricultural Universities (On-line Mode)	August 23-27, 2022	Dr Surya Rathore	77	Need-Based
59.	Training Workshop for Vigilance Officers of ICAR Institutes (In-person Mode)	August 24-26, 2022	Dr S Senthil Vinayagam	20	Workshop
60.	Off-Campus Training on Bio inoculants it's uses and Application in Agriculture for farmers under SC- Sub Plan at KVK, Gaddipally, Telangana State (Under SCSP Sub Plan)	August 25-26, 2022	Dr M Balakrishnan	60	Off-Campus
61.	BIRD-NAARM Collaborative Programme on "Interface of Producers Organization with Agri Startup" (In-person Mode)	September 1-3, 2022	Dr Ranjit Kumar	30	Need-Based
62.	MOOC on Digital Assessment and Evaluation Methodologies	September 1-30, 2022	Dr GRK Murthy Dr S Senthil Vinayagam	1731	MOOC
63.	Off-Campus Skill Development Programme on "Management of Horticulture Fruit Crops & Nurseries" SKLTSHU, KVK, Ramagirikhilla, Peddapalli Dist. (Under SCSP Sub Plan)	September 3-5, 2022	Dr M Ramesh Naik	205	Off-Campus
64.	Training Programme on ''Developing Winning Research Projects'' for faculty of Sri Konda Laxman Telangana State Horticultural University (On-line Mode)	September 5-9, 2022	Dr SK Soam	15	Need-Based
65.	MDP on Developing Winning Research Proposals (In-person Mode)	September 12-17, 2022	Dr SK Soam	33	Need-Based
66.	Competency Enhancement Training Programme on Motivation, Positive Thinking and Communication Skills for the Technical Officers (T-5 and above) of ICAR institute (In-person Mode)	September 13-16, 2022	Dr G Aneeja	23	Need-Based
67.	Training programme on Agricultural System Modelling for Natural Resource Management (In-person Mode)	September 19-24, 2022	Dr M Ramesh Naik	10	Need-Based
68.	Training Program on "Soft skills Development for Agri-Students" for Students of Sri Konda Laxman Telangana State Horticultural University (On-line Mode)	September 20-24, 2022	Dr VV Sumanth Kumar	64	Need-Based
69.	Off-Campus FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of RVSKVV, College of Agriculture, Indore, Madhya Pradesh (Under SCSP Sub Plan)	September 20-24, 2022	Dr M Balakrishnan	40	Off-Campus
70.	National Symposium on Mainstreaming of Agricultural Higher Education by Private Universities in India under NAHEP (In-person Mode)	September 29, 2022	Dr SK Soam	105	Symposium

Sl. No.	Title of the Programme	Period	Programme Directors	No. of Partici-pants	Programme Type
71.	Off-Campus Skill Development Training Programme on Eco-Friendly Bags making for Economic Empowerment of Rural Women at JAU-KVK Pipalia, Rajkot, Gujarat under SCSP	October 3-7, 2022	Dr M Balakrishnan	50	Off-Campus
72.	Training Workshop on Analysis of Multi-Environment Trials (On-line Mode)	November 3-8, 2022	Dr A Dhandapani	43	Workshop
73.	Training Programme on "Agri-Startups and Entrepreneurial Ecosystem:-An Indian Perspective" (In-person Mode)	November 14-19, 2022	Dr NA Vijay Avinashilingam	15	Need-Based
74.	Capacity Building Programme for "ICAR CJSC Members of ICAR Institutes" (In-person Mode)	November 15-19, 2022	Dr Vivek Purwar Mukul Raj Singh	42	Need Based
75.	Training on "Application of Bioinformatics in Agricultural Research and Education" Organized by (SKILLED-BIF, ICAR-NAARM) (In-person Mode)	November 15-24, 2022	Dr M Balakrishnan	17	Need-Based
76.	Off-Campus Skill Development training programme on "Backyard Poultry – A Sustainable Rural Livelihood" under SCSP in collaboration with Centre for Continuing Veterinary Education and Communication, Sri Venkateswara Veterinary University (SVVU), Tirupati, AP	November 26-28, 2022	Dr M Balakrishnan	30	Off-Campus
77.	Foundation Course for Faculty of Agricultural Universities (FOCFAU) (Online Mode)	December 1-21, 2022	Dr A Dhandapani Dr Surya Rathore	56	FOCFAU
78.	Collaborative Capacity Building Programme for "Young Agri-Business Professionals" (NEAT Phase-I)	December 12-16, 2022	Dr N Sivaramane	18	Need-Based
79.	MDP on Leadership Development (a Pre-RMP Programme) (In-person Mode)	December 12-23, 2022	Dr RVS Rao Dr Alok kumar	16	a pre-RMP
80.	Workshop on "Current Trends in Agricultural Bioinformatics" (Organized by SKILLED-BIF, ICAR-NAARM) (Hybrid Mode)	December 14-16, 2022	Dr M Balakrishnan	19	Workshop
81.	National Accreditation Board for Testing and Calibration Laboratories (NABL) Laboratory Assessor's Training Course (Accreditation Criteria ISO/IEC 17025:2017) (In-Person Mode)	December 19-23, 2022	Dr D Thammi Raju Dr Sanjiv Kumar	24	Need-Based
82.	Collaborative Capacity Building Programme for "Young Agri-Business Professionals" (NEAT Phase-I)	December 12-23, 2022	Dr PD Sreekanth	16	Need Based
83.	Training Programme on Analysis of Experimental Data (On-line Mode)	December 12-28, 2022	Dr S Ravichandran Dr P Supriya	64	Need-Based
83 Programmes with 5885 Participants					

5.3. Participation in Seminars/Symposium/ Conferences/Workshops (National)

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Presentation on Agricultural Research Management System (ARMS).	Virtual/ICAR	February 08, 2022
	National Level Consultations prior to the 36th FAO Regional Conference for Asia and the Pacific (APRC36)	Virtual/organized by the Office of the Representative in India, FAO of the United Nations	February 16, 2022
	FPO Summit	Virtual/ organised by CII and NCDEX jointly in partnership with NABARD and NABARD Consultancy Services Private Ltd (NABCONS)	February 28, 2022
	Webinar on COP26 and the Year Ahead	Virtual/ Oxford Climate Policy, Oxford	March 03, 2022
	ICAR Lecture Series #49 on Topic: Traditional Wisdom vis-à-vis Nutrition by Shri Atul Jain, General Secretary, Deendayal Research Institute under the Chairmanship of Secretary, DARE and DG, ICAR	Virtual/ICAR	March 21, 2022
	ICAR Lecture Series #52 under Azadi Ka Amrit Mahotsav on 'Saga of Science' by Dr S. Chandrasekhar, Secretary, DST, Govt of India.	Virtual/ICAR	April 11, 2022
	ICAR Lecture Series #53 under Azadi Ka Amrit Mahotsav on "Role of Youth for a secure and sustainable agriculture" by Dr R.S. Paroda, Chairman TASS and Former Secretary DARE and DG, ICAR.	Virtual/ICAR	April 21, 2022
	38th Foundation Day Celebrations of ICAR-CRIDA	Virtual/CRIDA	April 25, 2022
	ICAR Lecture Series #54 under Azadi Ka Amrit Mahotsav on Basics of Goal Setting by Dr Radha Shankar Narayanan, CEO, Smart Series	Virtual/ICAR	April 27, 2022
	Lecture #56 under Azadi Ka Amrit Mahotsav by Prof. Rattan Lal, World Food Laureate on "Managing Soil for Food and Climate Security and Advance SDGs of the UN" Chaired by Dr T. Mohapatra, Secretary (DARE) and DG (ICAR)	Virtual/ICAR	May 10, 2022
	Lecture #57 under Azadi Ka Amrit Mahotsav by Prof. Sanatanu Chaudhury, Director, IIT, Jodhpur on "AI and IOT: Enabling Sustainable Digital Agriculture" Chaired by Dr T. Mohapatra, Secretary (DARE) and DG (ICAR).	Virtual/ICAR	May 11, 2022
	Lecture #58 under Azadi Ka Amrit Mahotsav on "Issue of Child Rights in India - Challenges & Opportunities" by Mr Priyank Kanoongo, Chairperson, National Commission for Protection of Child Rights.	Virtual/ICAR	May 19, 2022
	National Symposium on Indian Agriculture after Independence to showcase the history, achievements and aspirations of Indian Agriculture in the last 75 years under the chairmanship of DG, ICAR.	Virtual/ICAR	May 24, 2022
	Foundation Day Programme of the National Academy of Agricultural Sciences	NAAS, New Delhi	June 05, 2022
	52nd & 53rd Combined Convocation of Acharya N.G. Ranga Agricultural University	Tirupati, A.P.	June 08, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Co-Chaired Session II : Agriculture 4.0 (Smart Farming) Ready Manpower: Stakeholder Perspective in the 14th National Symposium of IAUA "Creating Enabling Ecosystem in Agricultural Universities for Agri Tech Innovations"	PJTSAU, Hyderabad	June 09, 2022
	Inaugural Session of the Agri Info Hub Programme	PJTSAU, Hyderabad	July 01, 2022
	7th Interactive Meet of Nodal Officers of SAUs/DUs/CUs/CAUs	PJTSAU, Hyderabad	July 01, 2022
	94th Foundation Day of ICAR and Award Ceremony programme	A.P. Shinde Hall, New Delhi/ICAR	July 15-16, 2022
	National Symposium on Food, Nutrition and Environmental Security: Towards Achieving SDGs".	Virtual/TAAS, New Delhi, ICAR, NAAS and ISPGR in collaboration with Bioversity International and CIAT, ICRIST, CIMMYT and IRRI.	August 29-30, 2022
	Chief Guest in the Inauguration of the National Conference on Digital and Organic Interventions Towards Sustainable Agriculture Horticulture and Animal Husbandry (DOISHA-2022) on the occasion of World Food Day	ICAR-IIRR, Hyderabad/ organized by S&T SIRI Voluntary Organisation, Thorrur (Telangana).	October 16, 2022
	Chief Guest in the Orientation Day Programme for the third batch of UG, PG and Ph.D. Agriculture Courses for the academic year 2022-23	School of Agricultural Sciences, Malla Reddy University, Hyderabad.	October 20, 2022
	Presentation by Secretary, DARE & DG, ICAR on "Revitalizing ICAR: Aspirations and Action Plan".	Virtual/ICAR	November 11, 2022
	Presentation by DDG (CS), ICAR on Activities and Aspirations of ICAR	Virtual/ICAR	November 14, 2022
	Presentation by DDG (Agril. Engg), ICAR on Activities and Aspirations of ICAR.	Virtual/ICAR	November 17, 2022
	Presentation by DDG (AS), ICAR on Activities and Aspirations of ICAR.	Virtual/ICAR	November 22, 2022
	Presentation by DDG (Ag. Edn), ICAR on Activities and Aspirations of ICAR.	Virtual/ICAR	November 25, 2022
	Presentation by DDG (Horticulture), ICAR on Activities and Aspirations of ICAR.	Virtual/ICAR	December 05, 2022
	Opening Ceremony of International Year of Millets-2023 held at FAO Headquarters in Rome, Italy.	Virtual/FAO Hqrs	December 6, 2022
	Inaugural Session of International Conference on "Reimagining Rainfed Agro-ecosystems - Challenges & Opportunities"	Virtual/ ICAR-CRIDA, Hyderabad	December 22, 2022
	Celebration of KISAN DIWAS, 2022 held under the Chairmanship of Shri Narendra Singh Tomar, Hon'ble Minister of Agriculture & Farmers Welfare. Sushri Shobha Karandlaje & Shri Kailash Choudhary, Hon'ble Ministers of State for Agriculture & Farmers' Welfare.	Virtual/ICAR	December 23, 2022
	Chief Guest in the "FPO Start-up Immersion Program" organised by Centre of Excellence for Farmer Producer Organizations (COE-FPO) in collaboration with a-IDEA, NAARM	University of Horticultural Sciences Campus, Bengaluru.	December 30, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr S Senthil Vinayagam	National Conference on "NEP 2020 for Comprehensive Transformation in Agricultural Education – Initiatives, challenges and a way forward	UAS, Raichur	March 16, 2022
	National Conference on "Promoting Start-up Programme on Veterinary Entrepreneurship"	TANUVAS, Chennai	28 April 28, 2022
	National Conference on "Agri Start Ups: Prospects, Challenges, Technologies and Strategies" (AGRiPACTS 2022)	Agribusiness Incubation Centre, ICAR Research Complex for NEH Region, Meghalaya	May 26-27, 2022
	One-day international workshop on Science, Technology and Innovation for Sustainable Development Goals	Research and Information System for Developing Countries (RIS) jointly with the OPSA, New Delhi (Online)	October 11, 2022
	Ag Food Tech 360 International Conference	T-hub, Hyderabad – Organized by Lead Global Foundation	December 05, 2022
	State Credit Seminar	NABARD, Hyderabad	December 22, 2022
Dr G R Ramakrishna Murthy	6th I-LISS International Conference – 2022 on "Revitalizing the Libraries to the Android Society"	National Institute of Technology, Warangal	October 13 – 14, 2022
Dr D Thammi Raju	Brainstorming Session on Mainstreaming Agricultural Curriculum in School Education (MACE)	NASC, New Delhi	June 14, 2022
	State Level Workshop on "Bridging Research and Extension Gaps Among Multi-stakeholders for Sustainable Livestock Development in Karnataka State"	Veterinary College, KVAFSU, Bangalore	December 6-7, 2022
Dr V V Sumanth Kumar	International Conference on "Reimagining Rainfed Agro-ecosystems- Challenges & Opportunities (ICRA-2022)"	ICAR-CRIDA	December 22-24, 2022
	NAHEP sponsored ANGRAU Workshop for Organizing Session of Cloud-based Digital Agricultural Technology	Online for ANGRAU, Thirupathi Faculty	October 29, 2022
Dr Surya Rathore	National Seminar on 'Role of Agri-Entrepreneurship, Fisheries and in Atmanirbhar Bharat: Issues and challenges with special reference to Andaman and Nicobar Islands	Jawaharlal Nehru Rajkeeya Mahavidhyalaya, Port Blair	March 25, 2022
Dr Bharat S Sontakki	National Seminar on Comprehensive Extension Strategies for Sustainable Development of Farmer Producer Organization (FPOs): Issues and Challenges jointly organized by International Society of Extension Education, Nagpur and MANAGE Hyderabad	National Institute of Agricultural Extension Management (MANAGE), Hyderabad	April 22, 2022
	State Level Technical Programme of PJTSAU, Hyderabad as an invited expert in the area of Agricultural Extension.	PJTSAU, Hyderabad	May 11, 2022
	National Dialogue on Extension Services for Efficient Delivery of Horticultural Technologies: A way forward (in hybrid mode)	ICAR-IIHR, Bengaluru (ONLINE)	May 20, 2022
	National Seminar on Sustainable Food Production Systems for Self-Reliant and Climate-Resilient Agriculture	UAS, Dharwad	June 18, 2022
Dr P Venkatesan	10th National Seminar on Agriculture and More: Beyond 4.0	Society for Community Mobilization for Sustainable Development at SKUAST-Kashmir	May 26-28, 2022
Dr NA Vijay Avinashilingam	Agri Startup Conclave & Kisan Sammelan	IARI Pusa Ground, Dept. of Agriculture & Cooperation, Min. of Agriculture & Farmers Welfare	October 17 – 18, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr MB Dastagiri	Panel Discussion on "The Australia-India relationship: bilateral prospects in the context of the geopolitics and economic outlook of the Indo-Pacific.	Organized by The Centre for Policy Research (CPR) and Australian High Commission	August 04, 2022
	Wiley Training and Webinars on "Essentials of Data Sciences"	Wiley publishers	August 24, 2022
	"for Achieving Health Equity Together- Forging the Path Forward".	Springer Nature and Takeda	September 13, 2022
	FAO-SYMP-2022 "Asia-Pacific Symposium on Agrifood Systems Transformation"	FAO, Bangkok	October 07, 2022
	World Bank Live Development Events Brought "A conversation between IMF Managing Director Kristalina Georgieva and World Bank Group President David Malpass. The Way Forward: Addressing Multiple Crises in an Era of Volatility"	World Bank	October 10, 2022
	Webinar by ICC Annual Session with Dr V. Anantha Nageswaran, Chief Economic Advisor by Indian Chamber of Commerce	Indian Chamber of Commerce	November 07, 2022
	Webinar on "Debrief on the G-20 Summit"	FOREIGN POLICY Editor-in-Chief Ravi Agrawal	November 17, 2022
	"Regional Forum | CSA Technology for Sustainable Crop Production in Bangladesh: Adoption, Barrier and Way Forward"	IFPRI, Regional Forum Astana, Dhaka	December 13, 2022
	"Affordability of Healthy Diets in South Asia: Challenges and recommendations:	IFPRI, New Delhi	December 15, 2022
	Speaker - To commemorate Azadi Ka Amrit Mahotsav, the Centre for Policy Research (CPR): India and the World: A lecture series, Pravin Gordhan, Minister of Public Enterprises, Republic of South Africa	Centre for Policy Research (CPR), New Delhi	December 19, 2022
	VinFuture Prize Award Ceremony Week: 'Conversation with Members of VinFuture Prize Council and Pre-screening Committee' event in the VinFuture Prize Ceremony Week from December 17th to 21st, 2022. Hanoi,	VinFuture Prize Council, Hanoi, Vietnam	December 20, 2022
Dr M Ramesh Naik	GIZ India Consultancy Project on Human Wildlife Conflict (HWC) Mitigation in India in Agriculture and Veterinary Sectors	KVK, Kodagu, Karnataka.	August 07-08, 2022
	Advances in Agriculture and Food System Towards Sustainable Development Goals" (AAFS-2022)	University of Agricultural Sciences-Bangalore (UAS-B), Karnataka	August 22-24, 2022
Dr Umesh Hudedamani	Training of Rapid Response Teams under GIZ India Consultancy Project on Human Wildlife Conflict (HWC) Mitigation in India in Agriculture and Veterinary Sectors	Coorg, Karnataka, India	April 28-29, 2022
	Agri Udaan 5.0 Roadshow	VAMNICOM, Pune, Maharashtra.	September 06, 2022
	Entrepreneurship Development Program for the Agribusiness Management students	VAMNICOM, Pune, Maharashtra.	September 07, 2022
	Consultation roundtable on R&D statistics	Indian Institute of Science, Bangalore	November 28, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr S Ravichandran	73rd annual conference of Indian Society of Agricultural Statistics (Hybrid mode), Srinagar, India	SKUAST, Srinagar, India	November 14-16, 2022
	Webinar on "Inclusion of Specific Learning Disabilities (SLD) in Higher Education"	All India Council for Technical Education (AICTE)	September 07, 2022
	National webinar on 'Shikshak Parv'	All India Council for Technical Education (AICTE)	September 06, 2022.
	Save Soil: The world's largest people's Movement	ICAR Lecture Series #23, ICAR	August 25, 2022
	Science for the society: Agricultural imperatives	ICAR Lecture Series #21, ICAR	August 12, 2022
Dr S K Soam	Developing Winning Project Proposals	Online/ KVAFSU, Bidar	
Dr Yashavanth BS	Workshop on "Development of Implementer's Toolkits for HWC-NAP and Guidelines for HWC mitigation"	GIZ India	June 7-8, 2022
	Annual Conference of International India Statistical Association - 2022	Indian Institute of Science, Bengaluru organized by the International India Statistical Association	December 26-30, 2022
	R @ IISA2022 Workshop	IISc, Bengaluru organized by International India Statistical Association	December 30, 2022
Dr P Supriya	International Symposium on Advances in Plant Biotechnology and Nutritional Security (APBNS-2022)	Online/ ICAR-National Institute for Plant Biotechnology, New Delhi and Plant Tissue Culture Association-India	April 28-30, 2022
	International Conference on Plant Genetic Engineering and Genome Editing	Central University of Kerala	October 27-29, 2022
Dr Ranjit Kumar	Sunflower Field Day and Interaction with the Farmers	Siddipet, TS/ ICAR-IIOR, Hyderabad	March 05, 2022
	Expert Group Workshop of the AgroEco-2050 Foresight Study on Andhra Pradesh Community managed Natural Farming (APCNF)	Andhra Bhawan, New Delhi / RySS, Vijayawada	November 29-30, 2022
Dr Sanjiv Kumar	Sunflower Field Day and Interaction with the Farmers	Organized by ICAR-IIOR, Hyderabad at Chinnakodur village, Siddipet district, TS	March 05, 2022
Dr Alok Kumar	Mainstreaming of Agricultural Higher Education by Private Universities in India	NAHEP, Krishi Anusandhan Bhavan – II, New Delhi	March 23, 2022
Dr P Ramesh	Farmers' day (Kisan Diwas)	Agraharam potlapally	December 23, 2022

5.3.1 Participation in Meetings (other institutes)

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Review Meeting of Officers and Staff of ICAR Headquarters, Institutes of ICAR, ASRB and DARE under the Chairmanship of Secretary, DARE & DG, ICAR.	Virtual/ICAR	January 03, 2022
	ICAR SOC Meeting	Virtual/ICAR	January 06, 2022
	28th Board of Management Meeting of PJTSAU	Virtual/PJTSAU	January 08, 2022
	2nd Meeting of ICT Steering Committee	Virtual/ organized by Information and Communication Technology Unit, ICAR	January 12, 2022
	6th Meeting of Executive Council of NCCT	Virtual/NCCT, New Delhi	January 17, 2022
	NCCT Meeting to finalize parameters for deliverable targets and outcomes to review the training activities of NCCT institutes	Virtual/NCCT, New Delhi	January 21, 2022
	Meeting of the Institute Performance Review Committee	Virtual/NCCT, New Delhi	January 31, 2022
	123rd Meeting of the Executive Council of the National Academy of Agricultural Sciences	Hybrid mode/NAAS, New Delhi	January 18, 2022
	7th Meeting of Executive Council of National Council for Cooperative Training, New Delhi.	Hybrid mode/NCCT, New Delhi	February 05, 2022
	Annual Review Meeting with the CG Centres to discuss their ongoing projects/activities taken up during 2021 as per the approved Work Plan and future plan/activity to be taken in the year 2022	Virtual/ICAR	February 07, 2022
	ICAR SOC Meeting	Virtual/ICAR	February 08, 2022
	Interactive meeting by FA with Directors and Heads of Admin and Finance Officers of local ICAR Institutes	ICAR-NAAM	February 09, 2022
	Panelist in the Online Brainstorming Meeting on "Sustaining growth in pulses" in the Session on Conservation agriculture and natural farming on the occasion of 'World Pulses Day'	Virtual/ organized jointly by the ICAR-Indian Institute of Pulses Research, Kanpur and Indian Society of Pulses Research and Development.	February 10, 2022
	124th Executive Council Meeting of the National Academy of Agricultural Sciences	Virtual/NAAS, New Delhi	February 18, 2022
	First Meeting of the Steering Committee as Member to prepare Adaptation communication under UNFCCC under the Chairmanship of Additional Secretary, Ministry of Environment, Forest and Climate Change, New Delhi	Hybrid/MoEF& CC, New Delhi	February 18, 2022
	NCCT Committee Meeting for designing of course curriculum for launching Certificate Course on Agri-Logistics	Virtual/NCCT, New Delhi	February 26, 2022
	303rd Meeting of the Board of Management of Acharya N.G. Ranga Agricultural University.	S.V. Agricultural College, Tirupati/ANGRAU	March 04, 2022
	ICAR SOC Meeting	Virtual/ICAR	March 07, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	ICAR Committee on APAR	Virtual/ICAR	March 10, 2022
	Pre-Workshop Meeting of National Workshop on 'Agricultural Education in Private Universities in India: Let's Listen to Stakeholders' with Hon'ble Deputy Director General (Agril Education), ICAR	Education Division, ICAR, New Delhi	March 25, 2022
	93rd Annual General Meeting of the ICAR Society as Special Invitee	NASC Complex, New Delhi/ICAR	March 26, 2022
	26th Meeting of Food and Agriculture Division Council (FADC) of Bureau of Indian Standards (BIS), Govt. of India, under the Chairmanship of Secretary, DARE & DG, ICAR	Virtual/FADC, New Delhi	March 30, 2022
	29th Board of Management Meeting of PJTSAU, Hyderabad	Virtual/PJTSAU, Hyderabad	March 30, 2022
	ICAR SOC Meeting	Virtual/ICAR	April 07, 2022
	2nd Meeting of the Steering Committee to take stock of the work done by the Sectoral Working Group towards the preparation of India's Adaptation Communication, held under the Chairmanship of Ms. Richa Sharma, Additional Secretary, MOEF&CC, Govt. of India.	Virtual/ICAR	April 11, 2022
	57th Annual Rice Group Meetings	ICAR-IIRR, Hyderabad	April 25, 2022
	125th Executive Council Meeting of NAAS	NAAS, New Delhi	April 29, 2022
	Consultation Meeting on Glue Grant chaired by Shri Sanjay Murthy ji, Secretary, Department of Higher Education, Ministry of Education, Govt. of India.	Virtual/IIT Hyderabad	May 02, 2022
	Fourth Divisional Meeting of SMD	Virtual/Agricultural Education Division, ICAR	May 09, 2022
	Meeting to discuss the arrangements for the Prime Minister's Nationwide Interaction program with Farmers scheduled on 31.5.2022.	Virtual/CRIDA, Hyderabad	May 30, 2022
	Chaired the Meeting of the NCCT Committee for devising formats for output-outcome parameters	NCCT, New Delhi	June 02, 2022
	Conveners Meeting of NAAS Regional Chapters	NAAS, New Delhi	June 03, 2022
	126th Meeting of the Executive Council of the National Academy of Agricultural Sciences	NAAS, New Delhi	June 03, 2022
	29th General Body Meeting of **the** National Academy of Agricultural Sciences	NAAS, New Delhi	June 04, 2022
	304th Meeting of the Board of Management of ANGRAU	S.V. Agricultural College, Tirupati	June 07, 2022
	Online Meeting of ICAR Committee to revise APAR format.	Virtual/ICAR	June 23, 2022
	30th Meeting of the Board of Management of PJTSAU	PJTSAU, Hyderabad	June 24, 2022
	The valedictory session of the Training Programme on Communication and Documentation Skills for Professional Excellence of the Officers of the Department of Agriculture and Allied Sectors of the Southern States i.e. Andhra Pradesh, Telangana, Karnataka, Tamil Nadu, Kerala, Odisha, Puducherry, Andaman & Nicobar Islands and Lakshadweep.	EEI, Hyderabad	July 02, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Interaction Meet on the latest IPCC Report on Global Warming and the Upcoming COP-27, (Nominated by DG, ICAR),	Virtual/ organized by National Council for Climate Change Sustainable Development and Public Leadership (NCCSD), Ahmedabad	July 19, 2022
	Regional Advisory Group (RAG) Meeting of NABARD - First Meeting of Stakeholders for FY 2022-23 on the theme: Integrated Farming System Approach and Enterprise diversification for Enhanced Resilience of Small Holdings.	NABARD Regional Office, Hyderabad	August 03, 2022
	127th Meeting of the Executive Council of the National Academy of Agricultural Sciences (NAAS).	NASC Complex, New Delhi/NAAS, New Delhi	August 23, 2022
	Get-together Meeting by Shri G. Kishan Reddy, Hon'ble Minister of Culture, Tourism & Development of North Eastern Region (DoNER), Govt. of India with all Heads of Departments of Central Govt. situated in and around Hyderabad.	Hotel Taj Krishna, Hyderabad	August 26, 2022
	Meeting in the hybrid mode to discuss the creation, maintenance and operationalization of the Revenue Corpus Fund of ICAR under the Chairmanship of DG, ICAR	Virtual/ICARA August 29, 2022	August 29, 2022
	2nd Meeting of Expert Committee (EC) in Earth & Atmospheric Sciences for evaluation of N-PDF proposals by Science & Engineering Research Board (SERB), a Statutory body of the Department of Science and Technology, Govt. of India.	Indian Institute of Petroleum and Energy (IIPE), Visakhapatnam, Andhra Pradesh.	October 08-09, 2022
	ICAR SOC Meeting	Virtual/ICAR	October 12, 2022
	Chaired the 2nd Meeting of the NCCT Committee for devising formats for output outcome parameters.	NCCT, New Delhi	October 13, 2022
	ICAR SOC Meeting	Virtual/ICAR	November 16, 2022
	6th Divisional Meeting of SMD (Agricultural Education).	Virtual/ Agricultural Education Division, ICAR	November 18, 2022
	32nd Board of Management Meeting PJTSAU	Virtual/PJTSAU, Hyderabad	November 21, 2022
	NAAS Tech. Compendium Meeting	NAAS, New Delhi	November 29-30, 2022
	Meeting under the Chairmanship of DDG (Agric. Edn.) with the NAHEP Component 2 PIs/Co-PIs of IASRI, and NAARM.	NAHEP Committee Room, KAB-II, Pusa, New Delhi	December 01, 2022
	3rd ICT Steering Committee Meeting under the chairmanship of Secretary, DARE & DG, ICAR virtual mode	Virtual/ICAR	December 05, 2022
	ICAR SOC Meeting	Virtual/ICAR	December 6, 2022
	9th Meeting of Executive Council of NCCT, Ministry of Cooperation, Govt. of India.	Virtual/NCCT, New Delhi	December 8, 2022
	88th INSA Anniversary General Meeting	CSIR-National Institute of Oceanography, RC, Visakhapatnam.	December 14-16, 2022
	Meeting to review the procurement made through GeM and related matters held under the Chairmanship of AS&FA, DARE/ICAR.	Virtual/ICAR	December 20, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr Ch Srinivasa Rao	Chaired the Session on 'Sustainable soil management for resilient rainfed agro-ecosystem: conservation agriculture, organic farming, INM, soil-microorganisms-plant interactions' at the International Conference on "Reimagining Rainfed Agro-ecosystems - Challenges & Opportunities".	ICAR-CRIDA, Hyderabad	December 23, 2022
Dr S Senthil Vinayagam	Selection Committee meeting for conducting an interview for grant of advancement under NARS	MANAGE, Hyderabad	February 04, 2022
	The selection committee for promotion under CAS of ARS Scientist	ICAR-IIMR, Hyderabad	February 24, 2022
	The selection committee for assessing the promotion cases of ARS Scientists for the discipline of Agril. Extension	ICAR-IIRR, Hyderabad	March 02, 2022
	IMC meeting	ICAR-NRC Meat, Hyderabad	March 15, 2022
	Online BIRAC's Roundtable for Agri-based Incubation Centres	ICRISAT, Hyderabad	May 16, 2022
	Interaction meeting to set a framework for reviewing the BIRAC's 1st Nationwide call for E-YUVA proposals and finalized the interview modality as a knowledge partner	Tamil Nadu Agricultural University, Coimbatore, Tamil Nadu	May 06-07, 2022
	BIG 20 ESC Meeting	BIRAC BIG, DBT	July 12, 2022
	Roundtable Discussion G20 Presidency of India: Agriculture and Food Sustainability	Research and Information System for Developing Countries (RIS), New Delhi	July 15-16, 2022
	TQIMC Meeting	NIRD&PR, Hyderabad	July 18, 2022
	Regional Advisory Group Meeting	NABARD Andhra Pradesh Regional Office, Vijayawada	August 23, 2022
	Constitution of Selection Committee for the recruitment of Director, Monitoring and Evaluation (1 no) DDUGKY	NIRD&PR, Hyderabad	September 28, 2022
	Odisha Investors' Meet	Hotel Taj Krishna, Banjara Hills, Hyderabad – Organized by NABARD	October 17, 2022
	BIT Centre For Research (CFR) - Research Advisory Board Meeting	Bannari Amman Institute of Technology, Tamil Nadu.	November 19, 2022
Dr G R Ramakrishna Murthy	108th Meeting of the Academic Council of ANGRAU	ANGRAU, Guntur (AP)	July 14, 2022
Dr D Thammi Raju	Virtual Review meeting of Career Development Centers and Faculty Development Center	Online	January 07, 2022
	The selection committee for promotion under CAS of ARS Scientist	ICAR-IIMR, Hyderabad	February 24, 2022
	GIZ-Dale Carnegie Coaching Connect Meeting	Online	November 10, 2022
	NAHEP Review Meetings	Online	July 02, 2022 November 15, 2022
	Pre-Workshop Meeting on 'Agricultural Education in Private Universities in India: Let's Listen to Stakeholders	KAB, New Delhi	March 24, 2022
Dr V V Sumanth Kumar	Member of Selection Committee for evaluation of KRISHIBOOT Applicants	Online	March 07, 2022
	Orientation Meeting for DETM 2022 Students	Online	May 05, 2022

Name of staff	Name of the Programme	Venue / Organized by	Date
Dr Bharat S Sontakki	Attended DPC Meeting of Agricultural Extension Scientist at ICAR-NIANP, Bengaluru as Expert nominated by DG, ICAR	ICAR-NIANP, Bengaluru	January 06, 2022
	Online DPC Meeting for Promotion of Library (Technical) Staff of ICAR-CIFE, Mumbai	ICAR-CIFE, Mumbai (ONLINE)	March 10, 2022
	29th Meeting of Institute Management Committee of ICAR-Indian Institute of Millets Research, Hyderabad	Indian Institute of Millets Research, Hyderabad (Virtual Mode)	May 22, 2022
Dr S K Soam	Annual Review Meeting by DG, ICAR	Online/ ICAR, New Delhi	February 04, 2022
	Pre-conference meeting for the organization of "International Conference on Blended Learning Ecosystem for Higher Education in Agriculture, 10-12 October 2022" by ICAR under component 2 of NAHEP, New Delhi	Online/NAHEP, PIU, New Delhi	June 27, 2022
Dr N Srinivas Rao	Pre-conference meeting for the organization of "International Conference on Blended Learning Ecosystem for Higher Education in Agriculture, 10-12 October 2022" by ICAR under component 2 of NAHEP, New Delhi	Online/NAHEP, PIU, New Delhi	June 27, 2022
Dr Ranjit Kumar	NITI Aayog Agri-tech Committee meeting	Online (NITI Aayog)	January 06, 2022
	Project Management Committee meeting of the study on Co-Benefits of Largescale Organic Farming on Human Health (BLOOM)	The Online/ University of Edinburgh and Public Health Foundation of India (PHFI)	February 21, 2022, August 22, 2022, December 19, 2022
Dr B Ganesh Kumar	XXVI meeting of ICAR Regional Committee – II for the states of West Bengal, Odisha, Telangana, Andhra Pradesh and the UT of Andaman & Nicobar Islands	ICAR-NRRI, Cuttack in hybrid mode	October 14, 2022.
	XXVI meeting of ICAR Regional Committee – IV for the states of Uttar Pradesh, Bihar and Jharkhand	ICAR-IIVR, Varanasi in hybrid mode	November 07, 2022.
Dr Alok Kumar	The selection committee for assessing the promotion cases of ARS Scientists in the discipline of Agril. Extension	ICAR-DOGR, Pune	August 11, 2022
	Special guest in an online webinar on Career Counselling and Personality Development.	Indira Gandhi Krishi Visvidalya, Raipur, Chhattisgarh	March 29, 2022

Happenings at NAARM

6

6.1 Special Events at NAARM

6.1.1 24th Annual Conference of Society of Statistics, Computer & Applications (RASTA-2022)

The 24th Annual Conference of the Society of Statistics, Computer and Applications (SSCA) titled 'Recent Advances in Statistical Theory and Applications – RASTA 2022' was organized by the Academy during February 23-27, 2022. The conference aimed at bringing statisticians, economists, computer scientists, and researchers from other disciplines to a common platform and discuss the advances made, emerging issues and learn from one another. Dr RC Agrawal, DDG (Education), while inaugurating the conference and later giving a special lecture on the landscape of Agricultural Education, informed the gathering that the popularity of Agricultural Education is rising and the Council is committed for improving the quality of Agricultural Universities as well as increase the intake to meet the growing requirements of graduates from agricultural sciences. Dr Ch. Srinivasa Rao, Director, ICAR-NAARM and Chief-Patron of the conference emphasized on the emerging issues in Agricultural sciences, particularly climate change challenges and India's commitment to reducing greenhouse gases. He urged the statistical community to rise to provide solutions. Dr VK Gupta, former National Professor, ICAR and the president of SSCA, in his remarks, urged the young researchers to work harder and explore new avenues.

During the conference, 51 invited talk sessions, 68 Young Researchers' invited talk/contributory talks, 4 endowment lectures, one special session on Financial Statistics, one keynote address, one Memorial lecture in the name of late Prof. MN Das and two memorial sessions in the names of late Dr Hukum Chandra and late Dr Lal Mohan Bhar were conducted. Some of the key issues deliberated in the conference include: "Noval Differential Abundance Analysis of Gut Microbiota" by Dr Shyamal Peddada, "Alternatives to Global Index" by Prof. Arun Kumar Nigam, "Inequalities among Agricultural Households" by Dr K.J.S. Satyasai, advances in the area of Bayesian Data Analysis, Survival Analysis etc. A young researcher, Dr SM Patil from Shivaji University, Kolhapur, Maharashtra won the prestigious Prof. MN Memorial Young Scientist Award. More than 160 researchers, students, and experts from different parts of the country registered for the conference. The conference was convened by Dr G. Venkateshwarlu, Joint Director of the Academy and Dr A. Dhandapani and Dr S. Ravichandran of ICAR-NAARM were

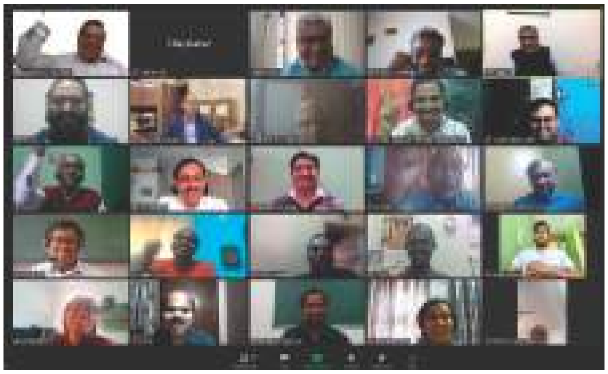


Figure 6.1.1: A snapshot of the participants of 24th Annual Conference of Society of Statistics, Computer & Applications (RASTA-2022)

the organizing Secretary and Co-Organizing Secretary of the conference, respectively.

6.1.2 4th Graduation Ceremony of PGDM-ABM

The Academy organized 4th Graduation Ceremony for its Post-Graduate Diploma in Management- Agri Business Management (PGDM-ABM) programme on May 14, 2022. As Chief Guest of the Ceremony, then Hon'ble Vice

Figure 6.1.2.a : Shri. M. Venkaiah Naidu, former Vice President addressing the gathering during the 4th Graduation Ceremony of PGDM-ABM

President Shri. M. Venkaiah Naidu awarded the degree to 144 students from four batches. One student from each batch was also awarded the Director's Medal for their outstanding overall performance and the Gold Medal for their best academic performance. In his Graduation Ceremony address, the Vice President advised the students to work on the problems faced by the farming community not just to increase their productivity but also to enhance overall welfare. He also told that India could become Atmanirbhar only when the agriculture sector continues to grow in the future as well. For this, the sectors need innovations and the use of frontier technologies that can be disseminated through agribusiness professionals.

On this occasion, Dr Trilochan Mohapatra, Secretary-DARE and DG-ICAR, in his Guest of Honour remarks, commended the PGDM-ABM programme of the Academy, which not only achieved 100 percent placement but also provided quality manpower to agribusiness industry and startups. He appreciated the Academy for its contributions of the capacity building of NARES and Think-Tank Research Policy Development in agriculture and food systems for the country.

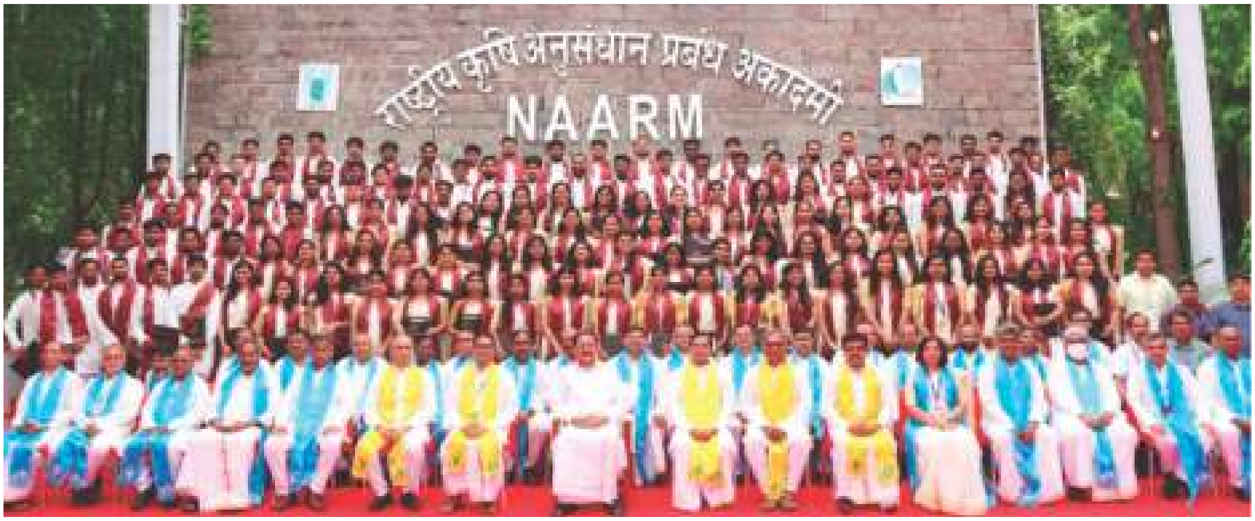


Figure 6.1.2.b: Degree recipients along with M. Venkaiah Naidu, former Vice President, and faculty of ICAR-NAARM

Dr Ch. Srinivasa Rao, Director, ICAR-NAARM, while presiding over the ceremony, presented overall achievements of skill development, education programmes, placement activities, international trainings and policy development covering all subsectors of agriculture. Dr G. Venkateshwarlu, Joint Director & Dean, ICAR-NAARM welcomed the distinguished guests and the degree recipients and anchored the proceedings.

6.1.3 MoU with Swarna Bharat Trust

The Academy signed an MoU with Swarna Bharat Trust (SBT), Muchintal on July 4, 2022. The objective of this MoU is to set up the general principles of collaboration between both parties, according to which the parties may jointly identify fields of mutual interest and create opportunities to develop programmes for cooperation in training, education and other knowledge-based activities based on reciprocity and mutual benefit. On this occasion, Dr Ch Srinivasa Rao mentioned that ICAR-NAARM will recommend technologies or skill development programmes best suited for rural youth and trainees to Swarna Bharat Trust to train for these programmes. He also mentioned the possible visiting of ICAR-NAARM trainees to the skill development center, organic farming block, vermicompost units, seed processing unit and custom hiring center of the Swarna Bharat Trust for better exposure to them. HoDs and the members of the PME Cell were present during this event.

Figure 6.1.3: Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, faculty and officers of ICAR-NAARM with the representatives of Swarna Bharat Trust

6.1.4 Lecture on Krishi becoming Rishi

Shri Kamlesh D. Patel, President, Sri Rama Chandra Mission, and Spiritual guide, Heartfulness Meditation (fondly known as Daaji) delivered a spiritual lecture on "Krishi becoming Rishi" as a part of "Azadi ka Amrit Mahotsav" ICAR Lecture Series organized at the Academy on July 8, 2022. Daaji's lecture is the 67th of the massive outreach campaign titled "75 lecture series" by ICAR as a part of the 75th year of India's independence.

While delivering the lecture, Daaji said food is a basic need of life, one can think of the integrity of mind, body and spirit only when well-nourished. He advised to tune the self-consciousness to higher consciousness through meditation which will result in creating great positive vibes to attain overall well-being of the mind and body besides greater wisdom. Dr RC Agrawal, DDG (Agril. Education), ICAR briefed the contribution of ICAR along with Agricultural Universities which have made the country self-reliant in food production. Speaking on the occasion, Dr Ch. Srinivasa Rao, Director said soil conservation, natural resource management, and self-peace with a focused mind are the key points to be followed from Daaji's lecture. He thanked Daaji for gracing the occasion, and delivering a heartfulness lecture in tune with agriculture, and practicing the meditation by the audience. The event was coordinated by Dr G. Venkateshwarlu, Joint Director and attended by Vice-Chancellors of Agricultural Universities, Deputy Director Generals of ICAR, Directors of

Figure 6.1.4: Shri Kamlesh D. Patel (Daaji) delivering the lecture on “Krishi becoming Rishi” as a part of “Azadi ka Amrit Mahotsav”

Figure 6.1.5.b: Team ICAR-IIHR, Bengaluru after winning the overall championship during ICAR-South Zone Sports Meet

various ICAR institutes, Staff of ICAR-NAARM, Senior Professors and faculty of Agricultural Universities, Teachers, scientists and students of NAARM and PJTSAU.

6.1.5 ICAR-South Zone Sports Meet

The zonal sports meet of 26 ICAR Institutes of South India was organized by the Academy during November 22-25, 2022. This was for the ninth time NAARM has hosted this tournament. The sports meet was inaugurated by international athlete Shri. N. Ramireddy at the South Central Railway - Railway Sports Complex, Secunderabad. The sports meet was held after a two-year gap due to the COVID pandemic. Over 750 sportspersons participated in these sports meet from 25 ICAR-Institutes. T10 cricket was introduced for the first time in this sports meet. ICAR-IIHR emerged as the overall champion with the maximum points in this tournament followed by CMFRI and IIRR. Mr. P. Rajendran of CMFRI emerged as the best male athlete while Dr Visha Kumari of CRIDA was adjudged as the best female athlete. Mr. A. Mohamed Kaleem from CMFRI was awarded with a special recognition for his all-around performance and for scoring a century in the T10 match in this tournament. Mrs. Alka Nangia Arora, IAS, Addl. Secretary DARE & FA, ICAR was the chief guest for the concluding ceremony while Shri. RK Purushottam, General Secretary, Indian Olympic Association - AP Chapter was the guest of honor.

Figure 6.1.5.a: Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM congratulates winners of an event during the ICAR-South Zone Sports Meet

6.1.6 National Workshop on Pathways for Effective Implementation of SC Sub-plan Scheme in ICAR

The Academy conducted a two-day National Workshop on Pathways for Effective Implementation of SC Sub-Plan Scheme in ICAR during August 18-19, 2022 in Hybrid mode. The objectives of the workshop were to deliberate on issues and challenges faced in the effective implementation of SCSP across ICAR institutions and to identify consensus-based operational reforms and good practices for the effective implementation of the SCSP scheme. A total of 62 participants representing 21 ICAR institutes implementing SCSP attended the programme of which 34 attended in person.

During the inaugural programme, Dr Ch. Srinivasa Rao, Director expressed that the guidelines which are being developed in discussion with the nodal officers of ICAR institutions will ensure smooth implementation of the SCSP scheme in ICAR and thereby ensure livelihood security of deprived sections of the society. Dr G. Venkateshwarlu, Joint Director stressed the need for identifying entrepreneurial modules in the agri-horti sector which greatly improves nutritional security and ensures equity among different sections of the society in Rural India.

During the workshop, brainstorming sessions were held on three thematic areas namely challenges faced in implementation; good practices; and operational reforms. The points emanated from the brainstorming sessions were synthesized in the plenary session held on Aug 19, 2022. Dr Ch. Srinivasa Rao stressed the urgent need to bring out the guidelines document for the effective implementation of the scheme at ICAR.

Figure 6.1.6: Participants of the National Workshop on Pathways for Effective Implementation of SC Sub-plan Scheme in ICAR

6.1.7 Sustainable Fishery for Food and Nutritional Security in India on World Fisheries Day

The Academy, in association with NAAS-Hyderabad Chapter, organized an online lecture by Dr M Vijay Gupta, World Food Prize laureate on November 21, 2022 on the occasion of World Fisheries Day. Dr Ch. Srinivasa Rao, Convener NAAS Hyderabad Chapter and Director, NAARM welcomed the dignitaries and guests on the occasion and briefly outlined the activities of the NAAS Regional Chapter-Hyderabad. He also highlighted the importance and significance of World Fisheries Day. Dr M Vijay Gupta, World Food Prize and Sunhak Peace Prize laureate delivered a lecture entitled "Sustainable Fishery for Food and Nutritional Security in India" wherein he briefed on the role of fish in food, nutrition and trade. He outlined the various challenges for meeting the demand for aquaculture and ways to address them. Dr P. Appa Rao, former Vice Chancellor, University of Hyderabad presided the program and stressed upon the need for the protection of the vulnerable coastal ecosystem. A total of 200 participants from SAUs, ICAR institutions, including NAAS, Regional Chapter-Hyderabad fellows, and associates joined the program virtually.

Figure 6.1.7: A snapshot of the participants of the Guest Lecture on Sustainable Fishery for Food and Nutritional Security in India on World Fisheries Day

6.1.8 Soil Management for Climate Change Mitigation on World Soil Day

The Academy, in association with NAAS-Hyderabad Chapter, organized an online Lecture on 'Soil Management for Climate Change Mitigation' by Dr Anil Kumar Singh, Vice President, NAAS and Former DDG (ICAR) on December 5, 2022 on the occasion of World Soil Day celebrations. In his lecture, Dr Singh gave a detailed account of the soil and outlined the status and importance of soil resources, the role of soil in supporting the 2030 Sustainable Development Goals (SDGs). The technologies for climate change mitigation and the impact of these mitigation technologies were also highlighted. Dr Ch. Srinivasa Rao, Convener of NAAS Regional Chapter-Hyderabad and Director, NAARM welcomed the dignitaries and guests on the occasion and briefly outlined the activities of the NAAS Regional Chapter-Hyderabad. He also highlighted the importance and significance of World Soil Day. Dr B Venkateswarlu, former Vice Chancellor, VNMKV, Parbhani & Fellow of NAAS stressed upon the need to work on the adaptation of soil in terms of drought management. A total of 100 participants from SAUs, ICAR institutions, including NAAS, Regional Chapter-Hyderabad fellows, and associates joined the program virtually.

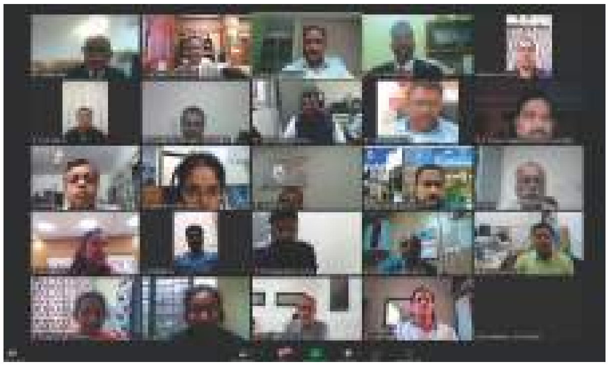


Figure 6.1.8: A snapshot of the participants of the Guest Lecture Soil Management for Climate Change Mitigation on World Soil Day

6.2 Visits and Interactions of Dignitaries

6.2.1 Interaction with Additional Secretary & FA, DARE/ICAR

Shri Sanjiv Kumar, Additional Secretary and Financial Advisor, DARE/ICAR interacted with Directors of Hyderabad based ICAR Institutes, Heads of Administration and Heads of Finance on February 9, 2022 at the Academy to review finance and administrative issues and budget utilization.

Figure 6.2.1: Shri Sanjiv Kumar, Additional Secretary and Financial Advisor, DARE/ICAR with Directors of Hyderabad based ICAR Institutes, Heads of Administration and Heads of Finance

During the interaction, he expressed his deep satisfaction with the working of the ICAR system and stressed the importance of knowing the working of institutes from field experience. He informed that the ICAR system is so vast that no activity of the Government can happen without the research undertaken by more than 100 different institutes under ICAR, and KVKs. He appreciated the green campus of ICAR-NAARM and termed it as the ideal place for learning. He stressed that the financial resources are limited and we have to generate more and more resources internally with the expert manpower the system has. Directors informed the house about the budget utilization as well as the achievement and activities of the respective institutes to him. He further interacted with administrative and finance heads on specific issues of the ICAR-Institutes.

In his welcome address, Dr Ch Srinivasa Rao, Director highlighted the importance of timely budget utilization and also internal resource generation and assured on behalf of all institutes that they will utilize 100% grants on priority activities of the Institutes.

6.2.2 Visit of NABARD Officials

Senior Officers from NABARD visited a-IDEA on March 15, 2022 to facilitate interaction with startups and acknowledge the progress of a-IDEA. In his presidential remarks, Dr. Ch. Srinivasa Rao, highlighted the importance of food production, shortage and wastage occurring in the country. He also expressed his views on how to reduce food losses through efficient & smart production with less carbon footprint. He mentioned about the challenges faced in the Agri Sector and the role of the National level adaptation fund by NABARD to turn those challenges into opportunities.

6.2.3 Shri. Shaji K V Deputy Managing Director, NABARD interacts with Agri-Startups of a-IDEA

Shri. Shaji K V, Deputy Managing Director, NABARD visited a-IDEA on April 18, 2022 and interacted with startups and acknowledged the progress of a-IDEA. During his visit, the incubatees of a-IDEA viz., Dhi Sathi Pvt Ltd (Farm Sathi), Isaayu Farms Pvt. Ltd., MLIT Solutions Pvt. Ltd. and RR Animal Health Care Pvt. Ltd presented their status of project and its benefits to the farming community.

Dr G Venkateshwarlu, Joint Director, ICAR-NAARM, in his address, highlighted about funding by DBT, 113+ ICAR institutions for networking and its synergies for upscaling of innovations in the startup ecosystem.

Shri. Shaji K V highlighted the need for strengthening the development of the digital platform for e-commerce and stressed the importance of technology as demand-driven rather than supply-driven to address the challenges in rural areas. Dr Ch. Srinivasa Rao, Director, ICAR-NAARM, in his presidential remarks, highlighted the existing technological support, mentoring support etc. by NARES system for entrepreneurs, students and startups.

6.2.4 Visit of Dr Ashok Dalwai, CEO, NRAA, New Delhi

Dr. Ashok Dalwai, CEO, National Rainfed Area Authority, New Delhi visited a-IDEA on April 23, 2022. Dr. Bharat S Sontakki, Head, XSM Division

Figure 6.2.2: Shri. Shaji K V, Deputy Managing Director, NABARD welcomed by Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM

Figure 6.2.3: Dr. Ashok Dalwai, CEO, National Rainfed Area Authority, New Delhi with team a-IDEA and NAARM faculty

welcomed him to the Academy & a-IDEA, while Dr. Senthil Vinayagam, CEO, a-IDEA made a brief presentation on various activities of a-IDEA towards strengthening startup & entrepreneurial ecosystem in agriculture & allied areas. Dr. Shobha Nagnur, Professor, College of Community Sciences, UAS Dharwad graced this visit.

6.2.5 Padma Bhushan Dr Krishna Ella visits a-IDEA, ICAR-NAARM

Dr Krishna Ella, awardee of Padma Bhushan and Chairman & Managing Director of Bharat Biotech International Limited (manufacturer of India's first indigenous Covid-19 vaccine, namely Covaxin) visited a-IDEA on June 03, 2022. The visit was aimed at having an understanding of the operations, programmes that are undertaken at and startups supported by a-IDEA. Team a-IDEA made a brief presentation on various activities under schemes viz., NIDHI PRAYAS, BIRAC-BIG, Bio-NEST and AGRIUDAAN supported by DST, DBT, NABARD of Govt. of India.

Figure 6.2.4: Dr Krishna Ella, Chairman & MD of BBIL with team a-IDEA and NAARM faculty

Dr Krishna Ella expressed his views on the role of startups in bringing a gradual transformation in the job sector with the younger generation moving towards being job providers instead of job seekers by assessing their growth in the startup ecosystem by quoting an example of Tasty Bites. He extended his continued support for the organization and also conveyed his regards to the faculty and team of a-IDEA for the commendable work put forth to the startup ecosystem.

6.2.6 Shri Giriraj Singh, Hon'ble Minster for Rural Development and Panchayat Raj visits ICAR-NAARM

Shri Giriraj Singh, Hon'ble Minister for Rural Development and Panchayat Raj visited the Academy on July 1, 2022 and addressed the faculty of the Academy. Hon'ble Minister expressed happiness in knowing about the Academy's activities in various dimensions. During his address, he emphasized the importance of agribusiness management and value addition for the self-help groups/ Farmer Producer Organizations. He opined that there is a strong need of developing protocols for complementing MGNAREGA activities in promoting Agroforestry technologies in the rural areas which would not only build the green cover in the country but also help in carbon sequestration. He released the Green Campus of ICAR-NAARM, a dossier on this occasion.

Dr Ch Srinivasa Rao, while welcoming the Hon'ble Minister, gave an overview of the Academy's activities in capacity building, Think Tank role

Figure 6.2.5: Shri Giriraj Singh, Hon'ble Minister for Rural Development and Panchayat Raj addressing NAARM & NIRDPR faculty and officers

in ICAR and also highlighted various strategies for enhancing the security in food, fodder and timber sectors. Dr Narendra Kumar, Director General, NIRDPR expressed his intention to have an MoU with the Academy in the areas of training different stakeholders, especially Self Help Groups through a-IDEA incubation center. Dr G.Venkateshwarlu, Joint Director; Faculty & Staff of ICAR-NAARM and NIRDPR were present.

6.2.7 Chairman, Capacity Building Commission visits ICAR-NAARM

Mr. Adil Zainulbhai, Chairman, and Mr. Hemang Jani, Secretary, Capacity Building Commission (CBC) visited the Academy on July 23, 2022 along with CBC team members and had an interactive summit with the faculty of the Academy. Mr. Adil Zainulbhai, elaborated on the Mission Karmayogi, the mandate of the Commission and India's capacity-building framework, the National Standards for Civil Service Training Institutions (NSCSTI) and its eight pillars of excellence. Mr. Hemang indicated that the Academy can contribute to various areas of the CBC, especially on MOOCs and developing common training courses as a central repository for leadership cases and training resources for all the training institutes. Dr Ch Srinivasa Rao, Director gave an overview of the Academy's achievements. Dr G. Venkateshwarlu and the faculty presented the capacity-building initiatives of the Academy. Commission also visited the TelAge lab and other facilities at the Academy.

Figure 6.2.6: Mr. Adil Zainulbhai, Chairman, CBC interacting with NAARM faculty

6.2.8 Dr Himanshu Pathak, Secretary, DARE & DG, ICAR Interacts with ICAR Directors

Dr Himanshu Pathak, Secretary DARE & DG, ICAR interacted with Directors of Hyderabad based ICAR Institutions and Heads of regional stations on September 24, 2022. He highlighted the recent policies of ICAR such as the five-day week, reviewing of ICAR Institute mandates in terms of output-outcomes, corpus fund formulation, transfer policy, filling of vacant positions and recruitment process of new ARS batch scientists, the delegation of powers to directors, revenue generation models and others. He emphasized the critical need for the system in terms of the efficiency of scientists and staff. Directors and senior officers of PDP, IIOR, IIRR, IIMR, NRC Meat, ATARI, CRIDA, NAARM and Heads of regional stations of IIMR and NBPGR participated in the interactive meeting.

Figure 6.2.7: Dr Himanshu Pathak, Secretary DARE & DG, ICAR with Directors of Hyderabad based ICAR institutes and NAARM faculty & officers

On this occasion, DG released two research policy publications of ICAR-NAARM; Guidelines for Incubation Centers and Strengthening Kisan Call Centers in India. He also interacted with CJSC members of the ICAR Institutes located in

Hyderabad. While addressing the faculty and staff of ICAR-NAARM, he highlighted the one-scientist one-product requirement for the overall mission of the Institute and ICAR. He also inaugurated the NAARM selfie point.

6.2.9 Interaction Meeting with Additional Secretary & FA, DARE/ICAR

Ms Alka Nangia Arora, Additional Secretary and Financial advisor, DARE/ICAR chaired an interactive meeting with the Directors, Heads of Administration, Heads of Finance & PME in charge of the Hyderabad based ICAR Institutes and ICAR-Regional Staff on November 25, 2022 at ICAR-NAARM. In the welcome address, Dr Ch. Srinivasa Rao, Director, ICAR-NAARM stressed on the timely and effective utilization of the budget including SCSP, and TSP programmes of ICAR. Directors of ICAR institutes located in Hyderabad and their representatives of IIRR, CRIDA, IIOR, IIMR, PDP, NRC Meat, ATARI-Hyderabad and NAARM presented the institute achievements and administrative/finance issues. FA reiterated the use of PMFS and TSA accounts as the funding agencies are using these platforms for efficient and effective tracking of the budgets provided to the institutes. She stressed that financial resources are limited and hence efforts at the institutes should be made towards resource and revenue generation. The Green Campus of ICAR-NAARM was well appreciated by the FA. She suggested for having a brainstorming session on resource mobilization strategies in ICAR and to bring these issues into the training curriculum of NAARM.

6.2.10 Visit of Expert Team from the National Board of Accreditation (NBA)

An Expert Team of three members from the National Board of Accreditation (NBA) visited the Academy during December 16-18, 2022 for consideration of granting accreditation for the PGDM-ABM program. During the visit, the expert team had interactions with the students, alumni, parents, and recruiters, besides the core faculty involved in teaching and guiding the students and other support staff. The team examined the activities and progress of the PGDM-ABM unit, including students.

Figure 6.2.8: Expert Team from National Board of Accreditation along with NAARM faculty

6.3 National Compaigns and Celebrations

6.3.1 Girl Child Day celebrations

As a part of the Celebration of National Girl Child Day on January 24, 2022 the Academy conducted a selfie photo contest under the theme "Selfie with Daughter". All NAARM employees, project staff, supporting staff and PGDM-ABM students actively participated in the contest. The best entries were awarded prizes at International Women's day celebrations.

Figure 6.3.1: Dr. G Venkateshwarlu, Joint Director, NAARM with dignitaries during the Celebration of National Girl Child Day

6.3.2 Republic Day Celebration

Dr Ch. Srinivasa Rao, Director, ICAR-NAARM hoisted the national flag in presence of the faculty, staff and students of the academy on the occasion of celebration of Republic Day on January 26, 2022. Cultural programmes were conducted by the PGDM-ABM students under the supervision of Dr B Ganesh Kumar, program coordinator.

Figure 6.3.2 Republic Day Celebrations

6.3.3 *Matrubhasha Diwas* Celebrations

The Academy celebrated "*Matrubhasha Diwas*" on February 21, 2022. As per the guidelines received from AICTE, Ministry of Education, GoI, the celebration was marked to promote and preserve Mother Languages across the country with the objectives of highlighting the linguistic diversity of our country; encouraging usage of not only the respective mother language but other Indian languages as well; to understand diverse cultures in India and various forms of literature, craft, performing arts, scripts and other forms of creative expression.

Figure 6.3.3: Dr. Ch. Srinivasa Rao, Director, NAARM addressing students during *Matrubhasha Diwas* Celebrations

While speaking on the occasion, Dr G Venkateswarlu, Joint Director expressed his desire to converse in his *matrubhasha* i.e., Telugu after serving in other states for more than 3 decades. He urged students to learn other Indian languages too for their own benefit and reap the advantage. Respecting other languages is another important aspect, he advocated. Dr Ch Srinivasa Rao, Director, in his address, elicited the advantage of conversing in *matrubhasha*, and shared his experiences while schooling. He advised the students to pick up a language other than *matrubhasha* to understand the diverse cultures in India. He added it will help to exposure of various forms crafts and arts, scripts and other forms of creative expression.

The PGDM (ABM) students, around 60, participated in the event enthusiastically and expressed their experiences with other Indian languages. Many of them spoke on the occasion while narrating the story of other language learning, the dialects variation within the language, and how important and easy it is to express in the mother tongue. Some students showcased their singing talent by singing in other than their mother tongue. Other students creatively expressed their journey of student life away from their own region, and how they picked up the local language. Besides students, faculty members and staff of the Academy also participated extempore in the event with much enthusiasm and interest.

6.3.4 National Science Day Celebrations

The Academy celebrated National Science Day on February 28, 2022 highlighting this year's main theme of 'Integrated Approach in Science and Technology for a Sustainable Future". The National Science Day is celebrated to mark the remarkable discovery of "the Raman Effect" by renowned Indian Scientist Sir Chandrasekhara Venkata Raman in 1928 who received the Nobel Prize in 1930. Besides, paying respectful tribute to all the scientists who have contributed significantly in enriching our scientific journey.

While speaking on the occasion, Dr Ch Srinivasa Rao, Director motivated and inspired the students by quoting eminent Indian scientists who contributed immensely in agriculture and other sectors. He explained how agricultural engineering, farm mechanization, and food processing is gaining importance in the country. While highlighting the importance of the day and how science and technology played a crucial role in addressing the recent worldwide covid pandemic, he elicited the example of Bharath Biotech's journey and its vaccine. There is no life without science, hence, he requested the students to develop a scientific temperament. Nutritional food requirements especially for women, less usage of plastic, and Swachch Bharath are other important aspects he stressed upon while addressing the students.

Figure 6.3.4: Group photo of the students from various Colleges during National Science Day celebrations

Dr G Venkateswarlu, Joint Director told that the integrated approach of science and technology is the mantra of research worldwide and the society cannot progress unless there are collective efforts from the scientific fraternity. He urged students to "learn to click; and click to learn" in science with a deep interest in order to reach their career goals.

During the valedictory celebrations, around 200 undergraduate and postgraduate Science and Engineering students from Gurunanak Institute of Technology, Sardar Patel College, Bharath Institute of Technology, Sidhartha Institute of Technology and Science, J.B. Institute of Engineering and Technology, Spoorthi College of Engineering, KPR Institute of Technology participated.

6.3.5 International Women's Day celebrations

The Academy celebrated International Women's Day on March 08, 2022. On this occasion, a workshop on 'Work-Life balance for Women' was organized. Around 60 participants including faculty members, staff, students and farm women attended the program. Dr G Venkateshwarlu, Joint Director highlighted the critical role of women in the workforce and family wellbeing

Figure 6.3.5: Dr. G Venkateshwarlu, Joint Director along with women employees of NAARM during International Women's Day celebrations

from ancient times to the present day. Smt Padmavathi, Prathinidhi, Kasturba Gandhi National Memorial Trust, Hyderabad, shared her journey and experiences, as well as the importance of the subject of Work-Life balance. She encouraged the attendees to practice self-care and dedicate some time for themselves. She also stated that many women are able to do excellent contributions in various professions because of strong support system available in different forms.

6.3.6 International Day of Yoga

The Academy celebrated the International Day of Yoga on June 21, 2021 with the theme "Yoga for Humanity". During the celebration, all the faculty members, staff and retired employees of ICAR located at Hyderabad were guided to perform yoga at the academy. Dr A Debnath, Consultant Physician and Yoga Practitioner conducted the Yoga session starting with warming-up exercises, *pranayama* and *asanas*.

Figure 6.3.6.a: Dr. Ch. Srinivasa Rao, Director addressing the audience on the occasion of International Day of Yoga

Figure 6.3.6.b: NAARM faculty, staff and students practicing Yoga on the occasion of International Day of Yoga

6.3.7 Independence Day Celebrations

Academy celebrated 76th Independence day on August 15, 2022. Dr Ch Srinivasa Rao, Director of ICAR-NAARM hoisted the national flag in presence of faculty, staff and students of the academy. Cultural programmes were conducted by the PGDM-ABM students under the supervision of Dr B Ganesh Kumar, program coordinator.

Figure 6.3.7: Independence Day Celebrations

6.3.8 Foundation Day Celebrations

The Academy celebrated 47th Foundation Day on September 1, 2022 in hybrid mode with fervor and enthusiasm. Dr Himanshu Pathak, Secretary DARE and DG, ICAR joined virtually and appreciated the activities and achievements of the academy in terms of expanded capacity building programmes, Think-tank policy research, a-IDEA, Technology Business

Incubator (TBI) and Post Graduate Diploma on Agri Business management. He opined that, as a Think Tank of the ICAR, there have been a lot of expectations from the Academy in offering solutions to the current-day problems of the agriculture fraternity of the nation. He also

Figure 6.3.8.a: Dr. Ch. Srinivasa Rao, Director, NAARM, Dr. G. Venkateshwarlu, Joint Director, NAARM and other dignitaries lighting the lamp during 47th Foundation Day

suggested that innovative ways and means to communicate the right thing to the right person at the right time need to be developed. On this occasion, he also assured that the required support from the Council will be provided to the Academy in its endeavors. Earlier in the day, Dr Pathak also joined virtually in the renaming ceremony of International Guest House I as "Haritha IGH I" to cherish Green Revolution and International Guest House II as "Sagarika IGH II" to cherish Blue Revolution. On behalf of the Chief Guest, the Director, ICAR-NAARM felicitated the awardees in various categories of staff. Besides, the innovative farmers from across India having close linkages with NAARM, best performing Start-ups, Farmer Producer Organizations (FPOs), collaborating institutes and Print and electronic media who have active engagement with NAARM have been felicitated on the occasion.

Figure 6.3.8.b: Dr Himanshu Pathak, Secretary DARE and DG, ICAR addressing the gathering during 47th Foundation Day

Dr Ch. Srinivasa Rao, Director welcomed all the guests and highlighted salient achievements of the Academy since the last Foundation Day in the areas of capacity building, research, policy support and Think Tank activities and acknowledged the roles of QRT, RAC and IMC members for their guidance and motivation. Dr G. Venkateswarlu, Joint Director, ICAR-NAARM proposed the words of thanks. The program was attended by former Directors of NAARM, Directors of various ICAR Institutes, Vice Chancellors from SAUs and retired NAARM staff.

6.3.9 राजभाषा पखवाड़ समापन समारोह

हिंदी पखवाडा समारोह अकादमी में 15-30 सितम्बर 2022 को आयोजित किया गया। इस अवसरपर, अधिकारियों / कर्मचारियों के लिए विभिन्न हिंदी प्रतियोगिताएँ आयोजित की गई। समापन समारोह 30 सितम्बर 2022 को संपन्न हुआ, जिसमें, डॉ. जे. रेणुका संयुक्त निदेशक, राजभाषा ने

Figure 6.3.9: Dr. Ch. Srinivasa Rao, Director, NAARM felicitating Dr. Ranjit Kumar during Hindi Pakhwada

सभा का स्वागत किया और कार्यक्रम का विस्तृत प्रतिवेदन प्रस्तुत किया। श्री विवेक पुरवार, प्रशासनिक प्रमुख एवं मुख्य प्रशासनिक अधिकारी (वरिष्ठ ग्रेड) ने कहा कि आज के वैज्ञानिक युग में कई डिजिटल सुविधाएँ प्राप्त हैं, जिनके प्रयोग द्वारा सभी कर्मचारी ई-ऑफिस में हिन्दी में कार्य कर सकते हैं। डॉ. सी. एच. श्रीनिवास राव, निदेशक, नार्म ने अपने अध्यक्षीय भाषण में कहा कि हिन्दी में कार्य करना हम सबका दायित्व है। उन्होंने अकादमी में भाषा अध्ययन प्रयोगशाला के माध्यम से 4 द्रविड़ भाषाओं को फोकार्स वैज्ञानिक प्रशिक्षणार्थियों को सिखाने की आवश्यकता को रेखांकित किया।

6.3.10 Gandhi Jayanthi celebrations

The academy celebrated Gandhi Jayanti on Oct 02, 2022 in which 120 staff of the Academy including Faculty Members, Technical Staff, Administrative Staff, Farm Staff and other Contractual workers took part. While inaugurating the event, Dr Ch. Srinivas Rao, Director paid floral tributes to Mahatma Gandhi on his 153rd birth anniversary and briefed the gathering about the importance of cleanliness at the workplace, villages, towns and other areas. As part of the celebration, 1 km walk was held within the campus. Dr G. Venkateshwarlu, Joint Director; Scientists; Admin & Finance staff also participated in the event. *Swachhata* Pledge was also administered on this occasion. Campus cleaning activities at various areas of the Academy were also carried out.

Figure 6.3.10: Dr. Ch. Srinivasa Rao, Director, NAARM leading the walk during Gandhi Jayanti celebrations

6.3.11 Mahila Kisan Diwas Celebration

The Academy celebrated *Mahila Kisan Diwas* on October 15, 2022 at Agraharam Pottapally Village, Mahabubnangar district, Telangana. On this occasion, women farmers were sensitized on the topic 'How to curb malnutrition among rural women?' Around 100 participants including women farmers, women agricultural laborers and other rural women attended the program. Dr Bharat Sontakki, Principal Scientist and Head (I/c); Dr T Supraja, Professor (Foods & Nutrition), PJTSAU addressed the gathering during the event.

Figure 6.3.11: Dr. Bharat S. Sontakki addressing farm women during Mahila Kisan Diwas Celebrations

6.3.12 Vigilance Awareness Week

The Academy organized vigilance awareness week during October 31 - Nov 6, 2022 by undertaking various activities on the theme "Corruption-free India for a Developed Nation". On Oct 31, 2022, Integrity Pledge was administered by Dr Ch. Srinivasa Rao, Director, NAARM. During the week-long celebration, several events were organized such as essay writing for college students, a painting competition for school students, elocution competition for PG students of this academy. In his remarks

Figure 6.3.12: Integrity Pledge being taken during the vigilance awareness week

on the closing day, Dr G. Venkateshwarlu, Joint Director, reiterated that preventive vigilance is more important for the organization to prevent corruption and also enhance transparency and organizational efficiency.

6.3.13 National Children's Day Celebration

The Academy, in association with NAAS-Hyderabad Chapter, organised a sensitization program on "Nutrition Literacy among Rural Children" on the occasion of Children's Day at Zilla Parishad Girls High School, Shamshabad, Rangareddy district, Telangana. Around 500 girl students and staff of the school participated in the program.

Figure 6.3.13: Students and staff of Zilla Parishad Girls High School, Shamshabad with Dr Ch. Srinivasa Rao, Director and faculty of NAARM

6.3.14 *Swachhta Pakhwada* Celebrations

Swachhta Pakhwada was celebrated from December 16, 2022 at the Academy by taking Swachhta Pledge. Activities like regular cleaning, sweeping of lawns, and main roads, moping of the floor of all buildings, removal of waste like stone boulders, and scrap, cutting of weeds in the campus and cleaning the surroundings of buildings and also residential quarters were taken up during this pakhwada. Regular fogging for the control of mosquitos and as a part of waste out of wealth, shifting of leaf litter & grass clippings to the vermicomposting unit for making vermicomposting was done during this period. Farm staff, outsourcing & housekeeping staff contributed to swachhta-related activities. On December 26, 2022, elocution, essay writing and quiz competitions were held on swachhta for the PGDM-ABM students. Around 120 students participated in the competitions.

Figure 6.3.14.a: Faculty and Students of NAARM during Swachhta Pakhwada Celebrations

Figure 6.3.14.b: Swachhta Pledge being taken by staff of NAARM during Swachhta Pakhwada Celebrations

6.3.15 *Rashtriya Kisan Diwas* Celebrations

Rashtriya Kisan Diwas (National Farmers' Day) was celebrated on December 23, 2022 by organizing an interactive meeting with farmers at Agraharam Potlapally Village, Rajapur Mandal, Mahabubnagar District of Telangana State. During the interaction, Dr Bharat S. Sontakki & Dr P. Venkatesan explained the importance of Rashtriya Kisan Diwas and the cultivation of medicinal Plants for augmenting farmers' health and economic status, respectively. Samplings of medicinal/herbal plants (*Costus igneus* and *Ocimum tenuiflorum*) were distributed among the farmer partners. Dr Laxman M. Ahire explained the health benefits of these plants to the farmers. Around 100 farmers actively participated in the program and expressed keen interest in growing these plants.

Figure 6.3.15: Dr. Bharat S. Sontakki addressing the participants on the occasion of National Farmers' Day

6.4 General Meetings

6.4.1 Fourth NAHEP External Advisory Panel Meeting

A meeting with Dr JC Katyal Member, External Advisory Panel was conducted on February 19, 2022 under the chairmanship of Dr Ch. Srinivasa Rao, Director, ICAR-NAARM. This meeting was organized to discuss the various activities and progress of the project Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education, NAHEP Component-2.

Figure 6.4.1: Dr JC Katyal, Member, External Advisory Panel along with Dr. Ch. Srinivasa Rao, Director, NAARM and NAHEP team

6.4.2 Interaction Meeting with PWC

As part of NAHEP, a study titled 'Assessment of human resources requirement in agriculture and allied sector for next 20 years' has been taken up under the leadership of the National Director, NAHEP and DDG (Agricultural Education). The Pricewaterhouse Coopers (PWC), a multinational consultancy firm has been appointed as a consultant to assess the human resource requirement in agriculture and allied sector in India. In this connection, the team from PWC, Delhi visited the Academy and interacted with the Joint Director and Heads of the Departments on February 02, 2022. During the interaction meeting, discussions were held on major driving & limiting factors of change in the coming 5/10/20 years in agriculture & allied sector, entrepreneurship/ startups, competency level of faculties; current growth rate and expected growth in the next 10 years and 20 years in different agriculture and allied sectors.

Figure 6.4.2: Team from PWC interacting with Dr. G Venkateshwarlu, Joint Director, NAARM and other senior faculty

6.4.3 Review meeting of NASF Project

A project review meeting of NASF-funded project on Farmer Led Innovations in the Plantation Sector was organized at IIPM, Bengaluru on January 6-7, 2022. During the meeting, methodologies for Agribusiness, Climate Analysis, Market Potential Analysis, Performance Efficiency Assessment and SWOT analysis of Innovations in the plantation sector were designed.

6.4.4 Pensioners Meeting

The 11th Pensioners' meeting was organized by PAU-NAARM on January 12, 2022 under the Chairmanship of Dr Ch. Srinivasa Rao, Director, ICAR-NAARM. A total of 57 participants including pensioners, representatives of the Pensioners Association, Bank, and representatives of institutes participated in the virtual meeting. In his opening remarks, Dr Ch. Srinivasa Rao, Director thanked all the sister institutes for conducting pensioners meeting at their institute level to resolve the grievances of the pensioners. He also highlighted various steps taken by the Academy during the pandemic for the health benefits of pensioners and staff including vaccination camps, free diabetic camps, and assistance in taking treatment at various hospitals at govt. rates etc. A total of 11 issues were raised by Pensioners Association namely, RICAREA, and CPWA and other individual pensioners.

6.4.5 Third Meeting of VIII Research Advisory Committee

The Third meeting of the VIII RAC was held at the Academy during September 13-14, 2022. The Meeting was chaired by Prof G Padmanaban, President NASI, Ahmedabad and Former Director, Indian Institute of Science (IISc), Bengaluru, wherein all the members, including Dr R.C. Agrawal, DDG (Edu.) participated.

Figure 6.4.3: Members of the VIII Research Advisory Committee with faculty of NAARM

Dr Ch Srinivasa Rao, Director, ICAR-NAARM introduced all the members of RAC and welcomed them to the meeting. Dr Tavva Srinivas, Principal Scientist and Member Secretary, VIII RAC of NAARM, presented the action taken report on the recommendations given during the 2nd meeting of the Committee held in June 2021. The Director presented a comprehensive overview of the NAARM activities related to research and think tank/policy advocacy including the number of projects currently underway, the number of research papers and policy papers published, copyrights registered, start-ups supported, MoUs signed, and the networks established during the preceding year. Dr G Venkateshwarlu, Joint Director, ICAR-NAARM presented the major activities related to capacity building and academic programmes of the Academy. All the heads of Divisions presented an overview of activities and projects going on in each division.

The Committee complimented the Academy, Director, and all the Scientists for the progress despite having a low number of faculty. The members expressed satisfaction regarding the action taken based on the recommendations. The committee recommended keeping more focus on the core mandate of the academy, to enhance technical training capacity and train the individuals towards future requirements. Also recommended to develop policies for reducing food wastage and keeping the focus on environmental and social governance, public-private partnerships and training of farmers in ICAR Institutes. The committee also suggested to upgrade the MOOCs on business mode to generate revenue. The committee visited various facilities of the academy such as the Biodiversity park, herbal garden and Telage Lab. CEO of a-IDEA, Dr S Senthil Vinayagam explained about the a-IDEA activities and ongoing startups under various schemes. Committee interacted with the startups and gave necessary suggestions.

6.4.6 NAHEP Review Meetings

A review meeting of NAHEP Component 2 was organized under the Chairmanship of Dr RC Agrawal, Deputy Director General (Agricultural Education) & National Director, NAHEP on July 5, 2022 at the Academy. The meeting was held to review the progress of the project activities during FY 2021-22 and action plans for the financial year 2022-23 of Component 2 of ICAR-NAARM.

Figure 6.4.4.a: Review meeting of NAHEP Component 2

Figure 6.4.4.b: A screengrab of the virtual review meeting of NAHEP Component 2

A virtual review meeting of NAHEP Component 2 was organized under the chairmanship of Dr Anuradha Agrawal, National Coordinator (Component 2), NAHEP on August 10, 2022. The meeting was held to review the progress of the project activities during FY 2021-22 and action plans for the financial year 2022-23 of Component 2 of ICAR-NAARM. Dr S K Soam, CCPI presented the progress and action plan of the NAHEP Component of ICAR-NAARM.

6.4.7 Review Meeting of Career Development Centers of NAHEP Component 2

The Review Meeting of the Career Development Centers under the World bank and ICAR-sponsored project "Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education (Under NAHEP Component 2A)" was held on September 28, 2022 under the chairmanship of Dr Anuradha Agrawal, National Coordinator, (Component 2 & CAAST), NAHEP and co-chaired by Dr Sudeep Marwaha, PI, NAHEP Component 2, IASRI, New Delhi, at ICAR-NAARM. This review meeting was held to discuss about the progress & expenditure of Financial Year 2020-21 & 2021-22 and the action plan for 2022-23 of five CDCs established at CAU, Imphal; IGKV, Raipur; SKNAU, Jobner; SVVU, Tirupati; UBKV, Coochbehar. The Nodal Officers and Coordinators of CDCs presented their progress. The project team from ICAR-NAARM & ICAR-

Figure 6.4.5: Dr. Ch. Srinivasa Rao, Director, NAARM and NAHEP team along with the representatives from CDCs

IASRI were present during this review meeting. Prior to this, Dr Ch Srinivasa Rao, Director and Dr G Venkateshwarlu, Joint Director interacted with Coordinators of CDCs and provided future directions for effective functioning.

6.4.8 Review Meeting of NAHEP Component-2 at NAHEP Project Implementation Unit, New Delhi

A Review meeting of the NAHEP Component 2 project titled “Investments in ICAR Leadership in Agricultural Higher Education” was held on December 01, 2022 at room no 511, KAB-II, ICAR, New Delhi under the chairmanship of Dr RC Agrawal, National Director, NAHEP & DDG (Agricultural Education), ICAR, New Delhi. This review meeting was conducted to discuss about the completed activities, foreign exposure visits and forthcoming programmes. Dr Ch Srinivasa Rao, Director, ICAR-NAARM and Dr SK Soam, CCPI, NAHEP Component 2, ICAR-NAARM attended this review meeting.

6.4.9 Academic Council Meetings (ACM)

ICAR-NAARM organizes monthly ACM meetings to serve as a platform to plan, review and monitor various activities of the Academy including the forthcoming training programmes, policy studies, educational programmes and administrative and financial issues. The meetings are chaired by the Director and attended by all faculty members, the Joint Director, the Head of Administration, CFAO and other officials of the academy.

6.4.10 Limited Departmental Competitive Examination

The academy organized a Limited Departmental Competitive Examination for the one UR post of Assistant Administrative Officer (AAO) and one UR post of Assistant in ICAR-NAARM, Hyderabad and published the result followed by the departmental promotion committee recommended placing two officials in these posts.

6.5. Other major activities

6.5.1 Govt school students' visit to NAARM on the occasion of National Science Day

First batch of 106 students of class 9th and 10th from Thorrur and Danthalapally High schools from the mahabubabad district visited NAARM on February 25, 2022. The second batch of 109 students of 9th and 10th classes from Zilla Parishad High School and Kodakandla model school from Jangaon district of Telangana visited NAARM on March 10, 2022. Dr Ch. Srinivasa Rao, Director, NAARM interacted and motivated the students towards pursuing science, esp. agricultural sciences, and careers related to them. He also enlightened the students about the Swachcha Bharath and the need for cleanliness in villages, and how the students can help in achieving the same. He also sensitized them about the harmful effects of the usage of plastic on soil and the environment. Earlier, Mrs G. Aneeja, CTO, NAARM briefed about NAARM and its activities. The third batch of 110 students studying 7th, 8th and 9th classes from ZP high schools of Peddavangara, Chityal, and Authapur villages of Telangana visited the Academy on

Figure 6.5.1: Students from Thorrur and Danthalapally High schools with Dr. Ch Srinivasa Rao, Director, NAARM and officers of NAARM

March 25, 2022. The students' visit to NAARM was facilitated by Madhumitha Foundation, Suryapet, Telangana with funding from DST.

6.5.2 Educational Tour-cum-Industrial Visit of PGDM-ABM Students

An Educational Tour-cum-Industrial Visit was organized for the students of PGDM-ABM 2020-22 Batch during March 16-22, 2022. The purpose of the tour was to expose the students to different academic & research institutions, and agribusiness companies in the states of Karnataka, Kerala and Tamil Nadu to understand the farm/agribusiness practices and commercialization of agricultural technologies. The organizations/agri-industries visited during the tour were: NSL Sugars Ltd., Srirangapatna, Mysore; Achoor Tea Factory, Vythri at Wayanad; ICAR-Indian Institute of Spices Research, Calicut; Kuttanad area in Aleppey; Outgrow Agricultural Research Station of WayCool Industries Pvt. Ltd., Padalur; etc.

Figure 6.5.2: PGDM (ABM) students at NSL Sugars Ltd., Srirangapatna, Mysore

6.5.3 Placement of 2020-22 Batch Students

All the students of the PGDM-ABM 2020-22 batch were placed in various agri-industry and agri-business companies through campus placement. They were placed in different sectors, viz. Agri-Input, Banking & Microfinance, FMCG & Retail, Supply Chain, Startup, Life Sciences, Social enterprises, Commodity and Consultancy. This year, 7 new companies visited the campus for placement of the students. Furthermore, 6 students got a pre-placement offer (PPO) based on their excellent performance during their Summer training. The average package of this batch was Rs. 8.32 lakh per annum and the maximum was Rs. 12.56 lakh per annum secured by a girl student.

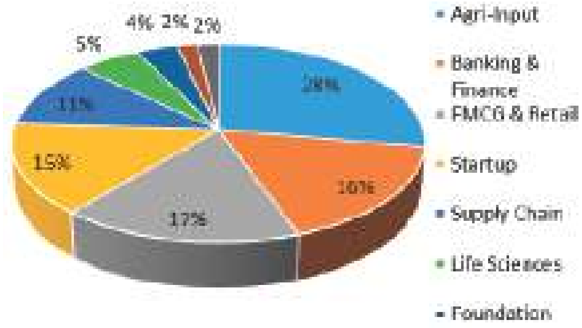


Fig 6.5.3: Industry-wise Placement Percentage for PGDM (ABM) students

6.5.4 Farewell Ceremony of PGDM-ABM 2020-22 Batch

The PGDM-ABM 2020-22 batch completed their 2-year programme. On this occasion, a farewell ceremony was organized for the students on April 02, 2022. The outgoing students shared

Figure 6.5.4: A group photo of the outgoing students with NAARM faculty

their journey of 2 years at NAARM. They also gave some valuable feedback for further improving the standard and reputation of the programme. Some of the senior NAARM faculty also wished them a bright future. Speaking on the occasion, Dr. Ranjit Kumar, Head of ABM Division congratulated the students for successfully completing the programme and leaving the campus with a job offer with them. He advised the students to always work honestly and sincerely, and help the Almamater in pushing forward the brand PGDM-ABM@NAARM.

6.5.5 Diploma in Technology Management in Agriculture (DTMA)

The contact classes of the DTMA - 2021 Batch of 29 students were conducted in online mode. The students also submitted the assignments and appeared in the term-end examinations. Evaluation of the semester exams and the project reports have been completed based on the procedures of the University of Hyderabad, in whose collaboration the program is being offered. In DTMA-2022 Batch, 20 students have been admitted. The Joint Admission Committee (JAC) meeting was conducted at the Academy on May 10, 2022 with the participation of officials and coordinators for DTMA from the academy and University of Hyderabad (UoH) in which issues pertaining to admission, and examination were deliberated. The aspects related to the alignment of education programmes in line with the New Education Policy (NEP) were also discussed. The contact session for the 2022 batch of DTMA was held between 14-16 June, 2022 with the participation of 19 newly admitted candidates.

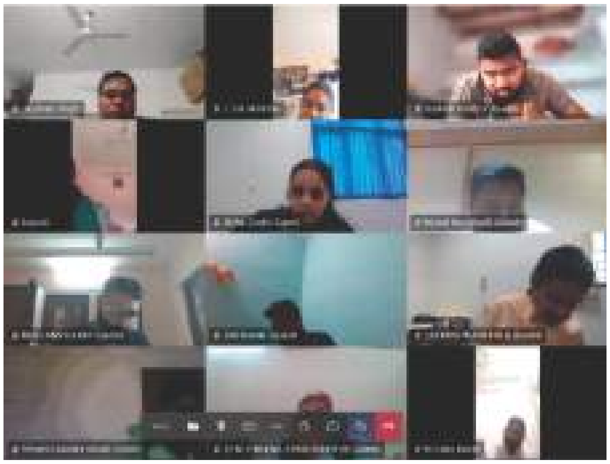


Figure 6.5.5.a: A snapshot of the online contact classes of the DTMA - 2021

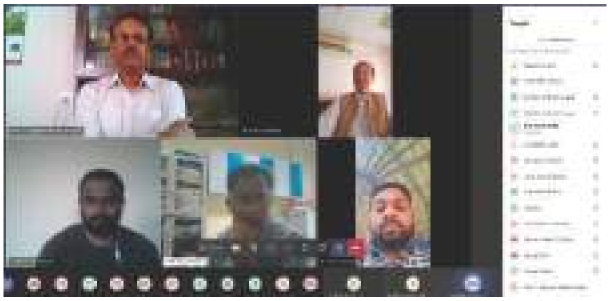


Figure 6.5.5.b: A screenshots of the Joint Admission Committee (JAC) meeting

6.5.6 Diploma in Educational Technology Management [DETM]

The DETM has received a total of 50 applications through online/offline mode. Out of 50 candidates, a total of 25 candidates paid the DETM course fee and confirmed their candidature in the said programme. The 11th JAC meeting was organized jointly by the Academy and the University of Hyderabad on May 10, 2022 at the Academy. JAC discussed about the appraisal of 1st and 2nd Semester Contact Classes, Internal Assessments and Semester-End Exams

Figure 6.5.6.a: Joint Admission Committee (JAC) meeting held at ICAR-NAARM

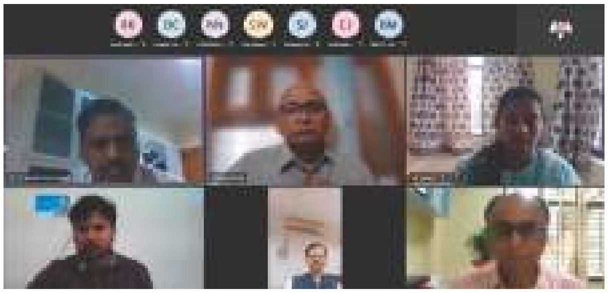


Figure 6.5.6.b: DETM programme online conact classes

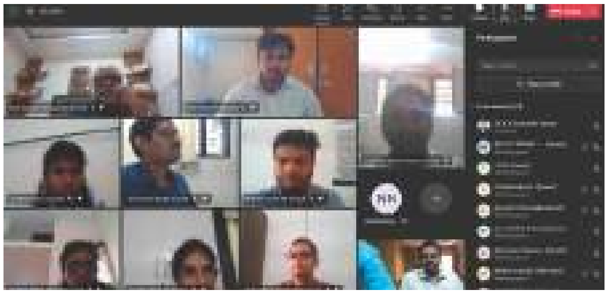


Figure 6.5.6.c: DETM programme online exmination

organized during Nov/Dec 2021 for the batches of 2019, 2020 and 2021 respectively. JAC also finalized the dates for DETM 2022 batch i.e 1st Semester Contact Classes & Semester-End Exams during June/July & Aug 2022. The profile of the candidates admitted into DETM 2022 was presented to members and the committee approved admissions and dates for DETM 2022 batch. Also, 1st Semester contact classes were organized from June 27 - July 03, 2022 in online mode. During the Informal Inaugural programme, the modalities of the DETM programme and the importance of attending contact sessions were clearly informed to all the participants.

6.5.7 REUNIRSE 2022

The PGDM-ABM alumni congregation event- “Reunirse 2022” was organized on May 14, 2022, in which about 150 alumni of PGDM-ABM of NAARM came from different parts of the country to participate in the mega event. Dr Trilochan Mohapatra then Secretary DARE and DG, ICAR was invited as the chief guest of the event. Alumni from almost every batch including the first batch (2009-11) marked their presence and shared their experiences and expectations during the event. Dr Ch Srinivasa Rao, Chief Patron of NPAA and Director, of NAARM and Dr G Venkateshwarlu, Patron, of NPAA and Joint Director, of ICAR-NAARM also graced the event. The Reunirse 2022 was co-sponsored by Aquaconnect. Mr Karthivelan, COO, of Aquaconnect gave a brief about his company and presented the “Aquaconnect Stellar Graduate Award” to Ms Amrit Kaur for her outreach activities and corporate relations. The General Body Meeting was also organized during the event. It was decided to have alumni erudition at least twice a month. Financial support or Rs. 5,000 per year will be provided to each alumni chapter to encourage chapter meetings. Dr Sanjiv Kumar was elected as treasurer and Dr B Ganesh Kumar as an honorary executive member of the association.

6.5.8 Sankalp 7.0

The PGDM-ABM students of ICAR-NAARM organized ‘Sankalp’, an annual business fest inviting students from different colleges across India. This year, the 7th edition of Sankalp was organized during September 16-17, 2022 with the support of the NAARM PG Alumni Association. The event received an overwhelming response as

Figure 6.5.7.a: Dr. Ch. Srinivasa Rao, Director, NAARM and guests awarding winner of an event during Sankalp 7.0

Figure 6.5.7.b: Group photo of NAARM faculty, guests, participants and students of NAARM during Sankalp 7.0

reflected by more than 9,100 registrations from more than 1,000 institutions which included all IIMs and IITs, several other leading B-schools and agricultural colleges. The B-fest comprised 3 flagship events (Samadhaan- The Case study competition, Advitiya -The B plan competition, Adhistatha-The best manager competition) and 4 sprint events (Analytica- The data analysis, Flashgun – The photography competition, Darpan – The business quiz competition and Ad-junction- the ad creation event). The registered participants for different events underwent several rounds of competition which were held online during the last 1 month before getting shortlisted as finalists. In all, around 60 finalists were selected and invited for the final round on campus during these 2 days. Total prize money worth Rs. 1.3 lakhs was given to the winners.

The event was sponsored by Fasal, a-IDEA, Bayer crop science, Rallis India, WayCool Foods, Synergy Technofin, Indofil Industries, Vegrow, Greenin Homes, NAAS- Hyderabad Chapter, and Kynetec.

The inaugural and valedictory sessions were graced by representatives from Fasal, Bayer, and Rallis. Dr Ranjit Kumar, Head, of ABM Division and Chairman of the organizing committee, briefed about the event and told how the event is attracting a greater number of participants each year. Dr G Venkateshwarlu, Joint Director, NAARM shared that events like Sankalp provides an excellent platform for the students to showcase their talent. Speaking on the occasion, Dr Ch Srinivasa Rao, Director, NAARM congratulated the winners of various events and suggested for bringing good farming solutions for agriculture in the country.

6.5.9 Exposure Visit of FDP participants from MGNCRE

NAARM has organized experiential field visits on November 29 & Dec. 6, 13, and 21, 2022 for four batches of Faculty Development Programmes (FDPs) organized by Mahatma Gandhi National Centre for Rural Education (MGNCRE). For all four batches, Mrs G. Aneeja, CTO, sensitized the participants about the NARES system and

Figure 6.5.8.a: Visitors from Mahatma Gandhi National Centre for Rural Education briefed by Mrs G. Aneeja, CTO

Figure 6.5.8.b: A group photo of the Visitors from the Mahatma Gandhi National Centre for Rural Education

briefed them about various activities of NAARM with an emphasis on agri-entrepreneurship at the village-level. The core areas covered were: Internship in the rural landscape: Opportunities, options and approaches, Participatory Rural appraisal and developmental work opportunities in rural areas by Dr B.S. Sontakki, Dr P. Venkatesan, and NAARM activities/programmes/activities initiated at village-level through various platforms by Dr Ramesh Naik, NAARM faculty members. Besides, all participants had a chance to witness the green initiatives implemented at NAARM, and also visited the gallery, herbal garden and nursery. Half campus tour by walk enabled them to witness the lush green, serene and sprawling campus of NAARM.

About 184 faculty members working in different Andhra Pradesh and Telangana Government Degree Colleges got benefitted from these experiential learning visits who are mandated to promote internship and apprenticeship of all degree college third-year students under the new education policy regime.

6.5.10 Guest Faculty invited from other institutions

Name of the faculty	Designation and place	Title of the lecture	Date
Dr. Debasis Chatterji	Founder and CEO, of Priceeagle.in	Leadership for Innovation Management & Design Thinking in Leadership	June 16, 2022
Dr. B. Venkateswarlu	Former Director, CRIDA & Former VC, VNMKV, Parbhani	Leadership Challenges – Experience Sharing	June 20, 2022
Dr. Vivek Modi	Life-Skills Corporate Trainer Corporate Training Services	Happiness & Well-Being & Coping Strategies for Leaders	June 21, 2022
Dr. N. Reddy	Former Regional Director, CBWE	Leadership Lessons from Epics	June 23, 2022
Dr. Debasis Chatterji	Founder and CEO, of Priceeagle.in	Design Thinking	December 12, 2022
Dr. Rama Rao	Former Director, ICAR-NAARM	Leadership Issues	December 12, 2022
Dr. B. Venkateswarlu	Former Director, CRIDA & Former VC, VNMKV, Parbhani	Leadership Challenges – Experience Sharing	December 14, 2022
Mr. Shitanshu Kumar	IIRR	Vigilance Management	December 21, 2022
Dr. N. Reddy	Former Regional Director, CBWE	Leadership Lessons from Epics	December 22, 2022

6.6 Visits of Students/Trainees/Farmers

Sl. No	Name of the University	Date of Visit	No. of students
1.	B. Sc. (Agri.) students of Dr Y S Parmar University of Horticulture and Forestry, Nauni, Solan (Virtual visit)	Jan. 19, 2022	185
2.	Students from Dr Y S Parmar University of Horticulture and Forestry, Nauni, Solan (Virtual visit)	Jan. 31, 2022	42
3.	B. Sc. (Agri.) students from Kumaraguru Institute of Agriculture, Nachimuthupuram, Erode (Virtual visit)	Mar. 22, 2022	54
4.	B. Sc. (Agri.) students from JNKKV, Jabalpur	Mar. 28, 2022	35
5.	Trainees of the Agro-based Entrepreneurship Development Programme of the National Institute for Micro, Small and Medium Enterprises (NI-MSME) representing five Agricultural colleges under Acharya N G Ranga Agricultural University	Mar. 30, 2022	40
6.	B. Sc. (Hons.) Community Science students from the College of Community Science, UAS, Dharwad	Mar. 31, 2022	70
7.	B. Sc. (Hons.) Community Science students from the College of Community Science, UAS, Dharwad	Apr.4, 202	24
8.	B. Sc. (Community science) students from the College of Agriculture, Hanumanamatti, Ranebennur, Haveri	Apr. 12, 2022	42
9.	B. Sc. (Agri.) students from Agriculture College and Research Institute, Madhurai, TNAU (Virtual visit)	Apr. 19, 2022	120
10.	Final Year B. Sc. (Hons. Agri.) students from College of Agriculture, University of Agricultural Sciences, Vijayapur	Apr. 19, 2022	66
11.	Students from the Department of Agricultural Extension, Nammazhwar College of Agriculture and Technology (affiliated to TNAU), Ramanathapuram	Apr. 20, 2022	46
12.	Final year B. Sc. (Hons. Agri.) students from Vanavayanar Institute of Agriculture (affiliated to TNAU), Manakkadavu, Pollachi	Apr. 20, 2022	121
13.	Final year B. Sc. (Horticulture) students from Adhiparasakthi Horticultural College G.B.Nagar, Kalavai, Ranipet (Virtual visit)	Apr. 21, 2022	100
14.	B. Sc. (Hons.) IV year students from Community Science College & Research Institute, TNAU, Madurai (Virtual visit)	Apr. 25, 2022	43
15.	Final year B. Sc. (Hons. Agri.) students from the Department of Agriculture, Kalasalingam School of Agriculture and Horticulture, Krishnankoil	Apr. 26, 2022	50
16.	Final year B. Tech. (Agri. Engg.) students from the Department of Agricultural Engineering, School of Bio, Chemical and Processing Engineering (SBCE), Kalasalingam Academy of Research Education, Srivilliputhur	Apr. 26, 2022	22
17.	M. Sc (Agri.) and Ph. D students from Dept. of Agril. Extension, College of Agriculture, GKVK, Bengaluru	May 5, 2022	43
18.	Final year B. Sc. (Horti.) students from the Department of Horticulture, Kalasalingam School of Agriculture and Horticulture, Kalasalingam Academy of Research and Education, Krishnankoil, Virudhunagar	May 7, 2022	44
19.	B. Sc. (Hons. Agri.) College of Agriculture, Ambalavayal, Wayanad	May 9, 2022	41
20.	Final year B. Tech. (Poultry Technology) students from the College of Poultry Production and Management, TNVASU, Mathigiri, Hosur	May 9, 2022	23

Sl. No	Name of the University	Date of Visit	No. of students
21.	B. Sc. (Hons. Agri.) students from College of Agriculture, Kerala Agriculture University Vellayani	May 10, 2022	129
22.	Final year B. Sc. (Horti.) students from College of Horticulture, Munirabad, University of Horticultural Science, Bagalkot	May 24, 2022	62
23.	Students from the College of Cooperation, Banking and Management, Kerala Agricultural University, Vellanikkara, Thrissur	Jun. 15, 2022	42
24.	Final year B. Sc. (Horti.) students from KRC College of Horticulture, UHS, Arabhavi	Jun. 20, 2022	74
25.	Final B. Sc. (Agri.) & ABM students from Agricultural College and Research Institute, TNAU, Coimbatore	Jul. 18, 2022	50
26.	Final B. Sc. (Agri.) students from Agricultural College and Research Institute, TNAU, Madurai	Jul. 19, 2022	130
27.	Students from College of Agro- Biotechnology, Dept. of Microbial and Environmental Biotechnology, MGM CABT, Gandheli, Maharashtra	Jul. 26, 2022	89
28.	Final year students from the College of Agriculture Business Management, Loni (Affiliated to MPKV, Rahuri, Maharashtra)	Jul. 27, 2022	45
29.	Final B. Sc. (Agri.) students from Agricultural College and Research Institute, TNAU	Aug. 3, 2022	72
30.	Final B. Sc. (Agri.) students from Agricultural College, Udgir, Latur, Maharastra	Aug. 11, 2022	100
31.	2nd year B. Sc. (Agri.) students from Sadguru College of Agriculture, Mirajgaon, Maharastra (Affiliated to MPKV, Rahuri)	Aug. 12, 2022	120
32.	Final B. Sc. (Biotechnology) students from College of Agricultural Biotechnology, Hatta, Hingoli Dt, Maharashtra (affiliated to MKV, Parbhani)	Aug. 23, 2022	50
33.	Final year B. Sc. (Agri. Engg.) students from Karunya Institute of Technology and Sciences, Coimbatore, Tamilnadu	Aug. 25, 2022	14
34.	Final year B. Sc. (Hons. Agri.) students from College of Agriculture, Annamalai University, Tamilnadu (Batch 1)	Sept. 5, 2022	101
35.	Final year B. Sc. (Hons. Agri.) students from College of Agriculture, Annamalai University, Tamilnadu (Batch 2)	Sept. 12, 2022	99
36.	Final year B. Sc. (Hons. Horti.) students from College of Agriculture, Annamalai University, Tamilnadu (Batch 3)	Sept. 19, 2022	94
37.	Final year B. Sc. (Hons.) students from Pandit Jawaharlal Nehru College of Agriculture and Research Institute, Kariakal, Puducherry	Sept. 19, 2022	110
38.	Final year B. Sc. (Hons.) Community Science students from the College of Community Science, Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani	Sept. 22, 2022	22
39.	Final year B. Tech, (Agri. Engg.) students from Kalasalingam Academy of Research and Education, Tamilnadu	Oct. 3, 2022	26
40.	Second Year B. Sc. (Agri.) students from D.Y. Patil College of Agriculture, Talsande (affiliated to MPKV, Rahuri)	Oct. 4, 2022	131
41.	Final year B. Arch. students from Aurora's Design Institute, Habsiguda, Hyderabad	Oct. 17, 2022	7
42.	Final year B. Sc. (Hons.) students from College of Agriculture, Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola	Nov. 21, 2022	138

Sl. No	Name of the University	Date of Visit	No. of students
43.	Final year B. Sc. (Agri.) students from Late Ambadasrao Warpukar College of Agriculture, Wadpur, Parbhani (Affiliated to V.N. Marathwada Agricultural University)	Dec. 1, 2022	52
44.	Final year B. Sc. (Hons.) students from Dept. of Agrl. Extension and Rural Sociology, Tamil Nadu Agricultural University, Coimbatore	Dec. 8, 2022	135
45.	Final year B. Sc. (Agri.) students from Adhiyamaan College of Agriculture and Research (affiliated to TNAU), Athimugam, Krishnagiri	Dec. 9, 2022	114
46.	Final years B.Sc (Hons) students from Sethubhaskara Agricultural College and Research Foundation (Affiliated to TNAU, Coimbatore) Karaikudi	Dec. 13, 2022	83
47.	Final year B.Sc (Hons) students from SRS Institute of Agriculture and Technology Vedasandur, Dindigul, Tamil Nadu	Dec. 13, 2022	82
48.	Final Year B. Sc. (Agri. Hons.) students from Nalanda College of Agriculture, M.R. Palayam, Sanamangalam, Trichy	Dec. 14, 2022	65
49.	Final year B.Sc (HonsAgri) students from Vanavarayar Institute of Agriculture (affiliated to TNAU, Coimbatore), Manakkadavu, Pollachi	Dec. 15, 2022	117
50.	Final year B.Sc (Hons. Hort.) students from Dept. of Fruit Science, College of Horticulture, Hiriyur, KSN Univ. of Agricultural and Horticultural Sciences, Shivamogga	Dec. 21, 2022	51
51.	Final B. Sc. (Hons. Hort.) students from College of Horticulture, Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences, Mudigere, Chikmagalur (Batch 1)	Dec. 22, 2022	44
52.	Final year B. Sc (ABM) students from Utkal University, Orissa	Dec. 23, 2022	35
53.	Final B.Sc (Horticulture) students from the College of Horticulture, Hiriyur Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences, Shivamogga	Dec. 23, 2022	50
54.	Final B.Sc (Hons. Hort.) students from College of Horticulture, Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences, Mudigere, Chikmagalur (Batch 2)	Dec. 23, 2022	41
55.	Final B. Sc. (Hons. Community Science) students from College of Community Science, UAS, Dharwad	Dec. 28, 2022	44
56.	Students from the College of Horticulture and Forestry, Dr Yashwant Singh Parmar University of Horticulture and Forestry Neri, Distt. Hamirpur	Dec. 28, 2022	37
57.	B. Sc. (Hons. Forestry) final year students from College of Horticulture and Forestry, Dr Y.S. Parmar University of Horticulture and Forestry, Neri	Dec. 29, 2022	31
58.	Final Year B. Tech. (Biotechnology) students from College of Horticulture and Forestry, Dr Y.S. Parmar University of Horticulture and Forestry, Neri	Dec. 29, 2022	30
59.	Final B. Sc. (Forestry) students from College of Forestry, Uttar Kannada, UAS, Dharwad	Dec. 30, 2022	66

Infrastructure and Facilities 7

The Academy is established in a green, serene and sprawling campus of about 50 hectares. It has various facilities like Faculty Center, Academic Block, Administrative Block, Library, Medical and Health Centre, Sports Complex, Children's Park, Guest houses and Trainees Hostel, Housing and Residential Quarters.

7.1 Library

The NAARM Library provides excellent information support services to the Institute's research and training activities. It has over 31,458 books, 11 international journals, 3 international online journals, and 6 online databases /software, including EBSCO Business Source Premier, Indiastat.com, Commodities, Economic Outlook, iTheticate License, and Orbit.com, among others. Many of the Library's publications are available in digital formats such as CD, VCD, and its digital repository (EPrints@ NAARM). The Academy is also a member of CeRA and has access to journals and databases via CSIRO, Springer, and Open J-Gate, among others. The open-source software "Koha" was used to digitize the library. In addition, the Academy has a large collection of online patent search providers, GIS software, and other statistical

Fig. 7.1: Library at ICAR-NAARM

software. Library services have been extended to students who undertake studies under the supervision of Institute scientists. They were instructed on how to use reference resources, CeRA, online databases, and foreign journals. On an official/payment basis, photocopying services were provided to Institute staff and other library users.

7.2 Health Center

ICAR-NAARM health center provides comprehensive health services consisting of preventive, promotive and curative health services to the employees and their dependents, trainees, PGDM students and more than 1500 pensioners and their dependent family members. The Academy's health centre at present has two experienced doctors viz. Dr. (Maj) A Debnath and Dr Himabindu are highly dedicated to their profession and are approachable anytime during the day and night for the sake of patients.

Apart from daily OPD service, the health center organizes various health camps, e.g. Diabetic camp, Lipid profile camp, Cardiology camp, vaccination camp etc. periodically for the benefit of the Academy's employees, trainees and pensioners and their family members. The health center organized a total 21 number of camps in the last year, which include 14 Diabetic camps, one Cardiac camp one Blood donation camp and 5 covid vaccination camps. During the Covid pandemic time last year, our doctors conducted 05 Covid vaccination camps in coordination with DHMO and Rajendranagar Civil Hospital. These facilities were free of cost. A total of 402 persons

Fig. 7.2: Covid Vaccination camp

including PGDM students were vaccinated with 2nd dose and the Booster dose. As part of our regular activity, the health center organized a Blood donation camp on 2nd September 2022 on the occasion of Academy's Foundation Day. This time the camp was specifically for the Thalassemia affected children, with the cooperation of the Telangana Red Cross Blood Bank. 47 units of Blood were collected from the employees, PGDM students and donated to the Thalassemia society. Director of the Academy Dr Ch Srinivasa Rao and Dr S. Ravichandran, Officer-In-charge were present and monitored the Blood donation camp.

Regular practicing Yoga is considered one of the best tools to remain healthy physically and mentally. ICAR-NAARM also made Yoga compulsory in every training program. Academy celebrated the International Day of Yoga on 21 June 2022 in a big way. All staff, trainees and PGDM students practiced yoga led by Dr Debnath. Dr Meeraji Rao, consultant cardiologist from Continental hospital was the chief guest. Honorable Minister of State for Agriculture and Farmers' Welfare Mr Kailash Choudhary also participated in the Yoga session in one of the EDP programmes and took an active role in practicing yoga with the trainees on 8th July 2022.

Fig. 7.3: Shri Kailash Choudhary, Honorable Minister of State for Agriculture and Farmers' Welfare with the participants of EDP after the Yoga session

All the staff of the health center including two doctors, one pharmacist (Mr Mahender), a Nurse (Mrs Sunitha), one supporting staff (Mr M S Ravi) and two Physiotherapists (Mrs Rama Sravanthi and Mr Hussain) are providing their best and dedicated services for the patients under the guidance of Dr S Ravichandran, Officer-In-Charge of the Health center. Patients from other ICAR institutes are also availing the facilities of physiotherapy and doctor's consultations at the Academy's health center. The ambulance vehicle is a boon to the people staying inside the campus and for the trainees and students during emergencies at odd hours to shift patients to hospitals. The total number of ICAR pensioners

Fig. 7.4: Blood donation camp

registered at the health center is 824. Hence, ICAR-NAARM health centre provides all the essential and emergency medical facilities to the entire ICAR family located in Hyderabad with full devotion and dedication. Academy's Director felicitated covid warriors Dr Debnath, Dr Himabindu and Dr Chandrakala of Rajendranagar Civil Hospital on 26th January 2022.

7.3 Sports Complex

The sports complex at NAARM contains a stadium for outdoor games such as cricket, football, lawn tennis, and a variety of sporting events. There is a badminton court, table tennis, billiards/snooker, and a gymnasium in the indoor complex. This facility is used by NAARM personnel and trainees, and the NAARM sports contingent has won prizes in zonal, inter-zonal, and inter-institutional sports tournaments as a result of their use of it.

Fig. 7.5: Sports Complex at NAARM

7.4 Agro-Technology Park

The Agro-technological demonstration blocks consist of Orchards, Floriculture Block (Jasmine, Rose, German flower & Tube rose), an Avenue plantation, Landscaping gardens, a nursery, a Poly house, Herbal Garden, an Agrobiodiversity park, Children's Park etc. The total area under the farm is around 31 Hectares. The agro-technology park is developed to demonstrate various agro-technologies in different aspects of the farm. These technologies will act as a show window of technologies developed by ICAR institutes as well as SAUs for the benefit of the commuters, visitors, trainee participants of various training programmes, students from various agricultural universities and farmers visiting the campus.

7.5 Orchard Block

The orchard block consists of Mango (2.0 ha), Sweet lime (1.5 ha), Sapota (0.25 ha), Aonla (0.25 ha), Custard apple (1.0 ha), Phalsa (0.5 ha) and many other miscellaneous fruit orchards grown in an around 6.5 hectares. Last year (2021) new orchards planted in the H-7 plot such as Drumstick, Papaya, Aonla, Rose apple, Dragon fruit, and Carambola are now well established. All the fruit orchards are pruned in time and fields are kept weed free as well as fertilizer application is done periodically.

Fig. 7.6: Weed-free mango orchard

Fig. 7.7: Sapota orchard after pruning

7.6 Floriculture Block

In the floriculture block, roses, jasmine, tube rose, German flower (commonly known as Chandani), golden rod etc. are cultivated in an area of 1.0 ha and available for demonstration purpose for the benefit of the farmers and other visitors. The cut flower of roses, tube rose and golden rod are used from time to time for flower arrangements and also used for flower bouquets during official functions.

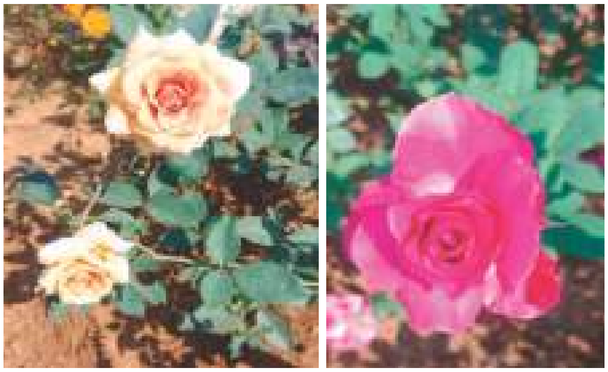


Fig. 7.8: Roses at full bloom in the Rose garden of the floriculture block

Fig. 7.9: Vadelia flowering creeper planted in the rose garden to stop soil erosion

Fig. 7.10: Chandani (German flower) flowers at the floriculture block

7.7 Agro-biodiversity park

The agro-biodiversity park is developed on 4.0 acres which consists of various native tree species such as *Azadiracta indica* (Neem Tree), Tamarind, *Ficus krishnae*, Flame of the forest, Bamboo, Tree jasmine, Gulmohor, Japanese fern tree, Kadamba, Orchid tree, Spathodia, Red shower, Pink shower etc. and several other ornamental plants such as Acalipha, Eranthimums, Euphorbia milli, Hibiscus etc. A few fruit plants such as mulberry, dragon fruit, and Karonda are also planted so that birds can also be attracted to the campus. Around 2000 new ornamental/flowering bushes are planted at the agro-biodiversity park to create the aesthetic value and beautification of the area and also attract the butterflies. In this new plantation Hibiscus, Jatropha, Acalipha, Arelia, Dressenia and Eranthimum plants are included.

Fig. 7.11: New plantation at Agro-biodiversity park

Fig. 7.12: Visit of RAC members at Agro-biodiversity park on September 16, 2022

7.8 New plantations at Hexagonal Park/ Nursery area

The Hexagonal park, Canteen area and Nursery to Guest hose road areas were brought under plantation with ornamental and flowering plants. These areas were planted with arelia (green), arelia (stripped) arelia (paper plate), Dressenia (Pink), Eranthimum (Yellow & Green), Brugmentia, Jatropha, Techoma (Yellow), Hibiscus, Ficus exotica, Areca palms and Difenbacia.

Fig. 7.13: Arelia (2 Varieties) & Dressenia planted at Hexagonal park

Fig. 7.14: Eranthimum plantation near nursery towards guest house road

Fig. 7.15: Spider lilly plantation behind Sagarika IGH-II

Fig. 7.16: Spider lilly plantation along with footpath from Health Centre to Hostels

7.9 Herbal Garden

The herbal garden is developed at the Academy in 875 sq mtr. area with the help of PJTSAU consisting of 50 species including medicinal, herbal and aromatic plants. Self-explanatory boards are placed in each bed of the plants displaying scientific names, local names and use of plant parts with purpose. This herbal garden is very informative for students, farmers and other visitors. The herbal garden is consisting of Citronela, Coleus, *Morinda Citrifoloa*, Vasa, Multivitamin plant, Insulin plant, *Vinca rosea*, Ashwagandha, Vettiver, Henna, Vavili, Saraswati, Machipatri, Vamaku, Jalabrahami Tippatiga, Lemongrass etc.

Fig. 7.17: Visit of RAC members at Herbal garden on September 16, 2022

Fig. 7.18: School Children visiting the Herbal garden

7.10 Plant propagation at Nursery

A nursery is developed in an area of 0.25 ha consisting of a plant propagation unit and potted plants unit. In the year 2022, around ten thousand Ornamental/ Flowering/ Herbal/ Medicinal/ Climbers are propagated and planted in various areas of the campus. The propagated plants are Hibiscus, Dressenia, Arelia, Brugmentia, Ribbon grass, money plant creepers, flowering creepers, Jatropha, Golden rod, Tube rose, Jasmine Plumeria, Insulin, Multivitamin, Tulasi, Ganesh creeper, Yellow Techoma, Jasmine species etc.

Fig. 7.19: Dressenia & Arelia plans propagated at the nursery

Fig. 7.20: Plant propagation at Nursery

Fig. 7.21: Potted plants/Hanging pots at the Nursery for decoration purposes in various buildings

Recognition and Publication

8

8.1 Awards and Recognitions

8.1.1 Institute Award

8.1.1.1 ICAR-NAARM bags Sardar Patel Best ICAR Institute Award

For the first time in its journey of 46 years, the Academy bagged the Sardar Patel Outstanding ICAR Institution Award 2021 (Large Institute Category) for its overall performance. This award is given once a year to recognize outstanding performance by the ICAR institutes, Deemed Universities of ICAR, Central Agricultural Universities and State Agricultural Universities; to provide an incentive for outstanding institutional/University performance; and to promote a sense of institutional pride in achievements. The award consists of Rs. 10,00,000 in cash, a citation and a plaque. Dr Ch Srinivasa Rao, Director received the award from the Union Minister of Agriculture and Farmers' Welfare Shri Narendra Singh Tomar in New Delhi. The awards were given away at a function to mark the 94th Foundation Day of the ICAR.

Figure 8.1.1. Dr Ch Srinivasa Rao, Director received the award from the Union Minister of Agriculture and Farmers' Welfare Shri Narendra Singh Tomar on the occasion of 94th Foundation Day of the ICAR.

8.1.2 Awards Received by the Faculty of NAARM

Name of Staff	Name of award	Date	Purpose	Awarding body
Ch. Srinivasa Rao, Director, ICAR-NAARM	Sardar Patel Best ICAR Institution Award 2021 (large institute category)	Jul 16, 2022	Best ICAR Institution Award 2021 (large institute category)	ICAR
G. Aneeja, CTO, NAARM	Padmasri Dr I.V. Subba Rao Rythunestham Award-2022	Nov 20, 2022	In agri journalism from former Vice President Sri N. Venkaiah Naidu during the 18th annual celebrations of Rythunestham at Swarna Bharathi Trust, Shamshabad	Swarna Bharathi Trust
Surya Rathore and V. V. Sumanth Kumar	MANAGE Best Agricultural Extension Book Award	Jun 11, 2022	Won Third Prize for the book titled "Digital Technologies in Agriculture"	MANAGE, Hyderabad

8.1.3 NAARM Foundation Day Awards

Award	Awardee
Annual Performance Awards (Period: 2021-22)	
Best Faculty Award	Dr Dhandapani, Principal Scientist, ICM Division
Best Trainer Award	Dr M.Balakrishnan, Principal Scientist, ICM Division
Best Promising Young Scientist	Dr B.S Yashavanth, Scientist, ICM Division
Best Research Paper	Spatiotemporal Analysis of Size and Equity in Ownership Dynamics of Agricultural Landholdings in India Vis-à-Vis the World. *Sustainability*. [K.Kareemulla, P. Krishnan, S. Ravichandran, B. Ganesh Kumar, Sweety Sharma, and Ramachandra Bhatta]
Best Policy Paper	Curriculum Development Framework for Agricultural Education, ICAR-NAARM. [Thammi Raju, D., Soam, S.K., Rao, N.S., Kumar, A., Kumar, V.V.S., Kumar, S., Rathore, S., Vinayagam, S.S., Balakrishnan, M., Yashavanth, B.S., Krishnan, P., Sudeep M., Prabhat Kumar., Venkateshwarlu, G., Srinivasarao, Ch. and Agrawal, R.C.]
Best Popular Articles	कोविड-19 महामारी का पशु धन क्त्र पर प्रभा षे व एव नं् यू नार्मल केलिये पहल, पशुधन प्रकाश [डी तम्मी राजू, श्रीकात खाडें, स्वीटी शर्मा एव एस कें सोम]
Administrative Staff (Officer)	Sri M.Dinesh, SAO

Administrative Staff (Non-Officer)	Sri M.Narsingh Rao
Technical (T-1 to T-5 Grade)	Sri N.Ashok
Technical (T-6 and Above)	Dr Laxman M Ahire
Skilled Support Staff (Male)	Sri Ratnaiah
Skilled Support Staff (Female)	Smt R.Prameela
Young Professionals	Dr G.Ranjith Kumar, YP II (DP Cell) Ms Veena U. Kambalimatha, YP II (PME Cell) Ms.Naga Jyothi, YP II, (Admin)
Project Staff	Sri B.Raghupathy, (NAHEP) & Sri Mahesh, (a-IDEA)
Best PG Student	Ms Alana Raphy & Mr Umang Kumar Shyam
Institution Building	Dr Tavva Srinivas, Head, PME Cell & Sri Mukul Raj Singh, SAO
Decadal Performance Awards (Period: 2011-22)	
Best Faculty	Dr RVS Rao, Head, HRM Division
Best Technical Staff	Mr. T Laxman (Driver)
Best Administrative Staff	Sri Telu Srinivas
Best SSS	Sri Kyasam Satnarayana (Despatcher)

8.1.4 Other Recognitions Received by the Faculty of NAARM

- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is Elected as a Fellow of the Indian National Science Academy (INSA).
- Dr D Thammi Raju, Principal Scientist acted as a member of the selection committee recruitment of RA under NAHEP.
- Dr D Thammi Raju, Principal Scientist acted as an expert for Ph.D. thesis evaluation for TANUVAS and SVVU.
- Dr. Surya Rathore served as a Member National Advisory Committee of National Seminar on ‘Role of Agri-Entrepreneurship, FPO and Fisheries in Atmanirbhar Bharat: Issues and challenges with special reference to Andaman and Nicobar Islands organized by Jawaharlal Nehru Rajkeeya Mahavidhyalaya, Port Blair, Andaman & Nicobar Islands in association with NABARD on March 25, 2022.
- Dr Surya Rathore served as a Member of the Selection Committee of Assistant Professors (Extension Education) at Dr Rajendra Prasad Central Agricultural University, Pusa, Samastipur (Bihar) on May 8, 2022.
- Dr Surya Rathore was a Panel Member for conducting an Online personal interview (PI) for the selection of candidates for Post Graduate Diploma in Management – Rural Management (PGDM-RM)–2022-24, Batch 5 and Post Graduate Diploma in Rural

Development Management (PGDRDM) – 2022-23, Batch 20 for the National Institute of Rural Development & Panchayati Raj, Hyderabad.

- Dr. Surya Rathore served as an expert for the National Level Stakeholders meeting on Restructuring Extension Education & Communication Management UG courses in view of National Education Policy 2020 of MPUA&T, Udaipur on July 8, 2022.
- Dr. Surya Rathore served as an expert to conduct an Oral preliminary examination of three Ph.D. students of Extension & Communication Management, College of Community & Applied Sciences, MPUA&T, Udaipur on July 18, 2022.
- Dr. Surya Rathore served as an External Examiner to evaluate the Ph.D. thesis and M. Sc. thesis of Banaras Hindu University, Varanasi and Indian Agricultural Research Institute, New Delhi respectively in the field of Agricultural Extension.
- Dr. Surya Rathore served as External Examiner for the evaluation of Ph.D. thesis in the subject Home Science Extension & Communication Management of Maharana Pratap University of Agriculture & Technology, Udaipur.
- Dr. Surya Rathore has been elected as Member – Scientific Category of Grievance Committee of ICAR – NAARM vide F. No. 2-306/22-23/GRC dated Dec. 23, 2022 [In pursuance of the Council Letter No. 5-2/87-Per. IV dated 10-2-1989].
- Dr. Surya Rathore has been Paper Reviewer for internationally NAAS rated Journal, 'Journal of Agricultural Science & Technology' (Article: Empowerment of Trainees in Agricultural Schools for the development of professional performance).
- Mrs G. Aneeja, CTO was nominated as a Vice Chairperson of the Public Relations Council of India, Hyderabad Chapter. PRCI has 45 chapters across the country and is part of the World Communicators' Council.

Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is nominated as Member/Chairman of the following Committees:

International

1) Member, Executive Board of The International Dryland Development Commission (IDDC), Cairo, Egypt. (2016 – till date).
2) Member, Asian Long Term Experimental Network for Agriculture (ALTENA), Tsukuba, Japan (2017 – till date).

National

1) Chairperson, Capacity Building Unit (CBU) for DARE/ICAR.
2) Member of the Council of National Academy of Sciences, India (NASI) for the year 2023.
3) Nodal Officer from ICAR/DARE for the Development of Tech Dome for the Central Vista Project.
4) Chairman, Committee for devising formats for output-outcome parameters, constituted by NCCT, Ministry of Cooperation, New Delhi.
5) Member, Steering Committee to prepare India's Adaptation Communication under UNFCCC, constituted by MoEF&CC, Govt. of India.
6) Chairman, Committee to recommend course curriculum for the Certificate Course on Agri-Logistics, constituted by NCCT, Ministry of Cooperation, Govt. of India.
7) Member, Food and Agriculture Division Council (FADC), Bureau of Indian Standards, Ministry of Consumer Affairs, Food and Public Distribution, Govt of India.

8) Member, Management Committee of Extension Education Institute (Southern Region), Department of Agriculture and Cooperation & Farmers' Welfare, Ministry of Agriculture & Farmers' Welfare, Govt. of India.

9) Member, Executive Council, National Council for Cooperative Training (NCCT), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt of India.

10) Member, Regional Advisory Group (RAG), NABARD.

11) Member, Board of Management, Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad, Telangana (as ICAR Nominee).

12) Member, Board of Management, Acharya N.G. Ranga Agricultural University, Guntur, Andhra Pradesh (as ICAR Nominee).

13) Member, Academic Council, Professor Jayashankar Telangana State Agricultural University, (PJTSAU), Hyderabad, Telangana.

14) Member, Academic Council, Acharya N.G. Ranga Agricultural University, Guntur, Andhra Pradesh.

15) Member, ICT Steering Committee of ICAR.

16) Member, NAAS Sectional Committee in Natural Resource Management.

17) Member, Section Committee on Life Sciences including Agricultural Sciences, Telangana Academy of Sciences (TAS), Hyderabad.

18) Member, ICAR Committee to revise and explore the possibility of filing APAR online.

19) Member of Chief Advisory Committee for Preparation of Integrated Horticulture Development Perspective Plan, Dept. of Horticulture and Sericulture, Govt. of Telangana.

20) Member, Selection-cum-Standing Committee for Emeritus Scientist Scheme of ICAR (2019-22).

21) Member, ICAR Steering committee to review and monitor the performance of the National Agricultural Innovation Fund (NAIF) Scheme.

22) Member, Advisory Committee, World Bank funded ICAR-NAHEP Component-2 Project on Investment in ICAR Leadership for Agriculture Higher Education.

23) Member, ICAR Committee on Agricultural Research Service for a period of two years (2021-23).

24) Member, ICAR Committee to review Career Advancement Scheme (CAS) and Score Card for Scientists of ICAR.

25) Member, Performance Review Committee, NCCT, Ministry of Cooperation, Govt. of India.

26) Member, Advisory Board, The Journal of Research – PJTSAU.

27) Member, Advisory Board, The Journal of Research – ANGRAU, Andhra Pradesh (April 2018 – till date).

28) Member, PGDM Committee for Post Graduate Diploma in Management - Agribusiness & Management of Vakunth Mehta National Institute of Cooperative Management (VAMNICOM), (Ministry of Agriculture & Farmers Welfare, Govt. of India), Pune, Maharashtra.

29) Member, ICAR Steering Committee for implementation of ICAR Research Data Management Guidelines.

30) Chairman of Expert Committee on Agro-biodiversity, National Biodiversity Authority, (MoEF&CC, Govt. of India), Chennai, Tamil Nadu for a period of one year from October 2021.

31) Member, Executive Council of NAAS, for a period of one year from 1-1-2022.

32) Accredited by PJTSAU for teaching and guiding its M.Sc. and Ph.D Students in the field of Soil Science and Agricultural Chemistry for a period of five years.

33) Editor, Journal of Climate Change & Environmental Sustainability.

34) Editor, Springer Journal.

35) Guest Editor, Journal Agronomy.

36) Member, Advisory Board, International Conference on “Agriculture Science and Technology: Challenges and Prospects in Future (AST-2022)” organized by National Environmental Science Academy (NESA), New Delhi in an association of RLBCAU, Jhansi; ICAR-CAFRI, Jhansi and ICAR-IGFRI, Jhansi from 28-30 January, 2022.

37) Member, Advisory Committee, International Conference on ‘Reimagining Rainfed Agro-ecosystems – Challenges & Opportunities’ (December 22-24, 2022) at ICAR-CRIDA, Hyderabad.

38) Member, Advisory Committee, XI Conference of India Meat Association (IMSACON-XI) and International Symposium on Novel Technologies and Policy Interventions for Sustainable Meat Value Chain (December 14-16, 2022) at ICAR-NRC on Meat, Hyderabad.

39) Member, Advisory Committee International Conference on Blended Learning Ecosystems for Higher Education in Agriculture (December 12-14, 2022) at NASC Complex, New Delhi.

8.1.5 ICAR-NAARM in Media


देश में कृषि उत्पादन को मजबूत करने की जरूरत : उपराष्ट्रपति

भाकृअनुप-नार्म का चौथा स्नातक समारोह आयोजित


వార్త

3/10

4th Graduation Ceremony of PGDM-ABM of ICAR-NAARM on May 14, 2022.

పశుపోషణకు పెరుగుతున్న ఆదరణ

News coverage about the Faculty Development Programme on Competency Enhancement for the Faculty of Sri Venkateswara Veterinary University (SVVU), Tirupati, organized by ICAR-NAARM under SCSP during May 24-28, 2022 at SVVU, Tirupati, in-person mode.

అత్యాశ, ధనాపేక్ష లేని జీవితంతోనే ఆనందం

Kamlesh Patel 'Daaji' Says 'India must be a harbinger of revolution in agriculture with a soul'

The Guide of Heartfulness was invited to deliver the 67th lecture at ICAR - NAARM to celebrate Azadi ka Amrit Mahotsav

News coverage about the spiritual lecture delivered by Shri Kamlesh D. Patel, President, Sri Rama Chandra Mission, and Spiritual guide, Heartfulness Meditation (fondly known as Daaji on "Krishi becoming Rishi" as a part of "Azadi ka Amrit Mahotsav" ICAR Lecture Series # 67, organized at NAARM on July 8, 2022.

'సర్దార్ పటేల్' అత్యుత్తమ అవార్డు 2021 నార్మ్ కైవసం

News coverage about Sardar Patel Outstanding ICAR Institution Award – 2021 received by Director, ICAR-NAARM on July 16, 2022, in New Delhi.

एनएआरएम का 47वां स्थापना दिवस समारोह आयोजित

NAARM 47 th Foundation day celebrations

News coverage about the 47th Foundation Day celebrations of ICAR-NAARM, Hyderabad held on September 1, 2022.

8.1.6 Invited Lectures Delivered by the Faculty and Staff

Name of Staff	Lecture	Programme	Organized by	Place	Date
A Dhandapani	e-Pest Surveillance & advisory for Pest Management	Awareness Programme on Furtherance in Integrated Pest Management (IPM) Approaches for Important Agricultural and Horticultural Crops of Delhi, Haryana and Rajasthan (Virtual mode)	ICAR-NCIPM, New Delhi	Online	January 20, 2022
Alok Kumar	Personality Development	Career Counselling and Personality Development	Indira Gandhi Krishi Visvidalya, Raipur Chhattisgarh	IGKV, Raipur	March 29, 2022
B Ganesh Kumar	Entrepreneurial Opportunities in Animal Husbandry Sector	Online Training Programme on 'Promotion of Agri-Preneurship and its Management in Agriculture and Allied Sectors' for the officers of Department of Agriculture and Allied Sectors of Southern states and UTs held at EEI, Hyderabad	EEI,	Hyderabad	January 21, 2022
	PPP in Animal Husbandry Sector - Challenges/ Issues; Working Models and Experiences	Online Training Programme in Collaboration with MANAGE on 'Building Public Private Partnership for Innovations in Agriculture Extension' for the officers of the Department of Agriculture and Allied Sectors of Southern States and UTs held at EEI, Hyderabad	EEI	Hyderabad	April 28, 2022
	Supply Chain Management in Agriculture	Online Training on 'Agrientrepreneurship Development for Women' organized by MANAGE, Hyderabad and University of Horticultural Sciences, Bagalkot	MANAGE	Hyderabad	July 13, 2022
	Livestock and Poultry Management during Disasters	Training Programme on 'Disaster Management and Mitigation Strategies in Agri and Allied Sectors' for the officers of Department of Agriculture and Allied sectors of Southern states and UTs held at	EEI	Hyderabad	January 6, 2022 and December 1, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
B Ganesh Kumar	Precision Dairy Farming	Online Training Programme on 'Promotion of Nutrition Management in Livestock and Dairy Sectors' for the officers of the Department of Animal Husbandry and Dairy Development Departments of Southern states and UTs held at EEI, Hyderabad	EEI	Hyderabad	December 9, 2022
	PPP in Animal Husbandry Sector - Challenges/ Issues; Working Models and Experiences	Training Programme on 'Public Private Partnership in Agri and Allied Sectors' for the officers of Department of Agriculture and Allied Sectors of Southern states and UTs held at EEI, Hyderabad	EEI	Hyderabad	December 23, 2022
B S Yashavanth	Data visualization through ggplot2	Advanced Statistical Techniques for Data Analysis using R	Indian Institute of Rice Research	Online	January 3, 2022
	Data Visualization using R; & Nonparametric tests	Data Analysis using R	CeL-KAU, Thrissur	Online	June 22 & 24, 2022
	Data visualization through ggplot2; Vector Autoregressive Model; & Co-Integration Analysis	DST-SERB sponsored high end workshop "KARYASHALA" on "Statistical and Machine Learning Techniques for Agricultural Systems Modeling and Forecasting using R"	Indian Institute of Rice Research	IIRR, Hyderabad	August 18 & 22, 2022
	Data Visualization using R	High-End Workshop (Karyashala) entitled "Hands-on-training using Q-GIS and R Programming: an integrated approach towards understanding the changing dynamics in Water Resources"	CWRDM, Kunnamangalam	Online	September 16, 2022
Ch Srinivasa Rao	Positive Thinking and Personality Development	ICAR Sponsored Winter School on "Obtaining globally recognized certifications for ensuring quality and safety of milk, meat, poultry and fish and establishing food testing laboratory"	ICAR-National Research Centre on Meat, Hyderabad	Hyderabad	February 24, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
Ch Srinivasa Rao	Innovations for Climate-Smart Agriculture	21-Days International Online Training-cum-Workshop Program on Agriculture 3.0- a futuristic approach towards Smart Agriculture and Sustainable Urban Farming (April 30 - May 20, 2022).	Just Agriculture - the Magazine and AEEF-WS, Society, Punjab	virtual mode.	May 5, 2022
	Solution and Public-Industry Partnership: Stubble Burning, Waste Management and Innovate Technology	Brainstorming Workshop on "National Air Quality Resource Framework for India (NARFI).	National Institute of Advanced Studies (NIAS), Bangalore with the support of the Office of the Principal Scientific Advisor to Govt. of India	Indian International Centre, New Delhi.	June 22, 2022
	Climate Smart Technologies and Resilient Practices: National Innovations in Climate Resilient Agriculture (NICRA)	Training Programme on "Role of technology in Community Level Disaster Mitigation for Scientists & Technologists" working in the Government Sector	LBSNAA, Mussoorie.	LBSNAA, Mussoorie	July 29, 2022
	Identification of Appropriate Agro-management Techniques towards Climate Change Adaptation in Tropical Ecosystems	RACC/IDDC/ICARDA/ALARI Hybrid Webinar on "Impact of Climate Change on Food Production in the Dry Areas" (a side event of COP-27 in Egypt during Nov 2022).	Ain Shams University, Cairo, Egypt	Virtual mode	September 3-5, 2022
	Building Climate-Resilient Villages in India	5-day Interdisciplinary Short Term Training Programme (STTP) under the DST-SERB Accelerated Vigyan Scheme on "Climate Change: Socio-Economic Impacts and Resilience Building	Visvesvaraya National Institute of Technology (VNIT), Nagpur	Virtual mode	September 30, 2022
	Soil Organic Carbon Sequestration for Sustainable Food Security and Climate Action	SP Roychoudary Memorial Lecture	Hyderabad Chapter of Indian Society of Soil Science at	EEI, Hyderabad	October 11, 2022.

Name of Staff	Lecture	Programme	Organized by	Place	Date
D Thammi Raju	Overview of Teaching Management	Executive Development Programme on Leadership Development	Dr. YSR Horticulture University	Online	
	Overview of Teaching Management	'FDP on Competency Enhancement in Agricultural Research and Education'	SVVU, Tirupati	Tirupati	May 24, 2022
	Innovative Teaching Methods for Quality Education	'FDP on Competency Enhancement in Agricultural Research and Education'	SVVU, Tirupati	Tirupati	May 24, 2022
	Teaching Management	FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of MPUAT	Udaipur, Rajasthan	Online	July 19, 2022
	Innovative Teaching Methods for Quality Education	FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of MPUAT	Udaipur, Rajasthan	Online	July 20, 2022
	Teaching Management	FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of RVSKVV	College of Agriculture, Indore	Online	September 20, 2022
	Innovative Teaching Methods for Quality Education	FDP on Competency Enhancement in Agricultural Research and Education for the Faculty of RVSKVV	College of Agriculture, Indore	Online	September 21, 2022
	Innovative Extension Strategies-Online Education System	ICAR sponsored Winter School on Advanced Extension & Communication strategies for sustainable livelihood through Animal Husbandry & Allied farming system" from February, 15th to 07th March, 2022	WBUAFS, West Bengal	Online	February 18, 2022
	Overview of Teaching Management	FDP on Competency Enhancement in Agricultural Research and Education organized by NAARM	College of Fisheries Mangaluru	Online	May 17, 2022
	Innovative Teaching Methods for Quality Education	FDP on Competency Enhancement in Agricultural Research and Education organized by NAARM	College of Fisheries Mangaluru	Online	May 19, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
G R Ramakrishna Murthy	E-Learning	NEP Workshop	MANUU	Hyderabad	May 5, 2022
	New vistas in TEL	Online lecture to SVVU participants	Veterinary College	Tirupati (SCSP)	May 27, 2022
	TEL	Faculty Induction Programme (online)	MANUU	Hyderabad	May 31, 2022
	TEL	FDP	MANUU	Hyderabad	July 26, 2022
	Technology Potential for Teaching and Learning	FDP	MANUU	Hyderabad	Septem-ber 15, 2022
	Digital Learning in Agriculture	NAHEP sponsored programme on "Capacity Building of Teachers in the advancement of Science and Technology"	ANGRAU	Virtual	October 29, 2022
	Open resources and E-portals and search engines for Bioinformatics	Application of Bioinformatics in Agricultural Research and Education	SKILL-BIF, ICAR-NAARM	Hyderabad	November 17, 2022
	New vistas in TEL and MOOCs - Orientation	Lectures to KVAFSU participants	College of Fisheries	Mangaluru	May 19, 2022
	e-Learning	Faculty Development Programme	MANAGE	Hyderabad	March 21, 2022
	MOOCs	VC Conference	ICAR	Delhi (virtual)	April 12, 2022
G Aneeja	"Etiquette, ethics and communication skills" and "Gender Sensitization"	Induction training for newly recruited stenographers	IT department	Hyderabad	January 17, 2022
	Soft Skills	Induction training for newly recruited Sericulture Officers	Andhra Pradesh State Sericulture Research and De-velopment Institute	Andhra Pradesh	February 3, 2022
	Communication skills" and "Gender Sensitization	Induction training for newly recruited Staff	Income Tax dept	Hyderabad	April 28, 2022
	Soft skills	Induction training for newly recruited Staff	Income Tax dept	Hyderabad	June 27, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
G Aneeja	Gender Sensitization and soft skills	Induction course for newly recruited staff of IT dept	Income Tax dept	Hyderabad	Septem-ber 5, 2022
K Kareemulla	Research Report Writing	Faculty Development Program	UGC-HRD, MANUU	Online	August 20, 2022
	Research Methods for Environmental Studies	Faculty Development Program	UGC-HRD, MANUU	Online	October 13, 2022
M Ramesh Naik	Organic Farming: Principles, Concepts, Status, Scope and Opportunities in India	For M.Sc (Ag) students	PJTSAU, Hy-derabad	College of Agriculture, Rajen-dranagar, Hyderabad	July 6, 2022
M Balakrishnan	Biological Databases and bioinformatics tools and software's	Online Faculty Development Programme (FDP "Advances in Bioinformatics"	BIT, Anna university Campus, Trichy	Tiruchirap-palli	August 28, 2022
	Bioinformatics in Agriculture – Challenges and Opportunities	Online Faculty Development Programme (FDP "Advances in Bioinformatics"	BIT, Anna university Campus, Trichy	Tiruchirap-palli	August 29, 2022
	Sequence Alignment and Phylogenetic study with hands-on session	Online Faculty Development Programme (FDP "Advances in Bioinformatics"	BIT, Anna university Campus, Trichy	Tiruchirap-palli	August 31, 2022
	Bioinformatics and its application in animal husbandry	Advanced biotechnological approaches to augment productivity in poultry for ensuring food and nutritional security" is being jointly organized by the International Livestock Research Institute (ILRI), South Asia, New Delhi and ICAR, New Delhi	ICAR-Di-rectorate of Poultry Research (DPR), Hy-derabad	Hyderabad	Septem-ber 21, 2022
	Information and communication technology (ICT) for smart decision-making in agriculture.	Keynote address on Technical Session III: Emerging opportunities for sustainable land & water use. In the National Symposium on Harmonizing Land Use for Emerging Challenges and Opportunities in the island and coastal region,	CIARI, Port Blair	Port Blair	October 13, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
N Sivaramane	Dummy-dependent models: Application of logit, probit, Tobit, multinomial logit models in Social Science Research.	Introduction to STATA, Data Acquisition, Processing, Descriptive analysis and Linear & Non-linear growth models	Tamil Nadu Agricultural University	Coimbatore	March 18, 2022
	Agri-Startups-Incubation facility	Online collaborative training on Agrientrepreneurship Development for Women, scheduled from 11-15 July 2022	MANAGE, Hyderabad and University of Horticultural Sciences, Bagalkot	Online	July 7, 2022
	Marketing Strategies for FPOs & Product Development	The interface of Producers Organization with Agri Startups	BIRD, Lucknow	Online	September 1-3, 2022
	Role of FPOs and Organizations towards enhancing the income and social conditions of the farming community	CII TS-FOODBIZ 2022	CII	Hyderabad	September 23, 2022
	Economic Impact Evaluation	Training Programme on Planning, Monitoring and Evaluation for Impact Assessment	Extension Education Institute	Hyderabad	November 26, 2022
	Rural/agricultural transformation through agri-business management	Recent advances in processing, storage and marketing of farm produce for agribusiness for small and marginal farmers.	ICAR-CIAE	Bhopal	December 22, 2022
	Presentation Skills	Sensitization of PG and PhD students of TNAU	Agricultural College and Research Institute, TNAU, Madurai	Madurai	September 26, 2022
	Thematic Analysis using NVivo-12	Webinar	Agricultural College, Bapatla ANGRAU, Bapatla	Bapatla, Andhra Pradesh	February 19, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
Ranjit Kumar	Importance of the business plan for the FPO	Workshop on business development for FPOs	ICAR-IIOR, Hyderabad	ICAR-IIOR, Hyderabad	February 22, 2022
	Role of Markets in Agri-Food System Transformation	Policy Roundtable on agri-food transformation and doubling farmers' income in Odisha	IFPRI and Govt. of Odisha	Mayfair Convention, Bhu-baneswar	October 20-21, 2022
	Value Chain Analysis: A case study of the Maize Seed Sector	CAFT on Concepts and Methods for Sustainable Food System Analysis	ICAR-Indian Agricultural Research Institute	ICAR-IARI, New Delhi	November 21, 2022
	Role of Markets in Transforming Agri-food System: Issues & Challenges	CAFT on Concepts and Methods for Sustainable Food System Analysis	ICAR-Indian Agricultural Research Institute	ICAR-IARI, New Delhi	November 21, 2022
S K Soam	Developing Winning Project Proposals	Online workshop on Thrust areas in Research and Developing Winning Project Proposals	Karnataka Veterinary, Animal and Fisheries Sciences University, Bidar	Online/ Bi-dar	Septem-ber 2, 2022
	Development and self-appraisal of the Research concept	One Week Workshop On Research Writing & Publishing during 12-18 October, 2022	Department Of Home Science, Mahila Mahavidy-alaya, Banaras Hindu University, Varanasi	Online/ Va-ranasi	October 12, 2022
	Intellectual Property Rights	Online Training Program on "Soft Skills & Entrepreneurship Development" during 21-26 February 2022	Dr. Balasa-heb Sawant Konkan Krishi Vidy-apeeth	Online/ Dapoli	February 23, 2022
S Ravichandran	Time Series Analysis	DST-SERB sponsored high-end workshop cum training programme on "Statistical and Machine Learning Techniques for Agricultural Systems Modeling and Forecasting using R"	ICAR-IIRR, Hyderabad	ICAR-IIRR, Hyderabad	July 21, 2022
	Data Science in Indian Agriculture - Scope, status, and the road ahead.	73rd Annual Conference of ISAS, Srinagar, India	ISAS, Sri-nagar, India	ISAS, Sri-nagar, India	November 14-16, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
S Ravichandran	Chairman of a technical session on Statistics and Machine Learning for Big Data Analytics	73rd Annual Conference of ISAS, Srinagar, India	ISAS, Srinagar, India	ISAS, Srinagar, India	November 14-16, 2022
S Senthil Vinayagam	Competency Enhancement for Digital Teaching	Refresher Course on Extension Management	CCSHAU	Hisar, Haryana	May 16, 2022
	Need of R-E-F linkages for convergence and SREP –An analysis of the existing schemes	Online Training Program on Integration of Market led Extension in the Preparation of Strategic Research Extension Plan for the District	MANAGE	Hyderabad	May 20, 2022
	Introduction to Entrepreneurship	Training program on “Entrepreneurial Skill Development	NDRI	Karnal	June 14, 2022
	Govt. Schemes for supporting Agri-Startups” and “Strategies for Promotion of Agri-Clinics and Agribusiness Centre	Training on Entrepreneurship Skill Development Programme	KAU	Thrissur	June 24, 2022
	Extension Strategies for enhancing market access	Training Program on Innovative Marketing Methods	MANAGE	Hyderabad	July 14, 2022
	Flagship Programmes of the Ministry of Agriculture & Farmers Welfare	2-day training cum workshop for 30Regional Coordinating Institutions (RCIs)	NIRDPR	Hyderabad	October 20, 2022
	Flagship Programmes of the Ministry of Agriculture & Farmers Welfare	2-day training cum workshop for 30Regional Coordinating Institutions (RCIs)	NIRDPR	Hyderabad	October 20, 2022
	Information Technology for Agriculture and Rural Development	International Training programme on “Good Governance for Management of Rural Development Programmes	NIRDPR	Hyderabad	November 3, 2022
	Opportunities in Bio business incubation	Biotech Industry Stakeholders Meet	TNAU	Coimbatore	November 7, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
S Senthil Vinayagam	Agri Startup Ecosystem	Officials of Karur Vysya Bank & AP Gramin Bank	State Bank Institute of Rural Banking	Hyderabad	November 22, 2022
	Core competencies in research, teaching and technology transfer	The fifth batch of Foundation Course on Research, Education and Extension Management	KAU	Kerala	November 25, 2022
	Challenges and Opportunities to Startups	UTKARSH – an Industry Institute Conclave	CVR College of Engineering	Ibrahimpatnam	December 27, 2022
Sanjiv Kumar	Linking farmers to market through FPO	Faculty Development Programme on Advances in Agriculture	School of Agriculture, MVN University, Haryana	Online	February 17, 2022
	Entrepreneurial opportunities in Agricultural Sector	Short training course on the Role of Agribusiness in the self-reliance of India	Department of Business Management, CCS Haryana Agricultural University	Online	February 22, 2022
	Digital Marketing: Emerging areas in Agribusiness Marketing	Short training course on the Role of Agribusiness in the self-reliance of India	Department of Business Management, CCS Haryana Agricultural University	Online	February 24, 2022
	Agripreneurs and Agri Start-ups	Workshop on SABAGRIS Ideation Hackathon for agripreneurs and agri startups	VKS College of Agriculture, Buxar, Bihar	Buxar, Bihar	May 5, 2022
	Strategic management of agribusiness (Principles & fundamentals)	Agripreneurship bootcamp cum Ideathon	NAU, Navsari, Gujarat	Navsari, Gujarat	September 5, 2022
	Role of ICT enabled marketing in dairy sector	Training programme on dairy management and extension	EEI, Hyderabad	Hyderabad	September 8, 2022
	Promotion of organic farming among farmers through FPOs	Training on Organic farming for sustainable agriculture	MANAGE, Hyderabad	Hyderabad	September 13, 2022

Name of Staff	Lecture	Programme	Organized by	Place	Date
Sanjiv Kumar	Principles and strategies of agribusiness management	Training programme on Innovative ideas for entrepreneurship development in the livestock sector	MANAGE, Hyderabad	Hyderabad	September 18, 2022
Surya Rathore	Motivating Girl Children for STEMM (Science, Technology, Engineering, Mathematics, Medicine) subjects	National Day of Girl Child	Indian Agricultural Research Institute, New Delhi	Online	January 24, 2022
	Strategies for promoting Women Agripreneurs	International workshop on 'Evidence-based Gender equitable Agriculture & Food systems	MANAGE, Hyderabad	Hyderabad	March 23, 2022
	Development of Soft Skills for Agripreneurship development among graduates: Strategic interventions under NAHEP	National Seminar on 'Role of Agri-Entrepreneurship, Fisheries and in Atmanirbhar Bharat: Issues and challenges with special reference to Andaman and Nicobar Islands	Jawaharlal Nehru Rajkeeya Mahavidhyalaya, Port Blair	Hybrid mode (Port Blair & Online)	March 25, 2022
	Nutrition-Sensitive Value Chain & Marketing Opportunities	MANAGE training on Nutrition-Sensitive Agriculture for Extension Advisory Services -FLS (Field Level staff)	MANAGE, Hyderabad	Hyderabad	October 11, 2022

8.2.10 Other Roles

Faculty	Other Roles
B Ganesh Kumar	• Dr. B. Ganesh Kumar is serving as Member, Training Quality Improvement Measures Committee (TQIMC) of NIRD & PR, Hyderabad. • Dr. B. Ganesh Kumar acted as Advisor on the Selection Committee for recruitment to the posts of Scientist in ICAR in the discipline of Agricultural Economics at ASRB, New Delhi on 20 December, 2022.
D Thammi Raju	• Expert for evaluation of Ph D thesis evaluation in the discipline of Veterinary and Animal Husbandry Extension Education TANUVAS, SVVU • Technical Expert for the Online Review of BIG proposals under BIRAC's • Member Secretary for the FOCARS Review Committee (ICAR)
G R Ramakrishna Murthy	• Evaluated BIRAC Proposals in the field of mechanization and IoT • Participation as Expert for the meet of Associate deans of ANGRAU with e-learning firm

Faculty	Other Roles
M Balakrishnan	• External Examiner Anna University, TN, IARI, New Delhi Ph.D. Thesis Evaluation • Academic Council Member Mall Reddy University, Hyderabad, Telangana For the Academic Activities
N Sivaramane	• External Examiner for reviewing PhD theses from ICAR-IARI, New Delhi, University of Agricultural Sciences, Bengaluru. • External Examiner for reviewing M.Sc (Ag) thesis from RPCAU, Pusa, Bihar.
N Srinivasa Rao	• Dr. N. Srinivasa Rao, Principal Scientist has been nominated as Nodal Officer, Transparency audit of Suo-moto disclosures of information by every public authority under RTI act for all ICAR institutions, ICAR HQ and DARE.
Ranjit Kumar	• Reviewer for the startup proposals under the BIRAC-BIG scheme. • Chaired Technical Session in 82nd Annual Conference of Indian Society of Agricultural Economics (ISAE) to be held at CAU, Imphal in Nov 2022. • External Member of Institute Technology Management Committee (ITMC) of ICAR-IIOR, Hyderabad. • External Examiner for reviewing the PhD theses from the Indian Institute of Foreign Trade (IIFT), New Delhi; Birla Institute of Technology (BIT), Mesra; and ICAR-IARI, New Delhi. • Member of the Editorial Board of the Indian Journal of Agricultural Economics, Mumbai.
S K Soam	• Guest of Honor during a two-day online Webinar on "Opportunities for Agricultural Students in Personality and Career Development" on 21-03-2022 organized by Indira Gandhi Krishi Vishwa Vidyalaya, Raipur • Panelist for a session on 'Prospects for employability in Agriculture by introducing agriculture as a mainstream subject in K-12' during National Brainstorming Workshop on "Mainstreaming Agriculture Curriculum in School Education (MACE), conducted by ICAR on 14-06-2022 at NASC, New Delhi. • Evaluated the business proposals of BIG-20 of BIRAC submitted to Society for Innovation & Entrepreneurship(SINE), Indian Institute of Technology Bombay. • Attended brainstorming meeting to discuss the draft to include Indian traditional agricultural practices in Traditional Knowledge Digital Library (TKDL), the online meeting was organized by the Council of Scientific and Industrial Research (CSIR), New Delhi on 7th June 2022, • Attended meeting to discuss the draft structure format for India's traditional agricultural practices and digitization of Indian traditional agricultural information towards inclusion into TKDL database conducted by CSIR-Traditional Knowledge Digital Library Unit, New Delhi on 08-08-2022 • External Expert Member-provided expert advice during the meeting of the Institute Technology Management Committee (ITMC)of IIMR, Hyderabad on 25-07-2022 • 'Panellist' in a brainstorming session on 'Mainstreaming agricultural curriculum in school education', organized by ICAR at NASC, New Delhi on 14th June 2022. • IPR Expert member of ITMC of ICAR- Indian Institute of Oil Palm Research, Pedavegi, provided expertise in the development of a bid for licensing of a patented technology 'Ablation Tool' and also finalizing the MoU with companies as non-exclusive license [during July-August 2022]. • External Examiner for reviewing the PhD theses from the Tamil Nadu Agricultural University, Coimbatore, Tamil Nadu.

Faculty	Other Roles
S K Soam	• Expert reviewer evaluated the business proposals submitted to the 20th call of Biotechnology Ignition Grant (BIG) under the Biotechnology Industry Research Assistance Council (BIRAC) scheme of the Department of Biotechnology, Govt of India. • Member of Organising Committee in the International Conference on Blended Learning Ecosystem for Higher Education in Agriculture 21-23 March 2023.
S Senthil Vinayagam	• Attended NAHEP Review meeting on 05 July 2022 under the Chairmanship of Dr. R.C. Agrawal National Director and DDG (Edn), ICAR, New Delhi. • Evaluated MSME IDEA HACKATHON 2.0 Applications (6 no.) organized by ICAR-NRC on Meat, Hyderabad on 07 Dec 2022.
Sanjiv Kumar	• External examiner to evaluate MBA (ABM) thesis of Department of Agribusiness & Rural Management, IGKV, Raipur, Chhattisgarh.
Tavva Srinivas	• Reviewer for the startup proposals under the BIRAC-BIG scheme. • Evaluator for M.Sc (Ag) thesis from Acharya NG Ranga Agricultural University, Lam, Guntur. • Co-guide for M.Sc (Ag) student from IGKV, Jabalpur.
V V Sumanth Kumar	• Treasurer, AMARA • Treasurer, a-IDEA (Till 24th January 2022) • Member, TAC • Member, Software Renewal Committee. • Member, LAC • Member of Prayas Management Committee for selection of Nidhi Prayas applicants. • Member Secretary for Guidelines for Incubators • Programme Coordinator for *Ek Bharat Shrest Bharat* at ICAR-NAARM • Member Secretary for the committee to prepare a-IDEA Guidelines book. • Member of team for management of Disaster Recovery Centre of ICAR. • Coordinator for the "Diploma in Educational Technology Management" Programme in collaboration with the University of Hyderabad in distance mode from 5th January 2022 onwards. • Member of Portfolio Monitoring & Mentoring Committee (PMMC) of a-IDEA. • Nodal officer for managing admin portal of a-Idea for adding and managing users of a-IDEA. • Member (Co-opted) of the Board of Studies (BOS) of the Academy. • Chairman of LPC committee for finalizing the firm for developing website of a-IDEA. • Chairman of "Committee for Certificates" for ICAR-South Zone Sports meet 22-25 Nov 2022 • Member of the "Admission committee" for DTMA and DETM 2023 batch. • Performed duty of Observer at Khammam for ICAR-IARI examination for recruitment of T-1 Positions on the basis of All India India Computer-based Examination during 28th February to 4th March 2022.

8.2 Publications

8.2.1 Research Papers in Refereed Journals

- Amuktamalyada, G., Supriya, K., Sreekanth, P.D., Rahul, P., Nirmala, B., Gabrijel, O., Meena, A., Gireesh, C., Madhyavenkatapura, S.A., Brajendra, P., Yadav, B.K., Raman, M.S., & Rathod, S. (2022). Characterization and Prediction of Water Stress Using Time Series and Artificial Intelligence Models. *Sustainability*, 14, 6690.
- Avinashilingam, V.N.A., Ola, C.M., Jat, N. K., & Tewari, P. (2022). Performance of Yield and Growth (*Pleurotus sajor-caju*) Mushroom on Wheat Straw at Arid Ecosystem: A Study in Jodhpur district of Rajasthan. *Environment and Ecology*, 40(2), 943-948.
- Balakrishnan, M., Sharma, T., Supriya, P., Soam, S.K., & and Srinivasarao, Ch. (2022). Next-generation sequencing technology: a boon to agriculture. *Genetic Resource Crop Evolution*, 1-20.. https://doi.org/10.1007/s10722-022-01512-5.
- Balakrishnan, M., Supriya, P., Soam, S.K., Srinivasarao, Ch., & Sumalatha, K. (2022). Comparative Molecular docking analysis of Target fruit ripening enzyme Tomato Beta-galactosidase (TBG-4). *Research Journal of Biotechnology*, 17(1), 29-39.
- Behera, S.K., Shukla, A.K., Patra, A.K., Prakash, C., Tripathi, A., Chaudhary, S.K., & Srinivasarao, Ch. (2022). Assessing farm-scale spatial variability of soil nutrients in central India for site-specific nutrient management. *Arabian Journal of Geosciences*, 15(9).
- Bhoomaiah, D., Krishna, P., Kantharajan, G., Agarwal, S., Hemalatha, M., Rajendrana, K.V., & Srinivasarao, Ch. (2022). Mapping the research impact of collaboration and networking of ICAR fisheries research institutes in India: A scientometric study. *Indian Journal of Fisheries*, 69(1), 1-21.
- Butail, N.P., Kumar, P., Shukla, A.K., Behera, S.K., Sharma, M., Kumar, P., Sharma, U., Takkar, P.N., Srinivasarao, Ch., Trivedi, V., Das, S., & Green, A. (2022). Zinc dynamics and yield sustainability in relation to Zn application under maize-wheat cropping on Typic Hapludalfs. *Field Crops Research*, 283(108525).
- Chouhan, L., Kaur, M., Chandra, S., & Rathore, S. (2022) Nexus between Knowledge of farmers related to *Paramparagat Krishi Vikas Yojana* and socio-personal variables. *Indian Journal of Extension Education*, 58(1), 59-65.
- Damale, R.D., Kushwaha, N.L., Rajput, J., Paparao, V., Deb, C.K., Sharma, R., Sontakki, B.S., Srinivasarao, Ch., Marathe, R.A., Dhinesh Babu, K., & Verma, R.K. (2022). Village Agricultural Development Action Plan through Participatory Rural Appraisal Techniques: A Case Study of Santa Village, Morena, Madhya Pradesh, *India. Journal of Community Mobilization and Sustainable Development*, Vol.2, 375-382.
- Dastagiri, M.B. (2022). Economics of India's agricultural domestic and international prices during WTO regime: signals and policies. *Indian Journal of Agricultural Sciences*, 92 (5), 587–92.
- Dastagiri, M.B. (2022). Nature, Natural, Organic Farming's for Sustainable Eco-Friendly Agriculture Development-A Global Perspective and Policies. *Acta Scientific Agriculture,* 6(9), 35-45.
- Dastagiri, M.B., Shailaja., & Lekhana, T. (2022). Geo-Political Economic Analysis of Agriculture and Farmers Welfare Under Economic Systems and WTO Global Corporate Era. *Acta Scientific Agriculture* 6 (11), 48-66.

- Dastagiri, MB., & Sindhuja, N.P.V (2022). Economic analysis of domestic and international food prices during WTO regime: Effects on Indian crop planning, production, poverty, inflation, and trade. *World Food Policy,* 8(2), 237-262.
- Dastagiri, MB., & Sindhuja, N.P.V. (2022). Global merchandise and agricultural trade developments during WTO regime: Commercial crops performance. *World Food Policy*, 8(1), 6-37.
- Dave, K.M., Raval, S.K., Nayak, J.B., Kanani, A.N., Shah, N., Dash, P.K., Shrivastava, N., Yashavanth, B.S., & Gandhale, P.N. (2022) Influence of herd structure on the prevalence of Crimean Congo Hemorrhagic Fever Virus antibodies among bovines in Gujarat. *Ruminant Science*, 11(1), 1-6.
- Devidas, D. R., Kushwaha, N.L., Rajput, J., Rao, V.P., Kumar, D.C., Sharma, R., Sontakki, B. S., Rao, Ch. S., Marathe, R.A., Babu, D. K., & Verma, R. K. (2022). Village Agricultural Development Action Plan Through Participatory Rural Appraisal Techniques: A Case Study of Santa Village, Morena, Madhya Pradesh, India. *Journal of Community Mobilization and Sustainable Development*, 2, 375-382.
- Jaiswal, A.K., Kumar, A., & Babu, B.A., (2022). Growth trends of lac production during XII plan *Vis-a-Vis* XI plan period in Chhattisgarh, India. *Environment Conservation Journal*, 23 (3), 335-342.
- Jaiswal, A.K., Kumar, A., Roy, S. & Sushil, S.N. (2022). Growth analysis of lac production during xii plan *vis-a-vis* xi plan period in Odisha, *India, Journal of Community Mobilization and Sustainable Development*, 17(1), 75-79.
- Kaur, M., Bhardwaj, N., & Rathore, S. (2022). Role Performance of Women Leaders in Panchayati Raj Institutions. *Journal of Community Mobilization and Sustainable Development*, 17(3), 989-996.
- Kumar, A., Ganesha, T.B., Mishra, B.P., Venkatesan, P., Kumar, S., Meena, P.C., Satya, P., & Gupta, B.K. (2022). Challenges of Agri-Startups in Post-Harvest Cold Storage Technologies. *Indian Journal of Extension Education*, 58(1), 89-92.
- Kumar, A., Jaiswal, A.K., Roy, S., & Sushil, S.N. (2022). Growth Analysis of Lac Production During XII Plan Vis-a-Vis XI Plan Period in West Bengal, India. *Agricultural Mechanization in Asia, Africa, and Latin America*, 53 (4), 6951-6957.
- Kumar, A., Jaiswal, A.K., Roy, S., & Sushil, S.N. (2022). Growth Analysis of Lac Production During XII Plan Vis-a-Vis XI Plan Period in Jharkhand, India. *Environment and Ecology*, 40 (2), 313-319
- Kumar, A., Praveen, P.S., Yogi, R.K., Kumar, S., Meena, P.C., Kumar, R., & Singh, R.K. (2022). Value chain analysis of Jatropha in tribal belt of Rajasthan. *The Pharma Innovation Journal*, SP-11(6), 1086-1089.
- Kumar, A., Rao, R.V.S., Ramesh, P., Rao K.H., & Srinivasa Rao, Ch. (2022). Developing Research Managers and Leaders through Management Development Programme in National Agricultural Research and Education System in India. *Indian Journal of Extension Education*, 58(4), 118–123.
- Kumar, N., M.V., Ramya, V., Govindaraj, M., Dhandapani, A., Maheshwaramma, S., Ganapathy, KN., Kavitha, K., Governdhan, M., &and Jagadeeshwar, R. (2022). India's rainfed sorghum improvement: Three decades of genetic gain assessment for yield, grain quality, grain mold and shoot fly resistance *Frontiers in Plant Sciences Sec. Plant Breeding*, 13.

- Kumar, R., Ritambhara, & Somashekhar, G. (2022). Waste to Wealth: India's case of Growing Consumption Demand for Mushrooms in Post-Pandemic Era. *Agricultural Mechanization in Asia*, 53(12), 11385-11401.
- Kumar, S., Kumar, R., Srinivas, K., & Kumar, A., (2022). Agritech Start-ups: Revolutionizing the Agricultural Value Chain in India. *Journal of Community Mobilization and Sustainable Development*, 17 (3), 701-706.
- Mrunalini, K., Behera, B., Jaraman, S., Abhilash, P.C., Dubey, P.K., Narayanaswamy, G., Prasad, J.V.N.S., Rao, K.V., Krishnan, P., Pratibha, G., & Srinivasarao, Ch. (2022). Nature-based solutions in soil restoration for improving agricultural productivity. *Land Degradation and Development*, 33(8), 1269-1289.
- Murthy, G.R.K., Senthil, V.S., Indradevi, T., & Seema, K.S., (2022). Exploring Teachers' Overall Competencies in Indian Agricultural Education System using TPACK Model, *Agricultural Research Journal* (PAU), 59 (1), 120-130.
- Naik, M.R., Hemalatha, S., Reddy, A.P.K., Madhuri, K.V.N., Umamahesh, V., and Rakesh, S. (2022). Efficient Need Based Nitrogen Management in Rabi Maize (*Zea mays* L.) using Leaf Colour Chart Under Varied Planting Density. *International Journal of Bio-resource and Stress Management*, 13(6), 586-594.
- Naorem, A., Jayaraman, S., Dalal, R.C., Patra, A., Srinivasarao, Ch., & Lal, R. (2022). Soil Inorganic Carbon as a Potential Sink in Carbon Storage in Dryland Soils – A Review. *Agriculture*, 12, 1256.
- Parimala, K.M., Devi, K.B.S., Naik, B.B., Jayasree, G., & Sreekanth, P.D. (2022). Performance of Super Early Pigeopea in Different Sowing Windows and Integrated Nutrient Management in Southern Telangana Zone, *The Journal of Research PJTSAU*, 50(1), 70-81.
- Pathak, H., Srinivasarao, Ch., & Jat, M.L. (2021). Conservation Agriculture for Climate Change Adaptation and Mitigation in India. *Journal of Agricultural Physics*, 21(1), 182-196.
- Prakash, S., Singh, A.K., Kumar, A., Kumar, M. & Das, A. (2022). Understanding the Level of Farmers' Explore the Extent of Adoption & Impact Over the socio-economic and Production Scenario on Vermicompost: A Post Training Behavioural Exploration of the Smallholder Farmers in Bihar State of India. *Biological Forum – An International Journal*, 14(2), 1510-1513.
- Raghuvanshi, R., Ansari, M. A., Rathore, S., & Verma, A. P. (2022). Adaptation: The only way to tackle the issues of Climate Change in Indian Himalayas. *Agricultural Mechanisation in Asia, Africa and Latin America*, 53(4), 7141-7148.
- Rajitha, B., Sontakki, B. S., Reddy, J.M., & Vidya Sagar, G.E.CH. (2022). Profile Characteristics of Farmers and their Overall Perception about Effectiveness of Agriculture Input Dealers in Telangana State. *Biological Forum – An International Journal*, 14(3), 1280-1285.
- Rakesh, J., Yashavanth, B.S., Supriya, K., & Sunandini, G.P. (2022) Spatio-Temporal Analysis of Agricultural Labour Wages Using Vector Error Correction Model: An Integrated Approach to Environment and Climate Change. *International Journal of Environment and Climate Change*, 12(11), 3778-3784.
- Ramesh, P., Yashavanth, B.S. & Rao, R.V.S. (2022) Does the Organisation have an Impact on the Happiness and Life Satisfaction of Agricultural Scientists in India? *The IUP Journal of Organisational Behaviour*, 21(4), 07-24.

- Ramesh, P., Yashavanth, B.S. & Rao, R.V.S. (2022) Role of Personality Traits and Wellbeing Attributes in Predicting the Academic Achievement of Post-Graduate Agri-Business Management Students, *Asian Journal of Agricultural Extension, Economics & Sociology*, 40(11), 569-578.
- Ramesh, P., Yashavanth, B.S., & Rao, R.V.S. 2021. Personality and well-being traits of agricultural scientists: Assessment, correlations and prediction. *Journal of Organisation and Human Behaviour*, 10 (3), 9-21.
- Rathore, S., & Kaur, M. (2022). Awareness and Usage of Information and Communication Technologies (ICTs) by the Faculty members in the academic ambiance of Indian State Agricultural Universities.*Journal of Community Mobilization and Sustainable Development*, 17(3), 878-882.
- Rathore, S., Poonam., Kaur, M., & Ravichandran, S., (2022). ICT in Agricultural Higher Education: Investigating the Flowers and Flaws. *Agricultural Mechanisation in Asia, Africa and Latin America*, 53(4), 471-478.
- Santosha, R., Gayatri, C., Nirmala, B., Gabrijel, O., Sundaram, R., & Raman, M.S. (2022). Modeling and Forecasting of Rice Prices in India during the COVID-19 Lockdown Using Machine Learning Approaches. *Agronomy*, 12, 2133.
- Sharan, B.R., Srinivasarao, Ch., Chandrasekhar, Chandrasekhar Rao, P., Lal, R., Rakesh, S., Kundu, S., Singh, R.N., Dubey, P.K., Abhilash, P.C., Rao, K.V., Abrol, V., & Somasundaram, J. (2022). Greenhouse Gas Emission and Agronomic Productivity as Influenced by Varying Levels of N Fertilizer and Tanksilt in Degraded Semi-Arid Alfisol of Southern India. *Land Degradation and Development*.
- Singh, S., Gupta, B.K., Mishra, B.P., Gaurav, S., Raghubanshi, S.S., Abhishek, K., & Kumar, A. (2022). Entrepreneurial Behaviour of skilled youths of Banda District of Bundelkhand Region, Uttar Pradesh. *Indian Journal of Extension Education*, 58(1), 188-191.
- Soam, S.K., Raghupathi, B., Sweety, S., Yashavanth, B.S., Balakrishnan, M., Srinivasa Rao, N., Thammi Raju, D., Prabhat, K., & Agrawal, R.C. (2022). Mind Mapping as a Learning and Teaching Tool in Agricultural Higher Education in India. *Indian Research Journal of Extension Education*, 22(3), 176–181.
- Soam, S.K., Venkatesan, P., Sivaramane, N., Sreekanth, P.D., Rakesh, S., Jagdish Tamak, Arunachalam, A., Pandey, S., & Srinivasarao, Ch. (2022). Evidence for the economic impacts of Eucalyptus and Subabul-based agroforestry in Andhra Pradesh and Telangana. *Indian Journal of Agroforestry*, 24(2), 52-60.
- Soujanya, L.P., Sekhar, J.C., Yathish, K.R., Chikkappa, Karjagi, G., Rao, K.S., Suby, S.SB., Jat, S.L., Kumar, B., Kumar, K., Vadessey, Vadessey, J., Subaharan, K., Patil, J., Vinay K.K., Dhandapani, A., & Sujay, R., (2022). Leaf Damage Based Phenotyping Technique and Its Validation Against Fall Armyworm, *Spodoptera frugiperda* (J. E. Smith), in Maize. *Frontiers in Plant Sciences Sec. Plant Breeding,* 13.
- Srinivasarao, Ch., Prasad, J.V.N.S., Rao, K.V., Kiran, B.V.S., Ranjit, M., Girija Veni, V., Priya, P., Abhilash, P.C., & Chaudhari, S.K. (2022). Land and water conservation technologies for building carbon positive villages in India. *Land Degradation & Development*, 33(3), 395-41.

- Srinivasarao, Ch., Naik, R., M., Ranjit, K, G., Manasa, R., Kundu, S., Narayana Swamy, G., Nataraj, K.C., & Prasad, J.V.N.S. (2022). Cover-crop technology for soil health improvement, land degradation neutrality, and climate change adaptation. *Indian Journal of Fertilizers*, 18(5), 440-460.
- Sumanth, K.V.V., Rathore, S., Raghuvanshi, R., & Padmaja, B. (2022). Personalized Advisory and Climate Information Service for Smallholders of South Central India, *Indian Journal of Dryland Agricultural Research and Development*, 2021 36(2), 37-42.
- Supriya, P., & Bhat, K. V. (2022). Transcriptome analysis of muskmelon (*Cucumis melo* L.) for moisture stress tolerance. *Indian Journal of Agricultural Sciences*, 92 (12), 1514–1516.
- Supriya, P., Kumar, A., Archak, S., & Bhat, K.V. (2022). Computational identification of microRNAs and their target genes in sesame (*Sesamum indicum* L.). *Indian Journal of Genetics and Plant Breeding*, 82(4), 469-473.
- Swetha, A., Indudhar Reddy, k., Rami Reddy, P.R., Yashavanth, B.S., & Yakadri, M. (2022) Effect of Rice Straw Mulching and Irrigation Regimes on Yield and Yield Attributes of Sunflower (*Helianthus annuus* L.) in Rice Fallow. *International Journal of Environment and Climate Change*, 12(11), 3113-3119.
- Thammi Raju, D., Ramesh, P., Raghuvanshi, R., Yashavanth, B.S., Sontakki, B.S., Krishnan, P., Raghupati, B., Soam, S.K. and Srinivasarao, Ch. (2022) Students' Approaches to Learning in Agricultural Higher Education, *Asian Journal of Agricultural Extension, Economics & Sociology*, 40(5), 19-29
- Venkatesan, P., Sivaramane, N., Sontakki, B.S., Burman, R.R., Srinivasarao, Ch., Chahal, V.P., Singh, A.K., Sethuraman, P., Sharma, J.P., Padaria, R.N., Chakravorthy, S., Sharma, N., Patel, N., Choudhary, H., Mondal, G., Singh, R., Kalyani, B., Sharma, S., & Kumar, R. (2022). Predicting adoption of agricultural technologies in Indo-Gangetic Region. *Indian Journal of Agricultural Sciences*, 92(6), 769-74.
- Venkatesan, P., Sundaramari, M., & Kumar, A. (2022). Scientific Rationality and Adoption of Indigenous Soil and Water Conservation in Kolli Hills, Tamil Nadu, India. *Indian Journal of Ecology*, 49(5), 1962-1964.
- Venkatesh, G., Gopinath, K.A., Reddy, K.S., Reddy, B.S., Prabhakar, M., Srinivasarao, Ch., Visha Kumari, V., & Singh, V.K. (2022). Characterization of Biochar Derived from Crop Residues for Soil Amendment, Carbon Sequestration and Energy Use. *Sustainability*, 14(4), 2295.
- Verma, P., Singh, A., Supriya, P., Bhat, K.V., & Lakhanpaul, S. (2022). Comparative DNA Methylome of Phytoplasma Associated Retrograde Metamorphosis in Sesame (*Sesamum indicum* L.). *Biology*, 11, 954.
- Yashavanth, B.S., & Sreekanth, P.D. (2022) Topic Modelling for Discovering Themes in the Queries Raised at Farmers' Call Center. *Journal of the Indian Society of Agricultural Statistics*, 76(1), 7-16.
- Yogi, R.K., Govind, P., Kumar, A., Singh, A.K., Nirmal, K., & Sharma K.K. (2022). Threat to the Sustainable Production of Natural Resins: A Case of *Rangeeni* Strain of *Kerria lacca* (Kerr) in Eastern Plateau & Hills Region of South East Asia. *International Journal of Environment and Climate Change,* 12(12), 1582–1599.

8.2.2 Books and Monographs

- Meena, R.S., Srinivasarao, Ch. & Kumar, A. (Eds.) (2022). *Plans and Policies for Soil Organic Carbon Management in Agriculture, Springer*. Singapore, pp221, ISBN 978-981-19-6178-6.

- Ramesh, P., Debnath, A., Bindu, T.H., Venkateshwarlu, G., & Srinivasa Rao, Ch. (2022). *A Manual on Stress Management and Well-Being in the Workplace*. ICAR – National Academy of Agricultural Research Management (NAARM), Hyderabad, India, pp48.
- Senthil, V.S., Sivaramane, N., Vijay, A., Kumar, V.V.S., Yashvanth, B.S., Umesh, H., Kareemulla, K., Shiva, K., Venkateshwarlu, G., & Srinivasa Rao, Ch. (2022). *Startup Panorama*, ICAR-National Academy of Agricultural Research Management, Hyderabad, India, pp56.
- Srinivasa Rao, Ch., Venkateshwarlu G, Sumanth Kumar V.V, Senthil Vinayagam S, Sivaramane N, Vijay Avinashilingam, NA & Yashavanth, B.S. (2022). *Guidelines for Incubators, a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture)*, ICAR – National Academy of Agricultural Research Management, Hyderabad, India, pp 36.
- Vinayagam, S.S., Sivaramane, N., Vijay, A., Sumanth Kumar, V.V., Yashavanth, B.S., Umesh, H., Kareemulla, K., Shiva, K., Venkateshwarlu, G., & Srinivasa Rao, Ch. (2022). *Startup Panorama*, ICAR-National Academy of Agricultural Research Management, Hyderabad, India, pp 56.

8.2.3 Book Chapters

- Meena, R.S., Kumar, S., Srinivasarao, Ch., Kumar, A., & Lal, R. (2022). Reforming the Soil Organic Carbon Management Plans and Policies in India. In: Meena, R.S., Rao, C.S., Kumar, A. (Eds.), *Plans and Policies for Soil Organic Carbon Management in Agriculture*. Springer, Singapore. https://doi.org/10.1007/978-981-19-6179-3_1.
- Ramesh, P. (2022). Emotional Intelligence in Extension Education. In: Goswamy, A., & Biswas, S. (Eds.), *Advanced extension & communication strategies for sustainable Livelihood through Animal Husbandry and Allied Farming system*. NIPA Genx electronic Resources & solutions Pvt. Ltd., New delh 110034, India.
- Rathore, S. (2022). Nurturing entrepreneurial ecosystem in agricultural Universities. In Gautam., U. S., Nain., M.S., Mishra., D. & Joshi., V. (Eds.), *Novel Extension Approaches for reshaping Indian Agriculture* (pp. 194-203). New Delhi: Biotech Books.
- Rathore, S., Rao, V.K.J., & Sontakki, B. S. (2022). Sprouting Startup culture among Agri-graduates: Way forward. In G. Kadirvel, S. Ghatak and V. K. Mishra (Ed.), *Agri-business Opportunities in North East India* (pp. 23-29). New Delhi: Today and Tomorrow's Printers and Publishers.
- Sontakki, B.S., Venkattakumar, R., & Pal, P. P. (2022). Dimensions of Social Sciences Research for Self Reliant Agriculture: Aligning Extension Research to Atma Nirbhar Bharat Abhiyan. In: S. K. Roy et al. (Eds.), *Building Self Reliant INDIA Through Techno-Rich Extension System in Agriculture and Allied Sectors* (pp. 77-93). New Delhi: International Books and Periodical Supply and Service.
- Srinivasa Rao, Ch., Sreekanth, P.D., & Kumar V.V.S. (2022). Technology Innovations for Sustainable Rainfed Farming in India in Souvenir of International Conference on Reimagining Rainfed Agro-ecosystems: Challenges and Oppurtunities held on 22-24 December 2022 (pp. 39-43).
- Srinivasarao, Ch., Krishnan, P., Thammi Raju, D., Srinivas, T., Sontakki, B.S. & Venkateswarlu, G. (2022). Leadership in Agricultural Research in Independent India. In: Pathak, H. et. al (Eds.) *Indian Agriculture after Independence*

(pp. 407-425). Indian Council of Agricultural Research, New Delhi.

- Srinivasrao, Ch., Rakesh, S., Kumar, R., Kiran, P., Manasa, R., Sahoo, S., Kundu, S., Prasad, J.V.N.S., Pratibha, G. & Swamy, G.N. (2022). Technologies, Programmes, and Policies for Enhancing Soil Organic Carbon in Rainfed Dryland Ecosystems of India. In: Meena, R.S., Rao, C.S., Kumar, A. (Eds.), *Plans and Policies for Soil Organic Carbon Management in Agriculture*. Springer, Singapore. https://doi.org/10.1007/978-981-19-6179-3_2.007/978-981-19-6179-3_2.
- Vanaja, M., Yadav, S.K., Maheswari, M. & Srinivasarao, Ch. (2022). Increasing Atmospheric Carbon Dioxide and Temperature-Threats and Opportunities for Rainfed Agriculture. In: *Agriculture under climate change: Threats, strategies and policies* (pp. 84-90). Allied Publishers Pvt. Ltd.

8.2.4 Popular and Newspaper Articles

- Dastagiri, M.B. (2022). 'Indian Farmers' Welfare: Policies, approaches, strategies, Prophecy. Indian Farming, 72(02), 30–33.
- Raghupathi, B., Rakesh, S., Divakar Reddy, P., & Soam, S.K. (2022, May). Agro-meteorology Services for Crop Planning and other Allied Aspects. *Agriculture World Magazine*, 10-15.
- Raghupathi, B., Rakesh, S., Balakrishnan, M., & Soam, S.K. (2022, January). Application of 5G in Agriculture. *Agriculture world*, *8(1)*, 9-12.
- Ravikishore, M., Supriya, P., & Rama Subbaiah., K. (2022, October) Food processing in India: Prospects and retrospect. *Kisan world*, *49(10)*, 34-37.
- Ramesh Naik, M., Umesh, H., Bhargav Reddy, M., Munni Manta., & Ramanjaneyulu, A.V. (2022). Status and Prospects of Organic Farming in India. *Chronicle of Bioresource Management, 6(2)*, 60-65.
- Rathore, S., & Kaur, M. (2022, June). 'सहभागी विधि से अनुश्रवण एवं मूल्यांकन' अभिनव कृषि, ४ (२), ३५.
- Srinivasa Rao, N., Nancy Celine, P., & Rakesh, S. (2022). Designing climate smart system: AI in precision agriculture. *Agriculture World*, *8(7)*, 16-20.
- Supriya, P., & Ravikishore, M. (2022, June). Urban Farming: The Saviour of Future Agriculture. *Kisan world*, *49(6)*, 34-39.
- Venkatesan, P., Srinivasarao, Ch. & Sivaramane, N. (2022, August). Doubling Farmers' Income by 2022 – Seeding the Change. *Just Agriculture: e-Magazine, 2(12)*.

8.2.5 Training Manuals/ Technical Bulletins

8.2.5.1 Training Manuals

- Balakrishnan, M., & Sujatha, K. (2022). Training Manual *competency enhancement in agriculture research and Education*, pp. 1-118.
- GRK Murthy (2022). *Course Management through MOOCs*, pp. 1-322.
- M Ramesh Naik. (2022). Achieving Sustainable Food Systems through Organic farming. Training manual on "*Organic farming for Sustainable Agriculture*". College of Agriculture, Rajendranagar, Hyderabad 500 030, Telangana, India. pp. 20-26.
- Ravichandran, S., & Sivaramane, N. (2022). "*Analysis of Experimental Data using R*".
- Senthil Vinayagam, S., Sivaramane, N., Vijay Avinashilingham, Sumanth Kumar, V.V., Yashvanth, B.S., Umesh, H., Kareemulla, K., Shiva, K., Venkateshwarlu, G., & Srinivasa Rao, Ch (2022). S*tartup Panorama*, ICAR-National Academy of Agricultural Research Management, Hyderabad, India, pp 56.
- Srinivasarao, Ch., Venkateshwarlu, G., Sumanth Kumar, V.V., Senthil Vinayagam,

S., Sivaramane, N., Vijay Avinashilingam, N.A. & Yashavant, B.S. (2022). *Guidelines for Incubators, a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture)*, ICAR-National Academy of Agricultural Research Management, Hyderabad, India, p24.

- Venkatesan, P., & Avinashilingam, V. N.A. (2022). *Management Development Programme for newly recruited Senior Scientists and Heads of Krishi Vigyan Kendras*, ICAR-NAARM, Hyderabad, India.

8.2.5.2 Technical Bulletins

- Avinashilingam, V. N.A., Sivaramane, N., Kumar, V.V.S., Yashavanth, B., Kumar, B.G., Sreekanth, P.D., Umesh, H., Vinayagam, S. S., Rao, N.S., Soam, S.K., Venkateshwarlu, G., & Rao, Ch. S. (2022). *Journey of a-IDEA in Shaping Agripreneurship Landscape in India*. ICAR-NAARM, Hyderabad, India.
- Sucharitha Devi. S., Supraja. T., Priya Sugadhi., Ramesh Naik, M., Balakrishna, M., & Srinivas Rao, Ch. (2022). *Value Addition to Fruits*. Published by ICAR-NAARM and College of Community Science PJTSAU.
- Sujatha, K., & Balakrishnan, M. (2022). Technical bulletins on *Backyard Poultry Rearing* (Telugu), pp1-41.
- Supraja, T., Janaki Srinath, P., Sarkar, S., Ramesh Naik, M., Murhari, D., Balakrishna, M., & Srinivas Rao Ch. (2022). *Bakery Products as an Enterprise*. Published by ICAR- NAARM and College of Community Science PJTSAU.

8.2.6 Papers Presented in Conferences/ Seminars

- Balakrishnan, M., Tamanna Sharma, Supriya, P., Soam, S.K., & Srinivasa Rao, Ch. (2022, April). *Homology Modeming of Photosystem I Iron-Sulfur Center (PsaC) from Citrullus Lanatus using Modeller*. Paper presented at the 43rd Annual Meeting of the Plant Tissue Culture Association & International Symposium on Advances in Plant Biotechnology and Nutritional Security (APBNS-2022), ICAR-National Institute for Plant Biotechnology (NIPB), New-Delhi, 28 – 30 April 2022, pp 344-345.
- Balakrishnan, M., Tamanna Sharma, Supriya, P., Soam, S.K., & Srinivasa Rao, Ch. (2022, February). *Homology Modelling of Capsid Protein from Mung Bean Yellow Mosaic India Virus using Modeller*. Paper presented at the 24th Annual Conference of SSCA (online) on Recent Advances in Statistical Theory and Applications (RASTA-2022).
- Dastagiri, M.B. (2022, April). *Geo-Political Economic Analysis of Agriculture & Farmers Welfare under Economic Systems and WTO Global Corporate Era*. Paper presented at the 26th Annual Western Hemispheric Trade Conference Proceedings, held on April 6-8, 2022 in Laredo, Texas, USA.
- Dastagiri MB (2022, April). *UNO, WORLD BANK, IMF, WTO Global Multilateral Institutions In Multipolar World: Geo-Political & Economic Policy Analysis*. Paper presented at the 26th Annual Western Hemispheric Trade Conference Proceedings, held on April 6-8, 2022 in Laredo, Texas, USA.
- Dhandapani, A. (2022, May). Panel Discussion on *Role of ICT in Plant Protection* held on 11-05-2022 in Training Program on Integrated Pest Management for Sustainable Agriculture organized by ICAR-NCIPM.
- Dhandapani, A. (2022, June). *Statistical modeling & data Science: Past, Present and Perceptions*. Invited talk in Two days National Seminar cum workshop on "Food for Thought: Applied Statistics and its Implications" on

the eve of 16th National Statistics day, 29-30 June, 2022 held at Navsari Agricultural University, Navsari, Gujarat.

- Dhandapani, A. (2022, November). HadamardR - *R package to generate Hadamard Matrices*. Invited talk presented in Lal Mohan Bhar Memorial Session held on 14-Nov-2022 in 73rd Annual Conference of Indian Society of Agricultural Statistics, SKUAST, Srinagar.
- Dhandapani, A. (2022, December). *Data Science and Agriculture*. Invited talk (online) in the International Conference on Knowledge Discoveries On Statistical Innovations & Recent Advances In Optimization held by the Department of Statistics, Andhra University, Visakhapatnam.
- Indradevi, T., Murthy, G.R.K., Seema Kujur, S., & Senthil Vinayagam (2022, October). *Feasibility of Traditional Evaluation System Through Online Mode: A Case Study.* published during the International Conference of "Revitalizing the Libraries to the Android Society" held on October 24- 14, 2022 at NIT Warangal, pp:600-604.
- Naik, R. & Hudedamani, U. (2022, August). *Sustainable Agri-Food Systems: Guide Towards Healthy Nation*. Paper presented at the International Conference on AAFS Aug. 22 - 24th, New Delhi Publishers, New Delhi – 110059, India, pp63.
- Rathore, S., Soam, S. K., & Raghuvanshi, R. (2022, March). *Development of Soft Skills for Agripreneurship development among graduates: Strategic interventions under NAHEP*. Lead Paper presented (in virtual mode) at National Seminar on Role of Agri-Entrepreneurship, Fisheries and in Atmanirbhar Bharat: Issues and challenges with special reference to Andaman and Nicobar Islands.
- Kumar, S., Kumar, R. Meena, P.C., & Yashavanth, B.S. (2022, November). *Natural Farming: Adoption and Perception among Farmers in Andhra Pradesh*. Paper presented in National Conference on Natural Farming for Sustainable Agriculture and National Prosperity at Sardarkrushinagar Dantiwada Agricultural University, Gujarat during Nov 11-13, 2022.
- Sontakki, B. S. (2022, April). *Farmer Producer Company as Farmer-led Extension Model*. Lead paper presented at National Seminar on Comprehensive Extension Strategies for Sustainable Development of Farmer Producer Organization (FPOs): Issues and Challenges, jointly organized by International Society of Extension Education, Nagpur and MANAGE, Hyderabad at MANAGE, Hyderabad.
- Sontakki, B. S. (2022, May). *Strengthening Extension Feedback Mechanism*. Lead paper presented (in virtual mode) at the National Dialogue on Extension Services for Efficient Delivery of Horticultural Technologies: A way forward (in hybrid mode) organized at ICAR-IIHR, Bengaluru.
- Sontakki, B. S. (2022, June). *Farmer Producer Organizations for Sustainable Secondary Agriculture: Strategic Options and Implications*. Lead paper presented in the National Seminar on Sustainable Food Production Systems for Self-Reliant and Climate Resilient Agriculture organized at UAS, Dharwad.
- Srinivasarao Ch. (2022, February). *Pulses for Agricultural Sustainability, Land Degradation Neutrality and Climate Change Mitigation*. In: Souvenir - International Conference on Pulses: Smart Crops for Agricultural Sustainability and Nutritional Security (ICPulses2023), at NASC Complex, New Delhi during 10-12 February, 2023.

- Srinivasarao, Ch. (2022, July). *Identification of Appropriate Agro-management Techniques towards Climate Change Adaptation in Tropical Ecosystems*. Paper presented in IDDC-Webinar on Developing Case Study related to Impact of Climate Change in Dry Areas, being organized by International Dryland Development Commission (IDDC) in association with RACC; STS, Kyoto, Japan; and ICARDA, July 23-24, 2022.
- Srinivasarao, Ch., Sreekanth, P.D., & Sumant Kumar, V.V. (2022, December). *Technology Innovations for Sustainable Rainfed Farming in India*. Paper presented in Souvenir - International Conference on Reimagining Rainfed Agroecosystems: Challenges & Opportunities at ICAR-CRIDA, Hyderabad during 22-24 December 2022.
- Supriya, P., Balakrishnan, M., Soam, S.K., & Srinivasa Rao, Ch. (2022, February). *Trends and Application of Bioinformatics and Data Science in Agriculture*. Paper presented in the 24th Annual Conference of SSCA (online) on Recent Advances in Statistical Theory and Applications (RASTA-2022).
- Supriya, P., Manish, S., Balakrishnan, M., Prakasam, V., Soam, S.K., Sundaram, R.M., & Satendra, K. M. (2022, April). *Identification and expression analysis of long non-coding RNAs induced during Rhizoctonia solani infection in rice*. Paper presented in 43rd Annual Meeting of the Plant Tissue Culture Association & International Symposium on Advances in Plant Biotechnology and Nutritional Security (APBNS-2022), ICAR-National Institute for Plant Biotechnology (NIPB), New-Delhi.
- Supriya, P., Balakrishnan, M., Tamanna Sharma, Soam, S.K., & Srinivasa Rao, Ch. (2022, October). *Computational tools for CRISPR Off-Target Detection*. Paper presented in International Conference on Plant Genetic Engineering and Genome Editing at CUK, Kerala.
- Venkatesan, P., Sundaramari, M., & Kumar, A. (2022, October*), Scientific Rationality and Adoption of Indigenous Soil and Water Conservation in Kolli Hills, Tamil Nadu, India*. Paper presented at International Conference on Sustainable Agricultural Innovations for Resilient Agri-Food Systems organized by Indian Ecological Society at Sher-e-Kashmir University of Agricultural Sciences & Technology of Jammu, India.
- Yashavanth, B.S. (2022, February). *Forecasting the number of animals insured under the Livestock Insurance Scheme using the Grey Model*. Paper presented at the 24th Annual Conference of SSCA (online) on Recent Advances in Statistical Theory and Applications (RASTA-2022).
- Yashavanth, B.S., & Rakesh, J. (2022, December). *The relationship of agricultural wage rate among states in India: A Vector Error Correction Model Analysis*. Paper presented at the Annual Conference of the International India Statistical Association at Indian Institute of Science, Bengaluru.

8.2.7. Research Project Report and Working Papers

- Balakrishnan, M., Thammi Raju, D., Soam S.K., Probodh, B., Mangrauthia, S.K., Pandey, M.K., Seema, J., & Srinivasa Rao, Ch. 2022. Bioinformatics in Agriculture. Status paper, ICAR-National Academy of Agricultural Research Management, Hyderabad, India, 24.
- Sivaramane, N., Sumanth Kumar, V.V., Avinashilingam, V., Yashavanth, B.S., Umesh, H., Ramesh Naik, M., Srinivas, K., Senthil Vinyagam, S., Soam, S.K., & Srinivasa Rao, Ch.

(2021). *Fostering Agri-based Entrepreneurship in India through aIDEA of NAARM and NABARD Technology Business Incubation Partnership*. ICAR-National Academy of Agricultural Research Management, Rajedranagar, Hyderabad – 500030, India, pp 40.

8. 2.8 Policy Briefs

- Balakrishnan, M., Thammi Raju, D., Soam, S.K., Probodh Borah, Mangrauthia, S.K., Manish K. Pandey, Seema Jaggi., & Srinivasa Rao, Ch. (2022). *Bioinformatics in Agriculture*. Status paper, ICAR-National Academy of Agricultural Research Management, Hyderabad, India, p24.
- Srinivasa Rao, Ch. & Ranjit Kumar (2022). *Agri-startups in India: Opportunities, Challenges and Way Forward*, NAAS Policy Paper 108. National Academy of Agricultural Sciences, New Delhi: pp. 17.
- Ranjit Kumar, Sanjiv Kumar, Pundir, R.S., Surjit, V., & Srinivasarao, Ch. (2022). *FPOs in India: Creating Enabling Ecosystem for their Sustainability*. ICAR-National Academy of Agricultural Research Management, Hyderabad, India. pp22.
- Thammi Raju, D., Krishnan, P., Soam, S.K., Venkateshwarlu, G., & Srinivasarao, Ch. (2022). *Attracting the Best Talent to Agricultural Education in India*, Indian Council of Agricultural Research (ICAR), New Delhi, pp22.
- Yashavanth, B.S., Bharat S Sontakki, Soam, S.K., Venkateshwarlu, G., & Srinivasa Rao, Ch. (2022). *Strengthening Kisan Call Centres in India*. ICAR-National Academy of Agricultural Research Management, Hyderabad, India. pp20.

8.2.9 Copyrights

Title	Registration Number and date	Authors
Career Development Centre Monitoring System, ICAR-NAARM.	SW-15437/2022. 24-12-2021	Soam, S.K., Srinivasarao, N.S., Kumar, A., Thammiraju, D., Raghupathi, B., Rathore, S., Sumanth Kumar, V.V., Yashavanth, B.S., Kumar S., Marwaha, S., Kumar, P., Srinivasarao, Ch. and Agrawal, R.C.
CXI-FOCARS 111 in New Normal: Standard Operating Procedures (SOPs) for COVID Pandemic	Copyright registration no L.31683/2021-CO/L under Literary category	Balakrishnan M, Sumanth Kumar VV, Krishnan P Soam SK, Srinivasa Rao Ch
Faculty Development Centre Monitoring System, ICAR-NAARM.	SW-15439/2022. 28-12-2021	Soam, S.K., Thammiraju, D., Srinivasarao, N., Kumar, A., Raghupathi, B., Krishnan, P., Vinayagam, S.S., Balakrishnan, M., Marwaha, S., Kumar, P., Srinivasarao, Ch. and Agrawal, R.C.
Vinutna: A Holistic Digital Ecosystem for Foundation Courses	SW-15694/2022 13-10-2021	Sumanth Kumar VV, M. Balakrishnan M, Krishnan P, Soam S.K, Srinivasa Rao Ch

9 Financials

9.1 Budget Allocation and Expenditure during Jan-Dec 2022 (Rs. in lakh)

S.No.	Head of Account	Budget	Expenditutre
A. Grant in Aid - Capital			
1	Works	471.16	471.16
2	Equipments	20.72	20.72
3	Information Technology	20.67	20.67
4	Library Books & Journals	0.00	0.00
5	Vehicles & Vessels	0.00	0.00
6	Furniture & Fixtures	3.65	3.65
Total Capital		**516.20**	**516.20**
Grants in Aid - Salaries (REVENUE)			
1	Establishment Expenses		
2	i. Establishment Charges	2320.02	2320.02
3	ii. Wages	4.92	4.92
4	iii. OTA	0.09	0.09
Total Salaries		**2325.03**	**2325.03**
B. Grant in Aid - General (REVENUE)			
1	B.Pension & Other Retirement Benefits	4453.30	4453.30
2	A.Domestice TA / Transfer TA	23.12	23.12
3	Research & Operational Expenses	298.87	298.87
4	Administrative Expenses	1348.13	1348.13
5	A.HRD	6.21	6.21
6	Miscellaneous Expenses	119.78	119.78
Total Grants in Aid - General		**6249.41**	**6249.41**
Grand Total (Capital+Salaries+Revenue)		**9090.64**	**9090.64**

9.2 Resource Generation

9.2.1 Sponsored Research Projects Budget (Rs. in lakhs)

Sl. No	Name of the Project	Fund received	Expenditure
1	IFPRI Sponsored Project - Strengthening & Integrating & Developing a National Dashboard for Evidence-Based Governance of Agril.Res. India	22.74	22.74
2	ICAR-NAARM-NIPHM Project"Plant Biosecurity through MOOC Platform"	1.06	1.06
3	ICAR-NAARM-MANAGE Project"Agri-Warehousing Management"	4.40	4.40
4	ICAR-NAARM-PJTSAU Digital Content for MOOCS	2.35	2.35

Sl. No	Name of the Project	Fund received	Expenditure
5	Profiling the Resources Devoted in Agril.R&D Dev.of National framework for Evidence-based S&T Governance in India - DST Scheme	14.00	14.00
6	Skill Development in Bioinformatics Infrastructure Facility on Agril. Bio Technology - DBT Scheme	11.42	8.56
7	ICAR-NAARM-NDRI, Karnal for developing MOOC Courses on Commercial Dairy farming & Milk processing & Value Edition	0.92	0.92
8	ICAR-NAARM-NIPHM Project developing online course on"Rodents & Household Pest Management"	0.67	0.67
9	International Consultancy Project -ICAR-NAARM-GIZ "Developing and Pilot testing Specialized Knowledge Products on Human Wildlife Conflict (HWC) Mitigation for Agriculture and Veterinary Sector in India	31.93	28.20
	Total	**89.49**	**82.90**

9.2.2 Off-Campus and Sponsored Programmes from Jan-2021 to Dec 2021 (Rs. in lakhs)

Sl. No	Name of the Programme	Total Budget received
1	Online training Programme for faculty of RAJUVAS,Bikaner from 4-11th Feb.22	4.50
2	Capacity Building Programmes for the faculty &students of College of Dairy Science, Kamadhenu University, Amerli for 5 Programmes	27.34
3	Training Programme on "Design Thinking in Research & Education" College of Horticulture, Rajendranagar from 15-19th February,2022	1.11
4	Induction training on "Project Management and Research Methodology" for the newly appointed Scientists of Forest Research Institute, Dehradun from 7-18th Feb.22	2.13
5	Capacity Building Progr.for Young Agri-Business Professional NEAT Phase-II, MORE, ZEAL Programmes - Coromandal Programmes	10.07
6	Capacity Building Programme for the Faculty of College of PJTSAU, Hyderabad on Statistical Data Analysis from 6-21st April,22	19.67
7	Foundation Programme for Newly Recruited Asst.Prof.of PVNRTVU, Rajendra Nagar, Hyderabad from 18-5-22 to 09-6-22	15.78
8	For conducting of 3 Programmes for the Faculty & Students of College of Horticulture, SKLTSHU, Rajendranagar	3.94
9	Soft Skills & Entre.Dev. For the students of col.of Agril,BSKVV,Dapoli from 21-26th Feb.2022	2.85
10	Foundation trg.progr. For the faculty of Veterinary and Fishery Science of PVNRTVU,Hyd	7.35

11	DWRP for the faculty of College of Agril., Jagityal, Aswaraopet, Warangal, Palem, Sangareddy, Rudrur, PJTSAU, Hyderabad	11.74
12	Entrepreneurial Skill Development for the 3rd & 4th year B.Tech Dairy Technology students of NDRI, Karnal from 02-06-2022 to 06-07-2022	24.80
13	FDP for "Improving Teachers Effectiveness" for the newly recruited Asst.Prof. of OUAT, Bhubaneswar from 17-30th June,2022	9.32
14	Interface of Producer Organizations with Rural/Agri Start ups for BIRD, Lucknow from 1-3rd September, 2022	3.68
15	Training Programme for R&D Team of RALLIS India Ltd. On Research Design & Statistical Analysis	5.94
	Total	**150.22**

9.2.3 Resource Generation from Educational Programmes (Rs. in lakhs)

Sl. No	Educational Programmes	Receipts (Rs. in Lakhs)
1	PGDM (ABM) 2020-22	18.72
2	PGDM (ABM) 2021-23	202.45
3	PGDM (ABM) 2022-24	233.37
4	DETM-2021	0.13
5	DTMA 2022	3.90
6	DETM 2022	4.72

9.2.4 Resource Generation from other Activities (Rs. in lakhs)

Sl. No	Net Resource Generation from other activities	Receipts (Rs. in Lakhs)
1	Receipts from Farm Produce	2.11
2	Receipts from Guest House and Quarters	69.10
3	Receipts from Institute Training Programmes	31.57
4	Miscellaneous Receipts	22.90
	Total	**125.68**

9.2.5 Net Receipts from Sponsored Research Projects (Rs. in lakhs)

Sl. No	Sponsored Research Projects	
1	Profiling the Resources Devoted in Agril.R&D Dev.of National framework for Evidence-based S&T Governance in India Scheme of DST	2.36
2	IFPRI-ASTI Project	21.75
3	NASF, New Delhi - Entrepreneurship Dev. Through Farmer Led Innovations	0.49
4	NIPHM-Rodents & Household Pest Mgmt.Project & Plant Bio Secutity through MOOC plat form Project	1.69
5	Development of MOOC Course, NDRI, Karnal	0.71
6	PJTSAU - Digital content for MOOCs	2.35
7	MANAGE- Agriwarehouse project	4.41
8	Net receipts from Educational Programmes	248.05
9	Net receipts from IRGS Programmes/Schemes	59.06
	Total	**340.87**

9.3 Plan Schemes (Rs. in lakhs)

Sl. No.	Name of the Scheme	Fund / Budget	Expenditure during (Upto Dec'21)
1	National Agril.Innovation Fund (Component-I)-ZTMC	11.00	9.13
2	National Agril.Innovation Fund (Component-II)-ABI	8.59	6.95
3	XII Plan EFC Main Scheme KRISHI Project	9.16	8.98
4	Management & Impact Ass. Of Farmer First Project	11.62	7.74
5	NASF, New Delhi - Entrepreneurship Dev. Through Farmer Led Innovations	6.30	6.30
6	Short Courses	10.50	10.50
7	Emeritus Scientist Scheme	6.46	5.21
8	NAHEP	341.15	103.84
	Total	**404.78**	**158.65**

Personnel

10

Human Resources at the Academy (as on 31.12.2022)

Category	Sanctioned Strength	In position	Vacant
Scientific – RMP	02	02	00
Scientific - Faculty	60	30	30
Technical	23	20	03
Administrative	45	32	13
Skilled Support Staff	49	44	05
Total	**179**	**128**	**51**

Research Management Positions

1. Dr Ch. Srinivasa Rao, Director

2. Dr G. Venkateshwarlu, Joint Director

Scientific Staff

Agribusiness Management Division

1. Dr Ranjit Kumar, Principal Scientist & Head (I/c)
2. Dr Tavva Srinivas
   Principal Scientist
3. Dr B. Ganesh Kumar
   Principal Scientist
4. Dr Prem Chand Meena
   Principal Scientist
5. Dr N. Sivaramane
   Principal Scientist
6. Dr Sanjiv Kumar
   Senior Scientist

Education Systems Management Division

1. Dr G. R. Ramakrishna Murthy
   Principal Scientist & Head (I/c)
2. Dr S. Senthil Vinayagam
   Principal Scientist
3. Dr D. Thammi Raju
   Principal Scientist
4. Dr V. V. Sumanth Kumar
   Principal Scientist

Extension Systems Management Division

1. Dr Bharat S. Sontakki
   Principal Scientist & Head (I/c)
2. Dr (Mrs.)Surya Rathore
   Principal Scientist
3. Dr P. Venkatesan
   Principal Scientist
4. Dr Vijay Avinashilingam N.A.
   Principal Scientist

Human Resource Management Division

1. Dr R. V. S. Rao
   Principal Scientist & Head (I/c)
2. Dr P. Ramesh
   Principal Scientist
3. Dr Alok Kumar
   Principal Scientist

Information and Communication Management Division

1. Dr S. K. Soam
   Principal Scientist & Head (I/c)
2. Dr A. Dhandapani
   Principal Scientist
3. Dr S. Ravichandran
   Principal Scientist
4. Dr M. Balakrishnan
   Principal Scientist
5. Dr N. Srinivasa Rao
   Principal Scientist
6. Dr P. D. Sreekanth
   Principal Scientist
7. Dr Yashavanth B.S.
   Scientist
8. Dr (Mrs.) Purru Supriya
   Scientist

Research Systems Management Division

1. Dr K. Kareemulla
   Principal Scientist & Head (I/c)

2. Dr M. B. Dastagiri
Principal Scientist
3. Dr P. Krishnan
Principal Scientist
(On Deputation)
4. Dr Umesh Hudedamani
Scientist
5. Dr M. Ramesh Naik
Scientist (Sr. Scale)

Administrative staff

Chief Admn. Officer (Sr. Grade)

Shri Vivek Purwar

Joint Director (Official Language)

Dr.(Mrs.) J. Renuka

Chief Finance & Accounts Officer

Shri Z.H. Khilji

Senior Administrative Officer

Shri Mukul Raj Singh

Principal Private Secretary

Shri K. Sanath Kumar

Finance & Accounts Officer

Shri Marisetti Naga Venkateswara Rao

Assistant Administrative Officer

1. Shri M. Dinesh
2. Smt. K.K. Rukmani Ammal
3. Dr K.R. Ghanshyam
4. Shri C.J. Samuel

Assistant Finance and Accounts Officer

5. Smt. N. Vijayalakshmi

Private Secretaries

1. Smt. A. Mercy
2. Smt. T. Vanisri
3. Smt. S. Shanthi

Assistants

1. Shri T. Srinivas
2. Shri G. Raj Reddy
3. Shri C. Phani Raj
4. Smt. B. Padma Saroja
5. Shri P. Srinivasu
6. Shri R. Chandra Babu
7. Shri M. Sridhar
8. Smt. Y. Gayathri
9. Shri K. Suryanarayana
10. Shri M.K. Samson

Personal Assistants

1. Shri T.V. Ramadas
2. Smt. Y. Anuradha
3. Smt. S. Sesha Sai
4. Smt. V. Shailaja

Upper Division Clerks

1. Smt. Rajashri Bokde
2. Shri P. Swamy
3. Shri M. Narsing Rao
4. Shri M. Ashok

Technical

Grade T-9 (Category-III)

1. Shri P. Namdev
Chief Technical Officer (Artist)
2. Shri P. Vijender Reddy
Chief Technical Officer
(Research Assistant)
3. Shri Sohail Ahmad Khan
Chief Technical Officer
(Junior Engineer-Civil)
4. Smt. G. Aneeja
Chief Technical Officer
(Assistant Editor)

Grade T(7-8) (Category-III)

1. Shri M. Shekhar Reddy
Assistant Chief Tech. Officer
(Dark Room Assistant)
2. Dr Ahire Laxman Maharu
Assistant Chief Technical Officer (Horticultural Technical Assistant)
3. Shri M. Ravi,
Assistant Chief Technical Officer
(Photographer-cum-Artist)
4. Shri Sham Bahadur
Asst. Chief Technical Officer
(Catering in-charge)
5. Shri P. Mohan Singh
Asst. Chief Technical Officer
(Computer Assistant)
6. Smt. Savithri Murali
Asst.Chief Technical Officer
(Catering in-charge)

Grade T-6 (Category-III)

1. Shri Kirttiranjan Baral
Sr. Technical Officer
(Farm Superintend)

Grade T-5 (Category-II)

1. Shri K. Shivaiah
Technical Officer (Technician)
2. Shri S. Rajukumar
Technical Officer
(Electronic Computer Operator)
3. Shri T. Laxman
Technical Officer (Driver)

Grade T-4 (Category-II)

1. Shri Pitla Srinivas
Sr. Technical Assistant
(Proof Reader)
2. Shri N. Ashok
Sr. Technical Assistant (Driver)
3. Shri N. Prabhakar Rao
Sr. Technical Assistant
(Plumber)
4. Shri M. Srinivasa Rao
Sr. Technical Assistant
(Pump Driver)
5. Shri B.K. Venkatram
Senior Technical Assistant
(Pump Driver)

Grade T-3 (Category-II)

1. Shri R. Siva Prasad
Technical Assistant (Driver)
2. Shri Danam Murahari
Technical Assistant
(A.V.Operator)

Skilled Support Staff

1. Shri Sirigiri Venkatesham
Xerox Machine Operator
2. Smt. S. Shakuntala
3. Shri Kumba Satyanarayana
4. Shri K. Pentaiah
5. Shri P. Yadaiah
6. Shri G. Pentaiah

7. Shri R. Sattaiah
8. Shri L. Satyanarayana
9. Smt. C. Kausalya
10. Shri M. Ganesh Kumar
11. Shri Chilumula Venkatesham
12. Shri Kyasam Satyanarayana
13. Shri B. Premdas
14. Smt. V. Saroja
15. Shri S. Nayab Rasool
16. Shri B. Ashok
17. Shri K. Daniel
18. Shri M. Satyanarayana
19. Shri D. Srisailam
20. Shri B. Ratnaiah
21. Shri D. Babaiah
22. Shri K. Samson
23. Shri.K. Satnarayana
24. Shri K. Narasimha
25. Smt.K. Andalu
26. Shri M.S. Ravi
27. Shri N. Krishna
28. Smt. P. Amrutha
29. Smt. K. Lakshmi
30. Smt. L. Chandramma
31. Shri G. Satyanarayana
32. Shri B. Raju
33. Shri Md. Sadath
34. Shri T. Srinivas
35. Smt. E. Yadamma
36. Smt. S. Lakshmamma
37. Shri P. Venkatesh
38. Shri G.Ramulu S/o.G.Laxmaiah
39. Shri P. Yadaiah
40. Smt. R. Pramila
41. Smt. M. Laxmamma
42. Shri G. Ramulu S/o. G. Danaiah
43. Smt. M. Sharada
44. Shri. N. Yadagiri

Appointments / New Joining

1. Shri Marisetti Naga Venkateswara Rao, has joined at this Academy on the forenoon of 26.05.2022 as Finance & Accounts Officer on his transfer (on Promotion) from ICAR-National Research Centre on Meat (ICAR-NRCM), Hyderabad.
2. Dr (Mrs.) J. Renuka, joined the Academy as Joint Director (OL)w.e.f. 01.08.2022 (F.N.) on her transfer on Promotion from ICAR-CIFT, Cochin.
3. Shri K. Sanath Kumar, joined the Academy as Principal Private Secretary w.e.f. 12.08.2022 (F.N.) on his transfer on Promotion from ICAR-IIMR, Hyderabad.
4. Shri N. Yadagiri, CLTS has been regularized as Skilled Support Staff (SSS) w.e.f. 01.09.2022.
5. Shri Kirttiranjan Baral, joined the Academy as T-6 Senior Technical Officer (Farm Superintendent) w.e.f. 02.09.2022 (F.N.)
6. Shri Vivek Purwar, joined the Academy as Chief Administrative Officer (Senior Grade) w.e.f. 27.09.2022 (FN) on his transfer from ICAR-NDRI, Karnal.
7. Dr Ch. Srinivasa Rao, joined the Academy as Director (2nd term) w.e.f. 17.12.2022 (FN)

Promotions / Financial upgradations under MACPS Promotions

1. Dr Sanjiv Kumar, Scientist has been promoted under CAS to the grades of Research Grade Pays of Rs.7000/- (Pay Level # 11) w.e.f. 02.07.2015 and Rs.8,000/- (Pay Level # 12) w.e.f. 02.07.2020, and designated as Senior Scientist.
2. Shri N. Prabhakar Rao, T-4 Sr. Technical Assistant (Plumber) has been promoted to the Grade of T-5 Technical Officer (Plumber) w.e.f. 29.06.2021.
3. Shri M. Srinivasa Rao, T-4 Sr. Technical Assistant (Pump Driver) has been promoted to the Grade of T-5 Technical Officer (Pump Driver) w.e.f. 29.06.2021.
4. Shri N. Ashok, T-4 Sr. Technical Assistant (Driver) has been promoted to the Grade of T-5 Technical Officer (Driver) w.e.f. 29.06.2021.
5. Shri R. Siva Prasad, T-3. Technical Assistant (Driver) has been promoted to the Grade of T-4 Sr. Technical Assistant (Driver) w.e.f. 20.10.2020
6. Dr Yashavanth, B.S., Scientist (Agricultural Statistics) has been promoted to the next higher grade under the revised CAS w.e.f. 05.07.2020 (Next higher grade / PB-3 : Rs.15,600-39,100 + RGP of Rs.7,000/- Revised Research Pay Level # 11 : Rs.68,900 – 2,05,500)
7. Shri C.J. Samuel, Assistant has been promoted to the post of Assistant Administrative Officer under LDCE exam w.e.f. 27.06.2022(AN) (Next higher gade/PB-3 : Rs.15,600-39,100 + RGP of Rs.7,000/- Revised Research Pay Level #11 : Rs.68,900-2,05,500)
8. Shri K. Suryanarayana, UDC has been promoted to the post of Assistant under LDCE exam w.e.f. 27.06.2022 (AN)
9. Shri K. Samson, UDC has been promoted to the post of Assistant w.e.f. 01.07.2022 (AN).
10. Shri M. Ashok, LDC has been promoted to the post of UDC w.e.f. 28.07.2022 (FN)
11. Shri C. Bikshapathi, LDC has been promoted to the post of UDC w.e.f. 28.07.2022 (FN)
12. Smt. S. Shanthi, Personal Assistant has been promoted to the post of Private Secretary w.e.f. 01.12.2022 (FN).

Retirement list from 1st January to 31st December, 2022

1. **Dr M.A. Basith,** T-9, Chief Technical Officer (Jr.Farm Superintendent), retired (on superannuation) w.e.f. 28.02.2022
2. **Dr K.H. Rao,** Principal Scientist , retired (on superannuation) w.e.f. 31.03.2022
3. **Shri A.C.P. Rama Nageswara Rao,** T-5, Technical Officer retired (on superannuation) w.e.f. 30.04.2022
4. **Shri. S. Narsimha,** SSS retired (on superannuation) w.e.f. 30.04.2022
5. **Shri. L. Narsimha,** SSS retired (on superannuation) w.e.f. 31.05.2022
6. **Shri M. Ravi Viswanathan,** T 7-8, Asstt.Chief Technical Officer(Editor-cum-Information Officer) retired (on superannuation) w.e.f. 31.07.2022
7. **Shri C. Bikshapathi,** Upper Division Clerk retired (on superannuation) w.e.f. 31.07.2022
8. **Shri D. Rajagopal Rao,** T-5 Technical Officer retired (on superannuation) w.e.f. 31.08.2022
9. **Shri N. Ashok,** T-5 Technical Officer retired (on superannuation) w.e.f. 30.09.2022
10. **Shri J. Chandraiah,** SSS retired (on superannuation) w.e.f. 31.10.2022
11. **Shri K. Shyamulu,** SSS retired (on superannuation) w.e.f. 31.10.2022

Financial Upgradations under MACPS:-

S. No.	Name	Designation	1st/2nd/3rd Financial Upgradation with GP/ Level of Pay	Effective Date
1.	**Smt. S. Sesha Sai**	PA	3rd / Pay Level # 8	15.03.2022
2.	**Smt. Rajashri Bokde**	UDC	3rd / Pay Level # 6	07.02.2022
3.	**Smt. K.K. Rukmani Ammal**	AAO	3rd / Pay Level # 8	02.11.2022

Clearance of Probation Period and Confirmation

S. No.	Name	Post	Date of Clearance of Probation	Date of Confirmation
1.	**Dr (Mrs.) P. Supriya**	Scientist	06.01.2022	07.01.2022
2.	**Shri K. Samson**	SSS	18.03.2022	19.03.2022
3.	**Shri L. Narsimha**	SSS	18.03.2022	19.03.2022

List of Committees

I. Higher Education Committee

1. Joint Director & Dean - Chairman
2. Heads of all Divisions - Members
3. CAO (SG) - Member
4. Chief Finance & Accounts Officer - Member
5. Principal Course Coordinator, PGDM – ABM - Member
6. Coordinators, PGDM-ABM - Member
7. Coordinators, DTMA - Member
8. Coordinators, DETM - Member
9. Controller of Examinations - Member
10. OIC, Hostels - Member
11. OIC, PGS Unit - Member Secretary

II. Board of Studies

1. Dr Ranjit Kumar, Head, ABM Division - Chairman
2. Dr S.K. Soam, Head, ICM Division - Member
3. Dr Bharat Sontakki, Head, XSM Division - Member
4. Dr A. Dhandapani, Pr. Scientist, ICM Division - Member
5. Dr P. Ramesh, Pr. Scientist & Controller of Examinations - Member
6. Dr D. Thammi Raju, Pr. Scientist ESM Division - Member
7. Dr B. Ganesh Kumar, Pr. Scientist ABM Division - Member

8. Shri Vivek Purwar, CAO(SG) - Member
9. Shri Z.H. Khilji, CFAO - Member
10. Coordinators of PGDM-ABM, DTMA, DETM - Co-opted Members
11. Dr PC. Meena, Pr. Scientist & OIC, P.G.S Unit - Member Secretary

III. Works Committee

1. Dr R.V.S. Rao, Pr. Scientist and I/C Head, HRM Divn - Chairman
2. Dr S. Ravi Chandran, Pr. Scientist - Member
3. Dr G.R. Ramakrishna Murthy, Pr. Scientist - Member
4. Dr M. Balakrishnan, Pr. Scientist - Member
5. Dr P.D. Sreekanth, Pr. Scientist - Member
6. CAO(SG) - Member
7. Chief Finance & Accounts Officer - Member
8. Dr P. Vijender Reddy, CTO - Member

IV. Anti-Ragging Committee

1. Joint Director / Dean - Chairman
2. Dr I. Sekar, Head, RSM Division - Member
3. CAO(SG) - Member
4. Academic Coordinators, PGDM (A) Programme - Member
5. Dr P.C. Meena, Pr.Scientist & OIC, P.G.S Unit - Member Secretary

V. Anti-Ragging Squad

1. Dr (Mrs.) Surya Rathore - Chairman
2. Dr B. Ganesh Kumar, Pr. Scientist, ABM Division - Member
3. Dr Umesh Hudedamani, Scientist - Member
4. Chief Finance & Accounts Officer - Member
5. Officer Incharge, Hostel - Member Secretary

VI. Grievance Committee

1. Dr G. Venkateswarlu, Joint Director - Chairman
2. Dr G.R.K. Murthy, Head I/C - Member
3. Shri Vivek Purwar, CAO (SG) - Member
4. Shri Z.H. Khilji, CFAO - Member
5. Shri Mukul Rajsingh, SAO - Member
6. Dr (Mrs) Surya Rathore, Principal Scientist - Elected Member – Scientific Category
7. Shri M. Ravi, ACTO - Elected Member – Technical Category
8. Shri C.Julius Samuel, AAO - Elected Member – Admn. Category
9. Shri Kyasam Satyanarayana, SSS - Elected Member – Skilled Supporting Category

VII. Farm Advisory Committee

1. Dr Mude Ramesh Naik - Chairman
2. Dr B.S. Yashavanth, Scientist - Member
3. Dr P. Vijender Reddy, CTO - Member
4. Dr L.M. Ahire, ACTO - Member
5. Dr P. Mohan Singh, ACTO - Member
6. Shri M. Shekhar Reddy, ACTO - Member
7. Shri T. Laxman - Member
8. Shri Danam Murahari, TA - Member
9. Dr P. Vijender Reddy, CTO - Member Secretary

VIII. Library Advisory Committee

1. Dr R.V.S. Rao, Pr. Scientist and I/C Head, HRM Divn - Chairman
2. All HoDs - Member
3. Dr V.V. Sumanth Kumar, Pr. Scientist - Member
4. CAO (SG) - Member
5. Chief Finance & Accounts Officer - Member
6. Dr N. Srinivasa Rao, Pr. Scientist - Member Secretary

IX. Newsletter Committee

1. Dr Ranjit Kumar Head-ABM Division - Chairman
2. Dr Tavva Srinivas, ABM Division - Member
3. Dr Sanjiv Kumar, Scientist ABM Division - Member
4. Dr B.S. Yashavanth, Scientist ICM Division - Member Secretary

X. Annual Report Committee

1. Joint Director - Chairman
2. Head, ICM Divn. - Member
3. Head, RSM Divn. - Member
4. Head, HRM Divn. - Member
5. Head, ESM Divn. - Member
6. Head, ABM Divn. - Member
7. Head, XSM Divn. - Member
8. Dr Tavva Srinivas, Pr. Scientist and Head, PME Cell - Member

9. Dr Sanjiv Kumar, Scientist
ABM Division - Member
10. Dr Yashavanth B. S., Scientist
ICM Division - Member
11. Dr Supriya S., Scientist, ICM Division - Member
12. Dr Alok Kumar, Pr.Scientist, HRM Divn. - Member Secretary

XI. Campus Residents Welfare Committee

1. Dr S.K. Soam, Head, ICM Division - Chairman
2. Dr Umesh Hudedamani, Scientist - Member
3. Dr Mude Ramesh Naik - Member
4. Shri Z.H. Khilji, CFAO - Member
5. Dr B.S. Yashavanth, Scientist - Member Secretary

XII. Examination Unit

1. Dr P. Ramesh, Pr. Scientist & COE - CoE
3. SAO - Member
4. Shri Pitla Srinivas, STA - Member
5. Smt. B. Padam Saroja, Assistant - Member
6. Shri B Ashok, SSS, Training Unit - Member Secretary

XIII. Hostel Management Committee

1. Shri Vivek Purwar, CAO(SG) - Chairman
2. Shri Z. H. Khilji, CF&AO - Member
3. Shri Mukul Rajsingh, SAO - Member
4. Dr. P. Vijender Reddy, CTO - Member
5. Shri Sohail Ahmad Khan, CTO - Member
6. Shri P. Mohan Singh, ACTO - Member
7. Smt. Savithri Murali, ACTO - Member
8. Shri K. Shivaiah, TO - Member
9. Shri M. Sridhar, Assistant - Member
10. Shri Sham Bahadur, ACTO - Member- Secretary
11. Dr M. BalaKrishnan, Pr. Scientist - Special invitee
12. Dr M. Ramesh Naik, Scientist (SS) -
Special invitee
13. Dr (Mrs.) P. Supriya, Scientist - Special invitee

XIV. Institute Technology Management Committee

1. Dr Ch. Srinivasa Rao, Director - Chairman
2. Joint Director - Member
3. Dr S. Senthil Vinayagam
Pr. Scientist, ESM Division - Member
4. Dr GRK Murthy, Pr.Scientist - Member
5. Dr V.V. Sumanth Kumar, Pr. Scientist - Member
6. CAO(SG) - Member
7. CFAO - Member
8. Dr RM Sundaram, Pr. Scientist
ICAR-IIRR.Hyd - Member
9. Dr Umesh Hudedemani
Scientist - Member Secretary

XV. Sports Complex Maintenance Committee

1. Dr M. Ramesh Naik - Chairman
2. Dr P. Vijender Reddy, CTO - Member
3. Shri Sohail Ahmad Khan, ACTO - Member
4. Dr Laxman M Ahire, ACTO - Member
5. Shri M. Shekar Reddy, ACTO - Member
6. Shri P. Mohan Singh, ACTO - Member
7. Shri Sham Bahadur, ACTO - Member
8. Shri M.K.Samson, UDC - Member
9. Dr Debnath, Medical Consultant - Member
10. Smt. K.K. Rukmani Ammal, AAO - Member
11. Dr. (Mrs.) P. Supriya - Member Secretary

XVI. Campus Canteen Committee

1. Dr. G.R. Ramakrishna Murthy
Pr. Scientist - Chairman
2. Dr. P. Vijender Reddy, CTO - Member
3. Shri S. Rajukumar
TO and Secretary, IJSC - Member
4. Shri M.Sridhar, Asst. And Member,
IJSC & CJSC - Member Secretary

XVII. Internal Complaints Committee

1. Dr. (Mrs.) Surya Rathore
Pr. Scientist - Chairman
2. Shri Mukul Rajsingh
SAO - Member
3. Smt. G. Aneeja
Asst. Chief Technical officer - Member
4. Smt. Vijayalaxmi, AF&AO - Member
5. Smt. P. Padmavathi, Kasturba Gandhi National Memorial Trust as 3rd Party Representative (representing either NGO or other body who is familiar with the issues of sexual harassment) - Member
6. Mrs. KK. Rukmani Ammal - Member Secretary

XVIII. Academic Committee Meeting Committee

1. Dr Ch. Srinivasa Rao, Director - Chairman
2. All Faculty - Members
3. Dr A. Dhandapani, Pr.Scientist - Member Secretary -1
4. Dr Alok Kumar
   Pr. Scientist - Member Secretary -2

XIX. Purchase Advisory Committee (Stores)

1. Dr A. Dhandapani, Pr. Scientist
   ICM Division - Chairman
2. Dr P. Ramesh, Principal Scientist - Member
3. Dr Umesh Hudedamani, Scientist - Member
4. Joint Director (Admn.) & Registrar - Member
5. CF&AO - Member
6. Dr Vijender Reddy, CTO - Member
7. Shri M. Sridhar, Assistant - Member
8. Shri Mukul Rajsingh, SAO - Member Secretary

XX. Technical Advisory Committee (Stores)

1. Dr GRK Murthy, Pr. Scientist - Chairman
2. Dr V.V. Sumanth Kumar, Pr. Scientist - Member
3. Shri S. Rajukumar, Technical Officer - Member Secretary

XXI. Academic Coordinators - DTMA

1. Dr K. Kareemulla, Pr. Scientist, RSM Division, Principal Course Coordinator
2. Dr Umesh Hudedamani, Scientist, RSM Division, Course Coordinator

XXII. Academic Coordinators - DETM

1. Dr V.V. Sumanth Kumar, Principal Course Coordinator
2. Dr (Mrs.) P. Supriya, Course Coordinator

XXIII. Admission Committee for PGDM-ABM (2023-25) Batch.

1. Dr Ranjit Kumar, Head, ABM Divn. - Chairman
2. All Head of Divisions - Members
3. Dr A. Dhandapani
   Pr. Scientist, ICM Divn. - Member
4. Dr B. Ganesh Kumar, Pr. Scientist & Principal Course Coordinator, PGDM - ABM - Member
5. Shri Vivek Purwar, CAO (Sr. Grade) - Member
6. Dr P.C. Meena, Pr.Scientist & OIC-PGS Unit - Member Secretary

XXIV. Applications Screening Committee for PGDM-ABM (2023-25) Batch.

1. Dr A. Dhandapani, Principal Scientist - Chairman
2. Dr N. Sivaramane, Principal Scientist - Member
3. Dr Sanjiv Kumar, Senior Scientist - Member
4. Dr P.C. Meena, OIC-PGS Unit - Member Secretary

XXV. Admission Committee for DTMA & DETM -2023 Batch.

1. Dr K. Kareemulla, Programme Coordinator, DTMA
2. Dr V.V. Sumanth Kumar, Programme Coordinator, DETM
3. Shri Pitla Srinivas, Senior Technical Assistant, PGS Unit

XXVI. RTI – Functionaries designated at ICAR NAARM under the provisions of RTI Act, 2005

1. Dr G. Venkateshwarlu
   Joint Director - Appellate Authority
2. Shri B.D. Phansal, Joint Director (Admin) & Registrar - Transparency Officer
3. Dr M. Balakrishnan, Principal Scientist - Central Public Information Officer (CPIO) for Scientific Matters
4. Dr P. Vijender Reddy, CTO - Central Public Information Officer (CPIO) for Technical Matters
5. Shri Mukul Rajsingh, AO - Central Public Information Officer (CPIO) for Admin. Matters
6. Shri M. Dinesh, AAO - Central Assistant Public Information Officer (CAPIO) for Scientific, Technical & Admin. Matters Nodal Officer

XXVII. Committee for organizing the meeting of retired scientists with Hon'ble Minister of Agriculture and farmers Welfare

1. Joint Director - Chairman
2. Dr Bharat S. Sontakki
   Head, XSM Divn. - Members
3. Dr Alok Kumar, Pr. Scientist - Member
4. Dr P. D. Sreekanth, Pr. Scientist - Member
5. Shri Vivek Purwar, CAO(SG) - Member
6. Shri Z. H. Khilji, CF&AO - Member
7. Dr A. Dhandapani
   Pr. Scientist - Member Secretary

XXVIII. Committee for allotment of residential quarters

1. Dr Bharat S. Sontakki, Head, XSM Divn. - Chairman
2. Dr M. Balakrishnan
   Principal Scientist - Members
3. CAO(SG) - Member
4. Chief Finance & Accounts Officer - Member
5. Dr M. Ramesh Naik, Scientist (SS) - Member
6. Shri Sohail Ahmad Khan, CTO - Member
7. Shri M. Dinesh, AAO - Member Secretary

XXIX. Institute Deputation Committee (IDC)

1. Head, Education System Management Divn. - Chairman
2. Dr. A. Dhandapani, Principal Scientist, ICM Divn. - Member
3. Officer Incharge, Training Unit - Member
4. Shri M. Dinesh
   AAO - Member Secretary

भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
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